

2024 Annual report of the National Poison Data System® (NPDS) from America's Poison Centers®: 42nd annual report

Michael C. Beuhler, Ryan Feldman, David D. Gummin, James B. Mowry, Laura J. Rivers, Kaitlyn Brown, Nathaniel P. T. Pham, Krys Johnson-O'Leary, Daniel A. Spyker & Alvin C. Bronstein

To cite this article: Michael C. Beuhler, Ryan Feldman, David D. Gummin, James B. Mowry, Laura J. Rivers, Kaitlyn Brown, Nathaniel P. T. Pham, Krys Johnson-O'Leary, Daniel A. Spyker & Alvin C. Bronstein (2025) 2024 Annual report of the National Poison Data System® (NPDS) from America's Poison Centers®: 42nd annual report, Clinical Toxicology, 63:12, 1029-1280, DOI: [10.1080/15563650.2025.2571299](https://doi.org/10.1080/15563650.2025.2571299)

To link to this article: <https://doi.org/10.1080/15563650.2025.2571299>



Published online: 23 Dec 2025.



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2024 Annual report of the National Poison Data System® (NPDS) from America's Poison Centers®: 42nd annual report

Michael C. Beuhler , Ryan Feldman , David D. Gummin , James B. Mowry ,
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ABSTRACT

Introduction: This is the 42nd Annual Report of America's Poison Centers® National Poison Data System®. As of 1 January, 2024, all 55 of the nation's Poison Centers uploaded case data automatically to NPDS.

Methods: We analyzed the case data, tabulating specific indices from the NPDS. The methodology was as in previous years. Where changes were introduced, the differences are identified. Cases with medical outcomes of death were evaluated by a team of medical and clinical toxicologists using an ordinal scale of 1-6 to assess the Relative Contribution to Fatality of the exposure.

Results: In 2024, 2,418,426 closed encounters were logged by the National Poison Data System: 2,092,689 human exposures, 34,919 animal exposures, 285,736 information requests, 5,054 human confirmed nonexposures, and 28 animal confirmed nonexposures. The upload interval was 4.97 [4.32, 9.32] (median [25%, 75%]) minutes, creating a near real-time national exposure and information database and surveillance system. Total encounters remained essentially constant with a 0.117% decrease from 2023 while human exposure cases increased by 0.578% and health care facility human exposure cases increased by 0.661%. All information requests decreased by 2.70%, medication identification (Drug ID) requests decreased by 17.0%, and medical information requests showed a 24.5% decrease. Drug Information requests showed a 0.218% increase and comprised 22.4% of all information contacts. Human exposures with less serious outcomes have decreased by 1.45% per year since 2008, while those with more serious outcomes (moderate, major or death) have increased by 4.08% per year since 2000.

Consistent with the previous year, the top 5 substance classes most frequently involved in all human exposures were analgesics (10.5%), household cleaning substances (6.94%), antidepressants (5.50%), cardiovascular drugs (5.12%) and cosmetics/personal care products (4.96%). As a class, stimulants and street drug exposures increased most rapidly, by 813 cases/year (4.54%/year) over the past 10 years for cases with more serious outcomes.

The top 5 most common exposures in children aged 5 years or less were household cleaning substances (9.96%), cosmetics/personal care products (9.15%), analgesics (8.85%), foreign bodies/toys/miscellaneous (7.92%), and dietary supplements/herbals/homeopathic (6.34%).

The National Poison Data System documented 2,809 human exposures resulting in death; 2,271 (80.9%) of these were judged as related (Relative Contribution to Fatality of 1-Undoubtedly responsible, 2-Probably responsible, or 3-Contributory).

Conclusions: These data support the continued value of Poison Center expertise and the need for specialized medical toxicology information to manage the increasing number of more serious exposures. Unintentional and intentional exposures continue to be a significant cause of morbidity and mortality in the US. The near real-time nature of the National Poison Data System represents a national public health resource for collecting and monitoring US exposure cases and information requests. The continuing mission of the National Poison Data System is to provide a nationwide infrastructure for surveillance for all types of exposures (e.g., foreign body, infectious, venomous, chemical agent, or commercial product), and the identification and tracking of significant public health events. The National Poison Data System is a model system for the near real-time surveillance of national and global public health.

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Introduction

This is the 42nd Annual Report of America's Poison Centers® National Poison Data System® (NPDS) [1]. Fifty-five regional Poison Centers (PCs), serving the entire population of the 50 United States, American Samoa, the District of Columbia, Federated States of Micronesia, Guam, Puerto Rico, and the US Virgin Islands, submitted information and exposure case data collected during the course of providing patient-specific exposure management and poison information for 2024.

NPDS is the data warehouse for the nation's PCs, which prioritizes exposure management, accurate data collection and coding, and responds to the continuing need for poison-related public and professional education. Consultation with the PCs' healthcare professionals is available free of charge to users, 24 hours a day, every day of the year. Poison Centers respond to questions from the public, healthcare professionals, and public health agencies. The number of exposures and information encounters has averaged over 3.25 million annually since the year 2000. Encounters involve either an exposed human or animal (Exposure case) or a request for information with no person or animal exposed to any foreign body, viral, bacterial, venom, chemical agent or commercial product (Information request). A unique feature of PC case management is the use of follow-up calls to monitor case progress, provide ongoing treatment recommendations, and determine the medical outcome of each case.

The NPDS products database

The NPDS products database contains over 490,000 products, ranging from viral and bacterial pathogens to commercial chemical and drug products. The products database is maintained and continuously updated by data analysts at Merative™ Micromedex® POISINDEX® System (Micromedex® [electronic version], Merative, Ann Arbor, MI, USA. Available at: <https://www.micromedexsolutions.com/>). A robust generic coding system categorizes the product data into 1,118 active America's Poison Centers Generic Codes®. These generic codes are grouped into Pharmaceutical (556) and Nonpharmaceutical (562) groups. These two groups are divided into Major (68) and Minor (189) categories. The generic coding schema undergoes continuous improvement through the work of the America's Poison Centers–Micromedex Joint Coding Committee. The group consists of America's Poison Centers members and Micromedex editorial and lexicon staff who work to follow best terminology practices. The generic code system provides enhanced report granularity as shown

in Appendix B (Table 22). The following generic codes were introduced in 2024:

Generic Codes Added in 2024*

1. Delta-9 Tetrahydrocannabinol in Combination with Other Psychoactive Substances

*Because the new codes were added during December 2024, the number of cases for this generic code in Appendix B (Table 22) does not reflect the entire year. For completeness, certain categories require customized data retrieval until they have been in place for more than 1 calendar year.

What's New in the 2024 NPDS Annual Report?

- Emerging trends: ECMO in poisoning
- Data snapshot: Processed mushrooms bars and psilocybin mushrooms
- New Figure 2: Health Care Facility (HCF) exposure cases and HCF information requests
- Removal of previous Figure 2: All Drug Identification and Law Enforcement Drug Identification Requests by Day since 1 January 2000
- New Figure 3A: Suspected Suicide and Unintentional General Exposures by Day since 1 January 2000
- New Figure 3B: Suspected Suicide and Unintentional General Exposures Adjusted for All Human Exposures by Day since 1 January 2000

PoisonHelp.org

Since the advent of the smartphone and tablet in 2006, consumers increasingly turn to web-based resources to obtain poisoning and other health-related information. In response to this trend, in 2015, America's Poison Centers devised an online tool for consumers to obtain simple, individualized poisoning recommendations. This tool provides an alternative to directly contacting a Poison Center and guides the consumer to a Poison Center when poison specialist input is deemed most appropriate. The platform launched during National Poison Prevention Week in 2016 with initial goals that included a system that would be credible, fast, easy-to-use, readily accessible from a variety of devices, and available free of charge to consumers. Data collection began in 2017, documenting web-based contacts which are defined as a completed case form submission on PoisonHelp.org. PoisonHelp.org features an interactive Get Help tool that provides critical triage recommendations on how to manage exposures to potentially toxic products, substances, and medications. Users are prompted to answer a few questions regarding a human exposure to any of the 490,000-plus unique products. Recommendations are vetted by specially trained physicians, pharmacists, and nurses from our nation's Poison Centers. The website can be accessed from any computer or mobile device and is free, confidential, and available 24/7.

Appendix D contains web-based contact data obtained during the years of PoisonHelp.org activity. Tables in Appendix D show the demographic information, and the most frequent substances queried for human exposure contacts in 2024.

Methods

Characterization of participating Poison centers and population served

Fifty-five US Poison Centers submitted data to NPDS on January 1, 2024. The Puerto Rico Poison Center closed on August 31, 2024 and without interruption starting September 1, 2024, cases from Puerto Rico were managed by the New York City Poison Control Center. The remaining fifty-four US Poison Centers submitted data to NPDS through 31 December 2024. The entire population of the 50 United States, American Samoa, the District of Columbia, the Federated States of Micronesia, Guam, Puerto Rico, and the US Virgin Islands was served by the US PC network in 2024 [2].

Encounter management – specialized poison exposure emergency providers

Poison Center Managing Directors are primarily responsible for patient care/information service operations, clinical education, and supervising staff instruction. Most have Doctor of Pharmacy (PharmD) or Registered Nurse (RN) degrees with additional American Board of Applied Toxicology (ABAT) certification in clinical toxicology. Medical direction is provided by Medical Directors who are board-certified physician medical toxicologists. At some PCs, the Managing and Medical Director roles are held by the same individual. Encounters with US PCs are managed by healthcare professionals who have received specialized training in toxicology to provide assessment, triage, management and monitoring of toxic emergencies. These personnel include medical and clinical toxicologists, RNs, pharmacists (PharmD or BS), physicians, and physician assistants. Most commonly, RNs and pharmacists make up the contingent of Specialists in Poison Information (SPIs) or Certified Specialists in Poison Information (CSPIs) in the US. SPIs triage lay public callers to the most appropriate level of care and provide health care professionals with the most up-to-date management recommendations to care for their poisoned/overdosed patients. For a SPI to become nationally certified as a CSPI, they must log, at minimum, 1,200 hours in a PC and handle 2,000 human exposure cases before being considered eligible to take the certification

examination. RNs, pharmacists, physicians, and physician assistants are the only individuals eligible to sit for the CSPI examination. Of note is the lack of appropriate core toxicology training within most graduate medical education curricula to allow these medical professionals to be prepared for PC patient management operations. These individuals must receive significant additional training beyond their degree programs to become CSPIs. Such training is only offered within the PCs. Poison Information Providers (PIPs) are allied healthcare professionals who are allowed to manage information requests and lower acuity cases while working under the supervision of a CSPI; they have been utilized to a lesser degree by PCs in recent years. Poison Centers undergo a rigorous accreditation process administered by America's Poison Centers and must submit an annual accreditation report and an extensive reaccreditation application every 7 years.

NPDS – near real-time data capture

Extensively enhanced over its predecessor, the Toxic Exposure Surveillance System (TESS), which began collecting data in 1983 and near real-time data in 2003, NPDS was launched on 12 April 2006. NPDS is the data repository for all US PCs and includes all case information collected by its predecessor. In 2024, all active US PCs uploaded case data automatically to NPDS in near real-time, making NPDS one of the few operational systems of its kind. Poison Center staff record cases contemporaneously in 1 of 6 electronic medical record systems.

The web-based NPDS software facilitates the detection, analysis, and reporting of surveillance anomalies. System software offers a myriad of surveillance uses allowing America's Poison Centers, its member centers, and public health agencies to utilize NPDS exposure data. Users can access regional data for their own areas and view national aggregate data. Custom surveillance definitions are available, along with ad hoc reporting tools. Information in the NPDS database is dynamic. Each year the database is locked prior to extraction of annual report data to ensure consistent, reproducible reports. Additional information including autopsy data on fatalities may be added after the lock date as an addendum to the fatality narrative. The 2024 database was locked on 4 August 2025 at 5:00PM EDT.

Note: NPDS entries are referred to as human or animal "exposure cases," rather than "calls," because a single call may describe more than one case and because the management of an NPDS case often requires more than a single telephone call or contact. The information in this report reflects only individual case entries that are not duplicates, and only those cases that have

been designated by the regional PC as being “closed.” A case is closed when the PC has determined that no further follow-up/recommendations are required or when no further information is available. Exposure cases are followed to obtain the most precise medical outcome possible, when appropriate to do so. Depending on the case specifics, most cases are “closed” within a few hours of the initial contact. Cases involving complex hospitalized patients or those resulting in death may remain open for months while data continue to be collected. Follow-up contacts provide a proven mechanism for monitoring the appropriateness of management recommendations, enabling continual updates of case information, augmenting patient guidelines, providing poison prevention education, and obtaining final medical outcomes to make the data collected as accurate and complete as possible.

Statistical methods

All tables except Tables 1A, 3B, and 17B, Tables A, B, C, D, E and Tables D-1, D-2, and D-3 were generated directly by the NPDS web-based application and can thus be reproduced by each PC. The analyses for Figures 1–5 were done using SAS JMP® Pro version 16.2.0 (SAS Institute, Cary, NC) with summary counts generated by the NPDS web-based application. The analyses for Figures 6–8, Tables 1A, 3B, 17B, A, B, C, D, E, D-1, D-2, D-3, and Figure D-1 were created using Microsoft Excel 365 (Microsoft, Redmond, WA) with summary counts and data generated by the NPDS web-based application.

NPDS surveillance

As previously noted, all active US PCs upload case data automatically to NPDS. This unique near real-time upload is the foundation of the NPDS surveillance system, making both spatial and temporal case volume and case-based surveillance possible. NPDS allows creation of volume and case-based definitions. Definitions can be applied to national, regional, state, or ZIP code coverage areas. Geocentric definitions can also be created, which use cases reported from a geographic location regardless of which PC managed the case. This functionality is available to every PC as well as to the America’s Poison Centers Toxicsurveillance Team. Poison Centers can also share NPDS near real-time surveillance technology with external organizations such as their state and local health departments or other regulatory agencies. Another NPDS feature is the ability to generate system alerts on adverse drug events and other drug or commercial products of public health interest,

such as contaminated food or product recalls. Thus, NPDS can provide near real-time adverse event monitoring, surveillance, response, and situational awareness.

Surveillance definitions can be created to monitor a variety of parameters (i.e., volume or case-based) on any desired substance or commercial product in the Micromedex products database; and/or set of clinical effects or other parameters. The products database contains over 490,000 entries ranging from viral and bacterial agents to commercial chemical and drug products. Surveillance definitions may be constructed using volume or case-based definitions with a variety of mathematical options and historical baseline periods from 1 to 5 years. NPDS surveillance tools include:

- Volume Alert Surveillance Definitions
 - Total Encounter Volume
 - Human Exposure Case Volume
 - Animal Exposure Case Volume
 - Information Request Volume
 - Clinical Effects Volume (signs and symptoms, or laboratory abnormalities)
 - Syndromic Surveillance Volume allows Boolean based definitions utilizing various NPDS data fields to be run based on historical trends for user defined periods of interest
- Case-Based Surveillance Definitions utilizing various NPDS data fields linked in Boolean expressions
 - Substance
 - Clinical Effects
 - Species
 - Medical Outcome and others

Incoming data are monitored continuously, and anomalous signals generate an automated, emailed alert to the America’s Poison Centers-designated PC or public health agency staff. These anomaly alerts are reviewed by the America’s Poison Centers Toxicsurveillance Team, the PC, or the public health agency that created the surveillance definition. When reports of potential public health significance are detected, additional information is obtained from reporting PCs through the NPDS surveillance correspondence system or by phone. Public health issues are brought to the attention of the Emerging Environmental Hazards and Health Effects Branch, Division of Environmental Health Science and Practice, Centers for Disease Control and Prevention (EEHHEB/DEHSP/CDC).

Clinical and medical toxicologists on the America’s Poison Centers Toxicsurveillance Team review surveillance definitions on a regular basis to fine-tune the queries. The CDC, as well as state and local health

departments with NPDS access granted by their respective PCs, can also create surveillance definitions for routine surveillance tasks or to respond to emerging public health events.

Emerging trends: ECMO in poisoning

Each year since the 2007 annual report, the authors have highlighted a trend in National Poison Data System (NPDS) data with public health significance. Past topics have included e-cigarette or vaping-associated lung injury (EVALI), adolescent suicides, cannabinoid exposures, fentanyl-related fatalities, and unregulated psychoactive substances such as phenibut, tianeptine, kratom, and alkyl nitrites. This year's report focuses on the use of extracorporeal membrane oxygenation (ECMO) in toxicologic emergencies. ECMO has gained increasing attention as a potentially life-saving intervention in poisoning-associated cardiac arrest, with guidelines suggesting an expanded role for ECMO in the management of critically poisoned patients [3]. This year NPDS was queried for all closed human exposure cases from 1 January 2019 (the first full year where ECMO was available to code as a therapy in NPDS), through 31 December 2024. Demographics, clinical characteristics, exposure intent, substances involved, and outcomes were analyzed. It should be noted that 2 primary types of ECMO are currently practiced: veno-arterial (VA) ECMO and veno-venous (VV) ECMO, and the indications for these 2 types of processes differ. NPDS does not differentiate between VA and VV ECMO having been performed. Because ECMO utilization is heavily influenced by available infrastructure, a nationwide analysis of ECMO use by state was also performed. Population-adjusted rates of ECMO utilization were modeled by U.S. state using mean U.S. Census population estimates for the study period [2].

Snapshot: Processed Mushroom Bars and Psilocybin Mushrooms

The Snapshot is intended to very briefly highlight an exposure of newly identified significance. Processed mushroom commercial product exposures reported to US PCs are examined in the 2024 Snapshot. These unregulated products have been associated with outbreaks of serious illness in the United States [4,5]. The NPDS database was queried for all closed human exposures for generic and product codes for Mushrooms, Processed Preparations: Amanita Containing (added 12/4/2023), Mushrooms, Processed Preparations: Psilocybin Containing (added 12/4/2023), and Unknown Stimulants or Street Drugs and Group 6 Mushrooms: Hallucinogenics (Psilocybin and Psilocin)

coded as an edible processed psychedelic mushroom from January 1, 2010 to June 30, 2025. Exposure rates were compared to naturally occurring mushrooms in Group 2 (*Amanita muscaria*) and Group 6 (Psilocybin & Psilocin) over the same time period.

Fatality case review and narrative selection

NPDS fatality cases are recorded as Death or Death (Indirect Report). A medical outcome of Death means that the PC was notified of the death in a case in which PC personnel were involved in some aspect of the management or responded to a specific case query. Deaths (Indirect Report) are deaths that the PC acquired, typically from medical examiners or media, but did not manage or answer any questions related specifically to that case.

The definitions used for the Relative Contribution to Fatality (RCF) classification are provided in [Appendix F](#), and the methods to select narratives for publications are described in [Appendix C](#). Although PCs may report death as an outcome, the death might not necessarily result from the exposure. We define exposure-related fatality as a death judged by the America's Poison Centers Fatality Review Team to be at least contributory to the exposure (RCF 1–3). The America's Poison Centers fatality review process was first published in detail in the 2008 NPDS Annual Report [1].

Pediatric fatality case review

A focused Pediatric Fatality Review team comprised of four toxicologists with pediatric expertise evaluated cases for patients under 19 years of age. The panel reviewed the documentation of all such cases, with specific focus on the conditions behind the poisoning exposure and on identifying commonalities which might inform efforts at prevention. When appropriate, this panel also solicited additional information from the authoring PC.

Results

Data upload frequency

The time to upload data for all PCs is 4.97 [4.32, 9.32] (median [25%, 75%]) minutes creating a near real-time national exposure database and surveillance system.

Information requests to poison centers

A total of 22,731,745 information requests have been logged into NPDS since the year 2000. Data from 285,736 information requests to PCs in 2024 ([Table 1C](#))

were entered into NPDS, including requests in optional reporting categories such as drug information (63,848), poison information (46,766), medical information (19,813), prevention/safety/education (16,359), drug ID (14,337) and environmental information (9,909). In 2024 56,702 informational calls were referred to other sources of expertise (e.g., animal poison control, health department, or medical advice lines). Total information requests to PCs decreased 2.70% from 2023 (293,663 requests) to 2024 (285,736 requests) (as shown in [Figure 1](#)).

Drug ID requests were the 8th most frequent information requests for 2024, comprising 14,337 requests for PCs during the year. Answering drug ID requests is optional for PCs, and some have stopped answering such requests from the public. Of these, 6,315 (44.0%) were identified as drugs with known abuse potential. However, these cases were categorized based on the drug's abuse potential without knowledge of whether abuse was intended.

Drug information requests increased 0.218% from 2023 (63,709) to 2024 (63,848). The most common drug information requests were drug-drug interactions (8.39%), other drug info (3.54%), and dosage (3.00%).

The number of medical information requests decreased 24.5% from 2023 (26,247) to 2024 (19,813). The most common medical information requests were other medical, diagnostic or treatment recommendations, general first aid, and explanation of disease states. Environmental inquiries comprised 3.47% of all information requests and decreased 1.15% from 2023 (10,024) to 2024 (9,909). Of these environmental inquiries, specific questions related to pesticides were most common, followed by carbon monoxide, potential toxicity of chemicals in the environment, air quality, and safe disposal of chemicals.

Of all the information requests, poison information comprised 16.4% of the requests with inquiries involving general toxicity the most common followed by questions involving other poisons, food preparation/handling practices, safe use of household products, and food poisoning.

Exposure cases logged at poison centers

In 2024, participating PCs logged 2,418,426 total encounters including 2,092,689 closed human exposure cases ([Table 1A](#)), 34,919 animal exposures ([Table 1B](#)), 285,736 information contacts ([Table 1C](#)), 5,054 human confirmed non-exposures, and 28 animal confirmed non-exposures. An additional 20 cases were still open at the time the database was locked. The cumulative America's Poison Centers database now contains over 83 million human exposure case records.

The average number of human exposure cases managed per day by all US PCs was 5,718. As in other years, higher volumes were observed in the warmer months, with a mean of 5,972 cases per day in May compared with 5,405 per day in December. On average, US PCs experienced a new encounter involving a human exposure every 15.1 seconds.

[Figure 1](#) shows the human exposures, information requests and animal exposures by day since 1 January 2000. Smoothing spline fit of these data shows departure from linearity (declining rate of cases since mid-2007) for human exposure cases with some flattening over the last 4 years. Animal exposure cases have been declining since mid-2005. Information requests had been declining until early 2020 when PCs response to COVID-19 showed a sustained increase in information requests which have since declined below pre-pandemic levels.

A hallmark of PC case management is the use of follow-up contacts to monitor case progress and medical outcome. PCs made over 2,442,000 follow-up contacts in 2024. (Note: there exists some variation in the way that this number is determined between the 6 data entry systems presently used by America's Poison Centers members). Follow-up was performed in 41.5% of human exposure cases. One follow-up contact was made in 18.9% of human exposure cases and multiple follow-ups were performed in 22.6% of cases.

[Figure 2](#) shows a graphic summary and analyses of Health Care Facility (HCF) exposure cases and HCF information requests. The trend in HCF exposure cases slightly departs from linearity but continues to increase. HCF information requests increased dramatically in 2020 and 2021 due to the COVID-19 pandemic before returning to a pre-pandemic rate in 2022. This increasing use of the PCs for the more serious cases (HCF exposures) is important in the face of the overall decline in exposure and information encounters.

[Tables 22A](#) (Nonpharmaceuticals) and [22B](#) (Pharmaceuticals) found in [Appendix B](#) provide exposure counts for all 2,092,689 human exposure cases, as well as summary demographic data for single substance exposures on patient age, reason for exposure, medical outcome, and use of an HCF presented by substance categories. The Pharmaceuticals category includes both licit and illicit drugs.

Column 1: Name of the major, minor generic categories and their associated generic substances. Note that for pharmaceuticals, the generic category or generic substance listed is for the initial FDA approved indication and may not reflect current indications or uses for the pharmaceutical.

Table 1A. America's Poison Centers population served and reported exposures (1983–2024)

Year	No. of participating centers	Population served (in millions)	Human exposures	Exposures per thousand population
1983	16	43.1	251,012	5.82
1984	47	99.8	730,224	7.32
1985	56	113.6	900,513	7.93
1986	57	132.1	1,098,894	8.32
1987	63	137.5	1,166,940	8.49
1988	64	155.7	1,368,748	8.79
1989	70	182.4	1,581,540	8.67
1990	72	191.7	1,713,462	8.94
1991	73	200.7	1,837,939	9.16
1992	68	196.7	1,864,188	9.48
1993	64	181.3	1,751,476	9.66
1994	65	215.9	1,926,438	8.92
1995	67	218.5	2,023,089	9.26
1996	67	232.3	2,155,952	9.28
1997	66	250.1	2,192,088	8.76
1998	65	257.5	2,241,082	8.70
1999	64	260.9	2,201,156	8.44
2000	63	270.6	2,168,248	8.01
2001	64	281.3	2,267,979	8.06
2002	64	291.6	2,380,028	8.16
2003	64	294.7	2,395,582	8.13
2004	62	293.7	2,438,643	8.30
2005	61	296.4	2,424,180	8.18
2006	61	299.4	2,403,539	8.03
2007	61	305.6	2,482,041	8.12
2008	61	308.5 ^b	2,491,049	8.07
2009	60	310.9 ^b	2,479,355	7.97
2010	60 ^a	313.3 ^b	2,384,825	7.62
2011	57 ^c	315.7 ^b	2,334,004	7.39
2012	57	318.0 ^b	2,275,141	7.15
2013	57 ^d	320.2 ^e	2,188,013	6.83
2014	56 ^d	322.9 ^e	2,165,142	6.71
2015	55 ^f	325.4 ^e	2,168,371	6.66
2016	55	327.0 ^e	2,159,032	6.60
2017	55	330.4 ^e	2,115,186	6.40
2018	55	330.0 ^e	2,099,751	6.36
2019	55	334.0 ^e	2,148,141	6.43
2020	55	336.4 ^e	2,128,198	6.33
2021	55	338.7 ^e	2,080,917	6.14
2022	55	337.0 ^e	2,064,875	6.13
2023	55	343.3 ^e	2,080,659	6.06
2024	55 ^g	339.9 ^e	2,092,689	6.16
Total			83,420,329	

^aAs of 1 July 2010 there were 60 Participating Centers.^bAmerica's Poison Centers Total as of 1 July Mid-Year US Census (2012 data for 50 United States, District of Columbia and Puerto Rico; 2011 data for Guam; 2010 data for American Samoa, Federated States of Micronesia, and the US Virgin Islands).^cAs of 1 July 2011 there were 57 Participating Centers.^dOne Participating Center closed in September 2013. Its data is included in the 2013 totals but not in the 2014 data.^eAmerica's Poison Centers Total as of 1 July Mid-Year US Census (data for 50 United States, District of Columbia and Puerto Rico, Guam, American Samoa, Federated States of Micronesia, Marshall Islands, Northern Mariana Islands, Palau, and the US Virgin Islands) [2].^fOne Participating Center closed in July 2014. Its data is included in the 2014 totals but not in the 2015 data.^gOne Participating Center closed in August 2024. Its data is included in the 2024 totals.**Table 1B.** Non-human exposures by animal type

Animal	N	%
Dog	30,962	88.67
Cat	3,466	9.93
Bird	103	0.29
Rodent/lagomorph	74	0.21
Horse	60	0.17
Sheep/goat	28	0.08
Aquatic	14	0.04
Cow	14	0.04
Other	198	0.57
Total	34,919	100.00

Column 2: Number of Case Mentions (all exposures) in grey shading, displays the number of times the specific generic code was

reported in any human exposure case. If a human exposure case has multiple instances of a specific generic code, it is only counted once.

Column 3: Number of Single Exposures displays the number of human exposure cases that identified only 1 substance (1 case, 1 substance).

The succeeding columns (Age, Reason, Treatment Site, and Outcome) show selected details from these single-substance exposure cases. Death cases include both cases that have the outcome of Death or Death

Table 1C. Distribution of information requests

Information request type	N	% of Info. requests
Drug identification		
Public inquiry: Drug sometimes involved in abuse	3,019	1.06
Public inquiry: Drug not known to be abused	3,184	1.11
Public inquiry: Unknown abuse potential	251	0.09
Public inquiry: Unable to identify	1,573	0.55
HCP inquiry: Drug sometimes involved in abuse	147	0.05
HCP inquiry: Drug not known to be abused	198	0.07
HCP inquiry: Unknown abuse potential	22	0.01
HCP inquiry: Unable to identify	122	0.04
Law Enf. inquiry: Drug sometimes involved in abuse	3,149	1.10
Law Enf. inquiry: Drug not known to be abused	1,461	0.51
Law Enf. inquiry: Unknown abuse potential	152	0.05
Law Enf. inquiry: Unable to identify	630	0.22
Other drug ID	429	0.15
Subtotal	14,337	5.02
Drug information		
Adverse effects (no known exposure)	3,800	1.33
Brand / generic name clarifications	198	0.07
Calculations	90	0.03
Compatibility of parenteral medications	130	0.05
Compounding	59	0.02
Contraindications	896	0.31
Dietary supplement, herbal, and homeopathic	310	0.11
Dosage	8,573	3.00
Dosage form/formulation	977	0.34
Drug use during breast-feeding	727	0.25
Drug-drug interactions	23,981	8.39
Drug-food interactions	1,435	0.50
Foreign drug	27	0.01
Generic substitution	74	0.03
Indications/therapeutic use	2,950	1.03
Medication administration	5,562	1.95
Medication availability	291	0.10
Medication disposal	569	0.20
Pharmacokinetics	666	0.23
Pharmacology	251	0.09
Regulatory	470	0.16
Stability/storage	1,501	0.53
Therapeutic drug monitoring	184	0.06
Other drug info	10,127	3.54
Subtotal	63,848	22.35
Environmental information		
Air quality	897	0.31
Carbon monoxide – no known patient(s)	531	0.19
Carbon monoxide alarm use	642	0.22
Chem/bioterrorism/weapons (suspected or confirmed)	15	0.01
Clarification of media reports of environmental contamination	9	0.00
Clarification of substances involved in a HAZMAT incident – no known victim(s)	313	0.11
General questions about contamination of air and / or soil	197	0.07
HAZMAT planning	53	0.02
Lead – no known patient(s)	357	0.12
Mercury thermometer cleanup	349	0.12
Mercury (excluding thermometers) cleanup	332	0.12
Notification of a HAZMAT incident - no known patient(s)	323	0.11
Pesticide application by a professional pest control operator	270	0.09
Pesticides (other)	1,224	0.43
Potential toxicity of chemicals in the environment	952	0.33
Radiation	48	0.02
Safe disposal of chemicals	751	0.26
Water purity/contamination	364	0.13
Other environmental	2,282	0.80
Subtotal	9,909	3.47
Medical information		
Dental questions	40	0.01
Diagnostic or treatment recommendations for diseases or conditions – non-toxicology	5,038	1.76
Disease prevention	348	0.12
Explanation of disease states	457	0.16
General first-aid	719	0.25
Interpretation of non-toxicology laboratory reports	222	0.08
Medical terminology questions	35	0.01
Rabies – no known patient(s)	273	0.10
Sunburn management	20	0.01
Other medical	12,661	4.43
Subtotal	19,813	6.93

(Continued)

Table 1C. Continued.

Information request type	N	% of Info. requests
Occupational information		
Occupational treatment/first-aid guidelines – no known patient(s)	38	0.01
Information on chemicals in the workplace	73	0.03
MSDS interpretation	16	0.01
Occupational MSDS requests	60	0.02
Routine toxicity monitoring	15	0.01
Safe handling of workplace chemicals	78	0.03
Other occupational	235	0.08
Subtotal	515	0.18
Poison information		
Analytical toxicology	540	0.19
Carcinogenicity	56	0.02
Food poisoning – no known patient(s)	1,268	0.44
Food preparation/handling practices	4,351	1.52
General toxicity	19,195	6.72
Mutagenicity	14	0.00
Plant toxicity	824	0.29
Recalls of non-drug products (including food)	393	0.14
Safe use of household products	4,094	1.43
Toxicology information for legal use/litigation	118	0.04
Other poison	15,913	5.57
Subtotal	46,766	16.37
Prevention/Safety/Education		
Confirmation of poison center number	11,454	4.01
General (non-poison) injury prevention requests	326	0.11
Media requests	219	0.08
Poison prevention material requests	3,201	1.12
Poison prevention week date inquiries	25	0.01
Professional education presentation requests	75	0.03
Public education presentation requests	131	0.05
Other prevention	928	0.32
Subtotal	16,359	5.73
Teratogenicity information		
Teratogenicity	307	0.11
Subtotal	307	0.11
Other information		
Other	39,499	13.82
Subtotal	39,499	13.82
Substance abuse		
Drug screen information	681	0.24
Effects of illicit substances – no known patient(s)	75	0.03
New trend information	43	0.02
Withdrawal from illicit substances – no known patient(s)	146	0.05
Other substance abuse	467	0.16
Subtotal	1,412	0.49
Administrative		
Expert witness requests	29	0.01
Faculty activities	15	0.01
Funding	6	0.00
Personnel issues	64	0.02
Poison center record request	151	0.05
Product replacement/malfunction (issues intended for the manufacturer)	292	0.10
Scheduling of poison center rotations	21	0.01
Other administration	15,691	5.49
Subtotal	16,269	5.69
Caller Referred		
Immediate referral – animal poison center or veterinarian	32,567	11.40
Immediate referral – drug identification	98	0.03
Immediate referral – drug information	173	0.06
Immediate referral – health department	4,904	1.72
Immediate referral – medical advice line	288	0.10
Immediate referral – pediatric triage service	239	0.08
Immediate referral – pesticide hotline	154	0.05
Immediate referral – pharmacy	682	0.24
Immediate referral – poison center	6,311	2.21
Immediate referral – private physician	1,421	0.50
Immediate referral – psychiatric crisis line	108	0.04
Immediate referral – teratology information program	90	0.03
Other call referral	9,667	3.38
Subtotal	56,702	19.84
Total	285,736	100.00

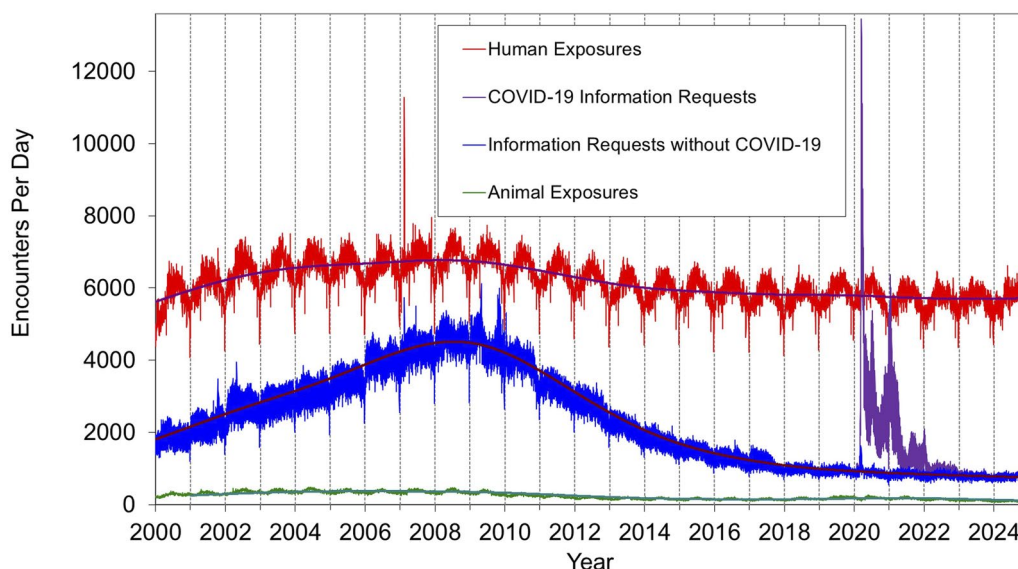


Figure 1. Human Exposure Cases, Information Requests, and Animal Exposure Cases by Day since 1 January 2000. Smoothing spline fits using $\lambda = 1200$ for human exposures had associated $RSqr = 0.520$, information requests $RSqr = 0.948$ (excluding COVID-19 information requests), and animal exposures $RSqr = 0.897$. COVID-19 information requests on the disease, vaccines, home tests, and monoclonal antibodies are plotted separately from other information requests.

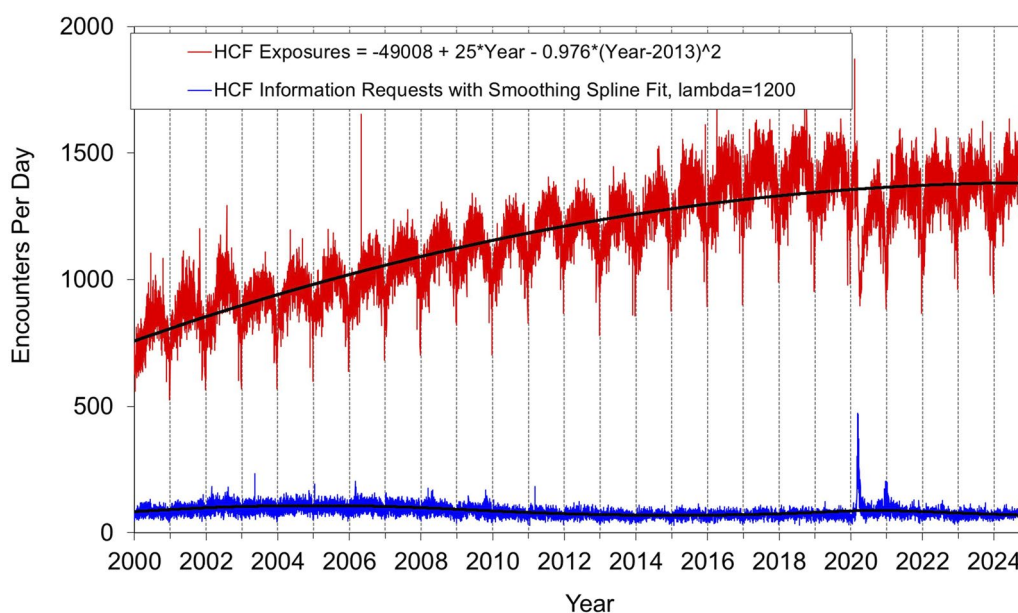


Figure 2. Health Care Facility (HCF) Exposure Cases and HCF Information Requests by Day since 1 January 2000. Both linear and second order (quadratic) terms were statistically significant ($p < 0.0001$) for regression of HCF human exposure with associated $RSqr = 0.699$ and $RSqr = 0.744$, respectively. The quadratic coefficient is negative meaning the case counts are increasing slower than linearly. Smoothing spline fit with $\lambda = 1200$ for HCF information requests had associated $RSqr = 0.260$.

(indirect report). These death cases are not limited by the RCF.

Tables 22A and 22B (Appendix B) restrict the breakdown columns to single-substance cases. Prior to 2007, when multi-substance exposures were included, a relatively innocuous substance could be mentioned in a death column when, for example, the death was attributed to an antidepressant, opioid, or cyanide. The

restriction of the breakdowns to single-substance exposures should increase precision and reduce misrepresentation of the results in this by-substance table. Single substance cases reflect the majority (87.5%) of all exposures. In contrast, only 46.2% of fatalities are single substance exposures (Table 5).

Table 22A and Table 22B (Appendix B) tabulate 2,531,739 case mentions, of which 1,831,206 were

single-substance exposures, including 900,071 (49.2%) nonpharmaceuticals and 931,135 (50.8%) pharmaceuticals. In 21.1% of single-substance exposures that involved pharmaceutical substances, the reason for exposure was intentional, compared to only 4.50% when the exposure involved a nonpharmaceutical substance. Correspondingly, treatment in an HCF was provided in a higher percentage of exposures that involved pharmaceutical substances (33.1%) compared with nonpharmaceutical substances (18.5%). Exposures to pharmaceuticals also had more severe outcomes. Of 1,274 single-substance exposure-related fatal cases, 968 (76.0%) were pharmaceuticals compared with 306 (24.0%) nonpharmaceuticals.

Age and gender distributions

The age and gender distribution of human exposures is outlined in Table 3A. Children younger than 3 years of age were involved in 27.4% of exposures and children ≤ 5 years accounted for 39.0% of human exposures. A male predominance was found among cases involving children ≤ 12 years, but this gender distribution was reversed in teenagers and adults, with females comprising most reported exposures. The overall rate of poison exposures reported to PCs is 616/100,000 population (Table 3B). The highest rates of poison exposures are in 1-year-old children (6,584/100,000 population) and 2-year-old children (6,353/100,000

population). Rates declined with age from 524/100,000 population in children 6–12 to 360/100,000 population in adults ≥ 20 years. Table 4 shows the age and gender distribution for the 1,301 exposure-related fatalities (excluding death, indirect).

Caller site and exposure site

As shown in Table 2, of the 2,092,689 human exposure cases reported, 67.2% of cases originated from a residence (own or other) while 92.1% of exposures occurred at a residence (own or other). Another 24.2% of cases originated from a HCF. Beyond residences, exposures occurred in the workplace (2.13% of cases), schools (1.78%), HCF (0.321%), and restaurants or food services (0.138%).

Table 2. Site of call and site of exposure, human exposure cases

Site	Site of caller		Site of exposure	
	N	%	N	%
Residence				
Own	1,383,393	66.11	1,890,525	90.34
Other	22,700	1.08	37,444	1.79
Workplace	23,347	1.12	44,559	2.13
Health care facility	505,868	24.17	6,718	0.32
School	12,865	0.61	37,275	1.78
Restaurant/food service	476	0.02	2,898	0.14
Public area	8,393	0.40	24,126	1.15
Other	129,373	6.18	27,171	1.30
Unknown	6,274	0.30	21,973	1.05

Table 3A. Age and gender distribution of human exposures

Age (y)	Male		Female		Unknown gender		Total		Cumulative total	
	N	% Of age group total	N	% Of age group total	N	% Of age group total	N	% Of total exposures	N	%
Children (<20)										
<1	48,895	51.91	44,784	47.55	510	0.54	94,189	4.50	94,189	4.50
1	125,323	51.77	116,099	47.96	639	0.26	242,061	11.57	336,250	16.07
2	123,372	51.96	113,298	47.72	772	0.33	237,442	11.35	573,692	27.41
3	70,809	55.03	57,354	44.57	507	0.39	128,670	6.15	702,362	33.56
4	38,732	56.65	29,219	42.74	414	0.61	68,365	3.27	770,727	36.83
5	24,770	57.69	17,858	41.59	307	0.72	42,935	2.05	813,662	38.88
Unknown ≤ 5	1,003	45.53	877	39.81	323	14.66	2,203	0.11	815,865	38.99
Child 6–12	83,490	56.36	62,849	42.42	1,810	1.22	148,149	7.08	964,014	46.07
Teen 13–19	66,206	37.26	110,531	62.20	956	0.54	177,693	8.49	1,141,707	54.56
Unknown Child	1,914	39.62	1,676	34.69	1,241	25.69	4,831	0.23	1,146,538	54.79
Subtotal	584,514	50.98	554,545	48.37	7,479	0.65	1,146,538	54.79	1,146,538	54.79
Adults (≥ 20)										
20–29	83,587	42.48	112,669	57.25	530	0.27	196,786	9.40	1,343,324	64.19
30–39	76,533	43.57	98,862	56.28	277	0.16	175,672	8.39	1,518,996	72.59
40–49	52,570	41.45	74,137	58.45	124	0.10	126,831	6.06	1,645,827	78.65
50–59	42,463	39.46	65,047	60.45	94	0.09	107,604	5.14	1,753,431	83.79
60–69	38,260	37.81	62,852	62.11	86	0.08	101,198	4.84	1,854,629	88.62
70–79	27,122	35.51	49,198	64.42	54	0.07	76,374	3.65	1,931,003	92.27
80–89	12,542	34.55	23,731	65.38	26	0.07	36,299	1.73	1,967,302	94.01
≥ 90	2,481	31.89	5,290	67.99	10	0.13	7,781	0.37	1,975,083	94.38
Unknown adult	38,595	37.01	63,381	60.77	2,320	2.22	104,296	4.98	2,079,379	99.36
Subtotal	374,153	40.11	555,167	59.51	3,521	0.38	932,841	44.58	2,079,379	99.36
Other										
Unknown age	3,969	29.82	5,861	44.03	3,480	26.15	13,310	0.64	2,092,689	100.00
Total	962,636	46.00	1,115,573	53.31	14,480	0.69	2,092,689	100.00	2,092,689	100.00

Table 3B. Population-adjusted exposures by age group

Age Group	Exposures/100k population	Number of exposures ^a	Population ^b
Children (<20)			
<1	2,569	94,189	3,667,029
1	6,584	242,061	3,676,619
2	6,353	237,442	3,737,755
3	3,531	128,670	3,644,443
4	1,824	68,365	3,748,053
5	1,131	42,935	3,796,789
Child 6–12	524	148,149	28,248,893
Teen 13–19	585	177,693	30,368,073
Subgroup	1,417	1,146,538	80,887,654
Adults (≥20)			
20–29	433	196,786	45,407,015
30–39	378	175,672	46,518,569
40–49	299	126,831	42,441,460
50–59	263	107,604	40,957,651
60–69	247	101,198	40,948,724
70–79	270	76,374	28,266,191
80–89	309	36,299	11,732,717
90+	282	7,781	2,759,044
Subgroup	360	932,841	259,031,371
Overall total	616	2,092,689	339,919,025

^aNumber of Exposures excludes unknown ages from the individual age categories, but includes them in the Subtotals and Overall Total (see Table 3A)

^bAmerica's Poison Centers Total as of 1 July 2023, 339,919,025 (see Table 1A) [2].

Exposures in pregnancy

Exposure during pregnancy occurred in 6,669 women (0.319% of all human exposures). Of those with known pregnancy duration ($n=6,254$), 43.8% occurred in the first trimester, 30.2% in the second trimester, and 26.0% beyond the second trimester. Most (75.4%) were unintentional exposures and 16.5% were intentional exposures. There were no deaths of pregnant females recorded in 2024.

Chronicity

Most human exposures, 1,767,814 (84.5%), were acute cases (single, repeated or continuous exposure occurring over 8 hours or less) although these comprised only 926 (33.0%) of the 2,809 fatalities. Chronic exposures (continuous or repeated exposures occurring over more than 8 hours) comprised 2.59% (54,121) of all human exposures. Acute-on-chronic exposures (single exposure that was preceded by a continuous, repeated, or intermittent exposure occurring over a period greater than 8 hours) numbered 236,510 (11.3%).

Reason for Exposure

The reason category for most human exposures was unintentional (77.8%), including: unintentional general (44.2%), therapeutic error (17.3%), and unintentional misuse (9.10%) (Table 6A).

Figure 3A shows suspected suicide and unintentional general human exposures by day since 1 January 2000, while Figure 3B shows the same data normalized for the number of human exposure cases per day. Smoothing spline fit of Figure 3A shows departure from linearity with a declining number of cases since mid-2007 for unintentional general and a modest increase in suspected suicide cases until 2020. When normalizing for daily human exposure case volumes, the decrease in unintentional general becomes linear, while a smoothing spline fit for suspected suicides continues to show a declining number of cases since 2020 (Figure 3B).

Scenarios

Of the total 361,402 cases involving therapeutic errors (Table 6A) the most common scenarios for all ages included: inadvertent double-dosing (31.3%), other incorrect dose (16.5%), wrong medication taken or given (14.6%), doses given/taken too close together (11.3%), and inadvertent exposure to someone else's medication (8.33%). The types of therapeutic errors observed are different for each age group and are summarized in Table 6B.

Reason by age

Intentional exposures accounted for 17.7% of human exposure cases. Suicidal intent was suspected in 12.4% of cases, intentional misuse in 2.67%, and intentional abuse in 2.01%. Unintentional exposures outnumbered intentional exposures in all age groups with the exception of ages 13–19 years (Table 7). In contrast, of the 1,301 reported fatalities with RCF 1–3, the major reason reported for children ≤5 years was unintentional, while most fatalities in teens (13–19 years) and adults (≥20 years) were intentional (Table 8).

Route of exposure

Ingestion was the route of exposure in 82.0% of cases (Table 9), followed in frequency by inhalation/nasal (7.97%), dermal (6.89%), and ocular routes (4.15%). For the 1,301 exposure-related fatalities, ingestion (82.6%), and inhalation/nasal (7.46%) were the predominant known exposure routes. Each exposure case may have more than one route.

Clinical effects

The NPDS database allows for the coding of up to 169 individual clinical effects (signs, symptoms, or

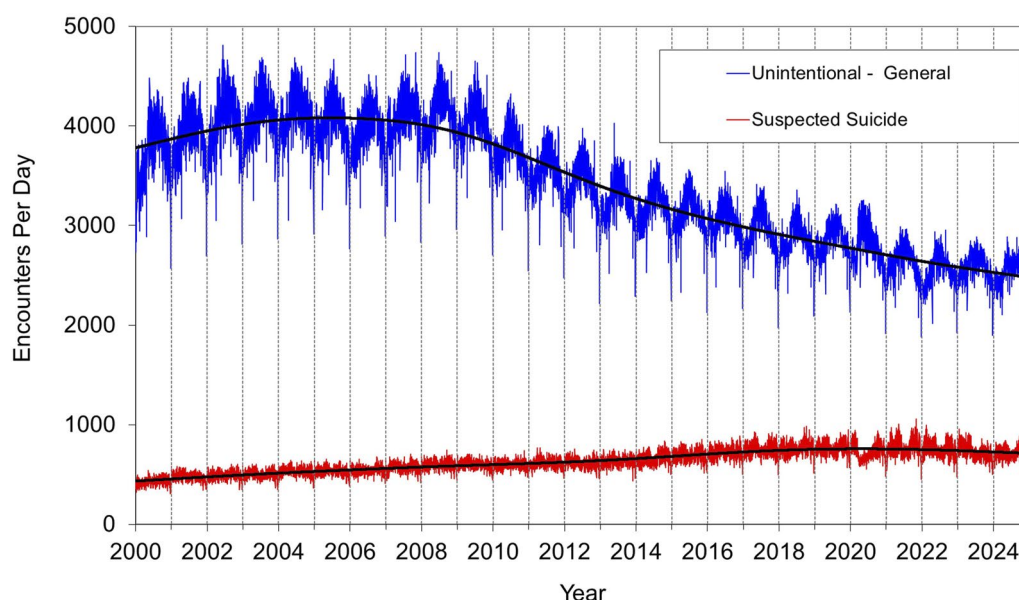


Figure 3A. Suspected Suicide and Unintentional General Exposures by Day since 1 January 2000. Smoothing spline fit used $\lambda = 8000$, suspected suicide had associated $RSqr = 0.736$ and unintentional-general $RSqr = 0.851$.

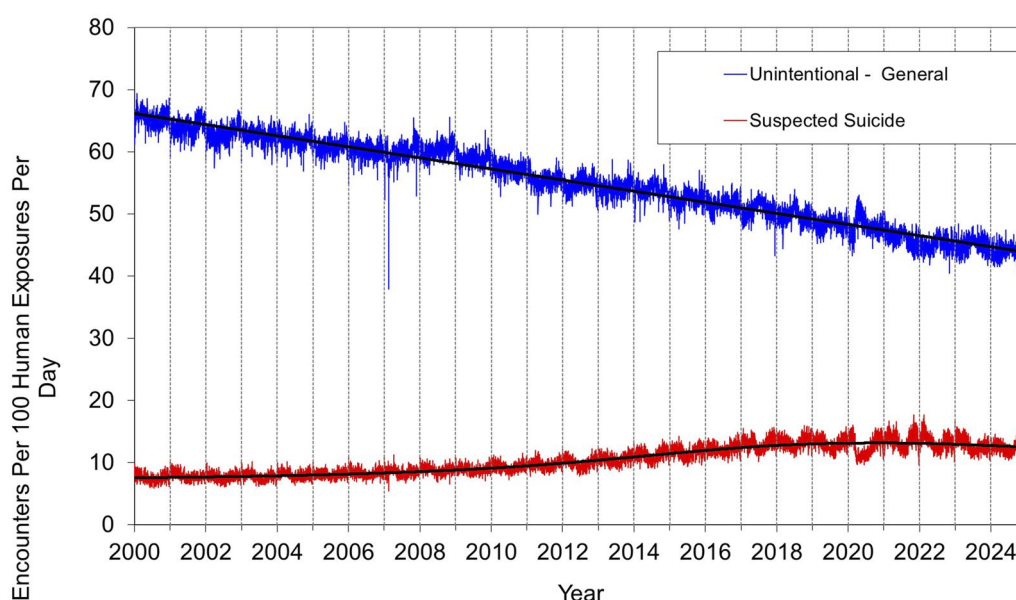


Figure 3B. Suspected Suicide and Unintentional General Exposures Adjusted for All Human Exposures by Day since 1 January 2000. Encounters adjusted per 100 human exposures per day. Linear regression of unintentional-general had an associated $RSqr = 0.948$, while smoothing spline with $\lambda = 8000$ for suspected suicides had an associated $RSqr = 0.840$.

laboratory abnormalities) for each case. Each clinical effect can be further determined as related, not related, or unknown if related. Clinical effects were coded in 815,914 (39.0%) cases (16.9% had 1 effect, 9.84% had 2 effects, 5.42% had 3 effects, 2.87% had 4 effects, 1.50% had 5 effects, and 2.49% had >5 effects coded). Of clinical effects coded, 75.7% were deemed related to the exposure, 9.95% were considered not related, and 14.3% were coded as unknown if related.

Case Management site

Most cases reported to PCs were managed outside of an HCF (66.6%), usually at the site of exposure, primarily the patient's own residence (Table 10). Evaluation or treatment in an HCF occurred in 30.6% of cases. Only 1.19% of cases were referred to an HCF but refused referral.

Of the 640,220 cases managed in an HCF, 315,743 (49.3%) were treated and released, 69,628 (10.9%) were admitted to a critical care unit, 82,789 (12.9%)

were admitted to a noncritical unit, and 90,262 (14.1%) were admitted to a psychiatric facility.

The percentage of patients treated in an HCF varied considerably with age. Only 13.9% of children ≤ 5 years and 20.2% of children between 6 and 12 years were managed in an HCF compared to 64.9% of teenagers (13–19 years) and 44.4% of adults (age ≥ 20 years).

Medical outcome

Table 11 displays the medical outcome of human exposure cases distributed by age. Older age groups exhibited a greater number of more serious medical outcomes (death, major effect and moderate effect) with 20.7% of teens (13–19 years) and 17.9% of adults with more serious medical outcomes. Table 12 compares medical outcome and reason for exposure and shows a greater frequency of serious outcomes in intentional exposures.

The duration of effect is required for all cases which report at least 1 clinical effect and have a medical outcome of minor, moderate or major effect ($n=520,500$; 24.9% of exposures). Table 13 demonstrates an increasing duration of the clinical effects observed with more severe outcomes.

Decontamination procedures and specific antidotes

Tables 14 and 15 outline the use of 112 decontamination procedures, specific physiological antagonists (antidotes), and measures to enhance elimination in the treatment of patients reported in the NPDS database.

Table 16 details the decontamination trends for all ages and pediatric exposures (≤ 5 years) for the last decade detailing low rates of ipecac use ($<0.002\%$) and declining activated charcoal use (1.51%).

Top substances in human exposures

Table 17A presents the 25 most common substance categories listed by frequency of human exposure. This ranking provides an indication where prevention efforts might be focused, as well as the types of more serious exposures PCs regularly manage (resulting in death, major effect or moderate effect). It is relevant to know whether exposures to these substances are increasing or decreasing.

To better understand these relationships, we examined exposures with more serious outcomes (death, major effect and moderate effect) per year over the last 10 years for the change over time for each of the

68 major generic categories by least squares linear regression. More serious outcome exposure cases per year over this period were increasing for 39, static for 5, and decreasing for 24 of the 68 categories with data for the entire time period. The change over time for the 10 yearly values was statistically significant ($P<0.05$) for 39 of the 68 categories with data for the entire time period. Table 17B shows the 25 categories which increased the most rapidly over the past 10 years. Statistical significance of the linear regressions can be verified by noting the 95% confidence interval on the rate of increase excludes zero for all but 7 of the 25 categories. Figure 4 shows the change over time and linear regressions for the top 4 increasing categories in Table 17B for the 10-year period.

Tables 17C and 17D present exposure results for children and adults, respectively, and show the differences between substance categories involved in pediatric and adult exposures.

Tables 17E reports the 25 categories of substances most frequently involved in pediatric (≤ 5 years) fatalities in 2024.

Table 17F reports the 25 Drug ID categories most frequently queried in 2024, highlighting the value of Drug ID information to America's Poison Centers, public health, public safety, and regulatory agencies. The top 10 substance categories include antibiotics, drugs with widespread use and abuse potential such as opioids and benzodiazepines, and drugs with the potential for serious toxicity.

Table 17G reports the 25 substance categories most frequently reported in exposures involving pregnant patients.

Changes over time

Total encounters peaked in 2008 at 4,333,012 including 2,491,049 human exposure cases and 1,703,762 information contacts. Total encounters decreased 0.117% from 2,421,251 in 2023 to 2,418,426 in 2024. Information requests decreased by 2.70% from 293,663 in 2023 to 285,736 in 2024, with a 24.5% decrease in medical information requests. There was a 1.84% increase in HCF information requests and a 0.218% increase in drug information requests from any source. Human exposures increased by 0.578% from 2,080,659 to 2,092,689 cases over the same time period.

Figure 5 shows the year-to-year change through 2024 as a percentage of year 2000 for human exposure cases broken down into cases with more serious outcomes (death, major effect, and moderate effect) and less serious outcomes (minor effect, no effect, not followed (non-toxic), not followed (minimal toxicity)

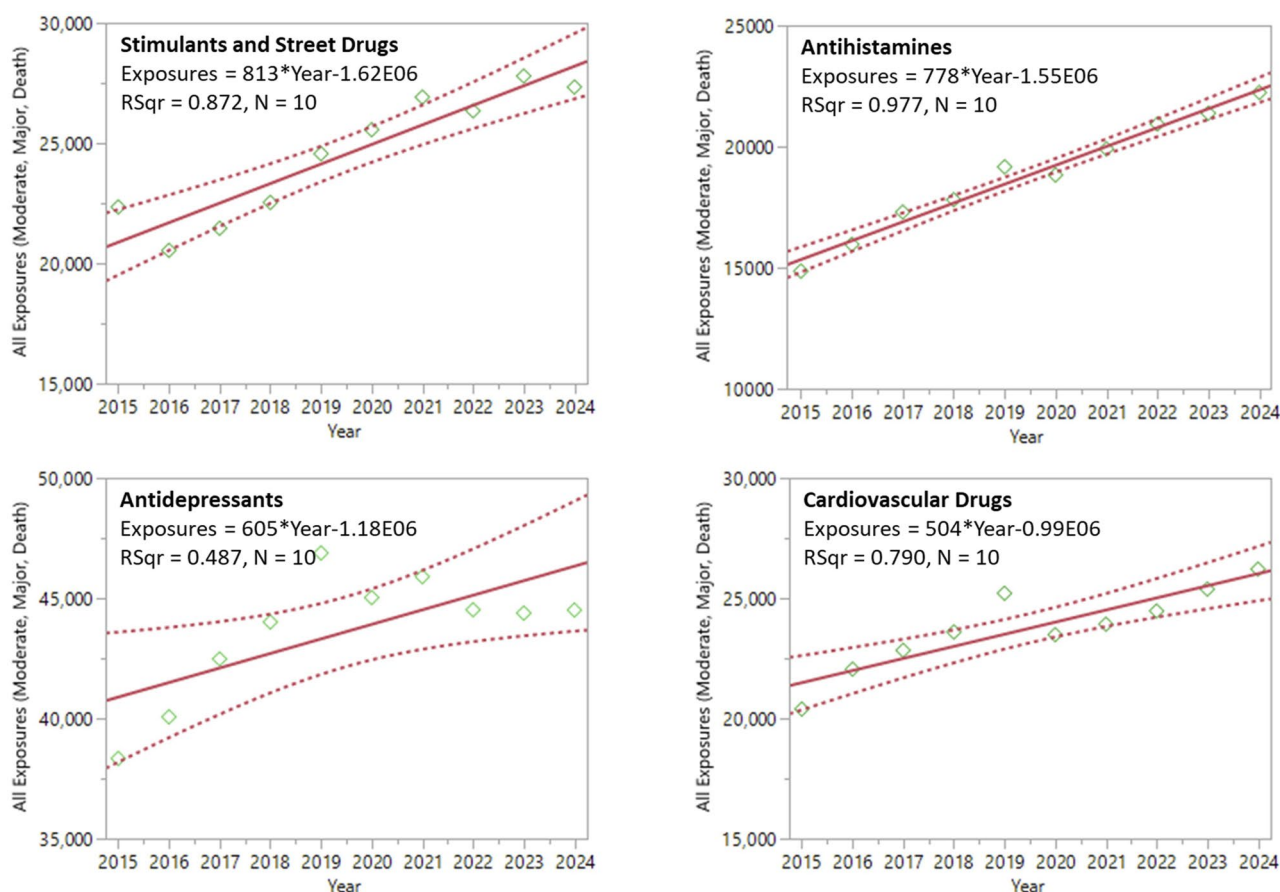


Figure 4. Substance Categories with the Greatest Rate of Exposure Increase since 1 January 2015 for More Serious Outcomes (Top 4). Solid lines show least-squares linear regressions for the human exposure cases per year for that category (◊). Broken lines show 95% confidence interval on the regression.

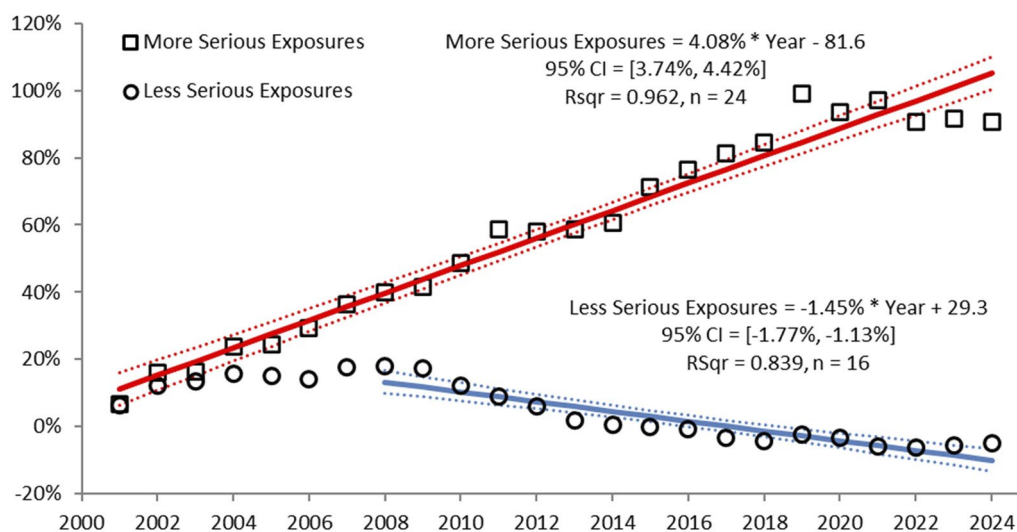


Figure 5. Change in Human Exposures by Outcome from Year 2000.

The percent change from baseline (year 2000) for human exposure cases divided among the 10 medical outcomes. The more serious exposures (moderate, major and death) increased. The less serious exposures (no effect, minor effect, not followed (non-toxic), not followed (minimal toxicity possible), unable to follow (potentially toxic) and unrelated effect) decreased after 2008. Solid lines show least-squares linear regressions for the change in more serious exposures per year (◻) and less serious exposures (◊). Broken lines show 95% confidence intervals on the regression.

possible), unable to follow (potentially toxic), and unrelated effect). Since 2000, cases with more serious outcomes have increased by 4.08% (95% CI [3.74%,

4.42%]) per year from 108,148 cases in 2000 to 206,166 cases in 2024. Cases with less serious outcomes have decreased since 2008 by 1.45% [-1.77%, -1.13%] per

year from 2,339,460 in 2008 to 1,885,511 cases in 2024. The decrease in less serious exposures has largely driven the overall decrease in human exposures since 2008; however, there has been a small increase (1.50%) in cases with less serious exposures from 2022 (1,857,589) to 2024 (1,885,511). America's PCs have seen a consistent increase in exposure cases from HCFs (Figure 2), as well as for the most severe exposures (Figure 5), while there is a continued decreasing trend in cases involving less severe exposures.

Emerging trends: ECMO in poisoning

Between 2019 and 2024, U.S. Poison Centers recorded 965 cases where ECMO was utilized in the management of poisoning (ECMO therapy coded as "performed"). Note that NPDS does not differentiate VA from VV ECMO. Annual ECMO mentions increased steadily from 127 cases in 2021 to 217 in 2024 (Figure 6). Most exposures involved polysubstance exposures ($n=554$; 57.5%), while 411 cases involved a single substance. The most common route of exposure was ingestion ($n=790$) followed by unknown ($n=112$), inhalational/nasal ($n=78$), parenteral ($n=38$), and aspiration ($n=19$).

Most patients were female ($n=523$; 54.3%) and 3 were pregnant. The age group most frequently reported was 13–19 years ($n=241$; 25.0%) (Figure 7). Suspected suicide accounted for the majority of ECMO-associated exposures (63.4%), followed by unknown intent (9.12%) and intentional abuse (7.36%). Single substance exposures were more common amongst pediatric patients <13 years ($n=64$ single substance vs. 17 polysubstance).

Overall mortality among patients receiving ECMO was 27.3%. Mortality was 26.1% in single-substance exposures and 28.2% in polysubstance exposures.

Across both single and polysubstance categories, calcium channel blockers were the most frequently implicated substance. The next most commonly involved agents in polysubstance exposure were beta blockers, ethanol, bupropion, atypical antipsychotics, and benzodiazepines. In single-substance exposures, the second most common agent was "unknown drugs," followed by bupropion, antiarrhythmics, non-drug substances, and beta blockers. See Table A for the top 20 substances in each category, and Table B for substances associated with pediatric (age < 6) ECMO occurring at least twice.

The ability to employ ECMO is dependent on the systems of care in place, which can vary by region. Population-adjusted ECMO use varied widely by state. The District of Columbia had the highest rate at 20.5 cases per million residents, followed by Minnesota (14.8), North Dakota (8.9), Nebraska (7.59) and New Mexico (6.6).

In contrast, several states had no ECMO cases recorded at all, such as Alaska, Mississippi, Rhode Island, New Hampshire, Vermont, and Wyoming. While these states have lower populations, other states with even lower populations, such as North Dakota (population 783,875) had the 3rd highest rate compared to all states or districts, potentially reflecting differences in regional ECMO availability, toxicology consultation access, or reporting practices. Table C presents state level incidence data.

The increase in ECMO use for poisoning between 2019 and 2024 reflects both growing clinical awareness and expanded access to advanced critical care

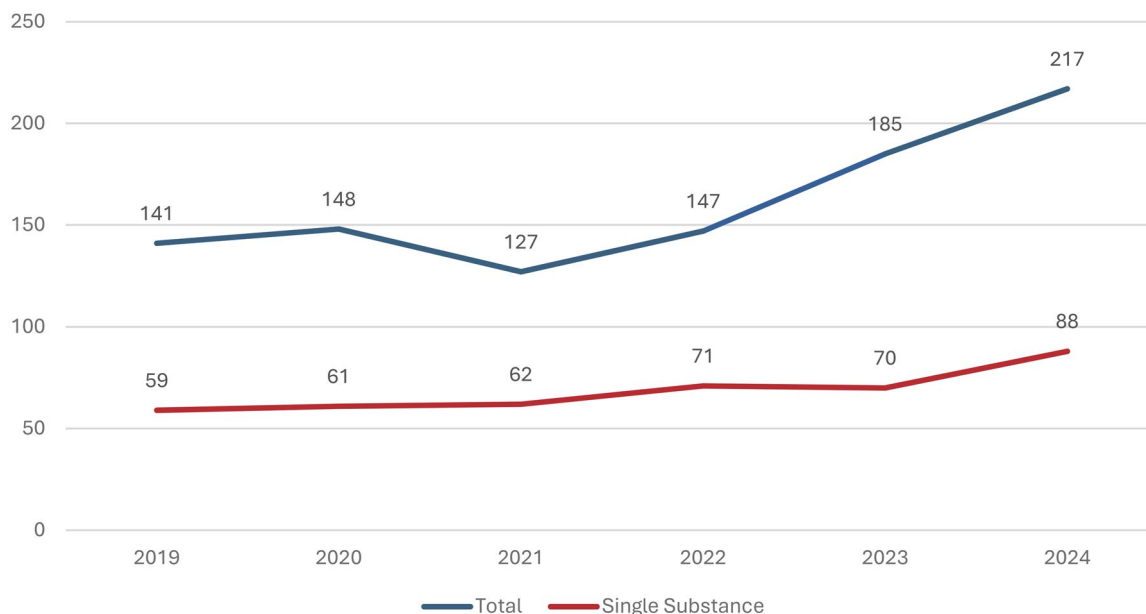


Figure 6. Human Exposures Receiving Extracorporeal Membrane Oxygenation (ECMO) from 1 January 2019 to 31 December 2024.

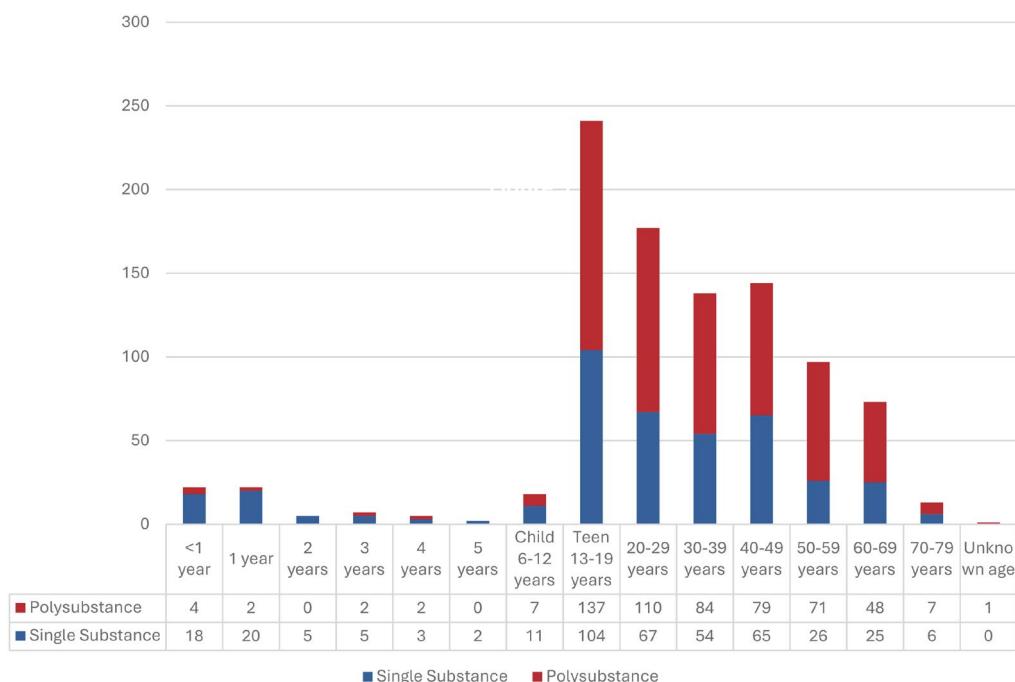


Figure 7. Distribution of Age Ranges in Human Exposures Receiving Extracorporeal Membrane Oxygenation (ECMO) from 1 January 2019 to 31 December 2024.

Table A. Substances most frequently involved in cases receiving extracorporeal membrane oxygenation (ECMO) from 1 January 2019 to 31 December 2024 (Top 20)

Polysubstance exposures	N	Single substance exposures	N
Calcium antagonist, alone	238	Calcium antagonist, alone	76
Beta blocker, alone	119	Miscellaneous unknown drugs	38
Ethanol (beverages)	110	Bupropion	26
Bupropion	80	Antiarrhythmics	12
Atypical antipsychotics	62	Other non-drug substances	12
Benzodiazepines	62	Beta blocker, alone	11
Angiotensin converting enzyme inhibitor, alone	45	Diphenhydramine alone	10
Acetaminophen alone	44	Acetaminophen alone	10
Miscellaneous unknown drugs	36	Freon and other propellants	7
Fluoxetine	34	Atypical antipsychotics	7
Ibuprofen	32	Other chemicals	6
Venlafaxine	30	Other suspicious substances (non-powder)	6
Angiotensin receptor blocker, alone	29	Oral hypoglycemics: biguanides	6
Hydroxyzine	29	Methamphetamines	6
Trazodone	29	Other or unknown opioids	6
Gabapentin	28	Nitrates and nitrites	5
Escitalopram	22	Fentanyl (prescription)	5
Clonidine	20	Ibuprofen	5
Sertraline	20	Disc batteries: lithium	4
Citalopram	19	Non-prescription fentanyl	4

Table B. Substances most frequently involved in pediatric cases receiving extracorporeal membrane oxygenation (ECMO) from 1 January 2019 to 31 December 2024

Polysubstance exposures	N	Single substance exposures	N
Calcium antagonist, alone	5	Miscellaneous unknown drugs	7
Beta blocker, alone	2	Disc batteries: lithium	4
Carbon monoxide	2	Calcium antagonist, alone	2
Citalopram	2	Cardiac glycosides	2
Miscellaneous unknown drugs	2	Iron and iron salts (excluding vitamins with iron)	2
		Kerosenes	2
		Other chemicals	2
		Other non-drug substances	2
		Other types of hydrocarbon	2
		Other types of sedative/hypnotic/anti-anxiety or anti-psychotic drug	2
		Unknown chemicals	2

ECMO-capable centers and toxicology expertise. These findings underscore the need for expanded access to ECMO facilities, and proactive public health strategies to address rising severity and complexity of poisonings. The use of ECMO will likely continue to play a growing role in the management of life-threatening poisoning.

Snapshot: processed mushroom bars and psilocybin mushrooms

2024 saw a surge in exposures to products marketed as psychedelic mushrooms, including those containing psilocybin and ibotenic acid/muscimol (e.g., *Amanita muscaria*). These products are often sold as chocolates,

technologies in toxicology. Most ECMO cases involved polysubstance ingestion in adolescents and young adults, with suspected suicide as the predominant exposure intent. Calcium channel blockers, bupropion, and beta blockers were among the most frequently implicated agents. Substantial geographic variability in ECMO use highlights inequities in access to

gummies, or syrups, under branding that may obscure their risk. Many are marketed as “legal” highs or micro-dose supplements, despite containing hallucinogenic or sedative compounds. In 2024 the Centers for Disease

Table C. Geographic variation in cases receiving extracorporeal membrane oxygenation (ECMO) from 1 January 2019 to 31 December 2024

State	Cases	Rate per million	State	Cases	Rate per million
Alabama	13	2.6	Montana	5	4.5
Alaska	0	0.0	Nebraska	15	7.6
Arizona	12	1.6	Nevada	2	0.6
Arkansas	13	4.3	New Hampshire	0	0.0
California	46	1.2	New Jersey	17	1.8
Colorado	22	3.8	New Mexico	14	6.6
Connecticut	14	3.9	New York	51	2.6
Delaware	6	5.9	North Carolina	19	1.8
District of Columbia	14	20.5	North Dakota	7	8.9
Florida	28	1.3	Ohio	21	1.8
Georgia	51	4.7	Oklahoma	20	5.0
Hawaii	1	0.7	Oregon	20	4.7
Idaho	2	1.0	Pennsylvania	59	4.5
Illinois	26	2.0	Rhode Island	0	0.0
Indiana	38	5.6	South Carolina	10	1.9
Iowa	11	3.4	South Dakota	1	1.1
Kansas	11	3.7	Tennessee	24	3.4
Kentucky	14	3.1	Texas	62	2.1
Louisiana	2	0.4	Utah	17	5.0
Maine	7	5.1	Vermont	0	0.0
Maryland	17	2.7	Virginia	29	3.3
Massachusetts	19	2.7	Washington	29	3.7
Michigan	42	4.2	West Virginia	4	2.2
Minnesota	75	13.1	Wisconsin	21	3.6
Mississippi	0	0.0	Wyoming	0	0.0
Missouri	30	4.8			

Control actively monitored adverse health effects associated with *Diamond Shrooms*™ brand products which included products labeled as “Microdosing Chocolate Bars,” “Infused Cones,” “Micro-Dose” and “Mega-Dose Extreme Gummies” that were found to contain a number of psychoactive compounds including, O-acetylsilicidin, psilocin, kavalactones, pregabalin, and muscimol [4,5]. The 2024 Snapshot examines human exposures to processed mushroom-related products reported to U.S. Poison Centers.

Figure 8 depicts natural psychoactive mushroom exposures (psilocybin and ibotenic acid/muscimol containing) and processed mushroom product exposures from January 2010 to June of 2025. Note the decrease in both processed and natural mushrooms depicted after the *Diamond Shrooms* outbreak in June of 2024. Overall, from 2010 to 2024 a total of 1,195 exposures were reported, reflecting widespread access and growing use across a range of age groups and settings.

- 749 (62.8%) of exposures occurred in males.
- 750 (62.8%) of exposures involved patients aged 13–39 years.
- 258 (21.6%) occurred in children under 13 years old.

Intentional use accounted for most exposures (819, 68.5%):

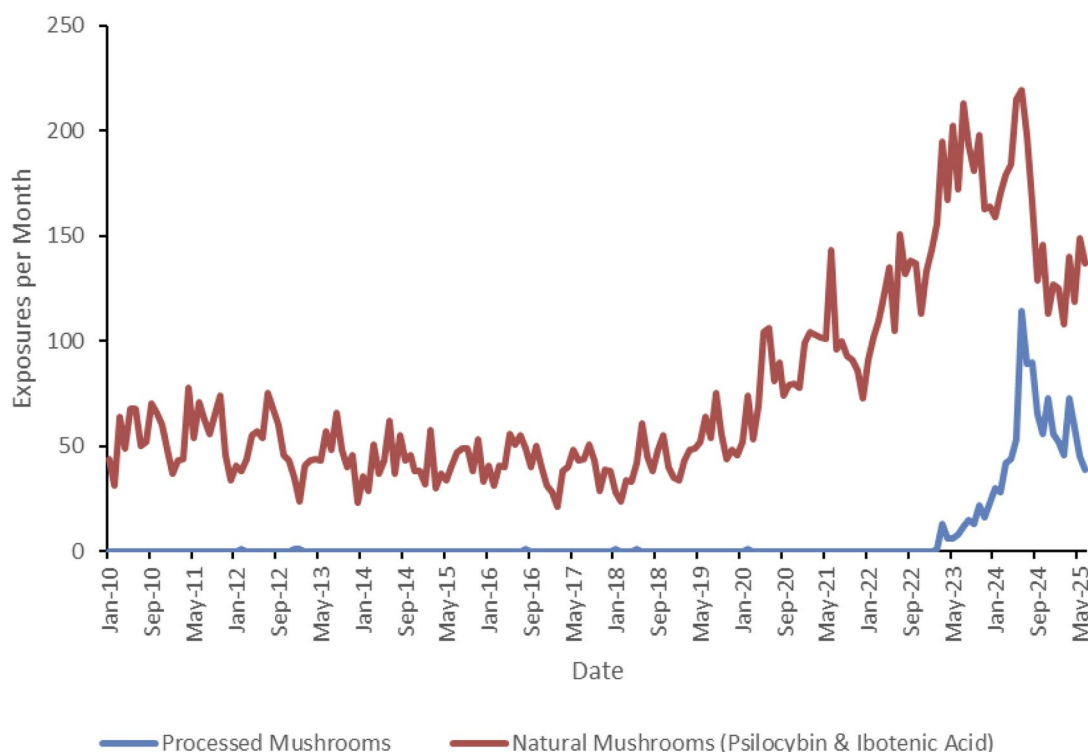


Figure 8. Natural and Processed Mushroom Human Exposures from 1 January 2010 to 30 June 2025.

- 660 (55.2%) were classified as intentional abuse.
- 73 (6.10%) were suspected suicide or intentional misuse.
- 246 (20.6%) were unintentional general exposures.

Clinical effects were seen in 751 (81.1%) of single substance exposures, the most commonly reported are listed in [Table D](#).

Healthcare resource utilization was substantial:

- 564 (60.9%) of single-substance exposures were already at or en route to a healthcare facility when the Poison Center was contacted.
- 432 (46.7%) were treated and released from the ED.
- 145 (15.7%) were admitted to a hospital, including 58 (6.3%) to a critical care unit.

Medical outcomes in single substance exposures can be seen in [Table E](#).

Products involved were diverse, including chocolate bars or gummies, and 267 (22.4%) exposures were linked to branded or labeled mushroom products.

Table D. Clinical effects reported in single substance processed mushroom product exposures from 1 January 2023 to 30 June 2025.

Clinical effect (N=926)	N	% of Cases
Agitation	207	22.4
Hallucinations/delusions	191	20.6
Tachycardia	174	18.8
No effect	164	17.7
Other – neurological	155	16.7
Vomiting	149	16.1
Nausea	135	14.6
Confusion	125	13.5
Hypertension	114	12.3
Mydriasis	106	11.4
CNS depression (mild)	102	11.0

Table E. Medical outcomes in single substance processed mushroom product exposures from 1 January 2023 to 30 June 2025.

Medical outcome (N=926)	N	% of Cases
No effect	71	7.70
Minor effect	251	26.8
Moderate effect	325	35.0
Major effect	45	4.73
Death, indirect report	2	0.22
Not followed, judged as nontoxic exposure (clinical effects not expected)	5	0.55
Not followed, minimal clinical effects possible (no more than minor effect possible)	89	9.79
Unable to follow, judged as a potentially toxic exposure	126	13.9
Unrelated effect, the exposure was probably not responsible for the effect(s)	12	1.32

The increasing number of exposures, severity of effects, and growing product diversity suggest a need for improved regulation, packaging standards, and public education. Poison Centers remain essential for surveillance and clinical guidance in these complex and evolving exposures.

Distribution of suicides

[Table 19A](#) shows modest variation in the distribution of suicides and pediatric deaths over the past 15 years as reported to the NPDS database. Within the last decade, the percentage of exposures determined to be suspected suicides ranged from 22.6 to 45.8% and the percentage of pediatric cases has ranged from 0.779 to 2.29%. The relatively large changes seen for 2017, 2018, 2020, 2021 and 2022 reflect the large increase in indirect death reports in those years (peaking at 2,482 in year 2020). Analyses of suicides and pediatric deaths for direct and indirect reports are shown in [Table 19B](#). For a more detailed review of trends in adolescent suicides, the reader is referred to the 2018 NPDS Annual Report [1].

Plant exposures

[Table 20](#) provides the number of times a specific plant was reported to NPDS ($n=54,556$). The 25 most commonly involved plant species and categories account for 44.1% of all reported plant exposures. Four of the top 10 categories in the table are essentially synonymous for unknown plant and comprise 31.0% (5,097) of all plant exposures. For various reasons, it was impossible to make a precise identification in these 4 groups. The most frequent plant exposures where positive plant identification was made were (descending order): Pokeweed (*Phytolacca americana*), Cherry pit or Cherry (not specified), Poison ivy, Elephant's ear (Family: Araceae), *Ilex* species, and *Solanum dulcamara*.

Deaths and exposure-related fatalities

A listing of cases is found in [Table 21 \(Appendix A\)](#). A summary of cases ([Tables 4, 5, 8, 9, and 18](#)) are provided for fatal cases with reasonable confidence that the death was a result of the exposure (exposure-related fatalities; RCF of 1, 2 or 3 and excluding death, indirect cases). [Tables 11, 12, 17E, 19A and 22](#) consider all deaths (direct and indirect), irrespective of the RCF. Beginning in 2010, fatalities recorded as death, indirect report were no longer reviewed by the America's Poison Centers fatality review team and the RCF is determined by the reporting PC. [Table 19C](#) indicates which cases are included in 12 tables reporting fatalities.

Table 4. Distribution of age^a and gender for fatalities^b

Age (y)	Male	Female	Unknown	Total (%)	Cumulative total (%)
<1 year	1	0	0	1 (0.08%)	1 (0.08%)
1 year	2	3	0	5 (0.38%)	6 (0.46%)
2 years	3	1	0	4 (0.31%)	10 (0.77%)
3 years	4	0	0	4 (0.31%)	14 (1.08%)
4 years	0	0	0	0 (0.00%)	14 (1.08%)
5 years	1	0	0	1 (0.08%)	15 (1.15%)
Child 6–12 years	6	4	0	10 (0.77%)	25 (1.92%)
Teen 13–19 years	11	47	0	58 (4.46%)	83 (6.38%)
20–29 years	62	55	0	117 (8.99%)	200 (15.37%)
30–39 years	87	78	1	166 (12.76%)	366 (28.13%)
40–49 years	85	94	0	179 (13.76%)	545 (41.89%)
50–59 years	125	114	0	239 (18.37%)	784 (60.26%)
60–69 years	94	133	0	227 (17.45%)	1,011 (77.71%)
70–79 years	72	89	0	161 (12.38%)	1,172 (90.08%)
80–89 years	36	58	0	94 (7.23%)	1,266 (97.31%)
≥90 years	14	16	0	30 (2.31%)	1,296 (99.62%)
Unknown adult	4	0	1	5 (0.38%)	1,301 (100.00%)
Unknown age	0	0	0	0 (0.00%)	1,301 (100.00%)
Total	607	692	2	1,301 (100.00%)	1,301 (100.00%)

^aAge includes cases with both actual and estimated ages as shown in Table 21.

^bIncludes cases with RCF of 1–Undoubtedly responsible, 2–Probably responsible, or 3–Contributory. This excludes reports with outcome of Death, indirect.

Table 5. Number of substances involved in human exposure cases

No. of substances	Human exposures		Fatal exposures ^a	
	N	%	N	%
1	1,831,206	87.50	601	46.20
2	165,194	7.89	319	24.52
3	52,353	2.50	172	13.22
4	21,645	1.03	104	7.99
5	10,193	0.49	48	3.69
6	4,961	0.24	26	2.00
7	2,782	0.13	16	1.23
8	1,705	0.08	6	0.46
≥9	2,650	0.13	9	0.69
Total	2,092,689	100.00	1,301	100.00

^aIncludes cases with RCF of 1–Undoubtedly responsible, 2–Probably responsible, or 3–Contributory. This excludes reports with outcome of Death, indirect.

There were 1,797 deaths, and 1,012 indirect deaths. Of these 2,809 cases, 2,271 were judged exposure-related fatalities (RCF = 1 – Undoubtedly responsible, 2 – Probably responsible, or 3 – Contributory). The remaining 538 cases were judged as follows: 117 as RCF 4 – Probably not responsible, 13 as RCF 5 – Clearly not responsible, and 408 as RCF 6 – Unknown. Deaths are sorted in Table 21 (Appendix A) according to the category, then substance deemed most likely responsible for the death (Cause Rank), and then by patient age. The Cause Rank permits the PC to judge 2 or more substances as indistinguishable in terms of cause, for example, 2 substances which appear equally likely to have caused the death could have Substance Rank of 1,2 and Cause Rank of 1,1. Additional agents implicated in the death are listed below the primary agent in the order of their contribution to the fatality.

As shown in Table 5, a single substance was implicated in 87.5% of reported human exposures, and

Table 6A. Reason for human exposure cases

Reason	N	% Human exposures
Unintentional		
Unintentional – General	925,442	44.20
Unintentional – Therapeutic error	361,402	17.30
Unintentional – Misuse	189,728	9.10
Unintentional – Environmental	59,977	2.90
Unintentional – Occupational	35,365	1.70
Unintentional – Bite/sting	34,769	1.70
Unintentional – Food poisoning	16,275	0.80
Unintentional – Unknown	5,062	0.20
Subtotal	1,628,020	77.80
Intentional		
Intentional – Suspected suicide	258,441	12.30
Intentional – Misuse	55,971	2.70
Intentional – Abuse	42,049	2.00
Intentional – Unknown	14,799	0.70
Subtotal	371,260	17.70
Adverse Reaction		
Adverse reaction – Drug	37,148	1.80
Adverse reaction – Other	9,266	0.40
Adverse reaction – Food	6,139	0.30
Subtotal	52,553	2.50
Unknown		
Unknown reason	21,373	1.00
Subtotal	21,373	1.00
Other		
Other – Malicious	8,252	0.40
Other – Contamination / tampering	8,161	0.40
Other – Withdrawal	3,070	0.10
Subtotal	19,483	0.90
Total	2,092,689	100.00

12.5% of patients were exposed to ≥2 substances. The exposure-related fatalities involved a single substance in 601 cases (46.2%), 2 substances in 319 cases (24.5%), 3 in 172 cases (13.2%), and 4 or more in the balance of cases.

In Table 21 (Appendix A), the Annual Report ID number [bracketed] indicates that the abstract for that case is included in Appendix C. The letters following the Annual Report ID number indicate: i=death,

Table 6B. Scenarios for therapeutic errors^a by age^b

Scenario	N	≤5 y (Row %)	6–12 y (Row %)	13–19 y (Row %)	≥20 y (Row %)	Unknown child (Row %)	Unknown adult (Row %)	Unknown age (Row %)
Inadvertently took/ given medication twice	113,299	11.80	10.01	5.53	65.96	0.07	6.37	0.26
Other incorrect dose	59,633	24.82	10.42	6.88	52.90	0.07	4.62	0.29
Wrong medication taken/given	52,634	15.56	10.51	5.86	62.19	0.05	5.54	0.29
Medication doses given/taken too close together	40,996	12.45	8.30	6.14	66.28	0.08	6.51	0.24
Inadvertently took/ given someone else's medication	30,107	11.61	16.73	6.65	59.77	0.05	4.97	0.22
Other/unknown therapeutic error	23,524	16.12	10.42	7.29	58.58	0.07	6.96	0.55
Incorrect dosing route	19,586	7.07	3.25	7.98	69.36	0.07	11.42	0.86
More than 1 product containing same ingredient	7,640	7.74	13.19	12.98	58.72	0.10	6.95	0.31
Health professional/ iatrogenic error (pharmacist/nurse/ physician)	6,892	19.04	10.53	6.54	59.14	0.16	4.16	0.42
Confused units of measure	6,838	47.12	12.52	4.01	32.71	0.12	3.33	0.19
Drug interaction	6,716	4.66	6.80	6.76	67.32	0.15	13.58	0.73
Incorrect formulation or concentration given	6,404	48.17	17.88	4.26	26.97	0.08	2.48	0.16
10-fold dosing error	3,108	52.32	5.73	2.03	37.87	0.13	1.77	0.16
Dispensing cup error	3,006	62.31	15.37	3.03	17.60	0.07	1.63	0.00
Incorrect formulation or concentration dispensed	1,442	44.31	13.80	4.92	32.66	0.21	3.68	0.42
Exposure through breast milk	229	93.89	0.00	0.00	3.06	1.31	1.31	0.44

^aAll cases with a scenario category of therapeutic error regardless of reason^bOf the human exposure cases reported to U.S. Poison Centers in 2024, 512,200 (24.5%) were coded to 1 or more of 54 scenarios.

indirect report (occurred in 970, 42.7% of cases), p=prehospital cardiac and/or respiratory arrest (occurred in 373, 16.4% of cases), h=hospital records reviewed (occurred in 1,061, 46.7% of cases), a=autopsy report reviewed (occurred in 1,126, 49.6% of cases). The distribution of NPDS RCF was: 1 = Undoubtedly responsible in 1,303 cases (57.4%), 2 = Probably responsible in 742 cases (32.7%), 3 = Contributory in 226 cases (9.95%). The denominator for these Table 21 percentages is 2,271.

All fatalities – all ages

Table 4 presents the age and gender distribution for the 1,301 exposure-related fatalities (excluding death, indirect). The age distribution of reported fatalities showed a decrease in deaths among children (<20 years old) compared to 2023, with 83 cases representing 6.38% of fatalities. This was an absolute decrease of 9 fatalities (9.78%) in that age group. The age distribution of reported fatalities in adults (≥20 years) was similar to prior years with 1,218 of 1,301 (93.6%) fatal cases occurring in that age group. While children ≤5

years old were involved in most exposures, the deaths in this group comprised just 1.15% of the exposure-related fatalities. The number of deaths in this age group decreased by 1 from 2023. Most (53.9%) of the fatalities occurred in 20 to 59-year-old individuals, in line with prior years, although 39.4% occurred in older individuals (>60 years old). Exposures causing death in children ≤5 years of age were mostly coded as Unintentional general, while those in ages ≥13 years were coded mostly as Intentional suspected suicide.

Table 21 (Appendix A) lists 2,271 human fatalities (RCF 1, 2, 3; including death, indirect) along with all the substances involved for each case. The headers in column 1 are the major categories of nonpharmaceutical or pharmaceutical the exposure substance belongs to. The substance listed in column 3 of Table 21 (alternate name) was chosen to be the most specific generic name based upon the Poisindex product name and generic code selected for that substance (for example, column 1 may have the major category of alcohols and column 3 will list the specific substance, methanol). Alternate names are maintained in NPDS for each substance

Table 7. Distribution of reason for exposure by age

Reason	≤5 y		6–12 y		13–19 y		≥20 y		Unknown child		Unknown adult		Unknown age		Total	
	N	Row %	N	Row %	N	Row %	N	Row %	N	Row %	N	Row %	N	Row %	N	%
Unintentional	809,285	52.7	123,627	8.1	63,177	4.1	530,937	34.6	4,280	0.3	87,796	5.7	8,918	0.6	1,628,020	77.8
Intentional	39	0.0	18,707	5.1	107,002	29.3	236,877	64.9	217	0.1	6,092	1.7	2,326	0.6	371,260	17.7
Adverse reaction	3,000	6.5	2,708	5.9	3,684	8.0	36,226	78.4	124	0.3	6,191	13.4	620	1.3	52,553	2.5
Unknown	2,244	11.2	1,436	7.2	2,390	11.9	13,076	65.2	102	0.5	1,219	6.1	906	4.5	21,373	1.0
Other	1,297	7.9	1,671	10.2	1,440	8.8	11,429	69.8	108	0.7	2,998	18.3	540	3.3	19,483	0.9
Total	815,865	41.1	148,149	7.5	177,693	9.0	828,545	41.8	4,831	0.2	104,296	5.3	13,310	0.7	2,092,689	100.0

Table 8. Distribution of reason for exposure and age for fatalities^a

Reason	≤5 y	6–12 y	13–19 y	≥20 y	Unknown child	Unknown adult	Unknown age	Total
Unintentional								
Unintentional – General	7	1	0	20	0	0	0	28
Unintentional – Environmental	4	5	0	28	0	1	0	38
Unintentional – Occupational	0	0	0	5	0	0	0	5
Unintentional – Therapeutic error	0	0	0	46	0	0	0	46
Unintentional – Misuse	0	0	0	20	0	0	0	20
Unintentional – Bite/sting	0	0	0	6	0	0	0	6
Unintentional – Unknown	1	0	0	11	0	0	0	12
Subtotal	12	6	0	136	0	1	0	155
Intentional								
Intentional – Suspected suicide	0	2	47	659	0	0	0	708
Intentional – Misuse	0	0	0	61	0	0	0	61
Intentional – Abuse	0	1	4	105	0	2	0	112
Intentional – Unknown	0	0	1	68	0	1	0	70
Subtotal	0	3	52	893	0	3	0	951
Other								
Other – Malicious	2	0	0	1	0	0	0	3
Other – Withdrawal	0	0	0	3	0	0	0	3
Subtotal	2	0	0	4	0	0	0	6
Adverse reaction								
Adverse reaction – Drug	0	1	0	45	0	0	0	46
Adverse reaction – Food	0	0	0	1	0	0	0	1
Adverse reaction – Other	0	0	0	2	0	0	0	2
Subtotal	0	1	0	48	0	0	0	49
Unknown								
Unknown reason	1	0	6	132	0	1	0	140
Subtotal	1	0	6	132	0	1	0	140
Total	15	10	58	1,213	0	5	0	1,301

^aIncludes cases with RCF of 1–Undoubtedly responsible, 2–Probably responsible, or 3–Contributory. This excludes reports with outcome of Death, indirect.

Table 9. Route of exposure for human exposure cases

Route	Human exposures			Fatal exposures ^a		
	N	% of All routes	% of All Cases	N	% of All routes	% of All cases
Ingestion	1,716,059	77.92	82.00	1,075	75.70	82.63
Inhalation/nasal	166,794	7.57	7.97	97	6.83	7.46
Dermal	144,121	6.54	6.89	13	0.92	1.00
Ocular	86,874	3.94	4.15	1	0.07	0.08
Bite/sting	34,829	1.58	1.66	5	0.35	0.38
Parenteral	24,058	1.09	1.15	39	2.75	3.00
Unknown	21,530	0.98	1.03	174	12.25	13.37
Other	3,446	0.16	0.16	2	0.14	0.15
Otic	1,534	0.07	0.07	0	0.00	0.00
Vaginal	1,322	0.06	0.06	0	0.00	0.00
Rectal	923	0.04	0.04	1	0.07	0.08
Aspiration (with ingestion)	747	0.03	0.04	13	0.92	1.00
Total number of routes	2,202,237	100.00	105.23^b	1,420	100.00	109.15^b

^aIncludes cases with RCF of 1–Undoubtedly responsible, 2–Probably responsible, or 3–Contributory. This excludes reports with outcome of death, indirect.

^bEach exposure case may have more than one route.

involved in a fatality. The cross-references at the end of each major category section in Table 21 list all cases that identify the substance as other than the primary

substance. This alternate name may not agree with the America's Poison Centers generic categories used in the summary tables (including Appendix B (Table 22)).

Table 10. Management site of human exposures

Site of management	N	%
Managed on site, non-health care facility	1,393,810	66.60
Managed in healthcare facility		
Treated/evaluated and released	315,743	15.09
Admitted to psychiatric facility	90,262	4.31
Admitted to noncritical care unit	82,789	3.96
Patient lost to follow-up / left AMA	81,798	3.91
Admitted to critical care unit	69,628	3.33
Subtotal (managed in HCF)	640,220	30.59
Other	20,737	0.99
Refused referral	24,947	1.19
Unknown	12,975	0.62
Total	2,092,689	100.00

Table 18 lists the top 25 minor generic substance categories associated with reported fatalities and the number of single substance exposure fatalities for that category. The 10 substance categories associated most frequently with fatalities were acetaminophen alone, miscellaneous sedative/hypnotics/antipsychotics, miscellaneous alcohols, calcium antagonist, miscellaneous stimulants and street drugs, pharmaceutical and illegal opioid preparations, beta blockers, miscellaneous antidepressants, single agent hypoglycemic, and miscellaneous unknown drug. Note that Table 18 is sorted by all substances to which a patient was exposed (i.e., a patient exposed to an opioid may have also been exposed to 1 or more other products) and shows single substance exposures in the right-hand column.

Table 21 contains 2,271 fatalities, of which the first ranked substances were pharmaceutical for 1,953 (86.0%). The text below is representative of the most common groups of pharmaceuticals included; the complete list is in Table 21:

948 analgesics (492 fentanyl (non-prescription), 210 acetaminophen, 58 opioid, N.O.S., 30 oxycodone, 24 salicylate, 20 acetaminophen/hydrocodone, 17 fentanyl (prescription), 14 heroin, 12 acetaminophen/oxycodone, 12 morphine, 10 methadone, 8 acetaminophen/diphenhydramine, 6 colchicine, 6 tramadol, 4 hydrocodone, N.O.S., 3 fentanyl analogs (para-fluoro), 3 ibuprofen).

335 stimulants/street drugs (241 methamphetamine, 72 cocaine, 5 amphetamine, 4 *Mitragyna speciosa korthals*, 3 stimulant/street drug, N.O.S., 2 amphetamine/dextroamphetamine, 2 lysergic acid diethylamide (LSD), 1 amphetamine (hallucinogenic), 1 3,4-methylenedioxymethamphetamine (MDMA), 1 lisdexamfetamine (ER), 1 marijuana N.O.S., 1 methylphenidate, 1 mushroom (processed, N.O.S.), 1 street drug, N.O.S.).

Table 11. Medical outcome of human exposure cases by patient age^a

Outcome	≤5 y		6–12 y		13–19 y		≥20 y		Unknown child		Unknown adult		Unknown age		Total	
	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%
No effect	164,935	20.22	24,143	16.30	31,047	17.47	103,308	12.47	607	12.56	9,154	8.78	1,220	9.17	334,414	15.98
Minor effect	64,221	7.87	20,209	13.64	49,251	27.72	169,921	20.51	330	6.83	10,963	10.51	1,236	9.29	316,131	15.11
Moderate effect	10,530	1.29	5,862	3.96	32,225	18.14	118,766	14.33	52	1.08	1,903	1.82	297	2.23	169,635	8.11
Major effect	1,497	0.18	524	0.35	4,555	2.56	27,946	3.37	3	0.06	161	0.15	48	0.36	34,734	1.66
Death	27	0.00	14	0.01	73	0.04	1,674	0.20	0	0.00	7	0.01	2	0.02	1,797	0.09
No follow-up, nontoxic	103,855	12.73	16,671	11.25	6,187	3.48	40,739	4.92	722	14.95	11,434	10.96	960	7.21	180,568	8.63
No follow-up, minimal toxicity	444,249	54.45	73,809	49.82	40,349	22.71	292,800	35.34	2,597	53.76	57,671	55.30	5,283	39.69	916,758	43.81
No follow-up, potentially toxic	16,496	2.02	3,925	2.65	9,578	5.39	40,312	4.87	452	9.36	9,358	8.97	3,859	28.99	83,980	4.01
Unrelated effect	10,041	1.23	2,985	2.01	4,410	2.48	32,136	3.88	67	1.39	3,632	3.48	389	2.92	53,660	2.56
Death, indirect report	14	0.00	7	0.00	18	0.01	943	0.11	1	0.02	13	0.01	16	0.12	1,012	0.05
Total	815,865	100.00	148,149	100.00	177,693	100.00	828,545	100.00	4,831	100.00	104,296	100.00	13,310	100.00	2,092,689	100.00

^aTotal number of cases where Death was an outcome (1,797 + 1,012) is greater than the number of fatalities (1,301) judged to be exposure-related (RCF of 1–Undoubtedly responsible, 2–Probably responsible, or 3–Contributory).

Table 12. Medical outcome by reason for exposure in human exposures^a

Outcome	Unintentional		Intentional		Other		Adverse reaction		Unknown		Total	
	N	%	N	%	N	%	N	%	N	%	N	%
No effect	267,232	16.41	61,360	16.53	1,915	9.83	2,403	4.57	1,504	7.04	334,414	15.98
Minor effect	193,537	11.89	105,325	28.37	3,596	18.46	11,331	21.56	2,342	10.96	316,131	15.11
Moderate effect	48,268	2.96	107,464	28.95	1,652	8.48	7,589	14.44	4,662	21.81	169,635	8.11
Major effect	4,375	0.27	26,299	7.08	304	1.56	1,217	2.32	2,539	11.88	34,734	1.66
Death	208	0.01	1,157	0.31	11	0.06	111	0.21	310	1.45	1,797	0.09
No follow-up, nontoxic	173,805	10.68	3,920	1.06	978	5.02	1,397	2.66	468	2.19	180,568	8.63
No follow-up, minimal toxicity	859,484	52.79	28,830	7.77	6,848	35.15	17,709	33.70	3,887	18.19	916,758	43.81
No follow-up, potentially toxic	47,136	2.90	27,130	7.31	2,660	13.65	3,506	6.67	3,548	16.60	83,980	4.01
Unrelated effect	33,927	2.08	8,852	2.38	1,516	7.78	7,290	13.87	2,075	9.71	53,660	2.56
Death, indirect report	48	0.00	923	0.25	3	0.02	0	0.00	38	0.18	1,012	0.05
Total	1,628,020	100.00	371,260	100.00	19,483	100.00	52,553	100.00	21,373	100.00	2,092,689	100.00

^aTotal number of cases where Death was an outcome (1,797+1,012) is greater than the number of fatalities (1,301) judged to be exposure-related (RCF of 1-Undoubtedly responsible, 2-Probably responsible, or 3-Contributory).

Table 13. Duration of clinical effects by medical outcome

Duration of effect	Minor effect		Moderate effect		Major effect	
	N	%	N	%	N	%
≤2 Hours	84,179	26.63	5,213	3.07	586	1.69
>2 Hours, ≤8 hours	99,061	31.34	39,430	23.24	3,032	8.73
>8 Hours, ≤24 hours	59,162	18.71	62,488	36.84	7,899	22.74
>24 Hours, ≤3 days	16,306	5.16	31,337	18.47	11,492	33.09
>3 Days, ≤1 week	4,262	1.35	8,172	4.82	6,143	17.69
>1 Week, ≤1 month	1,257	0.40	2,022	1.19	1,800	5.18
>1 Month	398	0.13	351	0.21	180	0.52
Anticipated permanent	110	0.03	114	0.07	403	1.16
Unknown	51,396	16.26	20,508	12.09	3,199	9.21
Total	316,131	100.00	169,635	100.00	34,734	100.00

261 cardiovascular drugs (118 amlodipine, 18 digoxin, 18 propranolol, 17 metoprolol, 16 diltiazem, 12 carvedilol, 11 verapamil, 7 flecainide, 5 diltiazem (ER), 4 beta blocker, N.O.S., 4 calcium antagonist, N.O.S., 4 clonidine, 4 metoprolol (ER), 3 nifedipine (ER), 3 verapamil (ER)).

111 antidepressants (36 bupropion, 16 amitriptyline, 13 bupropion (ER), 8 lithium, 6 venlafaxine, 5 trazodone, 3 doxepin, 3 escitalopram, 3 tianeptine, 2 citalopram, 2 cyclic antidepressant, N.O.S., 2 duloxetine (ER), 2 lithium (ER), 2 paroxetine, 2 sertraline, 2 venlafaxine (ER)).

62 sedative/hypnotic/antipsychotics (16 quetiapine, 14 alprazolam, 5 clonazepam, 3 benzodiazepine, N.O.S., 3 lorazepam, 2 diazepam, 2 haloperidol, 2 propofol, 2 temazepam, 2 zolpidem).

54 unknown drug (54 drug, N.O.S.).

50 hormones and hormone antagonists (33 metformin, 5 insulin, N.O.S., 3 insulin (lispro), 3

Table 14. Decontamination and therapeutic interventions

Therapy	N	%
Decontamination only	949,321	45.36
Therapeutic intervention only	251,051	12.00
Decontamination and therapeutic intervention	96,734	4.62
Not coded	795,583	38.02
Total	2,092,689	100.00

metformin (ER), 2 semaglutide, 1 glimepiride, 1 insulin (detemir) (ER), 1 insulin (regular), 1 metformin/sitagliptin).

The exposure was acute (A) in 708 (31.2%), acute on chronic (A/C) in 368 (16.2%), chronic (C) in 124 (5.46%), and unknown (U) in 1,071 (47.2%) of fatalities.

A total of 1,440 tissue concentrations for 1 or more related analytes were reported in 736 cases. Most of these (1,310) involved fatalities with RCF of 1-3 and are listed in [Appendix A \(Table 21\)](#). The most frequent of the 141 analytes included: 378 acetaminophen (APAP), 125 ethanol, 58 salicylate, 45 carboxyhemoglobin, 31 lithium, 26 ethylene glycol, 24 fentanyl, 24 gabapentin, 23 methanol, 22 methamphetamine, 22 amphetamine, 21 bupropion, 20 amlodipine, 20 digoxin.

Route of exposure was ingestion only in 1,000 cases (44.0%), inhalation/nasal only in 113 cases (4.98%), and parenteral only in 23 cases (1.01%). Most other exposures recorded a combination of routes or an unknown route.

The Intentional exposure reason was abuse in 944 cases (41.6%), suspected suicide in 773 cases (34.0%), unknown in 74 cases (3.26%), and misuse in 66 cases (2.91%). Unintentional exposure reasons were environmental in 56 cases (2.47%), therapeutic error in 48 cases (2.11%), general in 43 cases (1.89%), misuse in 22 cases (0.969%), unknown in 13 cases (0.572%), bite/sting in 6 cases (0.264%), and occupational in 5 cases (0.220%). Adverse drug reaction was the reason in 49 cases (2.03%).

Table 15. Therapy provided in human exposures by age

Therapy	≤5 y	6–12 y	13–19 y	≥20 y	Unknown child	Unknown adult	Unknown age	Total
Decontamination								
Cathartic	137	68	315	719	0	11	0	1,250
Charcoal, multiple doses	27	17	193	413	0	2	0	652
Charcoal, single dose	3,416	1,014	9,602	16,881	5	73	16	31,007
Dilute/irrigate/wash	346,114	49,566	27,184	195,352	1,626	34,634	2,294	656,770
Food/snack	105,166	13,141	6,714	38,480	665	5,923	352	170,441
Fresh air	6,248	3,743	4,773	49,208	656	14,194	1,577	80,399
Ipecac	7	0	2	28	0	0	1	38
Lavage	17	4	53	193	0	1	2	270
Other emetic	6,209	873	1,591	8,302	11	817	58	17,861
Whole bowel irrigation	104	29	187	691	0	0	1	1,012
Other therapies								
2-PAM	4	1	4	46	0	0	0	55
Alkalinization - Systemic	48	53	855	4,807	0	12	1	5,776
Alkalinization - Urinary	24	34	505	1,353	0	1	1	1,918
Amyl nitrite	0	0	0	2	0	0	0	2
Anthrax vaccine	0	0	1	0	0	0	0	1
Antiarrhythmic	9	9	87	892	0	1	0	998
Antibiotics	1,582	809	1,505	15,311	7	308	42	19,564
Anticonvulsants ^a	222	59	564	2,984	0	6	2	3,837
Antiemetics	1,869	1,483	12,123	24,789	3	100	14	40,381
Antifungals	6	2	12	95	0	1	0	116
Antihistamines	1,304	997	1,807	9,239	23	684	52	14,106
Antihypertensives	28	20	192	3,328	0	6	1	3,575
Antipsychotics	10	68	1,315	7,497	0	29	6	8,925
Antivenom (Immune Fab fragment) – Not Specified	133	249	280	2,296	0	2	3	2,963
Antivenom (Immune Fab) - Centruroides (Scorpion)	106	33	10	170	0	1	1	321
Antivenom (Immune Fab) - Latrodectus	0	1	0	15	0	0	0	16
Antivenom - Elapidae	0	2	2	25	0	0	0	29
Antivenom - Latrodectus	3	4	1	32	0	0	0	40
Antivenom/antitoxin (Non-Fab) – not specified	7	8	31	175	0	1	0	222
Antivirals	21	5	12	141	0	1	0	180
Atropine	150	67	268	1,926	0	5	0	2,416
BAL	1	0	2	3	0	0	0	6
Benzodiazepines	1,211	844	9,805	35,593	1	94	34	47,582
Blood products	56	21	92	1,216	0	6	1	1,392
Botulinum antitoxin	11	0	3	34	0	0	0	48
Bronchodilators	309	168	375	4,515	6	154	21	5,548
Calcium	2,573	367	685	5,343	0	45	6	9,019
Cardioversion	6	2	38	309	0	1	1	357
Colony stimulating factors	0	0	0	16	0	0	0	16
Continuous renal replacement therapy (CRRT)	12	7	96	1,033	0	2	1	1,151
CPR	120	29	202	1,613	0	10	2	1,976
Deferoxamine	8	4	58	32	0	0	0	102
Digoxin immune fab	4	0	5	351	0	0	0	360
Direct-acting oral anticoagulant reversal agents	1	0	1	18	0	0	0	20
DMPs	0	0	0	1	0	0	0	1
DTPA – calcium	0	0	0	1	0	0	0	1
ECMO	9	4	45	159	0	0	0	217
EDTA	24	3	5	5	0	0	0	37
Ethanol	0	0	1	22	0	2	0	25
Extracorp. procedure (other)	3	1	11	46	0	1	0	62
Fluids, IV	6,521	3,839	32,108	102,437	0	190	55	145,150
Flumazenil	28	11	74	634	0	1	0	748
Folate	7	4	64	3,269	0	3	1	3,348
Fomepizole	99	35	363	2,686	0	4	2	3,189
Glucagon	30	14	152	2,017	0	1	0	2,214
Glucarpidase	0	0	1	7	0	0	0	8
Glucose, > 5%	456	75	489	5,178	0	4	2	6,204

(Continued)

Table 15. Continued.

Therapy	≤5 y	6–12 y	13–19 y	≥20 y	Unknown child	Unknown adult	Unknown age	Total
Granulocyte stimulating growth factors	0	0	0	8	0	0	0	8
Hemodialysis	6	3	97	2,212	0	2	3	2,323
Hemoperfusion	0	0	2	15	0	0	0	17
High dose insulin/glucose	2	7	46	704	0	0	0	759
Hydroxocobalamin	12	17	14	274	0	2	3	322
Hyperbaric oxygen	27	32	41	314	0	7	1	422
Hypothermia Protocol	18	8	18	231	0	0	0	275
Insulin	17	11	165	2,370	0	3	1	2,567
Intubation	466	150	1,775	15,501	1	60	27	17,980
L-Carnitine	8	11	79	476	0	0	0	574
Leucovorin	6	2	19	228	0	0	0	255
Lipid emulsion therapy	14	6	69	400	0	1	0	490
Magnesium	47	179	2,872	14,354	0	12	1	17,465
Methylene blue	17	11	53	551	0	2	2	636
Molecular adsorbant recirculating system (MARS)	1	0	5	9	0	0	0	15
NAC, IV	346	622	8,207	16,847	0	18	4	26,044
NAC, PO	6	21	274	596	0	1	2	900
Nalmefene	1	0	2	12	0	0	0	15
Naloxone	1,427	327	2,249	18,122	2	125	36	22,288
Neuromuscular blocker	59	17	254	1,787	0	3	0	2,120
Octreotide	78	5	41	539	0	0	0	663
Opioid analgesia	352	261	1,020	10,463	0	30	9	12,135
Other	17,704	4,677	7,893	48,579	48	3,020	401	82,322
Oxygen	2,089	804	3,808	40,496	7	163	64	47,431
Pacemaker	1	1	6	262	0	1	0	271
Penicillamine	0	0	0	4	0	1	0	5
Physostigmine	2	11	76	103	0	0	0	192
Phytonadione	29	13	171	643	0	1	0	857
Plasmapheresis	4	3	13	42	0	0	0	62
Potassium	398	459	5,675	22,051	0	16	9	28,608
Potassium iodide	0	2	12	46	0	0	0	60
Propofol	98	56	1,082	10,410	0	30	14	11,690
Prussian blue	0	0	0	5	0	0	1	6
Pyridoxine	13	1	28	530	0	1	1	574
Rabies immune globulin	49	69	57	358	0	15	4	552
Rabies vaccine	61	85	67	453	0	20	6	692
Raxibacumab	1	2	0	5	0	0	0	8
Rivastigmine	0	1	4	4	0	0	0	9
Sedation (other)	398	187	1,923	11,723	0	34	15	14,280
Silibinin	1	0	0	8	0	0	0	9
Smallpox vaccine	1	0	0	1	0	0	0	2
Sodium bicarbonate - metabolic acidosis	50	26	446	3,112	0	5	0	3,639
Sodium bicarbonate - nebulized	5	3	24	187	0	2	1	222
Sodium nitrite	2	1	1	9	0	0	0	13
Sodium thiosulfate	2	1	1	27	0	0	0	31
Steroids	508	324	514	5,116	5	289	29	6,785
Succimer	183	16	12	45	0	1	0	257
Surgical intervention	193	100	106	793	0	6	0	1,198
Thiamine	20	5	159	6,002	0	8	4	6,198
Transplantation	1	3	8	16	0	0	0	28
Vasopressors	179	120	862	10,558	2	23	10	11,754
Ventilation, Non-invasive (CPAP, BiPAP)	104	29	142	2,291	0	11	1	2,578
Ventilator	496	153	1,907	16,477	1	63	24	19,121

^aExcludes benzodiazepines.**Pediatric fatalities – age ≤5 years**

Although children younger than 6 years were involved in most exposures, they comprised only 41 (1.50%) of the 2,809 fatalities. These numbers reflect those reported since 2010 (Table 19A, all RCFs and include death, indirect reports). Table 8 (RCF 1, 2 or 3, excludes

death, indirect reports) shows the percentage of fatalities in children ≤5 years related to total pediatric exposures was 15/815,865 (0.00184%). By comparison, 1,218/932,841 (0.131%) of all adult exposures involved a fatality. Of the 15 pediatric fatalities in Table 8, 12 (80.0%) were reported as unintentional, 2 (13.3%) had an other-malicious reason, and 1 (6.67%) had an

Table 16. Decontamination trends for human and pediatric exposures ≤5 years: last decade

	Total exposures			Activated charcoal			Cathartic			Ipecac			Lavage			Other emetic			Whole bowel irrigation		
	Human		Children	Human		Exposures children	Human		Exposures children	Human		Exposures children	Human		Exposures children	Human		Exposures children	Human		Exposures children
	N	%	N	N	%	N	N	%	N	N	%	N	N	%	N	N	%	N	N	%	N
2015	2,168,371	1.017	369	42,712	1.97	6,965	6,510	0.30	495	0.05	0.05	105	0.00	0.00	29	0.00	0.00	14,310	1,683	0.08	53
2016	2,159,032	1.002	344	40,633	1.88	6,333	4,985	0.23	376	0.04	0.04	88	0.00	0.00	22	0.00	0.00	14,974	1,653	0.08	61
2017	2,115,186	956	871	39,985	1.89	5,743	5,134	0.24	327	0.03	0.03	64	0.00	0.00	12	0.00	0.00	14,534	1,640	0.08	60
2018	2,099,751	927	487	36,807	1.75	5,006	3,964	0.19	275	0.03	0.03	63	0.00	0.00	21	0.00	0.00	15,300	1,562	0.07	60
2019	2,148,141	920	148	35,541	1.65	4,633	2,731	0.13	202	0.02	0.02	61	0.00	0.00	14	0.00	0.00	15,836	1,517	0.07	85
2020	2,128,198	886	342	33,654	1.58	4,007	2,045	0.10	125	0.01	0.01	70	0.00	0.00	16	0.00	0.00	17,941	1,131	0.05	70
2021	2,080,917	846	482	35,819	1.72	3,936	1,958	0.09	152	0.02	0.02	47	0.00	0.00	8	0.00	0.00	18,389	1,105	0.05	82
2022	2,064,875	30	754	33,194	1.61	3,466	1,805	0.09	125	0.02	0.02	50	0.00	0.00	3	0.00	0.00	17,836	1,006	0.05	73
2023	2,080,659	828	669	33,021	1.59	3,514	1,414	0.07	129	0.02	0.02	37	0.00	0.00	8	0.00	0.00	17,751	1,027	0.05	100
2024	2,092,689	815	865	31,659	1.51	3,443	1,250	0.06	137	0.02	0.02	38	0.00	0.00	7	0.00	0.00	17,861	1,012	0.05	104

Table 17A. Substance categories most frequently involved in human exposures (top 25)

Substance (Major Generic Category)	All substances	% ^a	Single substance exposures	% ^b
Analgesics	268,217	10.53	171,957	9.39
Cleaning substances (household)	176,818	6.94	152,155	8.31
Antidepressants	140,007	5.50	63,078	3.44
Cardiovascular drugs	130,329	5.12	52,014	2.84
Cosmetics/Personal care products	126,478	4.96	120,164	6.56
Antihistamines	118,006	4.63	75,789	4.14
Sedative/Hypnotics/Antipsychotics	111,264	4.37	39,904	2.18
Foreign Bodies/Toys/Miscellaneous	99,916	3.92	96,367	5.26
Stimulants and Street Drugs	80,028	3.14	50,098	2.74
Dietary supplements/Herbals/Homeopathic	78,277	3.07	66,535	3.63
Pesticides	76,997	3.02	71,977	3.93
Alcohols	68,660	2.70	21,229	1.16
Hormones and hormone antagonists	68,001	2.67	45,604	2.49
Anticonvulsants	64,209	2.52	24,981	1.36
Vitamins	61,122	2.40	48,129	2.63
Topical preparations	60,538	2.38	58,725	3.21
Plants	54,556	2.14	52,229	2.85
Cold and cough preparations	52,395	2.06	33,900	1.85
Chemicals	51,918	2.04	43,256	2.36
Antimicrobials	51,001	2.00	40,019	2.19
Gastrointestinal preparations	50,348	1.98	34,776	1.90
Fumes/Gases/Vapors	47,596	1.87	44,298	2.42
Bites and envenomations	39,229	1.54	38,503	2.10
Electrolytes and minerals	39,155	1.54	30,673	1.68
Miscellaneous foods	30,753	1.21	27,514	1.50

^aPercentages are based on the total number of substances reported in all exposures (N=2,547,434)^bPercentages are based on the total number of single substance exposures (N=1,831,206)**Table 17B.** Substance categories with the greatest rate of exposure increase (top 25)

Substance (major generic category)	Increase in serious exposures per year ^a		All substances serious exposures in 2024
	Mean	95% CI ^b	
Stimulants and street drugs	813	[559, 1068]	27,337
Antihistamines	778	[681, 876]	22,237
Antidepressants	605	[99, 1111]	44,512
Cardiovascular drugs	504	[292, 715]	26,214
Alcohols	425	[126, 725]	27,029
Unknown drug	375	[273, 476]	11,575
Analgesics	122	[-671, 916]	49,943
Fumes/Gases/Vapors	112	[71, 152]	4,714
Mushrooms, naturally occurring	110	[63, 157]	1,616
Hormones and hormone antagonists	97	[13, 182]	7,858
Dietary supplements/Herbals/Homeopathic	94	[48, 140]	2,957
Miscellaneous drugs	86	[52, 119]	3,083
Anesthetics	67	[43, 90]	1,182
Electrolytes and minerals	63	[52, 73]	1,643
Cosmetics/Personal care products	56	[-3, 115]	2,786
Infectious and toxin-mediated diseases	53	[-109, 215]	603
Other/Unknown nondrug substances	50	[-18, 117]	1,447
Chemicals	43	[4, 82]	4,304
Vitamins	40	[30, 50]	1,366
Opioid antagonists	39	[30, 48]	555
Gastrointestinal preparations	38	[-11, 86]	3,593
Serums, toxoids, vaccines	30	[-59, 119]	180
Veterinary drugs	25	[13, 38]	324
Miscellaneous foods	24	[11, 36]	625
Diuretics	17	[-2, 37]	1,656

^aSerious exposures have outcomes of moderate, major or death.^bIncrease and confidence intervals are based on least squares linear regression of the number of cases per year for 2015–2024.

Table 17C. Substance categories most frequently involved in pediatric (≤ 5 years) exposures (top 25)^a

Substance (major generic category)	All substances	% ^b	Single substance exposures	% ^c
Cleaning substances (household)	84,861	9.96	80,899	10.23
Cosmetics/Personal care products	77,923	9.15	76,067	9.62
Analgesics	75,366	8.85	68,844	8.71
Foreign Bodies/Toys/Miscellaneous	67,499	7.92	65,859	8.33
Dietary Supplements/Herbals/Homeopathic	54,054	6.34	51,011	6.45
Vitamins	41,173	4.83	36,229	4.58
Antihistamines	38,556	4.53	34,580	4.37
Topical preparations	33,860	3.97	33,121	4.19
Pesticides	31,286	3.67	30,510	3.86
Plants	30,044	3.53	29,146	3.69
Gastrointestinal preparations	23,438	2.75	21,123	2.67
Electrolytes and minerals	20,503	2.41	18,346	2.32
Antimicrobials	18,349	2.15	17,182	2.17
Cardiovascular drugs	17,549	2.06	11,500	1.45
Stimulants and street drugs	17,229	2.02	15,573	1.97
Cold and cough preparations	15,755	1.85	14,234	1.80
Arts/Crafts/Office supplies	15,466	1.82	14,901	1.88
Tobacco/Nicotine/eCigarette products	13,484	1.58	13,414	1.70
Deodorizers	13,147	1.54	12,983	1.64
Hormones and hormone antagonists	12,883	1.51	10,331	1.31
Essential oils	11,341	1.33	10,766	1.36
Miscellaneous foods	11,195	1.31	10,425	1.32
Chemicals	10,859	1.27	9,843	1.24
Antidepressants	10,469	1.23	7,617	0.96
Alcohols	7,292	0.86	7,041	0.89

^aIncludes all children with actual or estimated ages ≤ 5 years old. Results do not include "Unknown Child" or "Unknown Age."

^bPercentages are based on the total number of substances reported in pediatric exposures ($N=851,928$).

^cPercentages are based on the total number of single substance pediatric exposures ($N=790,612$).

unknown reason. The 26 fatalities in children ≤ 5 years detailed in (Table 21) (includes death, indirect reports and RCF 1-3) included 14 nonpharmaceuticals and 12 pharmaceuticals. The first ranked substances associated with these fatalities included: fumes/gases/vapors (11), analgesics (6), antihistamines (2), chemicals (2), cardiovascular drugs (1), cleaning substances (household) (1), cold and cough preparations (1), stimulants and street drugs (1), and unknown drug (1).

Pediatric fatalities – ages 6–12 years

In the age range 6–12 years, 17 fatalities are listed in Appendix A (Table 21) (includes death, indirect reports and RCF 1-3) included: fumes/gases/vapors (9), chemicals (2), analgesics (1), antidepressants (1), cardiovascular

Table 17D. Substance categories most frequently involved in adult (≥ 20 years) exposures (top 25)^a

Substance (major generic category)	All substances	% ^b	Single substance exposures	% ^c
Analgesics	132,016	10.37	66,088	8.86
Cardiovascular drugs	96,554	7.58	32,157	4.31
Antidepressants	93,287	7.33	37,195	4.98
Sedative/Hypnotics/Antipsychotics	88,040	6.91	28,597	3.83
Cleaning substances (household)	74,838	5.88	57,302	7.68
Alcohols	53,863	4.23	11,328	1.52
Hormones and hormone antagonists	49,139	3.86	31,460	4.22
Anticonvulsants	48,947	3.84	16,550	2.22
Antihistamines	47,044	3.69	21,532	2.89
Pesticides	39,152	3.07	35,430	4.75
Stimulants and street drugs	38,230	3.00	18,265	2.45
Fumes/Gases/Vapors	36,429	2.86	33,706	4.52
Chemicals	33,324	2.62	26,728	3.58
Cosmetics/Personal care products	31,133	2.44	28,103	3.77
Bites and envenomations	25,961	2.04	25,447	3.41
Antimicrobials	24,622	1.93	17,242	2.31
Cold and cough preparations	22,891	1.80	12,114	1.62
Topical preparations	21,670	1.70	20,827	2.79
Gastrointestinal preparations	20,974	1.65	9,924	1.33
Hydrocarbons	19,446	1.53	18,082	2.42
Miscellaneous drugs	18,269	1.43	9,396	1.26
Muscle relaxants	17,703	1.39	6,391	0.86
Unknown drug	16,535	1.30	9,921	1.33
Plants	15,601	1.23	14,628	1.96
Other/Unknown nondrug substances	14,736	1.16	13,264	1.78

^aIncludes all adults with actual or estimated ages ≥ 20 years old. Results also include "Unknown Adult" but do not include "Unknown Age."

^bPercentages are based on the total number of substances reported in adult exposures ($N=1,273,419$).

^cPercentages are based on the total number of single substance adult exposures ($N=746,213$).

drugs (1), hydrocarbons (1), plants (1), and sedative/hypnotics/antipsychotics (1). In the 10 deaths with RCF 1-3 (no death, indirect reports), the exposure reasons were: 5 unintentional – environmental, 2 intentional–suspected suicide, 1 unintentional–general, 1 intentional–abuse, and 1 adverse reaction–drug (Table 8). The relatively small number of pediatric fatality cases in this age range highlights the bimodal distribution of pediatric fatalities.

Adolescent fatalities – ages 13–19 years

In the age range 13–19 years, there were 58 reported fatalities with RCF 1–3 (no death, indirect reports), a decrease of 7 (10.8%) from 2023. Reasons included 52 intentional and 6 unknown (Table 8). The 74 fatalities listed in Appendix A (Table 21) (includes death, indirect reports and RCF 1–3) included 68 pharmaceuticals and

Table 17E. Substance categories most frequently involved in pediatric (≤ 5 years) deaths^a

Substance (Major Generic Category)	All substances	% ^b	Single substance exposures	% ^c
Fumes/Gases/Vapors	16	34.04	14	40.00
Analgesics	6	12.77	6	17.14
Stimulants and Street Drugs	6	12.77	4	11.43
Antihistamines	3	6.38	3	8.57
Chemicals	3	6.38	0	0.00
Unknown Drug	3	6.38	3	8.57
Cardiovascular Drugs	2	4.26	0	0.00
Cosmetics/Personal Care Products	2	4.26	0	0.00
Bites and Envenomations	1	2.13	1	2.86
Cleaning Substances (Household)	1	2.13	1	2.86
Cold and Cough Preparations	1	2.13	1	2.86
Heavy Metals	1	2.13	1	2.86
Miscellaneous Drugs	1	2.13	0	0.00
Serums, Toxoids, Vaccines	1	2.13	1	2.86

^aIncludes all children with actual or estimated ages ≤ 5 years old. Results do not include "Unknown Child" or "Unknown Age." Includes death and death, indirect regardless of RCF.

^bPercentages are based on the total number of substances reported in pediatric fatalities ($N=47$).

^cPercentages are based on the total number of single substance pediatric fatalities ($N=35$).

Table 17F. Substance categories most frequently identified in drug identification requests (top 25)

Substance (major generic category)	All substances	% ^a
Analgesics	4,593	29.04
Sedative/Hypnotics/Antipsychotics	2,344	14.82
Unknown Drug	1,577	9.97
Cardiovascular Drugs	1,472	9.31
Anticonvulsants	811	5.13
Stimulants and Street Drugs	724	4.58
Antidepressants	706	4.46
Antimicrobials	585	3.70
Antihistamines	564	3.57
Muscle Relaxants	551	3.48
Hormones and Hormone Antagonists	402	2.54
Gastrointestinal Preparations	352	2.23
Information Calls	252	1.59
Diuretics	251	1.59
Miscellaneous Drugs	160	1.01
Cold and Cough Preparations	90	0.57
Anticoagulants	62	0.39
Other/Unknown Nondrug Substances	56	0.35
Electrolytes and Minerals	46	0.29
Asthma Therapies	36	0.23
Vitamins	30	0.19
Anticholinergic Drugs	21	0.13
Antineoplastics	16	0.10
Chemicals	15	0.09
Dietary Supplements/Herbals/Homeopathic	13	0.08

^aPercentages are based on the total number of substances reported in all drug identification requests ($N=15,814$).

6 nonpharmaceuticals. The first ranked pharmaceuticals associated with these fatalities included: analgesics

Table 17G. Substance categories most frequently involved in pregnant exposures^a (top 25)

Substance (major generic category)	All substances	% ^b	Single substance exposures	% ^c
Analgesics	942	12.23	602	10.17
Cleaning Substances (Household)	646	8.39	436	7.36
Fumes/Gases/Vapors	609	7.91	574	9.70
Pesticides	396	5.14	362	6.11
Antidepressants	318	4.13	192	3.24
Antihistamines	270	3.51	169	2.85
Electrolytes and Minerals	268	3.48	217	3.67
Miscellaneous Foods	249	3.23	222	3.75
Vitamins	234	3.04	153	2.58
Chemicals	216	2.80	171	2.89
Hormones and Hormone Antagonists	215	2.79	189	3.19
Sedative/Hypnotics/Antipsychotics	213	2.77	98	1.66
Bites and Envenomations	195	2.53	195	3.29
Plants	168	2.18	161	2.72
Cosmetics/Personal Care Products	160	2.08	152	2.57
Antimicrobials	158	2.05	107	1.81
Gastrointestinal Preparations	155	2.01	105	1.77
Foreign Bodies/Toys/Miscellaneous	153	1.99	144	2.43
Stimulants and Street Drugs	136	1.77	64	1.08
Hydrocarbons	136	1.77	132	2.23
Alcohols	133	1.73	66	1.11
Cardiovascular Drugs	124	1.61	79	1.33
Topical Preparations	118	1.53	114	1.93
Other/Unknown Nondrug Substances	114	1.48	100	1.69
Cold and Cough Preparations	107	1.39	66	1.11

^aIncludes all patient classified as pregnant and all female patients with a 'duration of pregnancy' greater than 0.

^bPercentages are based on the total number of substances reported in pregnant exposures ($N=7,702$).

^cPercentages are based on the total number of single substance pregnant exposures ($N=5,920$).

(28), antidepressants (15), stimulants and street drugs (7), antihistamines (3), electrolytes and minerals (3), anticonvulsants (2), sedative/hypnotics/antipsychotics (2), anesthetics (1), cold and cough preparations (1), and hormones and hormone antagonists (1). The first ranked nonpharmaceuticals associated with these fatalities included: chemicals (4), automotive/aircraft/boat products (1), and fumes/gases/vapors (1).

Adult fatalities – ages ≥ 20 years

In the age range ≥ 20 years including adults of unknown age, there were 1,218 reported fatalities with RCF 1–3 (no death, indirect reports), a decrease of 110 (8.3%) from 2023. Reasons included 896 intentional, 137 unintentional, 48 adverse reactions, 4 other and 133 unknown (Table 8). The 2,141 fatalities listed in Appendix A (Table

Table 18. Categories associated with largest number of fatalities (top 25)^a

Substance (minor generic category)	All substances	% ^b	Single substance exposures	% ^c
Acetaminophen Alone	259	9.16	127	21.13
Miscellaneous Sedative/Hypnotics/Antipsychotics	226	8.00	20	3.33
Miscellaneous Alcohols	205	7.25	26	4.33
Calcium Antagonist	191	6.76	39	6.49
Miscellaneous Stimulants and Street Drugs	164	5.80	37	6.16
Pharmaceutical and Illegal Opioid Preparations	147	5.20	21	3.49
Beta Blockers	130	4.60	13	2.16
Miscellaneous Antidepressants	99	3.50	12	2.00
Hypoglycemic, Single Agent	89	3.15	26	4.33
Miscellaneous Unknown Drug	85	3.01	16	2.66
Sedating Antihistamines	78	2.76	9	1.50
Selective Serotonin Reuptake Inhibitors (SSRI)	74	2.62	0	0.00
Miscellaneous Cardiovascular Drugs	72	2.55	17	2.83
Miscellaneous Fumes/Gases/Vapors	65	2.30	48	7.99
Acetaminophen Combinations	61	2.16	21	3.49
Anticonvulsants: Gamma Aminobutyric Acid and Analogs	58	2.05	1	0.17
Nonsteroidal Anti-inflammatory Drugs	56	1.98	6	1.00
Miscellaneous Muscle Relaxants	48	1.70	5	0.83
Serotonin Norepinephrine Reuptake Inhibitors (SNRI)	47	1.66	3	0.50
Acetylsalicylic Acid Alone	45	1.59	12	2.00
Miscellaneous Chemicals	43	1.52	28	4.66
Cannabinoids and Analogs	36	1.27	1	0.17
Miscellaneous Anticonvulsants	35	1.24	2	0.33
Tricyclic Antidepressants (TCA)	35	1.24	10	1.66
Automotive Products	26	0.92	17	2.83

^aNumbers represent total exposures associated with 1,301 fatalities (with RCF of 1-Undoubtedly responsible, 2-Probably responsible, or 3-Contributory and no death, indirect cases); each fatality may have had exposure to more than one substance.

^bPercentages are based on the total number of substances reported in fatal exposures ($n=2,826$).

^cPercentages are based on the total number of single substance fatal exposures ($n=601$).

Table 19A. Comparisons of death data (2010–2024)^a

Year	Total fatalities		Suicides		Pediatric deaths ^b	
	N	% of cases	N	% of deaths	N	% of deaths
2010	1,730	0.072	779	45.03	55	3.18
2011	2,765	0.118	865	31.28	42	1.52
2012	2,937	0.129	890	30.30	46	1.57
2013	2,477	0.113	785	31.69	51	2.06
2014	1,835	0.085	790	43.05	34	1.85
2015	1,831	0.084	814	44.46	42	2.29
2016	1,977	0.091	906	45.83	44	2.23
2017	3,208	0.151	954	29.74	25	0.78
2018	3,111	0.148	953	30.63	51	1.64
2019	2,619	0.122	909	34.71	31	1.18
2020	4,488	0.210	1,013	22.57	57	1.27
2021	4,497	0.216	1,042	23.17	56	1.25
2022	3,255	0.157	1,016	31.21	61	1.87
2023	3,272	0.157	963	29.40	41	1.30
2024	2,809	0.134	904	32.20	41	1.50

^aHuman exposures with medical outcome of death or death, indirect regardless of RCF.

^bIncludes all children with actual or estimated ages ≤ 5 years old. Results do not include "Unknown Child" or "Unknown Age". Includes death and death, indirect regardless of RCF.

21) (includes death, indirect reports and RCF 1-3) included 1,860 pharmaceuticals and 281 nonpharmaceuticals. The first ranked pharmaceuticals associated with these fatalities included: analgesics (910), stimulants and street drugs (321), cardiovascular drugs (254), antidepressants (95), sedative/hypnotics/antipsychotics (59), unknown drug (53), hormones and hormone antagonists (49), antihistamines (29), anticonvulsants (20), muscle relaxants (18) and 14 pharmaceutical classes with less than 10 fatalities each. The first ranked nonpharmaceuticals associated with these fatalities included: alcohols (79), fumes/gases/vapors (69), chemicals (40), automotive/aircraft/

boat products (24), household cleaning substances (19), hydrocarbons (13), pesticides (13) and 11 nonpharmaceutical classes with less than 7 fatalities each.

Pregnancy and fatalities

There were no deaths in pregnant women reported to NPDS in 2024 (includes death, indirect reports). A total of 65 deaths of pregnant women have been reported between 2000 and 2024. The majority (55 of 65, 84.6%) were intentional exposures (misuse, abuse or suspected suicide).

Table 19B. Comparisons of direct and indirect death data (2010–2024)^a

Year	All deaths			Suicides					Pediatric deaths				
	Total	Direct	Indirect	Total	% of deaths	Direct	% of direct	Indirect	Total	% of deaths	Direct	% of direct	Indirect
2010	1,730	1,455	275	779	45.03	732	50.31	47	55	3.18	47	3.23	8
2011	2,765	1,503	1,262	865	31.28	758	50.43	107	42	1.52	31	2.06	11
2012	2,937	1,507	1,430	890	30.30	759	50.36	131	46	1.57	30	1.99	16
2013	2,477	1,552	925	785	31.69	698	44.97	87	51	2.06	43	2.77	8
2014	1,835	1,559	276	790	43.05	757	48.56	33	34	1.85	23	1.48	11
2015	1,831	1,670	161	814	44.46	784	46.95	30	42	2.29	34	2.04	8
2016	1,977	1,852	125	906	45.83	885	47.79	21	44	2.23	37	2.00	7
2017	3,208	1,832	1,376	954	29.74	821	44.81	133	25	0.78	19	1.04	6
2018	3,111	1,821	1,290	953	30.63	823	45.19	130	51	1.64	32	1.76	19
2019	2,619	1,923	696	909	34.71	855	44.46	54	31	1.18	22	1.14	9
2020	4,488	2,006	2,482	1,013	22.57	880	43.87	133	57	1.27	37	1.84	20
2021	4,497	2,234	2,263	1,042	23.17	941	42.12	101	56	1.25	35	1.57	21
2022	3,255	2,072	1,183	1,016	31.21	953	45.99	63	61	1.87	44	2.12	17
2023	3,272	1,947	1,325	963	29.43	898	46.12	65	41	1.25	29	1.49	12
2024	2,809	1,797	1,012	904	32.18	827	46.02	77	41	1.46	27	1.50	14

^aHuman exposures with medical outcome of death or death, indirect regardless of RCF.**Table 19C.** Detail of cases included in fatality tables

Table	Fatalities included	RCF	N
4	Death only	1,2,3	1,301
5	Death only	1,2,3	1,301
8	Death only	1,2,3	1,301
9	Death only	1,2,3	1,301
11	Death and Death (indirect report)	All	2,809
12	Death and Death (indirect report)	All	2,809
17E	Pediatric Death and Death (indirect report)	All	41
18	Death only	1,2,3	1,301
19A	Death and Death (indirect report)	All	2,809
19B	Death and Death (indirect report)	All	2,809
21	Death and Death (indirect report)	1,2,3	2,271
22	Death and Death (indirect report) - Single substance deaths only	All	968

America's poison centers surveillance results

Key components of the NPDS surveillance system include the automated monitoring tools available to the NPDS user community. Since Surveillance Anomaly 1, generated at 2:00pm EDT on 17 September 2006, over 499,000 anomalies have been detected and reported. Over 3,100 were confirmed as representing public health significance with PCs working collaboratively with local health departments and, in some instances the CDC, on the identified issues.

At the time of this report, 187 surveillance definitions run continuously, monitoring case and clinical effects volumes and a variety of case-based definitions from food poisoning to nerve agents. These definitions represent the surveillance work by many individual PCs, health departments, America's Poison Centers, and the Emerging Environmental Hazards and Health Effects Branch/Division of Environmental Health

Science and Practice/National Center for Environmental Health/CDC. NPDS has also been used for surveillance during mass gathering events, such as the Super Bowl, and its use is planned for future events.

The methodology for automating surveillance continues to be improved in efforts to detect the index case of any relevant public health event. Algorithms for identifying the index case vary greatly regarding the substance to be identified. No individual algorithm works for every application. The magnitude and pene- trance of NPDS are critical to epidemiologic surveil- lance and to the ability to substantiate situational awareness for clinicians, policymakers, and public health officials nationwide. Typically, NPDS surveillance detects the response to an event, rather than predict- ing an event. This fosters situational awareness that is undoubtedly beneficial to public health activities.

Discussion

The exposure cases and information requests reported by PCs in 2024 do not reflect the full extent of PC efforts, which include poison prevention activities, pub- lic health entity partnerships, research endeavors, and public and health care professional education programs.

NPDS exposure data may be considered "numera- tor data" in the absence of a true denominator. We do not know the number of actual exposures that occur in the population. NPDS is a passive reporting system. NPDS data covers only those exposures that are reported to PCs since most poison exposures and poisoning deaths are not universally reportable events.

NPDS 2000–2024 encounter volume data clearly demonstrate an overall trend toward decreasing expo- sure cases. This decline has been apparent since mid-2007 and reflects the decreasing use of the PC for

Table 20. Frequency of plant exposures (top 25)^a

	Botanical name or category	America's Poison Centers generic code name	N
1	<i>Phytolacca americana</i> Pokeweed	Plants: Gastrointestinal irritants (excluding oxalate containing plants)	4,067
2	Cherry pit/Cherry (species unspecified)	Plants: Amygdalin and/or cyanogenic glycosides	2,461
3	Plants-cyanogenic glycosides (unknown botanical name)	Plants: Amygdalin and/or cyanogenic glycosides	1,903
4	Poison ivy	Plants: Skin irritants (excluding oxalate containing plants)	1,666
5	Unknown botanical name	Plants: Unknown toxic types or unknown if toxic	1,298
6	Elephant's ear	Plants: Oxalates	1,098
7	<i>Ilex</i> species	Plants: Gastrointestinal irritants (excluding oxalate containing plants)	1,068
8	Plants-oxalates (unknown botanical name)	Plants: Oxalates	1,017
9	<i>Solanum dulcamara</i>	Plants: Solanine	978
10	Berry	Plants: Unknown toxic types or unknown if toxic	879
13	Plants-cardiac glycosides (unknown botanical name)	Plants: Cardiac glycosides (excluding drugs)	863
14	Plants-general-unknown	Plants: Unknown toxic types or unknown if toxic	855
15	<i>Epipremnum aureum</i>	Plants: Oxalates	837
16	<i>Conium maculatum</i>	Plants: Nicotine (excluding tobacco products)	767
17	Apple (Seed/Fruit)	Plants: Amygdalin and/or Cyanogenic glycosides	661
18	Plants-solanine (unknown botanical name)	Plants: Solanine	650
19	<i>Monstera deliciosa</i> (Liebm)	Plants: Oxalates	499
18	<i>Zamioculcas zamiifolia</i>	Plants: Oxalates	479
19	<i>Spathiphyllum species</i>	Plants: Oxalates	477
20	<i>Euphorbia tirucalli</i>	Plants: Gastrointestinal irritants (excluding oxalate containing plants)	460
21	<i>Solanum nigrum</i>	Plants: Solanine	441
22	Apricot pit kernels	Plants: Amygdalin and/or Cyanogenic glycosides	438
23	Unknown botanical name	Plants: Oxalates	436
24	Unknown botanical name	Plants: Amygdalin and/or Cyanogenic glycosides	430
25	<i>Solanum tuberosum</i>	Plants: Solanine	415

^aNumber of substances related to a human exposure with a Major Generic Category of Plant. Unknown Botanical Name represents substances with a Major Generic Category of Plant and a NULL substance code. Total = 56,997.

less serious exposures. However, during this same period, exposures with a more serious outcome (death, major, moderate) and HCF cases have continued to slightly increase. Possible contributors to declining PC utilization include declining US birth rate (especially since exposure rates are much higher in children ≤ 5 years of age), increasing use of text rather than voice communication (although some PCs offer text services and these communications become NPDS cases), lack of familiarity with Poison Center services and increasing use of and reliance on internet resources. To meet our public health goals, PCs will need to understand and provide access via the public's 21st-century communication preferences. We are concerned that failure to respond to these changing preferences may result in a regression in the pattern of seeking care, with more people seeking medical care at HCFs for exposures that could have been managed on-site by a PC, or more worrisome if they seek medical advice through unverified and unmonitored internet resources. Minor exposures may progress to more serious morbidity and mortality because of incorrect internet-based information or the absence of PC management. The net effect could be more serious poisoning outcomes because fewer people have taken advantage of PC services, with a resultant increased burden on the national healthcare infrastructure as may be reflected in the increased number of cases managed in an HCF. Nevertheless, the public appears to be increasing their internet use for exposure questions. PoisonHelp.org (Appendix D) was created for this reason to provide

appropriate management and triage of poisoning cases by members of the public who do not initially contact a Poison Center.

The vast majority of reported fatalities in 2024 were due to pharmaceuticals, with analgesics accounting for the greatest burden; this category encompasses both opioids (such as fentanyl) and other common causes for drug overdose death like acetaminophen. Stimulants and street drugs, most notably methamphetamine, and cardiovascular drugs such as amlodipine and beta blockers also figured prominently, underscoring the lethality of both illicit substances and commonly prescribed medications.

Pediatric fatalities remain rare compared to the volume of exposures, with most in young children related to unintentional environmental or nonpharmaceutical exposures, while adolescent fatalities were overwhelmingly intentional and linked to pharmaceuticals. These age-stratified findings reinforce that prevention strategies must be tailored: safe storage and environmental controls in young children, and enhanced suicide prevention, substance use treatment, and prescribing vigilance in adolescents and adults. As has been true for several years, the circumstances of the case are often not captured in the reason code or the narrative. The pediatric fatality review team encourages that additional detail be obtained from law enforcement or child protective services, postmortem investigation, regarding how the child accessed the substances responsible for the fatality, in order to further identify areas for public health intervention to aid in safeguarding our children.

Unlike many other data sources, NPDS offers a near real-time view of these public health issues without data source extrapolations. A limitation of NPDS data has been the perceived lack of fatality case volume compared to other reporting sources. However, when change over time is studied, NPDS is consistent with other public health fatality analyses. One of the issues leading to this concern is the fact that medical record systems seldom have common output streams. This is particularly apparent with the various electronic medical record systems available. It is important to build a federated approach like the one modeled by NPDS to allow data sharing, for example, between hospital emergency departments and other medical record systems, including medical examiner offices, nationwide. Enhancements to NPDS such as a harmonized clinical effect nomenclature would promote interoperability between NPDS and electronic medical records systems to better trend poison-related morbidity and mortality in the US and internationally.

Summary

Unintentional and intentional exposures continue to be a significant cause of morbidity and mortality in the US. The near real-time status of NPDS represents a national public health resource to collect and monitor US exposure cases and information requests. Changes in 2024 encounters from 2023 are shown in Figures 1, 3, and 4 include:

- Total encounters (all exposure cases and information requests) remained essentially constant, decreasing by 0.117%.
- Information requests decreased 2.70%, Drug ID requests decreased 17.2%, Drug information requests increased 0.218%, and medical information requests decreased 24.5%.
- HCF information requests increased 1.84%.
- Exposure cases reported from an HCF increased 0.661%, reflecting the steady increase since 2000.
- Human exposures with less serious outcomes increased 0.719%.
- Human exposures with more serious medical outcomes (moderate, major or death) had a decrease of 0.538% compared to 2023, while serious outcome cases overall have increased 4.08% annually since 2000.
- The most rapidly increasing substance categories resulting in more serious outcomes over the past decade are stimulants and street drugs, antihistamines, antidepressants, and cardiovascular drugs.

The data support the continued value of PC expertise and need for specialized medical toxicology information to manage more severe exposures, despite the slight decrease in total exposures. In addition to telephonic services, PCs must consider newer communication approaches that match current and future public preferences. The continuing mission of NPDS is to provide a nationwide infrastructure for public health surveillance for all types of exposures, public health event identification, resilience, response and situational awareness tracking. NPDS is a model system for the nation and global public health.

Disclaimer

America's Poison Centers (<https://poisoncenters.org/>) maintains the national database of information logged by the country's current 53 regional Poison Centers (PCs) serving all 50 United States, Puerto Rico, the US territories of Guam, American Samoa, Federated States of Micronesia, US Virgin Islands, and the District of Columbia. The records reflect only information provided when the public or healthcare professionals report an actual or potential exposure to a substance (e.g., an ingestion, inhalation, or topical exposure, etc.) or request information/educational materials. Exposures do not necessarily represent a poisoning or overdose. America's Poison Centers cannot independently verify the accuracy of every report made to member centers. Additional exposures may go unreported to PCs and data referenced from America's Poison Centers should not be construed to represent the complete incidence of US exposures to any substance(s).

Disclosure statement

The authors report nothing to declare.

Funding

The author(s) reported there is no funding associated with the work featured in this article.

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Appendix A

Table 21. Listing of fatal nonpharmaceutical and pharmaceutical exposures.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
Non-pharmaceutical exposures										
Alcohols										
1ai	20 y F	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
2h	25 y M	Methanol	1	1	A	Ingst	Int-U	1	Methanol	222 mg/dL In Blood (unspecified) @ Unknown
3ai	27 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
4ai	29 y M	Methanol	1	1	U	Unk	Int-A	1		
		Sertraline	2	2						
5ai	31 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
6ph	34 y F	Ethanol (beverage)	2	1	U	Ingst + Unk	Unk	3	Ethanol	198 mg/dL In Serum @ Unknown
		Carbamazepine	3	2						
		Gabapentin	4	3						
		Cough and cold preparation, N.O.S.	1	4						
		Barbiturate (long acting)	5	5						
		Marijuana, N.O.S.	6	6						
7pha	34 y F	Methanol	1	1	A	Ingst	Int-U	1	Methanol	67 mg/dL In Blood (unspecified) @ Unknown
		Drug, N.O.S.	2	2						
8ai	35 y F	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
9pha	35 y M	Ethanol (beverage)	1	1	U	Ingst	Int-S	3	Ethanol	350 mg/dL In Blood (unspecified) @ Unknown
		Diazepam	2	2					Nordiazepam	200 ng/mL In Blood (unspecified) @ Unknown
		Diazepam	2	2					Diazepam	88 ng/mL In Blood (unspecified) @ Unknown
		Carbamazepine	3	3					Carbamazepine-10, 11-epoxide	0.54 mcg/mL In Blood (unspecified) @ Unknown
		Carbamazepine	3	3					Carbamazepine	4.7 mcg/mL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Gabapentin	4	4					Gabapentin	9.9mcg/mL In Blood (unspecified) @ Unknown
10ai	35 y F	Ethanol (beverage)	1	1	C	Unk	Int-A	1		
11ha	35 y M	Methanol	1	1	A	Ingst	Unk	1	Methanol	390mg/dL In Serum @ 1 h (pe)
		Methanol	1	1					Methanol	487mg/dL In Serum @ 0 h (pe)
12h	36 y M	Ethanol (beverage)	1	1	A	Ingst	Unk	2		
13h	37 y M	Ethanol (beverage)	1	1	A/C	Ingst	Int-U	2	Ethanol	250mg/dL In Serum @ Unknown
14	37 y F	Ethanol (beverage)	1	1	C	Ingst + Inhal	Int-A	3	Ethanol	298mg/dL In Blood (unspecified) @ Unknown
15h	38 y F	Insecticide, N.O.S.	2	2						
		Ethanol (beverage)	1	1	A/C	Ingst	Int-A	3		
16ai	38 y M	Ethanol (beverage)	1	1	A	Unk	Int-A	1		
17h	39 y F	Methanol	1	1	A/C	Inhal + Unk	Int-A	1	Methanol	0.055g/dL In Blood (unspecified) @ Unknown
18pha	39 y M	Nail polish remover	2	2						
		Ethanol (beverage)	1	1	U	Ingst	Int-S	1	Ethanol	129mg/dL In Serum @ Unknown
		Amitriptyline	2	2					Amitriptyline	77.3 ng/mL In Serum @ Unknown
19ai	39 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
20ph	40 y M	Ethanol (beverage)	1	1	A	Ingst	Int-A	2	Ethanol	117 mg/dL In Blood (unspecified) @ Unknown
21ai	40 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
22ai	40 y M	Ethanol (beverage)	1	1	C	Unk	Int-A	1		
23ai	41 y F	Ethanol (beverage)	1	1	C	Unk	Int-A	1		
24h	41 y F	Ethanol (beverage)	1	1	U	Ingst	Int-A	3	Ethanol	0.117 g/dL In Blood (unspecified) @ Unknown
25ai	41 y M	Acetaminophen	2	2	U	Unk	Int-A	1		
26h	41 y M	Ethanol (beverage)	1	1						
		Ethanol (beverage)	3	1	A/C	Ingst	Int-S	2	Ethanol	120 mg/dL In Blood (unspecified) @ 0 h (pe)
		Oxycodone	2	1						
		Phenobarbital	1	1					Phenobarbital	114mcg/mL In Blood (unspecified) @ 42 h (pe)
		Phenobarbital	1	1					Phenobarbital	117.5 mcg/mL In Blood (unspecified) @ 18 h (pe)
		Phenobarbital	1	1					Phenobarbital	29mcg/mL In Blood (unspecified) @ 10 d (pe)
		Phenobarbital	1	1					Phenobarbital	93.8mcg/mL In Blood (unspecified) @ 0 h (pe)
27ai	41 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
28ph	41 y M	Ethanol (beverage)	1	1						
		Ethanol (beverage)	1	1	U	Ingst	Int-U	1	Ethanol	552 mg/dL In Blood (unspecified) @ Unknown
29ai	42 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
		Nitrite (isobutyl)	2	2						
		Hydroxyzine	3	3						
30ai	42 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
31ai	44 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
32ai	44 y M	Ethanol (beverage)	1	1	C	Unk	Int-A	1		
33ai	44 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
34h	45 y F	Ethanol (beverage)	1	1	A	Unk	Int-A	2	Ethanol	175 mg/dL In Blood (unspecified) @ 30 m (pe)
35ai	45 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
36pha	46 y F	Ethanol (beverage)	1	1	A/C	Ingst	Int-A	1	Ethanol	0.113 % (wt/Vol) In Vitreous @ Autopsy
		Ethanol (beverage)	1	1					Ethanol	0.181 % (wt/Vol) In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	1	1					Ethanol	233 mg/dL In Serum @ 1 m (pe)
		Haloperidol	2	2						
		Lithium	3	3						
		Metoprolol	4	4						
		Quetiapine	5	5						
37ai	47 y F	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
38ai	47 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
39h	48 y F	Ethanol (beverage)	1	1	C	Ingst + Unk	Int-S	3	Ethanol	120 mg/dL In Blood (unspecified) @ Unknown
40h	50 y F	Drug, N.O.S.	2	2	A	Ingst	Int-A	1	Methanol	20 mg/dL In Blood (unspecified) @ 6 h (pe)
		Methanol	1	1					Methanol	309 mg/dL In Blood (unspecified) @ 1 h (pe)
41ha	50 y M	Methanol	1	1	A	Ingst + Unk	Int-A	1	Methanol	52 mg/dL In Blood (unspecified) @ Unknown
42ph	50 y F	Methamphetamine	2	2	U	Ingst	Unk	2	Ethanol	253 mg/dL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	1	1						
43p	50 y M	Drug, N.O.S.	2	2	U	Ingst	Oth-W	2		
44h	52 y F	Ethanol (beverage)	1	1	U	Ingst + Unk	Unk	3		
		Acetaminophen	2	2					Acetaminophen (apap)	2.2 mcg/mL In Blood (unspecified) @ Unknown
45h	52 y F	Methanol	1	1	A	Ingst	Int-S	1	Methanol	15 mg/dL In Blood (unspecified) @ Unknown
46ai	53 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
47	53 y F				A	Ingst	Int-U	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
48ai	53 y M	Methanol	1	1					Methanol	337 mg/dL In Serum @ Unknown
		Ethanol (beverage)	1	1	U	Unk	Int-A	1		
49pa	54 y F	Methamphetamine	2	2	A	Ingst	Int-A	1		
		Ethanol (beverage)	1	1					Ethanol	439 mg/dL In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	1	1					Ethanol	491 mg/dL In Serum @ Unknown
50h	54 y M	Ethanol (beverage)	1	1	A/C	Ingst	Int-A	2		
		Ethanol (beverage)	1	1					Ethanol	34 mg/dL In Blood (unspecified) @ Unknown
51h	54 y M	Ethanol (beverage)	1	1	A/C	Ingst	Int-A	2		
		Ethanol (beverage)	1	1					Ethanol	199 mg/dL In Blood (unspecified) @ 15 m (pe)
52ha	55 y F	Drug, N.O.S.	2	2	A/C	Ingst + Inhal + Par	Int-A	2		
		Ethanol (beverage)	1	1					Ethanol	111 mg/dL In Plasma @ Unknown
		Ethanol (beverage)	1	1					Ethanol	167 mg/dL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	2	2					Delta-9-thc	2.1 ng/mL In Plasma @ Unknown
53ai	56 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
		Isopropanol	2	2						
		Trazodone	3	3						
54p	56 y M	Alcohol, N.O.S.	1	1	A	Ingst	Int-S	2		
		Drug, N.O.S.	2	2						
55ai	57 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
56h	57 y M	Ethanol (beverage)	1	1	U	Unk	Unk	3		
		Ethanol (beverage)	1	1					Ethanol	118.3 mg/dL In Blood (unspecified) @ Unknown
57ha	57 y F	Methanol	2	2	A	Ingst	Int-S	1		
		Isopropanol	1	1					Isopropanol	200 mg/dL In Blood (unspecified) @ 5 m (pe)
		Isopropanol	1	1					Acetone	25 mg/dL In Blood (unspecified) @ 5 m (pe)
		Diazepam	2	2					Diazepam	130 ng/mL In Blood (unspecified) @ 5 m (pe)
		Diazepam	2	2					Nordiazepam	190 ng/mL In Blood (unspecified) @ 5 m (pe)
58h	57 y F	Ethanol (beverage)	1	1	U	Ingst	Unk	2		
		Ethanol (beverage)	1	1					Ethanol	316 mg/dL In Blood (unspecified) @ Unknown
59a	57 y M	Ethanol (beverage)	1	1	C	Ingst	Int-A	2		
60ph	57 y M	Isopropanol	1	1	U	Ingst	Unk	3		
		Ethanol (beverage)	2	2					Isopropanol	105 mg/dL In Serum @ Unknown
		Ethanol (beverage)	2	2					Ethanol	12 mcg/dL In Serum @ Unknown
61ha	59 y M	Alcohol, N.O.S.	1	1	A	Unk	Unk	3		
62ha	60 y M	Methanol	1	1	A	Ingst	Unk	3		
		Isopropanol	2	2					Methanol	11 mcg/dL In Blood (unspecified) @ Unknown
		Isopropanol	2	2					Isopropanol	10 mcg/dL In Blood (unspecified) @ Unknown
63h	60 y F				C	Ingst	Unk	3		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Ethanol (beverage)	1	1					Ethanol	233 mg/dL In Blood (unspecified) @ Unknown
64h	61 y M	Ethanol (beverage)	1	1	A/C	Ingst	Int-A	3	Ethanol	10 mg/dL In Blood (unspecified) @ Unknown
65h	62 y F	Ethanol (beverage)	1	1	A	Ingst	Int-S	2	Ethanol	816 mg/dL In Blood (unspecified) @ Unknown
66	63 y M	Ethanol (beverage)	1	1	C	Ingst	Int-A	2	Ethanol	555 mg/dL In Blood (unspecified) @ Unknown
67ha	63 y F	Ethanol (beverage)	1	1	U	Ingst + Unk	Oth-W	2		
		Duloxetine (ER)	2	2					Duloxetine	440 ng/mL In Blood (unspecified) @ Unknown
		Marijuana	3	3					Delta-9-thc	1.3 ng/mL In Blood (unspecified) @ Unknown
68ai	64 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
69ha	64 y M	Methanol	1	1	A	Ingst	Int-U	1	Methanol	382 mg/dL In Serum @ Unknown
		Methanol	1	1					Methanol	85 mg/dL In Serum @ Unknown
70ai	65 y M	Ethanol (beverage)	1	1	C	Unk	Int-A	1		
71ai	65 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
72h	67 y M	Isopropanol	1	1	U	Ingst	Int-S	2		
		Substance (non-drug), N.O.S.	2	2						
73ai	68 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
74a	73 y F	Methanol	1	1	A	Ingst	Int-S	1	Methanol	634 mg/dL In Blood (unspecified) @ Unknown
75h	74 y F	Methanol	1	1	U	Ingst	Unk	1	Methanol	113 mg/dL In Serum @ 11 h (pe)
		Methanol	1	1					Methanol	16 mg/dL In Serum @ 20 h (pe)
		Methanol	1	1					Methanol	455 mg/dL In Serum @ 1 h (pe)
76h	75 y M	Ethanol (beverage)	1	1	A/C	Ingst	Int-A	2	Ethanol	209 mg/dL In Serum @ 1 m (pe)
77	78 y F	Methanol	1	1	U	Unk	Unk	2		
78ai	89 y M	Ethanol (beverage)	1	1	U	Unk	Int-A	1		
79ph	40+ y M	Ethanol (beverage)	1	1	A	Ingst	Int-M	2		
		Energy drink (caffeine/ethanol)	2	2						
		Vitamin (ascorbic acid/caffeine/cyanocobalamin)	3	3						
		Androgen or precursor	4	4						
		Androgen or precursor	5	5						

See Also case 84, 94, 99, 158, 162, 165, 167, 171, 172, 183, 203, 216, 226, 235, 244, 282, 283, 302, 304, 305, 323, 324, 347, 356, 358, 364, 380, 385, 389, 406, 407, 408, 409, 414, 426, 428, 432, 441, 450, 457, 459, 460, 471, 481, 482, 488, 491, 493, 494, 497, 498, 501, 511, 513, 517, 527, 539, 545, 549, 551, 557, 571, 575, 578, 584, 593, 598, 607, 610, 625, 629, 633, 635, 638, 643, 644, 645, 647, 652, 654, 657, 666, 670, 671, 674, 675, 680, 682, 687, 689, 691, 699, 700, 706, 708, 715, 716, 723, 725, 727, 728, 732, 738, 739, 746, 750, 751, 752, 755, 761, 763, 778, 786, 788, 791, 795, 815, 816, 818, 820, 822, 824, 826, 832, 836, 840, 841, 842, 846, 854, 866, 869, 873, 888, 901, 916, 938, 940, 953, 956, 959, 961, 962, 963, 978, 979, 980, 983, 994, 996, 997, 1000, 1003, 1018, 1020, 1036, 1039, 1054, 1061, 1074, 1085, 1095, 1103, 1104, 1128, 1134, 1144, 1145, 1149, 1168, 1172, 1173, 1179, 1191, 1213, 1251, 1253, 1279, 1285, 1326, 1332, 1334, 1338, 1345, 1364, 1367, 1371, 1372, 1380, 1381, 1386, 1390, 1416, 1424, 1425, 1431, 1433, 1435, 1441, 1444, 1445, 1478, 1486, 1488, 1490, 1494, 1495, 1499, 1504, 1512, 1521, 1526, 1527, 1532, 1536, 1538, 1544, 1547, 1551, 1556, 1559, 1560, 1573, 1576, 1579, 1596, 1607, 1619, 1623, 1630, 1634, 1648, 1649, 1661, 1667, 1670, 1714, 1727, 1737, 1740, 1753, 1759, 1763, 1765, 1832, 1833, 1834, 1843, 1845, 1852, 1856, 1876, 1890, 1901, 1910, 1912, 1914, 1916, 1931, 1934, 1939, 1940, 1948, 1957, 1966, 1968, 1977, 1990, 1998, 2008, 2013, 2021, 2022, 2027, 2034, 2048, 2051, 2054, 2069, 2071, 2075, 2087, 2088, 2091, 2092, 2101, 2103, 2109, 2112, 2121, 2129, 2130, 2137, 2140, 2151, 2156, 2159, 2167, 2170, 2178, 2185, 2191, 2206, 2225, 2227, 2228, 2229, 2241, 2252, 2261, 2265, 2269

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
Arts/Crafts/Office Supplies										
80h	57 y M	Ink	1	1	A	Ingst	Int-S	2		
		Quetiapine	2	2						
		Lisinopril	3	3						
		Sertraline	4	4						
		Drug, N.O.S.	5	5						
Automotive/Aircraft/Boat Products										
81ph	19 y F	Glycol ethers/ diethylene glycol	1	1	A	Ingst	Int-S	2		
		Diphenhydramine	2	2						
82ha	28 y M	Ethylene glycol (antifreeze)	1	1	A	Unk	Int-S	1	Ethylene glycol	61 mg/dL In Serum @ 0.5 h (pe)
83h	29 y F	Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	1	Ethylene glycol	217 mg/dL In Blood (unspecified) @ Unknown
84a	34 y M	Methanol	1	1	A	Ingst	Int-A	1	Methanol	46 mg/dL In Serum @ Unknown
		Ethanol (beverage)	2	2						
85h	37 y M	Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	1	Ethylene glycol	3062.7 mg/L In Serum @ Unknown
		Drug, N.O.S.	2	2						
86pa	42 y M	Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	1	Ethylene glycol	581 mg/L In Blood (unspecified) @ Unknown
87ha	43 y M	Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	1		
88h	43 y M	Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	3	Ethylene glycol	16.9 mg/dL In Blood (unspecified) @ Unknown
89ha	49 y M	Ethylene glycol (antifreeze)	1	1	U	Ingst	Int-S	1	Ethylene glycol	12 mg/dL In Blood (unspecified) @ Unknown
		Olanzapine	2	2						
		Mirtazapine	3	3						
		Sertraline	4	4						
		Atypical antipsychotic, N.O.S.	5	5						
		Atypical antipsychotic, N.O.S.	6	6						
90ha	50 y M	Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	1		
91a	54 y M	Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	1	Ethylene glycol	160 mg/dL In Blood (unspecified) @ Unknown
92a	54 y M	Methanol	1	1	A	Ingst	Int-S	1	Methanol	125 mg/dL In Serum @ Unspecified
		Methanol	1	1					Methanol	500 mg/dL In Serum @ Unknown
93ha	56 y F	Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	2		
94ph	56 y F	Ethylene glycol (antifreeze)	3	1	A	Ingst	Int-S	2		
		Alprazolam	1	2						
95h	58 y M	Ethanol (beverage)	2	2	U	Unk	Unk	1		
		Methanol	1	1					Methanol	248 mg/dL In Serum @ Unknown
96h	59 y M				U	Ingst	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Ethylene glycol (antifreeze)	1	1					Ethylene glycol	120 mg/dL In Blood (unspecified) @ Unknown
		Ethylene glycol (antifreeze)	1	1					Ethylene glycol	161 mg/dL In Blood (unspecified) @ Unknown
		Ethylene glycol (antifreeze)	1	1					Ethylene glycol	206 mg/dL In Blood (unspecified) @ Unknown
		Ethylene glycol (antifreeze)	1	1					Ethylene glycol	866 mg/dL In Blood (unspecified) @ Unknown
97h	60 y M	Acetaminophen/diphenhydramine	2	2						
		Ethylene glycol (antifreeze)	1	1	A	Ingst	Int-S	1		
98ha	60 y M				A	Ingst	Int-S	1		
		Ethylene glycol (antifreeze)	1	1					Ethylene glycol	530 mg/dL In Blood (unspecified) @ Unknown
99	60 y M	Ibuprofen	2	2	A	Ingst	Int-S	1		
		Ethylene glycol (antifreeze)	1	1						
100ha	64 y F	Ethanol (beverage)	2	2	A	Ingst	Int-S	1		
		Methanol	1	1					Methanol	345 mg/dL In Blood (unspecified) @ Unknown
101pha	67 y F				A	Ingst	Int-S	1		
		Ethylene glycol (antifreeze)	1	1					Ethylene glycol	110 mg/dL In Serum @ 30 m (pe)
102pha	78 y M				A	Ingst + Aspir	Int-S	1		
		Ethylene glycol (antifreeze)	1	1					Ethylene glycol	640 mg/dL In Blood (unspecified) @ Unknown
103h	84 y M				A	Oc + Derm	Oth-M	1		
104h	86 y M	Diethylether/heptane	1	1	A	Ingst	Int-S	2		
		Ethylene glycol (antifreeze)	1	1						
105ha	94 y F				A	Ingst	Int-S	1		
		Ethylene glycol (antifreeze)	1	1					Ethylene glycol	110 mg/dL In Blood (unspecified) @ Unknown
See Also case 1504										
Bites and Envenomations										
106	29 y M				A	B-S	Unt-B	2		
		Envenomation (snake, N.O.S.)	1	1						
[107ha]	34 y M				A	Derm	Unt-B	1		
		Bite (<i>Heloderma suspectum</i>)	1	1						
108ph	34 y M				A	B-S	Unt-B	1		
		Envenomation (Hymenoptera)	1	1						
109p	69 y M				A	Inhal + B-S	Unt-B	2		
		Envenomation (Hymenoptera)	1	1						
110pha	82 y F				A	B-S	Unt-B	1		
		Envenomation (Hymenoptera)	1	1						
[111ph]	87 y M				A	B-S	Unt-B	1		
		Envenomation (Hymenoptera)	1	1						
Building and Construction Products										
112ph	34 y M				C	Inhal	Unt-O	3		
		Calcium hydroxide	1	1						
[113h]	51 y M				A	Inhal	Unt-M	2		
		Calcium hydroxide	1	1						
Chemicals										
114ph	2 y M				A	Inhal	Unt-E	1		
		Cyanide	1	1						
		Carbon monoxide	2	2						
[115h]	2 y M				A	Ingst	Unt-G	2		
		Hydrofluoric acid	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
116pai	9 y M	Phosphoric acid	2	2						
		Glycerin	1	1	A	Ingst + Aspir	Unt-G	1		
117ph	12 y M				A	Inhal	Unt-E	1		
		Cyanide	1	1						
		Carbon monoxide	2	2						
[118ha]	17 y M				A	Ingst	Int-S	2		
		Ethylene glycol	1	1					Ethylene glycol	56.1 mg/dL In Blood (unspecified) @ Unknown
119ph	17 y F				A	Ingst	Int-S	1		
		Nitrate (sodium)	1	1					Methemoglobin	30 % In Blood (unspecified) @ Unknown
120	19 y F				A	Ingst	Int-S	2		
		Nitrate, N.O.S.	1	1						
121pha	19 y F				A	Ingst	Int-S	1		
		Nitrate (sodium)	1	1					Nitrite	1000 µmol/L In Blood (unspecified) @ Unknown
		Ondansetron	2	2						
122pha	20 y M				A	Unk	Int-S	1		
		Nitrite (sodium)	1	1						
123h	20 y M				A	Ingst	Int-S	1		
		Ethylene glycol	1	1					Ethylene glycol	29 mg/dL In Serum @ 36 h (pe)
		Diphenhydramine	2	2						
124	21 y M				A	Ingst	Int-S	2		
		Nitrate (ammonium)	1	1						
125ai	22 y F				U	Unk	Int-S	1		
		Nitrite (sodium)	1	1						
		Bromazolam	2	2						
126ai	22 y M				U	Unk	Int-S	1		
		Nitrite (sodium)	1	1						
		Bromazolam	2	2						
127ph	22 y F				A	Unk	Int-S	2		
		Nitrate (sodium)	1	1						
128p	22 y M				A	Ingst	Int-U	2		
		Nitrite (sodium)	1	1						
129ha	22 y F				A	Ingst	Int-S	1		
		Nitrite (sodium)	1	1						
130pha	25 y M				A	Ingst	Int-S	1		
		Nitrate, N.O.S.	1	1					Nitrite	1000 mmol/L In Blood (unspecified) @ Unknown
		Nitrate, N.O.S.	1	1					Methemoglobin	12 % In Blood (unspecified) @ Unknown
131h	25 y M				A	Ingst	Int-S	2		
		Nitrate (sodium)	1	1						
132ph	25 y M				A	Ingst	Int-S	1		
		Nitrate, N.O.S.	1	1						
133pha	26 y M				A	Ingst	Int-S	1		
		Nitrite (sodium)	1	1					Nitrite	1000 µmol/L In Blood (unspecified) @ Autopsy
134p	26 y M				A	Ingst	Int-S	1		
		Nitrate (sodium)	1	1						
[135h]	26 y M				A	Ingst	Int-S	1		
		Hydrochloric acid	1	1						
136ha	26 y M				A	Ingst	Int-S	2		
		Nitrite (sodium)	1	1					Nitrates/nitrites	1000 µmol/L In Blood (unspecified) @ Autopsy
		Pseudoephedrine	2	2						
[137h]	28 y M				A	Ingst	Int-S	1		
		Sodium azide	1	1						
138pha	28 y M				A	Ingst	Int-S	1		
		Nitrate (sodium)	1	1					Methemoglobin	19 % In Serum @ 1 h (pe)
139p	32 y F				A	Ingst	Int-U	1		
		Nitrate, N.O.S.	1	1						
140p	37 y M				A	Inhal	Unt-O	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
141ph	37 y M	Methyl pyrrolidone	1	1	A	Ingst	Int-S	1		
		Nitrate (potassium)	1	1						
142h	41 y F	Nitrate (sodium)	1	1	U	Ingst	Int-S	1	Nitrite	1000 mmol/L In Blood (unspecified) @ Autopsy
[143ha]	44 y M	Cyanide	1	1	A	Ingst	Int-S	1	Cyanide	0.5 mcg/mL In Blood (unspecified) @ 1 h (pe)
144ph	46 y M	Ethylene glycol	1	1	A	Ingst	Unk	1	Ethylene glycol	94 mg/dL In Serum @ Unknown
145	46 y F	Ethylene glycol	1	1	U	Ingst	Int-S	2	Ethylene glycol	208 mg/dL In Blood (unspecified) @ 1 h (pe)
146pha	50 y F	Sodium azide	1	1	A	Ingst	Int-S	1		
		Duloxetine (ER)	2	2						
		Prochlorperazine	3	3					Duloxetine	580 ng/mL In Whole Blood @ Autopsy
									Prochlorperazine	18 ng/mL In Whole Blood @ Autopsy
147p	51 y F	Nitrite (sodium)	1	1	A	Ingst	Int-S	1		
[148ha]	54 y M	Potassium thiocyanate	1	1	A	Ingst	Int-S	1	Cyanide	0.21 mcg/mL In Blood (unspecified) @ Unknown
		Potassium thiocyanate	1	1					Thiocyanate	260 mcg/mL In Blood (unspecified) @ Unknown
149ph	55 y F	Ethylene glycol	1	1	A	Ingst	Int-S	2	Ethylene glycol	7 mg/dL In Blood (unspecified) @ Unknown
150ai	60 y M	Ethylene glycol	1	1	U	Unk	Int-S	1		
151p	60 y M	Ethylene glycol	1	1	A	Ingst	Int-U	1		
152ha	61 y M	Ethylene glycol	1	1	U	Ingst	Int-S	1	Ethylene glycol	91 mg/dL In Blood (unspecified) @ Autopsy
		Zolpidem	3	2						
		Metformin	2	3						
		Amlodipine	4	4						
153h	68 y M	Hydrochloric acid	1	1	A	Derm	Unk	3		
154hi	68 y M	Ethylene glycol	1	1	A	Ingst	Unt-M	2		
155h	68 y M	Hydrochloric acid	1	1	A	Ingst + Aspir	Int-S	1		
156	73 y M	Ethylene glycol	1	1	A	Ingst	Int-S	2	Ethylene glycol	54 mg/dL In Serum @ Unknown
		Ethanol (non-beverage)	2	2					Ethanol	26 mg/dL In Serum @ Unknown
157h	74 y F	Ethylene glycol	1	1	A	Ingst	Int-S	1	Ethylene glycol	14 mg/dL In Serum @ 68 h (pe)
		Ethylene glycol	1	1					Ethylene glycol	195 mg/dL In Serum @ 22 h (pe)
158h	76 y F	Nitrate (sodium)	1	1	A	Ingst	Int-S	2		
		Ethanol (beverage)	2	2						
		Mirtazapine	3	3						
159	78 y F	Acid (acetic)	1	1	A	Ingst	Int-S	2		
160ph	20+ y M	Nitrate (sodium)	1	1	A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
161ai	Unknown age U	Nitrate (sodium)	1	1	U	Vag	Int-S	1	Nitrite	1000 µmol/L In Blood (unspecified) @ Unknown
See Also case 29, 168, 197, 239, 946, 1628, 1948, 2009, 2183										
Cleaning Substances (Household)										
162h	34 y M	Cleaner (household, N.O.S.)	1	1	A	Ingst	Int-S	2		
		Ethanol (non-beverage)	2	2					Ethanol	119mg/dL In Serum @ Unknown
163p	36 y M	Disinfectant (pine oil)	3	3	A	Ingst	Int-S	2		
		Cleaner (anionic/nonionic)	1	1						
		Ethanol	2	2						
164h	41 y M	Drain cleaner, N.O.S.	1	1	A	Ingst + Aspir	Int-S	1		
165h	48 y M	Drain cleaner (sulfuric acid)	1	1	A	Ingst	Int-S	1		
		Ethanol (beverage)	2	2					Ethanol	253mg/dL In Serum @ Unknown
166h	48 y M	Cleaner (hypochlorite)	1	1	A	Ingst	Int-S	2		
		Cleaner (anionic/nonionic)	2	2						
167a	59 y M	Disinfectant (pine oil)	1	1	A	Ingst	Unk	3		
		Ethanol (beverage)	2	2					Ethanol	103mg/dL In Blood (unspecified) @ Unknown
[168h]	66 y M	Cleaner (phosphoric acid)	1	1	A	Ingst	Int-S	1		
		Paint additive	2	2						
169	70 y F	Hydrofluoric acid	1	1	A	Ingst	Unt-M	2		
170ha	72 y M	Laundry detergent	1	1	A	Ingst + Aspir	Int-S	3		
171h	74 y M	Drain cleaner (sulfuric acid)	1	1	A	Ingst	Int-S	1		
		Ethanol (beverage)	2	2						
172ha	75 y M	Drain cleaner (sodium hydroxide/sodium hypochlorite/sodium silicate)	1	1	A	Ingst	Int-S	1		
		Ethanol (beverage)	2	2					Ethanol	227.2mg/dL In Blood (unspecified) @ Unknown
173pi	75 y M	Cleaner (anionic/nonionic)	1	1	A	Ingst	Unt-M	3		
174h	81 y M	Disinfectant (cationic)	1	1	A	Ingst	Unt-M	1		
175	85 y F	Cleaner (anionic/nonionic)	1	1	A	Ingst + Aspir	Unt-G	2		
176ha	86 y F	Fabric softener (cationic)	1	1	A	Ingst	Unt-M	3		
177hai	91 y M	Automatic dishwasher detergent, N.O.S.	1	1	A	Ingst	Unt-G	3		
178ha	93 y F	Laundry detergent (pod)	1	1	A	Ingst	Unt-G	3		
[179ha]	94 y F	Cleaner (acid/cationic)	1	1	A	Ingst	Unt-G	1		
180h	98 y F				A	Ingst + Aspir	Unt-G	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
[181ha]	14 m F	Cleaner (anionic/nonionic)	1	1						
		Cleaner (heptamethylnonane)	1	1	A	Ingst + Aspir	Unt-G	1		
Cosmetics/Personal Care Products										
182h	42 y M	Hand sanitizer (methanol)	1	1	A	Ingst	Int-U	1	Methanol	202 mg/dL In Blood (unspecified) @ Unknown
[183h]	47 y F	Hydrogen peroxide	1	1	A	Ingst	Unt-M	1		
184ha	62 y F	Ethanol (beverage)	2	2						
		Ethanol	1	1	A	Ingst	Int-S	1	Ethanol	24 mg/dL In Serum @ Unknown
		Aminomethyl propanol/benzophenone-4/ethanol)	2	2						
See Also case 17, 156, 163, 166										
Fumes/Gases/Vapors										
185pi	1 y F				A	Ingst	Unt-E	2		
186ph	2 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
		Carbon monoxide	1	1						
187phi	2 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
188ph	3 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
		Carbon monoxide	1	1					Carboxyhemoglobin	4.9 % In Blood (unspecified) @ Unknown
189pi	4 y M				A	Inhal	Unt-E	1		
190pi	4 y M	Carbon monoxide	1	1	A	Ingst	Unt-E	2		
		Carbon monoxide	1	1						
191pi	5 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
192hi	5 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
193ph	5 y M	Carbon monoxide	1	1	A	Inhal + Par	Unt-E	1		
		Carbon monoxide	1	1					Carboxyhemoglobin	59 % In Serum @ Unknown
194ph	6 y M	Hydroxocobalamin	2	2						
		Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	8.4 % In Blood (unspecified) @ 3 h (pe)
195phi	6 y F				A	Inhal	Unt-E	1		
196ph	7 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
		Carbon monoxide	1	1					Methemoglobin	5.1 % In Blood (unspecified) @ Unknown
197ph	7 y F				A	Inhal	Unt-E	1		
		Carbon monoxide	1	1					Carboxyhemoglobin	29.4 % In Blood (unspecified) @ Unknown
198pi	9 y F	Cyanide	2	2						
199ph	9 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
		Carbon monoxide	1	1					Carboxyhemoglobin	52 % In Blood (unspecified) @ Unknown
200pi	10 y M				A	Inhal	Unt-E	1		
201phi	11 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
		Carbon monoxide	1	1					Carboxyhemoglobin	5.48 % In Blood (unspecified) @ 3 d (pe)
		Carbon monoxide	1	1					Carboxyhemoglobin	61 % In Blood (unspecified) @ 1 h (pe)
202hi	11 y F				U	Inhal	Unt-E	1		
203ph	14 y M	Carbon monoxide	1	1	A	Ingst + Inhal	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Butane	1	1						
		Ethanol (beverage)	2	2						
		Marijuana, N.O.S.	3	3						
		Hydroxyzine	4	4						
		Sertraline	5	5						
		Buspirone	6	6						
		Venlafaxine	7	7						
204p	20 y M				U	Inhal	Int-U	2		
		Carbon monoxide	1	1						
205p	21 y M				A	Inhal	Unt-O	1		
		Methane	1	1						
[206pha]	21 y F				A	Inhal + Derm	Int-S	1		
		Hydrogen sulfide	1	1					Thiosulfate	11 mcg/mL In Blood (unspecified) @ Unknown
207p	23 y M				A	Ingst + Inhal	Int-S	1		
		Methane	1	1						
		Ibuprofen	2	2					Ibuprofen	300 mcg/mL In Serum @ Autopsy
208ai	23 y F				U	Inhal	Unk	1		
		Carbon monoxide	1	1						
209ph	23 y M				A	Inhal	Unt-G	1		
		Carbon monoxide	1	1					Carboxyhemoglobin	45.2 % In Blood (unspecified) @ Unknown
210ph	23 y M				A	Inhal	Unt-E	1		
		Carbon monoxide	1	1						
		Hydrogen sulfide	2	2						
211p	24 y M				A	Inhal	Int-S	2		
		Hydrogen sulfide	1	1						
212pha	24 y F				U	Inhal	Unt-E	1		
		Carbon monoxide	1	1					Carboxyhemoglobin	35 % In Blood (unspecified) @ Unknown
		Carbon monoxide	1	1					Carboxyhemoglobin	41 % In Blood (unspecified) @ Unknown
[213pha]	27 y M				A	Inhal	Unt-O	1		
		Hydrogen sulfide	1	1					Thiosulfate	4.2 mcg/mL In Blood (unspecified) @ Autopsy
214ai	32 y M				U	Inhal	Unk	1		
		Carbon monoxide	1	1						
215h	32 y M				A	Inhal	Int-A	2		
		Nitrous oxide	1	1						
216ai	33 y F				U	Unk	Int-S	1		
		Carbon monoxide	1	1						
		Carisoprodol	2	2						
		Ethanol (beverage)	3	3						
217p	33 y M				A	Inhal	Int-S	2		
		Carbon monoxide	1	1						
218ph	36 y F				A	Inhal	Int-S	1		
		Carbon monoxide	1	1						
219ai	39 y M				A	Inhal	Int-S	1		
		Carbon monoxide	1	1						
220ai	40 y M				U	Inhal	Int-S	1		
		Carbon monoxide	1	1						
221ai	41 y M				A	Inhal + Unk	Int-A	1		
		Carbon monoxide	1	1						
		Methamphetamine	2	2						
222pi	41 y M				A	Inhal	Int-S	1		
		Carbon monoxide	1	1						
223ai	44 y M				U	Inhal + Unk	Unk	1		
		Carbon monoxide	1	1						
		Methamphetamine	2	2						
224h	44 y F				U	Unk	Unk	3		
		Carbon monoxide	1	1					Carboxyhemoglobin	12.9 % In Blood (unspecified) @ Unknown
225ph	45 y F				A	Inhal	Unt-G	2		
		Carbon monoxide	1	1						
		Opioid, N.O.S.	2	2						
226h	47 y F				A	Inhal	Unt-E	3		
		Carbon monoxide	1	1					Carboxyhemoglobin	6.9 % In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
[227ph]	47 y M				A	Inhal + Derm	Unt-O	3		
228ph	50 y F	Acetylene	1	1	C	Inhal	Unt-E	1		
229ph	50 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	40 % In Blood (unspecified) @ Unknown
230ai	51 y F				A	Inhal	Int-S	1		
231ph	52 y M	Carbon monoxide	1	1	A	Inhal	Unt-M	1	Carboxyhemoglobin	59.2 % In Serum @ Unknown
232ph	56 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	10.8 % In Blood (unspecified) @ 1 h (pe)
233ph	56 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	3	Carboxyhemoglobin	42 % In Blood (unspecified) @ Unknown
		Carbon monoxide	1	1					Carboxyhemoglobin	5.5 % In Blood (unspecified) @ 6 h (pe)
234ph	56 y F				A	Inhal	Unk	2		
235pha	58 y M	Chloramine	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	61.6 % In Blood (unspecified) @ Unknown
		Carbon monoxide	1	1						
		Ethanol (beverage)	2	2					Ethanol	160 mg/dL In Blood (unspecified) @ Unknown
236h	58 y F				A	Inhal	Int-S	2	Carboxyhemoglobin	8.2 % In Serum @ Unknown
237ph	59 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	35 % In Blood (unspecified) @ Unknown
238p	60 y M	Carbon monoxide	1	1	C	Inhal	Int-S	1	Carboxyhemoglobin	29.6 % In Blood (unspecified) @ Unknown
239ph	60 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	0.3 % In Blood (unspecified) @ Unknown
240h	61 y M	Cyanide	2	2	A	Inhal	Unt-M	2		
241ha	62 y M	Chlorine gas	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	38.9 % In Blood (unspecified) @ Unknown
		Carbon monoxide	1	1						
242ph	62 y M	Carbon monoxide	2	2	A	Inhal	Unt-E	1	Carboxyhemoglobin	20.3 % In Blood (unspecified) @ Unknown
		Carbon monoxide	1	1						
243ai	62 y F				A	Inhal	Unk	1		
244ha	64 y M	Carbon monoxide	1	1	A	Inhal + Derm	Unt-E	3		
		Smoke, N.O.S.	1	1						
		Metformin	2	2						
		Ethanol (beverage)	3	3					Ethanol	16 mg/dL In Serum @ Unknown
		Amphetamine/dextroamphetamine	4	4						
245pha	66 y F				A	Inhal	Unt-E	1	Carboxyhemoglobin	1.8 % In Blood (unspecified) @ 14 h (pe)
		Carbon monoxide	1	1					Carboxyhemoglobin	15.6 % In Blood (unspecified) @ 2 h (pe)
		Carbon monoxide	1	1					Carboxyhemoglobin	41.6 % In Blood (unspecified) @ 1 m (pe)

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
246ph	67 y F	Methane	2	2						
		Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	58 % In Blood (unspecified) @ Unknown
247pha	71 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	6.5 % In Whole Blood @ Unknown
248	71 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	21.5 % In Whole Blood @ Unknown
249	72 y M	Carbon monoxide	1	1	A	Inhal	Unt-G	1	Carboxyhemoglobin	43 % In Blood (unspecified) @ Unknown
250p	73 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	3	Carboxyhemoglobin	35 % In Blood (unspecified) @ Unknown
251ph	78 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	3	Carboxyhemoglobin	29.7 % In Whole Blood @ 1 h (pe)
252h	79 y F	Carbon monoxide	1	1	A	Inhal	Unt-G	3	Carboxyhemoglobin	15.8 % In Blood (unspecified) @ Unknown
253ph	80 y M	Carbon monoxide	1	1	U	Inhal	Int-S	3	Carboxyhemoglobin	37.2 % In Blood (unspecified) @ Unknown
[254ph]	82 y M	Nitrogen	1	1	A	Inhal	Int-S	2		
255ph	82 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	2	Carboxyhemoglobin	15.5 % In Whole Blood @ 1 h (pe)
256pi	83 y M	Carbon monoxide	1	1	A	Inhal	Unt-G	2		
257ph	84 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	39.5 % In Blood (unspecified) @ Unknown
258ph	86 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	2		
259h	87 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	16 % In Blood (unspecified) @ 1 h (pe)
		Carbon monoxide	1	1					Carboxyhemoglobin	16 % In Blood (unspecified) @ 30 m (pe)
260ph	88 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	52 % In Blood (unspecified) @ Unknown
261pai	90 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
262ph	92 y M	Carbon monoxide	1	1	U	Inhal	Int-S	1	Carboxyhemoglobin	21.1 % In Blood (unspecified) @ Unknown
263ha	92 y F	Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	8 % In Blood (unspecified) @ Unknown
264pai	95 y M	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
265ph	98 y M	Carbon monoxide	1	1	A	Inhal + Derm	Unt-E	1	Methemoglobin	0.3 % In Blood (unspecified) @ Unknown
		Carbon monoxide	1	1					Carboxyhemoglobin	8.6 % In Blood (unspecified) @ Unknown
266pi	17 m F	Carbon monoxide	1	1	A	Inhal	Unt-E	1		
267phi	≤5 y U				A	Inhal	Unt-E	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
268ph	50+ y M	Carbon monoxide	1	1						
		Carbon monoxide	1	1	A	Inhal	Unt-E	1	Carboxyhemoglobin	19.4 % In Blood (unspecified) @ Unknown
		Carbon monoxide	1	1					Carboxyhemoglobin	47.7 % In Blood (unspecified) @ Unknown
269pi	50+ y M				A	Inhal	Unt-U	1		
270pi	Unknown adult (≥ 20 yrs) M	Carbon monoxide	1	1	A	Inhal	Int-S	1		
271p	Unknown adult (≥ 20 yrs) M	Carbon monoxide	1	1	U	Inhal	Unt-E	2		
272pi	Unknown adult (≥ 20 yrs) M	Carbon monoxide	1	1	A	Inhal	Int-S	1		
273pi	Unknown adult (≥ 20 yrs) F	Carbon monoxide	1	1	A	Inhal	Unt-E	2		
274pi	Unknown adult (≥ 20 yrs) M	Carbon monoxide	1	1	A	Inhal	Int-S	1		
275pi	Unknown age M	Carbon monoxide	1	1	A	Inhal	Unt-G	1		
276pi	Unknown age U	Carbon monoxide	1	1	A	Inhal	Unt-E	2		
277phi	Unknown age U	Carbon monoxide	1	1	A	Inhal	Unt-E	2		
See Also case 114, 117, 857										
Heavy Metals										
[278ha]	39 y M				A	Ingst	Int-S	1		
		Thallium	1	1					Thallium	5000 ng/mL In Serum @ Unknown
		Thallium	1	1					Thallium	68000 mcg/L In Urine (quantitative only) @ Unknown
Hydrocarbons										
[279ph]	12 y M				A	Ingst	Int-A	2		
		Carbon tetrachloride	1	1						
		Calcium magnesium carbonate	2	2						
		Beta-cyclodextrin	3	3						
280ph	22 y M				A/C	Inhal	Int-A	2		
		Hydrocarbon (fluorinated)	1	1						
281ph	22 y M				A	Inhal	Int-U	1		
		Hydrocarbon (fluorinated)	1	1					1,1-difluoroethane	35 mcg/mL In Blood (unspecified) @ 5 m (pe)
282ai	28 y M				U	Inhal + Unk	Int-A	1		
		Hydrocarbon (fluorinated)	1	1						
		Ethanol (beverage)	2	2						
		Substance (non-drug), N.O.S.	3	3						
283ai	29 y M				U	Inhal + Unk	Int-S	1		
		Hydrocarbon (fluorinated)	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
284pha	32 y M	Promethazine	2	2	A	Inhal	Int-A	1		
		Ethanol (beverage)	3	3						
		Hydrocarbon (fluorinated)	1	1						
		Marijuana, N.O.S.	2	2						
		Marijuana, N.O.S.	2	2					Delta-9-thc	2.2 ng/mL In Blood (unspecified) @ Unknown
									Delta-9-carboxy-thc	41 ng/mL In Blood (unspecified) @ Unknown
285ai	32 y F				U	Inhal	Int-A	1		
		Hydrocarbon (fluorinated)	1	1						
286ph	32 y M				A	Inhal	Unk	2		
		Hydrocarbon (fluorinated)	1	1						
287h	33 y M				A	Inhal	Int-A	2		
		Hydrocarbon (fluorinated)	1	1						
288ph	34 y F				A	Inhal	Int-A	1		
		Hydrocarbon (fluorinated)	1	1						
289ai	37 y F				U	Inhal	Int-A	1		
		Hydrocarbon (fluorinated)	1	1						
290h	37 y M				A/C	Inhal	Int-S	2		
		Hydrocarbon (fluorinated)	1	1						
291h	40 y M				C	Inhal + Derm	Int-A	2		
		Hydrocarbon (fluorinated)	1	1						
292h	49 y F				U	Inhal	Int-A	2		
		Hydrocarbon (fluorinated)	1	1						
See Also case 626, 822										
Industrial Cleaners										
293	85 y M				A	Ingst + Aspir	Unt-M	2		
		Industrial cleaner, NOS	1	1						
		Cleaner (acid)	2	2						
Miscellaneous Foods										
294ai	46 y F				U	Unk	Int-A	1		
		Water	1	1						
		Amlodipine	2	2						
		Bupropion	3	3						
295pai	64 y M				A/C	Ingst	Int-U	1		
		Vanilla extract	1	1						
296	78 y M				A	Ingst	Int-U	1		
		Sodium bicarbonate	1	1						
Mushrooms, Naturally Occurring										
297h	35 y M				A	Ingst	Unt-M	2		
		Mushroom (unknown)	1	1						
[298h]	69 y F				A	Ingst	Unt-M	1		
		Mushroom (cyclopeptide)	1	1						
See Also case 1888, 1889, 2036										
Other/Unknown Nondrug Substances										
299h	67 y F				A	Ingst + Unk	Unk	2		
		Substance (non-drug), N.O.S.	1	1						
		Acetaminophen	2	2						
		Cocaine	3	3						
		Alprazolam	4	4						
See Also case 46, 72, 279, 282, 331, 370, 430, 464, 465, 485, 486, 547, 575, 619, 634, 676, 697, 717, 737, 758, 792, 810, 893, 1047, 1080, 1270, 1507, 1911, 1913, 1940, 1958, 1965, 1973, 1979, 1989, 1994, 1997, 2003, 2010, 2011, 2018, 2023, 2026, 2027, 2045, 2054, 2063, 2077, 2095, 2097, 2122, 2139, 2163, 2173, 2179, 2180, 2187, 2199, 2200, 2202, 2208, 2210										
Pesticides										
[300h]	26 y F				A	Ingst + Par	Int-S	1		
		Diquat	1	1					Diquat dibromide	23000 ng/mL In Serum @ Unknown
[301ha]	30 y F				A/C	Ingst	Int-S	2		
		Bromethalin	1	1						
302	36 y M				A	Ingst	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Glyphosate	1	1						
		Ethanol (beverage)	2	2					Ethanol	192 mg/dL In Blood (unspecified) @ Unknown
303h	39 y F	Pyrethroids, N.O.S.	1	1	A	Inhal	Int-M	2		
304	41 y F	Boric acid	1	1	A	Ingst	Int-S	2		
		Ethanol (beverage)	2	2						
305h	42 y M	Glyphosate	1	1	A	Ingst	Int-S	2		
		Ethanol (beverage)	2	2						
[306h]	43 y M	Dinitrophenol	1	1	A/C	Ingst	Int-S	1		
[307h]	59 y M	Paraquat	1	1	A	Ingst	Int-S	1		
308	65 y M	Glyphosate	1	1	A	Ingst	Int-S	3		
[309h]	68 y M	Phosphine	1	1	A	Ingst	Int-S	2		
310h	82 y F	Paraquat	1	1	A	Ingst	Unt-M	3		
[311h]	82 y M	Malathion	1	1	A	Ingst	Int-S	2		
312h	87 y M	Glyphosate	1	1	A	Ingst	Int-S	2		
See Also case 14, 562										
Plants										
[313]	8 y M	<i>Taxus canadensis</i>	1	1	A	Ingst	Unt-G	3		
[314ph]	49 y M	Ibogaine	1	1	A	Ingst	Int-M	2	Benzoylcognine	2340 ng/mL In Serum @ Unknown
		Ibogaine	1	1					Ibogaine	5330 ng/mL In Serum @ Unknown
[315p]	62 y F	<i>Cerbera odallam</i>	1	1	A	Ingst	Int-S	2		
Tobacco/Nicotine/eCigarette Products										
[316pha]	31 y F	Nicotine (liquid)	1	1	A	Ingst	Int-S	1	Nicotine	700 ng/mL In Blood (unspecified) @ Unknown
		Nicotine (liquid)	1	1					Cotinine	9700 ng/mL In Blood (unspecified) @ Unknown
Pharmaceutical Exposures										
Analgesics										
317ai	1 y M	Fentanyl (non-prescription)	1	1	U	Unk	Unk	1		
318ai	1 y M	Fentanyl (non-prescription)	1	1	U	Unk	Unk	1		
[319ph]	3 y M	Fentanyl (non-prescription)	1	1	A	Unk	Unt-G	1	Fentanyl	0.15 ng/mL In Serum @ 21 h (pe)
		Fentanyl (non-prescription)	1	1					Norfentanyl	1.33 ng/mL In Serum @ 21 h (pe)
320pi	10 y M	Fentanyl (non-prescription)	1	1	A	Inhal	Int-A	1		
321h	13 y F	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	142 mcg/mL In Blood (unspecified) @ Unknown
322p	14 y F	Fentanyl (non-prescription)	1	1	A	Inhal	Int-A	2		
		Fentanyl analogs (despropionyl)	2	2						
323	15 y F				A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
324ph	15 y F	Acetaminophen	1	1					Acetaminophen (apap)	59 mcg/mL In Blood (unspecified) @ Unknown
		Diphenhydramine	2	2						
		Ethanol (beverage)	3	3					Ethanol	17 mg/dL In Blood (unspecified) @ Unknown
		Fentanyl (non-prescription)	1	1	A	Ingst + Unk	Unk	1		
		Xylazine	2	2						
[325pha]	15 y F	Ethanol (beverage)	3	3					Ethanol	97 mg/dL In Blood (unspecified) @ Unknown
		Fentanyl (non-prescription)	1	1	A	Unk	Int-S	1	Fentanyl	208.9 ng/mL In Urine (quantitative only) @ Unknown
		Fentanyl (non-prescription)	1	1					Norfentanyl	316.2 ng/mL In Urine (quantitative only) @ Unknown
326h	15 y F	Methamphetamine	2	2						
		Acetaminophen	1	1	A/C	Ingst	Int-S	1	Acetaminophen (apap)	85 mcg/mL In Blood (unspecified) @ Unknown
327h	15 y F	Iron, N.O.S.	2	2						
		Acetaminophen	1	1	A/C	Ingst	Int-S	2	Acetaminophen (apap)	35 mcg/mL In Blood (unspecified) @ Unknown
328	15 y F	Acetaminophen	1	1	A	Ingst	Int-S	1		
329i	16 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
330ai	16 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
331ai	16 y M	Substance (non-drug), N.O.S.	3	3						
		Acetaminophen	1	1	U	Ingst	Unk	1		
332	16 y F	Acetaminophen	1	1	A	Unk	Int-A	1		
333pha	16 y F	Fentanyl (non-prescription)	1	1	U	Ingst	Int-U	1		
		Oxycodone	1	1						
334ph	16 y M	Citalopram	2	2					Citalopram	45 mg/mL In Urine (quantitative only) @ Unknown
		Energy drink, N.O.S.	3	3	U	Ingst	Int-S	2	Norfentanyl	4.4 ng/mL In Blood (unspecified) @ Unknown
335pha	16 y F	Fentanyl (non-prescription)	1	1					Fentanyl	6.9 ng/mL In Blood (unspecified) @ Unknown
		Fentanyl (non-prescription)	1	1					Fluoxetine	100 ng/mL In Serum @ Unknown
		Fluoxetine	2	2					Norfluoxetine	40 ng/mL In Serum @ Unknown
		Fluoxetine	2	2						
		Acetaminophen/oxycodone	3	3						
336ai	17 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
337h	17 y F	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	18 mcg/mL In Serum @ Unknown
338	17 y F	Acetaminophen	1	1	U	Ingst	Unk	2	Acetaminophen (apap)	149 mcg/mL In Serum @ Unknown
339ai	17 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
340h	17 y F	Acetaminophen	1	1	A	Ingst	Int-S	1		
341pha	17 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1	4-anpp (despropionyl fentanyl)	0.8 ng/mL In Blood (unspecified) @ Unknown
		Fentanyl (non-prescription)	1	1					Norfentanyl	1.9 ng/mL In Blood (unspecified) @ Unknown
		Fentanyl (non-prescription)	1	1					Fentanyl	7.9 ng/mL In Blood (unspecified) @ Unknown
		Marijuana	2	2					11-oh-thc (11-hydroxy-delta-9-tetrahydrocannabinol)	21 ng/mL In Blood (unspecified) @ Unknown
		Marijuana	2	2					Delta-9-thc	27 ng/mL In Blood (unspecified) @ Unknown
		Marijuana	2	2					Delta-9-carboxy-thc	270 ng/mL In Blood (unspecified) @ Unknown
342ai	19 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
343ai	19 y F	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
344ai	19 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
345ai	19 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (carfentanyl)	2	2						
346ai	19 y M	Methamphetamine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
347h	19 y F	Dextromethorphan	3	3	A	Ingst	Int-S	2		
		Naproxen	3	1						
		Atenolol	2	2						
		Amlodipine	1	3						
		Loratadine	4	4						
		Ibuprofen	5	5						
		Ethanol (beverage)	6	6						
348h	19 y F	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	71 mcg/mL In Serum @ Unknown
		Quetiapine	2	2						
		Ibuprofen	3	3						
349i	20 y M	Fentanyl (prescription)	1	1	A/C	Inhal	Int-A	2		
350ai	20 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (despropionyl)	2	2						
351ai	20 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
352ai	20 y M	Fentanyl (non-prescription)	1	1						
		Morphine	2	2						
		Marijuana, N.O.S.	3	3						
[353ha]	20 y F	Salicylate	1	1	U	Unk	Int-S	1		
		Amphetamine	2	2						
		Colchicine	1	1	A	Ingst	Int-S	1	Colchicine	510 ng/mL In Blood (unspecified) @ Autopsy
354ai	20 y M	Prednisone	2	2						
		Ethinyl estradiol/norgestimate	3	3						
					U	Unk	Int-A	1		
355ai	21 y F	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2	U	Unk	Int-A	1		
356ha	21 y M	Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (despropionyl)	2	2						
		Acetaminophen	1	1	U	Ingst	Int-S	1	Acetaminophen (apap)	9 mcg/mL In Blood (unspecified) @ Unknown
357ai	21 y F	Isopropanol	2	2						
		Ethanol (beverage)	3	3						
		Iron, N.O.S.	4	4					Iron	326 mcg/dL In Blood (unspecified) @ Unknown
358h	21 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
359ai	21 y M	Acetaminophen	1	1	U	Ingst	Unk	1	Acetaminophen (apap)	47 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2					Ethanol	18 mg/dL In Blood (unspecified) @ Unknown
360pa	21 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
361h	21 y M	Oxycodone	3	3						
		Fentanyl (prescription)	1	1	A	Par	Int-A	1	4-anpp (despropionyl fentanyl)	0.32 ng/mL In Whole Blood @ Unknown
									Norfentanyl	4 ng/mL In Whole Blood @ Unknown
362ph	21 y F	Fentanyl (prescription)	1	1					Fentanyl	5.1 ng/mL In Whole Blood @ Unknown
		Methamphetamine	2	2					Methamphetamine	170 ng/mL In Whole Blood @ Unknown
									Amphetamine	36 ng/mL In Whole Blood @ Unknown
363h	22 y F	Methamphetamine	2	2						
		Acetaminophen	1	1	A/C	Ingst	Unt-M	1	Acetaminophen (apap)	48 mcg/mL In Serum @ Unknown
363h	22 y F				U	Ingst	Unk	1		
		Fentanyl (non-prescription)	1	1						
					A	Ingst + Unk	Int-S	2	Acetaminophen (apap)	19.6 mcg/mL In Blood (unspecified) @ 7.5h (pe)
363h	22 y F	Acetaminophen	1	1						
		Drug, N.O.S.	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
364ai	22 y M	Barbiturate (long acting)	3	3						
		Marijuana	4	4	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
365p	22 y F	Ethanol (beverage)	2	2						
		Marijuana, N.O.S.	3	3	U	Unk	Int-A	2		
		Fentanyl (non-prescription)	1	1						
366i	22 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Marijuana, N.O.S.	2	2						
367i	23 y F				A	Unk	Int-A	2		
		Opioid, N.O.S.	1	1						
					U	Unk	Int-A	1		
368ai	23 y F	Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
		Alprazolam	3	3						
369ai	23 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
370ai	23 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
371ai	23 y M	Substance (non-drug), N.O.S.	3	3						
					U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
372	23 y M	Methamphetamine	2	2						
					A	Inhal	Int-A	2		
		Fentanyl (prescription)	1	1						
373ai	24 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
374ai	24 y F	Amphetamine	3	3						
					U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
375ai	24 y M	Methamphetamine	2	2						
		Clonazepam	3	3						
					U	Unk	Int-A	1		
376ai	24 y F	Fentanyl (non-prescription)	1	1						
					U	Unk	Int-A	1		
		Cocaine	3	3						
377ai	24 y M	Fentanyl analogs (carfentanil)	2	2						
		Methamphetamine	3	3						
					U	Unk	Int-A	1		
378ai	24 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Bromazolam	3	3						
379ai	24 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
380ai	24 y F	Marijuana, N.O.S.	2	2						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
381ai	24 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Oxycodone	1	1						
		Cocaine	2	2						
382ai	24 y M	Ketamine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
383h	24 y F	Marijuana, N.O.S.	3	3						
		Ibuprofen	1	1	A	Ingst	Int-S	1		
384ai	25 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
385ai	25 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
386ai	25 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
387ai	25 y M	Methamphetamine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		<i>Mitragyna speciosa korthals</i>	2	2						
388ha	25 y M	Bromazolam	3	3	A	Ingst + Unk	Int-S	3		
		Fentanyl (non-prescription)	1	1						
389ai	25 y M	Fluoxetine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
390i	26 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
391ai	26 y M	Lorazepam	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3						
[392ha]	26 y M				A	Ingst	Int-S	1		
		Salicylate	1	1					Salicylate	54 mg/dL In Serum @ Unknown
		Salicylate	1	1					Salicylate	63 mg/dL In Serum @ Unknown
		Salicylate	1	1					Salicylate	79 mg/dL In Serum @ Unknown
		Salicylate	1	1					Salicylate	91 mg/dL In Serum @ Unknown
		Salicylate	1	1					Salicylate	98 mg/dL In Serum @ Unknown
393ai	26 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		<i>Mitragyna speciosa korthals</i>	2	2						
394ai	26 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
395h	26 y F	Fentanyl (non-prescription)	1	1	A/C	Unk	Int-U	2		
396ai	26 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
397ai	26 y M	Methamphetamine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
398ai	26 y M	Xylazine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
399	26 y M	Acetaminophen/hydrocodone	1	1	A	Ingst + Inhal	Int-S	1		
		Acetaminophen	2	2					Acetaminophen (apap)	13 mcg/mL In Blood (unspecified) @ Unknown
400pi	26 y F	Marijuana	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
401ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
402ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
403ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Bromazolam	3	3						
404ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
405ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Marijuana, N.O.S.	2	2						
406ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Alprazolam	2	2						
		Ethanol (beverage)	3	3						
407ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
		Ethanol (beverage)	3	3						
408ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
409ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Chlorpheniramine	2	2						
		Ethanol (beverage)	3	3						
410ai	27 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3						
411ai	27 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
412ai	27 y M	Xylazine	2	2	U	Unk	Int-A	1		
		Risperidone	3	3						
		Fentanyl (non-prescription)	1	1						
413h	27 y M	Methamphetamine	2	2	C	Ingst	Int-M	1	Acetaminophen (apap)	247 mcg/mL In Blood (unspecified) @ Unknown
		Dextromethorphan	3	3						
		Acetaminophen	1	1						
414h	27 y F	Acetaminophen	1	1	U	Ingst	Int-S	2	Acetaminophen (apap)	13 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	54 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2					Ethanol	208 mg/dL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2					Ethanol	80.7 mg/dL In Blood (unspecified) @ Unknown
		Metformin	3	3						
415h	27 y M	Prednisone	4	4	C	Ingst	Int-M	2	Acetaminophen (apap)	116.5 mcg/mL In Serum @ Unknown
		Omeprazole	5	5						
		Montelukast	6	6						
		Acetaminophen	1	1						
416ai	28 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
417i	28 y F	Marijuana, N.O.S.	2	2	A	Unk	Int-A	2		
418ai	28 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
419h	28 y F	Fentanyl (non-prescription)	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	6.7 mcg/mL In Serum @ 48 h (pe)
		Methadone	2	2						
		Alprazolam	3	3						
420ai	28 y M	Acetaminophen	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
421ai	28 y M	Methamphetamine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
422ai	28 y M	Cocaine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Oxcarbazepine	2	2						
423ai	28 y M	Trazodone	3	3	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	1	1						
		Cocaine	2	2						
424ai	28 y M	Marijuana, N.O.S.	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
425ai	28 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
426ai	28 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
		Ethanol (beverage)	3	3						
427ai	28 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
		Marijuana, N.O.S.	3	3						
428ai	28 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
429ai	28 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Alprazolam	3	3						
430ai	28 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
431ph	28 y F	Fentanyl (non-prescription)	1	1	A/C	Inhal	Int-A	2		
432h	28 y F	Acetaminophen	1	1	A	Ingst + Unk	Int-S	3	Acetaminophen (apap)	13 mcg/mL In Blood (unspecified) @ Unknown
		Diphenhydramine	2	2						
		Ethanol (beverage)	3	3						
		Salicylate	4	4					Salicylate	7.8mg/dL In Blood (unspecified) @ Unknown
433i	28 y M	Opioid, N.O.S.	1	1	A/C	Unk	Int-A	2		
434h	28 y F	Acetaminophen	1	1	A/C	Ingst	Int-S	2	Acetaminophen (apap)	51 mcg/mL In Blood (unspecified) @ Unknown
435ai	29 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
436ai	29 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (despropionyl)	2	2						
437ai	29 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaethylene	2	2						
		Tramadol	3	3						
438ai	29 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
439ai	29 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	2		
		Methamphetamine	2	2						
440ai	29 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Bromazolam	2	2						
		Methadone	3	3						
441ai	29 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
442ai	29 y M	Oxycodone	1	1	U	Unk	Unk	1		
		Morphine	2	2						
		Trazodone	3	3						
443h	29 y F				A	Ingst	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
444pha	29 y F	Acetaminophen	1	1					Acetaminophen (apap)	213 mcg/mL In Blood (unspecified) @ 9 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	23 mcg/mL In Blood (unspecified) @ 30 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	62 mcg/mL In Blood (unspecified) @ 19.5 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	8 mcg/mL In Blood (unspecified) @ 46 h (pe)
		Ibuprofen	2	2	U	Unk	Int-U	1		
445ha	29 y F	Fentanyl (prescription)	1	1						
		Methamphetamine	3	2						
		Marijuana, N.O.S.	2	3	A	Ingst	Int-S	1		
		Acetaminophen	1	1					Acetaminophen (apap)	107 mcg/mL In Blood (unspecified) @ 25 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	534 mcg/mL In Blood (unspecified) @ 1 h (pe)
446ai	29 y M	Iron	2	2					Iron	266 mcg/dL In Blood (unspecified) @ Unknown
		Iron	2	2					Iron	318 mcg/dL In Blood (unspecified) @ 10 h (pe)
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1	A/C	Unk	Int-A	2		
		Acetaminophen	1	1	A	Ingst	Int-S	2		
449ai	30 y M	Acetaminophen	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Alprazolam	3	3						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
450ai	30 y M	Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Tramadol	3	3						
451ai	30 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Tramadol	3	3						
		Opioid, N.O.S.	1	1	A	Par	Int-A	2		
		Fentanyl (non-prescription)	1	1	U	Inhal	Unk	2	Fentanyl	148 ng/mL In Urine (quantitative only) @ Unknown
453pha	30 y F	Fentanyl (non-prescription)	1	1					Norfentanyl	318.5 ng/mL In Urine (quantitative only) @ Unknown
		Methamphetamine	2	2						
		Cocaine	3	3						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
455ai	30 y F				U	Unk	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
456ai	30 y M	Morphine	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
457pha	30 y F	Acetaminophen/diphenhydramine	1	1	C	Ingst + Unk	Unk	1	Acetaminophen (apap)	143.7 mcg/mL In Blood (unspecified) @ Unknown
458h	30 y M	Drug, N.O.S.	2	2	A/C	Ingst	Int-M	2		
		Salicylate	3	3						
		Ethanol (beverage)	4	4						
		Acetaminophen	1	1						
		Ibuprofen	2	2						
459ai	30 y M				U	Unk	Int-A	1		
460ai	30 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-S	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
461pha	30 y M	Hydrocodone, N.O.S.	1	1	U	Unk	Unk	1		
		Tramadol	2	2						
		Ethanol (beverage)	3	3						
		Drug, N.O.S.	1	1					Fentanyl	1.8 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	2	1						
		Fentanyl analogs (para-fluoro)	3	1						
		Gabapentin	4	1						
462ai	30 y M	Morphine	1	1	A	Unk	Int-A	1		
		Chlorpheniramine	2	2						
		Dextromethorphan	3	3						
463ai	30 y F				U	Unk	Int-A	1		
464ai	30 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Fentanyl (non-prescription)	1	1						
465ai	30 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	3	3						
		Fentanyl (non-prescription)	1	1						
466h	30 y F	Methamphetamine	2	2	C	Ingst	Unt-M	2	Acetaminophen (apap)	12 mcg/mL In Serum @ Unknown
		Substance (non-drug), N.O.S.	3	3						
		Acetaminophen	1	1						
467	30 y F				A/C	Ingst	Int-U	2		
		Fentanyl (non-prescription)	1	1						
		Clonazepam	2	2						
468i	30 y M	Opioid, N.O.S.	1	1	A/C	Unk	Int-A	2		
469h	30 y M				C	Ingst	Int-M	2		
		Acetaminophen	1	1					Acetaminophen (apap)	34 mcg/mL In Blood (unspecified) @ 17.5 h (pe)
		Iron	2	2						
470ha	30 y F				C	Ingst	Unt-T	1		461 mcg/dL In Blood (unspecified) @ 3 d (pe)

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Acetaminophen	1	1					Acetaminophen (apap)	34 mcg/mL In Serum @ Unknown
		Salicylate	2	2					Salicylate	6.8 mg/dL In Serum @ Unknown
471ai	31 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
472i	31 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
473ai	31 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
474ai	31 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
475ai	31 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Trazodone	2	2						
		Hydroxyzine	3	3						
476ai	31 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
477ai	31 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Hydroxyzine	3	3						
478ai	31 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
479ai	31 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
480ai	31 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Fentanyl analogs (para-fluoro)	3	3						
481ai	31 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
482ai	31 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
483ai	31 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
484h	31 y F	Acetaminophen	1	1	U	Ingst	Int-S	1	Acetaminophen (apap)	207.5 mcg/mL In Serum @ Unknown
485ai	31 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						
486ai	31 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
487phi	31 y M	Acetaminophen/ oxycodone	1	1	A	Ingst + Unk	Int-A	2		
		Amphetamine, N.O.S.	2	2						
		Carisoprodol	3	3						
488ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
489ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Acetaminophen	2	2						
490h	32 y F	Acetaminophen	1	1	C	Ingst	Unt-T	2	Acetaminophen (apap)	34 mcg/mL In Serum @ Unknown
		Ibuprofen	2	2						
		Bismuth subsalicylate	3	3						
491ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
492ai	32 y F	Fentanyl analogs (para-fluoro)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
493ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Amphetamine	2	2						
		Ethanol (beverage)	3	3						
494ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Amphetamine	2	2						
		Ethanol (beverage)	3	3						
495ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
496ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
497ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
		Ethanol (beverage)	3	3						
498ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
499ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
500ai	32 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
501ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Buprenorphine	3	3						
502ai	32 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
503ai	32 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
504h	32 y F	Methamphetamine	2	2	U	Ingst	Unk	3		
		Marijuana, N.O.S.	3	3						
505ph	32 y F	Acetaminophen/ diphenhydramine	1	1	A	Ingst	Int-U	2		
		Fentanyl (non-prescription)	1	1						
506h	32 y M	Acetaminophen/ caffeine/ acetylsalicylate	1	1	A	Ingst	Int-S	2	Salicylate	87.8 mg/dL In Serum @ Unknown
		Haloperidol	2	2						
507ai	33 y F	Lorazepam	3	3	U	Unk	Int-A	1		
		Midazolam	4	4						
508i	33 y F	Caffeine	5	5	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
509ai	33 y M	Oxycodone	2	2	U	Unk	Int-A	2		
		Clonazepam	3	3						
510	33 y F	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
511ai	33 y M	Acetaminophen	1	1	C	Ingst	Unt-M	1	Acetaminophen (apap)	100 mcg/mL In Serum @ 46 h (pe)
512ai	33 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
513ai	33 y M	Marijuana, N.O.S.	3	3	U	Unk	Int-S	1		
		Fentanyl (non-prescription)	1	1						
514ai	33 y M	Fentanyl analogs (para-fluoro)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
515ai	33 y F	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
516ai	33 y M	Methamphetamine	2	2	A	Unk	Int-A	1		
		Marijuana, N.O.S.	3	3						
517ai	33 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Alprazolam	3	3						
518i	33 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
519ai	33 y M	Ethanol (beverage)	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
520ai	33 y M	Cocaine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Xylazine	3	3						
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
521ai	33 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Heroin	2	2						
		Methamphetamine	3	3						
522ai	33 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (despropionyl)	2	2						
523ai	33 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3						
524ph	33 y M	Fentanyl (non-prescription)	1	1	U	Unk	Unk	2		
		Drug, N.O.S.	2	2						
525h	33 y M	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	177 mcg/mL In Blood (unspecified) @ 15 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	377 mcg/mL In Blood (unspecified) @ 5 m (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	57 mcg/mL In Blood (unspecified) @ 30 h (pe)
526h	33 y U	Acetaminophen	1	1	U	Ingst	Int-S	1	Acetaminophen (apap)	14 mcg/mL In Serum @ Unknown
527ai	34 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
		Ethanol (beverage)	3	3						
528i	34 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Cocaine	3	3						
529ai	34 y F	Fentanyl analogs (para-fluoro)	1	1	U	Unk	Int-A	1		
530ai	34 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
531ai	34 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
532ai	34 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3						
533ai	34 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Olanzapine	3	3						
534ai	34 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Amphetamine	2	2						
		Marijuana, N.O.S.	3	3						
535i	34 y F	Acetaminophen/oxycodone	1	1	A	Unk	Int-A	2		
		Cocaine	2	2						
536ai	34 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
537ai	34 y M	Bromazolam	2	2	U	Unk	Int-A	1		
		Buprenorphine	3	3						
		Fentanyl (non-prescription)	1	1						
538i	34 y M	Methamphetamine	2	2	A	Unk	Int-A	2		
539ai	34 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
540h	34 y F	Fentanyl (non-prescription)	1	1	A	Ingst	Unk	2	Acetaminophen (apap)	61.8 mcg/mL In Serum @ Unknown
		Ethanol (beverage)	2	2						
		Acetaminophen	1	1						
541ai	34 y F				U	Unk	Int-A	1		
542ai	34 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Methadone	3	3						
543ai	34 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Fentanyl (non-prescription)	1	1						
544ai	34 y M	Cocaine	2	2	U	Unk	Int-A	1		
		Phenobarbital	3	3						
		Fentanyl (non-prescription)	1	1						
545ai	34 y M	Amphetamine	2	2	U	Unk	Int-A	1		
		Marijuana, N.O.S.	3	3						
		Fentanyl (non-prescription)	1	1						
546ai	34 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
547ai	34 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
548ai	34 y M	Substance (non-drug), N.O.S.	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
549ph	34 y M				A/C	Ingst + Unk	Int-A	2		
		Fentanyl (non-prescription)	1	1					4-anpp (despropionyl fentanyl)	0 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	1	1					Norfentanyl	1.7 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	1	1					Fentanyl	7.8 ng/mL In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	3	2					Amphetamine	27.9 ng/mL In Blood (unspecified) @ Autopsy
		Methamphetamine	2	2						
		Methamphetamine	2	2					Methamphetamine	80.8 ng/mL In Blood (unspecified) @ Autopsy
550ph	35 y F				U	Ingst + Unk	Int-S	2		
551ai	35 y F	Fentanyl (prescription)	1	1	U	Unk	Int-A	1		
		Opioid, N.O.S.	2	2						
		Clonazepam	3	3						
		Marijuana (edible)	4	4						
		Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
552ai	35 y M	Chlordiazepoxide	3	3						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
553ai	35 y F	Marijuana, N.O.S.	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
554ai	35 y F	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
555ai	35 y M	Diphenhydramine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
556ai	35 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Methadone	3	3						
557ai	35 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
558ai	35 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Methadone	1	1						
		Diphenhydramine	2	2						
		Alprazolam	3	3						
559	35 y F	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	304 mcg/mL In Blood (unspecified) @ Unknown
560i	35 y M				U	Unk	Int-A	2		
561ai	35 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
562h	35 y F	Acetaminophen/drugs, N.O.S.	1	1	A	Ingst	Unk	2	Acetaminophen (apap)	8.7 mcg/mL In Serum @ Unknown
		Rodenticide, N.O.S.	2	2						
		Topiramate	3	3						
		Marijuana	4	4						
563i	35 y M	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
564ai	35 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
565ai	35 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
566ai	35 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
567i	35 y F	Opioid, N.O.S.	1	1	A/C	Unk	Int-A	2		
568i	35 y F				U	Unk	Int-A	2		
		Opioid, N.O.S.	1	1						
		Fentanyl (prescription)	2	2						
569i	35 y M	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
570i	35 y M	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
571ai	35 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
572ai	35 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
573ai	35 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
574ai	35 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
575ai	35 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
576ai	35 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
577i	35 y M	Ethanol (beverage)	2	2	A/C	Unk	Int-A	2		
578ai	36 y F	Substance (non-drug), N.O.S.	3	3	U	Unk	Int-A	1		
579ai	36 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
580ai	36 y M	Cocaine	2	2	U	Unk	Int-A	1		
581ai	36 y M	Ethanol (beverage)	3	3	U	Unk	Int-A	1		
582i	36 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	2		
583i	36 y M	Marijuana, N.O.S.	2	2	A	Unk	Int-A	2		
584ai	36 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
585i	36 y M	Alprazolam	2	2	U	Unk	Int-A	1		
586pha	36 y F	Fentanyl (non-prescription)	1	1	A	Ingst + Par	Int-A	1	Norfentanyl	12 ng/mL In Blood (unspecified) @ 90 m (pe)
587ai	36 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1	Fentanyl	18 ng/mL In Blood (unspecified) @ 90 m (pe)
588ai	36 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1	4-anpp (despropionyl fentanyl)	9.2 ng/mL In Blood (unspecified) @ 90 m (pe)
		Cocaine	2	2					Benzoyllecognine	1300 ng/mL In Blood (unspecified) @ 90 m (pe)
		Cocaine	2	2					Ecgonine methyl ester	240 ng/mL In Blood (unspecified) @ 90 m (pe)
587ai	36 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
588ai	36 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
589ai	36 y M	Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3	U	Unk	Int-A	1		
590ai	36 y F	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Cocaine	3	3	U	Unk	Int-A	1		
591ai	36 y F	Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3	U	Unk	Unk	1		
592ai	36 y M	Oxycodone	1	1						
		Quetiapine	2	2						
					U	Unk	Int-A	1		
593ai	36 y M	Fentanyl (prescription)	1	1						
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3	U	Unk	Int-A	1		
594ai	36 y M	Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
		Ethanol (beverage)	3	3	U	Unk	Int-A	1		
595	36 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
					A	Ingst	Int-S	1		
596ph	36 y M	Acetaminophen	1	1					Acetaminophen (apap)	126 mcg/mL In Blood (unspecified) @ 4 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	15.8 mcg/mL In Blood (unspecified) @ 72 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	156 mcg/mL In Blood (unspecified) @ 3 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	26.2 mcg/mL In Blood (unspecified) @ 56 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	36.5 mcg/mL In Blood (unspecified) @ 37.5 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	37.6 mcg/mL In Blood (unspecified) @ 44 h (pe)
597ph	37 y F	Acetaminophen/hydrocodone	1	1	A/C	Ingst	Int-U	1		
		Alprazolam	2	2						
598ai	37 y F	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Cocaine	3	3	U	Unk	Int-A	1		
599pha	37 y F	Oxycodone	1	1						
		Tramadol	2	2						
		Ethanol (beverage)	3	3	U	Ingst + Unk	Int-U	2		
600ai	37 y M	Methadone	1	1						
		Cocaine	2	2						
		Benzodiazepine, N.O.S.	3	3						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Alprazolam	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
601ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
602i	37 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
603ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
604h	37 y F	Acetaminophen	1	1	A	Ingst	Int-S	2		
		Venlafaxine	2	2						
		Clozapine	3	3						
		Buspirone	4	4						
		Trazodone	5	5						
		Benzotropine	6	6						
605ai	37 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-S	1		
		Methamphetamine	2	2						
		Methadone	3	3						
606ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analog (acetyl)	2	2						
		Buprenorphine	3	3						
607ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
608ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
609ai	37 y F	Hydrocodone, N.O.S.	1	1	U	Unk	Int-S	1		
		Venlafaxine	2	2						
		Aripiprazole	3	3						
610ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Trazodone	2	2						
		Ethanol (beverage)	3	3						
611	37 y F	Salicylate	2	1	A	Ingst	Int-U	1		
		Acetaminophen	1	2					Acetaminophen (apap)	557 mcg/mL In Serum @ Unknown
		Acetaminophen	1	2					Salicylate	80.9 mcg/dL In Serum @ Unknown
612ai	37 y M	Oxycodone	1	1	U	Unk	Int-A	1		
		Morphine	2	2						
		Buprenorphine	3	3						
613i	37 y M	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
614ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
615ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Olanzapine	2	2						
616ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
617ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3						
618ai	37 y F				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
619ai	37 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Fentanyl (non-prescription)	1	1						
620h	37 y F	Methamphetamine	2	2	A	Ingst	Int-S	1	Acetaminophen (apap)	156.3 mcg/mL In Blood (unspecified) @ Unknown
		Substance (non-drug), N.O.S.	3	3						
		Acetaminophen	1	1						
621h	37 y F				C	Ingst	Int-M	1	Acetaminophen (apap)	125 mg/mL In Serum @ Unknown
		Acetaminophen	1	1						
622h	37 y M				U	Ingst	Int-A	3	Acetaminophen (apap)	53.2 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1						
623	37 y F				C	Ingst	Int-M	1	Acetaminophen (apap)	54 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
624ai	38 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
625	38 y F	Methamphetamine	2	2	C	Ingst	Int-M	1	Acetaminophen (apap)	22.9 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen (ER)	1	1						
626ai	38 y M	Ethanol (beverage)	2	2	U	Inhal + Unk	Int-A	1		
		Cocaine	3	3						
		Hydrocodone, N.O.S.	1	1						
627ai	38 y M	Hydrocarbon (fluorinated)	2	2	U	Unk	Int-A	1		
		Clonidine	3	3						
		Fentanyl (non-prescription)	1	1						
628h	38 y M	Fentanyl analogs (para-fluoro)	2	2	C	Ingst	Int-M	2		
		Cocaine	3	3						
		Acetaminophen	1	1						
629ai	38 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
630ai	38 y F	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
631ai	38 y M	Lorazepam	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
632ai	38 y M	Marijuana, N.O.S.	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
633ai	38 y M	Aripiprazole	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
634ai	38 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
635ai	38 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	3	3						
		Fentanyl (non-prescription)	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
636ai	38 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Marijuana, N.O.S.	3	3						
		Fentanyl (non-prescription)	1	1						
637ha	38 y F	Methamphetamine	2	2	A	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1					Fentanyl	2.3 ng/mL In Blood (unspecified) @ 6 h (pe)
		Fentanyl (non-prescription)	1	1					Norfentanyl	5.1 ng/mL In Blood (unspecified) @ 6 h (pe)
		Falsified pharmaceutical, N.O.S.	2	2						
		Methamphetamine	3	3					Methamphetamine	610 ng/mL In Blood (unspecified) @ 6 h (pe)
		Methamphetamine	3	3					Amphetamine	72 ng/mL In Blood (unspecified) @ 6 h (pe)
638ha	38 y M	Acetaminophen	1	1	A	Ingst	Int-S	3	Acetaminophen (apap)	569 mcg/mL In Blood (unspecified) @ 3.5 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	620 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	680 mcg/mL In Blood (unspecified) @ 35 m (pe)
		Ibuprofen	2	2						
		Ethanol (beverage)	3	3					Ethanol	0.31 g/dL In Blood (unspecified) @ 35 m (pe)
		Sertraline	4	4					Sertraline	220 ng/mL In Blood (unspecified) @ Unknown
639pi	38 y M	Heroin	1	1	A	Unk	Int-A	2		
640ai	39 y M	Heroin	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
641ai	39 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3						
642ha	39 y M	Acetaminophen/diphenhydramine	1	1	C	Ingst	Unt-U	1	Diphenhydramine	1100 ng/mL In Blood (unspecified) @ Unknown
		Acetaminophen/diphenhydramine	1	1					Acetaminophen (apap)	39 mcg/mL In Serum @ Unknown
643ai	39 y M	Ibuprofen	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Bromazolam	2	2						
644i	39 y F	Ethanol (beverage)	3	3	U	Ingst + Unk	Int-A	2		
		Opioid, N.O.S.	1	1						
645ai	39 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Heroin	1	1						
646ai	39 y F	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Alprazolam	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
647h	39 y F	Acetaminophen	1	1	C	Ingst	AR-D	2	Acetaminophen (apap)	28 mcg/mL In Serum @ Unknown
		Ethanol (beverage)	2	2					Ethanol	43 mg/dL In Serum @ Unknown
		Ethanol (beverage)	2	2					Salicylate	5.3 mg/dL In Serum @ Unknown
648h	39 y F				A	Ingst	Int-S	2		
		Methadone	1	1						
		Quetiapine	2	2						
		Clonidine	3	3						
		Gabapentin	4	4						
649i	39 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
650ai	39 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
651ai	39 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
652ai	39 y M				U	Unk	Int-A	1		
		Morphine	1	1						
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
653ai	39 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
654ai	39 y F				U	Unk	Int-A	1		
		Oxycodone	1	1						
		Ethanol (beverage)	2	2						
655ai	39 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
656ph	39 y F				C	Unk	Int-A	2		
		Fentanyl (non-prescription)	1	1						
657ai	40 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
658i	40 y F	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
659h	40 y F				U	Ingst	Int-S	2		
		Acetaminophen	1	1					Acetaminophen (apap)	35 mcg/mL In Plasma @ Unknown
660ai	40 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3						
661ai	40 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		<i>Mitragyna speciosa korthals</i>	2	2						
		Alprazolam	3	3						
662ai	40 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
663i	40 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
664ai	40 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3						
665h	40 y F				C	Ingst	Unt-T	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Acetaminophen	1	1					Acetaminophen (apap)	40 mcg/mL In Serum @ 2 d (pe)
666ai	40 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Morphine	2	2						
667ai	40 y M	Ethanol (beverage)	3	3	U	Unk	Int-A	1		
		Heroin	1	1						
		<i>Mitragyna speciosa korthals</i>	2	2						
668ai	40 y M	Alprazolam	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
669ai	40 y M	Marijuana, N.O.S.	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
670ai	40 y M	Buprenorphine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
671ai	40 y M	Ethanol (beverage)	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
672ai	40 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
673ai	40 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
674	40 y M	Methadone	3	3	A	Ingst	Int-U	1	Acetaminophen (apap)	115 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
		Ethanol (beverage)	2	2	U	Unk	Int-A	1		
675ai	40 y M	Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
676ai	40 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						
677h	40 y F	Acetaminophen	1	1	A/C	Ingst	Unt-M	2	Acetaminophen (apap)	5 mg/dL In Blood (unspecified) @ Unknown
		Acetaminophen/codeine	2	2						
		Acetaminophen/hydrocodone	3	3						
		Ibuprofen	4	4						
678h	40 y F	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	359 mcg/mL In Serum @ Unknown
		Trazodone	2	2						
679ph	41 y M	Fentanyl (non-prescription)	1	1	A	Ingst + Inhal	Int-A	1		
		Stimulant/street drug, N.O.S.	2	2						
680ai	41 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
681ai	41 y M	Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
		Ethanol (beverage)	3	3						
682ai	41 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3						
683ai	41 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
684ai	41 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
685ai	41 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Oxycodone	1	1						
		Methadone	2	2						
686i	41 y M	Alprazolam	3	3	A	Unk	Int-A	2		
687ai	41 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
688ai	41 y M	Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
		Marijuana, N.O.S.	3	3						
689ai	41 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
		Fentanyl (non-prescription)	1	1						
690i	41 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
691ai	41 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
692ai	41 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
693ph	41 y F	Ethanol (beverage)	3	3						
		Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
694ai	41 y F	Acetaminophen	1	1	U	Unk	Unk	2	Acetaminophen (apap)	63.3 mcg/mL In Blood (unspecified) @ Unknown
		Fentanyl (non-prescription)	1	1						
		Tizanidine	2	2						
695h	41 y F	Hydroxyzine	3	3	A/C	Ingst	Int-S	1		
696ai	41 y F	Acetaminophen	1	1	U	Unk	Int-S	1		
697ai	41 y M	Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3						
698pha	41 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
699h	41 y M	Fentanyl (non-prescription)	2	1					Fentanyl	2.2 ng/mL In Blood (unspecified) @ Unknown
		Cocaine	1	2					Ecgonine methyl ester	0.179 mg/L In Blood (unspecified) @ Unknown
		Cocaine	1	2					Cocaine	0.209 mg/L In Blood (unspecified) @ Unknown
		Cocaine	1	2					Benzoyllecognine	1.94 mg/L In Blood (unspecified) @ Unknown
700ha	41 y M	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	167 mcg/mL In Blood (unspecified) @ 5 d (pe)
701h	41 y M	Ethanol (beverage)	2	2	A	Ingst	Int-S	1		
		Salicylate	1	1					Salicylate	45 mg/dL In Blood (unspecified) @ Autopsy
		Salicylate	1	1					Salicylate	57 mg/dL In Blood (unspecified) @ Unknown
		Salicylate	1	1					Salicylate	69 mg/dL In Blood (unspecified) @ Unknown
		Cocaine	2	2					Benzoyllecognine	1800 ng/mL In Blood (unspecified) @ Autopsy
		Cocaine	2	2					Ecgonine methyl ester	250 ng/mL In Blood (unspecified) @ Autopsy
		Cocaine	2	2					Cocaine	27 ng/mL In Blood (unspecified) @ Autopsy
		Ethanol (beverage) Marijuana, N.O.S.	3 4	3 4					Delta-9-thc	1.6 ng/mL In Blood (unspecified) @ Unknown
702h	41 y M	Salicylate	1	1	A	Ingst	Int-S	1	Salicylate	143.9 mg/dL In Blood (unspecified) @ Unknown
		Salicylate	1	1					Salicylate	59.5 mg/dL In Blood (unspecified) @ Unknown
703ai	42 y F	Acetaminophen	1	1	U	Ingst	Int-S	3	Acetaminophen (apap)	62.2 mcg/mL In Serum @ Unknown
704ai	42 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Alprazolam	3	3						
705h	42 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
706ai	42 y M	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	460 mcg/mL In Blood (unspecified) @ Unknown
707ai	42 y M	Methylphenidate	2	2						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
707ai	42 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
708ai	42 y M	Marijuana, N.O.S.	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
709i	42 y M	Ethanol (beverage)	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
710h	42 y M	Methamphetamine	2	2	A	Ingst	Int-S	1	Acetaminophen (apap)	226.3 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
711ai	42 y M	Ibuprofen	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
712ai	42 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
713ai	42 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Amphetamine	3	3						
714ai	42 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Heroin	1	1						
715pha	42 y M	Fentanyl (non-prescription)	2	2	U	Ingst + Unk	Int-A	2	Para-fluorofentanyl	2.8 ng/mL In Serum @ Unknown
		Methamphetamine	3	3						
716ai	42 y M	Fentanyl (prescription)	1	1	U	Unk	Int-A	1	Ethanol	119 mg/dL In Serum @ Unknown
		Ethanol (beverage)	2	2						
717ai	42 y M	Methamphetamine	3	3	U	Unk	Int-A	1	Methamphetamine	32 ng/mL In Serum @ Unknown
		Fentanyl (non-prescription)	1	1						
718h	42 y F	Xylazine	2	2	A	Ingst	AR-D	3		
		Ethanol (beverage)	3	3						
719ai	42 y M	Substance (non-drug), N.O.S.	3	3	U	Unk	Int-A	1		
		Acetaminophen	1	1						
720h	42 y F	Fentanyl (non-prescription)	1	1	A/C	Ingst	Int-M	1	Acetaminophen (apap)	17 mcg/mL In Blood (unspecified) @ 2 d (pe)
		Methamphetamine	3	3						
721h	42 y F	Acetaminophen	1	1	A	Ingst	Int-U	2	Acetaminophen (apap)	150.3 mcg/mL In Blood (unspecified) @ 21 m (pe)
		Acetaminophen	1	1						
722ai	43 y F	Drug, N.O.S.	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Bromazolam	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
723pha	43 y M	Diphenhydramine	3	3	U	Unk	Unk	1		
		Fentanyl (non-prescription)	1	1					4-anpp (despropionyl fentanyl)	0.42 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	1	1					Norfentanyl	0.72 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	1	1					Fentanyl	3.33 ng/mL In Blood (unspecified) @ Autopsy
		Cocaine	4	2					Benzoyllecognine	184 ng/mL In Blood (unspecified) @ Autopsy
		Opioid, N.O.S.	3	3						
		Drug, N.O.S.	2	4						
724ai	43 y M	Ethanol (beverage)	5	5	U	Unk	Int-A	1		248 mg/dL In Blood (unspecified) @ Unknown
725p	43 y M	Fentanyl (non-prescription)	1	1	A	Unk	Int-U	2		
		Hydroxyzine	2	2						
726i	43 y M	Fentanyl (non-prescription)	1	1	A	Unk	Int-A	2	Ethanol	295 mg/dL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	2	2						
727h	43 y F	Ethanol (beverage)	3	3	C	Ingst	Int-M	2		
		Opioid, N.O.S.	1	1						
728ha	43 y F	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	115 mcg/mL In Blood (unspecified) @ Unknown
		Ibuprofen	2	2						
		Ethanol (beverage)	3	3					Ethanol	150 mg/dL In Blood (unspecified) @ Unknown
		Acetaminophen/caffeine/salicylate	2	1					Salicylate	117 mg/L In Whole Blood @ Unknown
		Acetaminophen/caffeine/salicylate	2	1					Acetaminophen (apap)	34 mcg/mL In Blood (unspecified) @ Autopsy
		Acetaminophen/caffeine/salicylate	2	1					Acetaminophen (apap)	36 mg/L In Whole Blood @ Unknown
		Hydroxychloroquine	1	1					Potassium	1.4 mEq/L In Blood (unspecified) @ Unknown
729	43 y F	Hydroxychloroquine	1	1	A	Unk	Int-A	2	Potassium	4.4 mEq/L In Blood (unspecified) @ Unknown
		Gabapentin	3	2					Gabapentin	140 mcg/mL In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	4	3					Ethanol	108 mg/dL In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	4	3					Ethanol	381 mg/dL In Blood (unspecified) @ Unknown
730ai	43 y F	Heroin	1	1	U	Unk	Int-A	1		
731ai	43 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Diphenhydramine	2	2						
		Alprazolam	3	3						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
732ai	43 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
		Ethanol (beverage)	3	3						
733ai	43 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
734ai	43 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3						
735i	43 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
		Alprazolam	3	3						
736i	43 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
737ai	43 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						
738ai	43 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
739ha	43 y M	Salicylate	1	1	A	Ingst	Int-S	1	Salicylate	103 mg/dL In Blood (unspecified) @ 3 h (pe)
		Ethanol (beverage)	2	2					Ethanol	370 mg/dL In Blood (unspecified) @ 3 h (pe)
740ph	43 y M	Acetaminophen (ER)	3	3	A	Unk	Int-A	2		
		Fentanyl (non-prescription)	1	1						
		Stimulant/street drug, N.O.S.	2	2						
741ai	44 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Oxycodone	2	2						
		Cocaine	3	3						
742h	44 y F	Acetaminophen/diphenhydramine	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	101 mcg/mL In Serum @ 26 h (pe)
		Acetaminophen/diphenhydramine	1	1					Acetaminophen (apap)	11 mcg/mL In Serum @ 51 h (pe)
		Acetaminophen/diphenhydramine	1	1					Acetaminophen (apap)	13 mcg/mL In Serum @ 45 h (pe)
		Acetaminophen/diphenhydramine	1	1					Acetaminophen (apap)	19 mcg/mL In Serum @ 40 h (pe)
		Acetaminophen/diphenhydramine	1	1					Acetaminophen (apap)	47 mcg/mL In Serum @ 33 h (pe)
		Acetaminophen/diphenhydramine	1	1					Acetaminophen (apap)	80 mcg/mL In Serum @ 16 h (pe)
		Acetaminophen/diphenhydramine	1	1					Acetaminophen (apap)	9 mcg/mL In Serum @ 54 h (pe)
		Diphenhydramine	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
743ai	44 y M	Oxycodone	1	1	U	Unk	Int-A	1		
		Morphine	2	2						
		Alprazolam	3	3						
744ai	44 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
745i	44 y M	Opioid, N.O.S.	1	1	U	Par	Int-A	2		
746ai	44 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Diphenhydramine	2	2						
		Ethanol (beverage)	3	3						
747ai	44 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
748ai	44 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
749i	44 y M	Fentanyl (non-prescription)	1	1	A	Inhal	Int-A	2		
750h	44 y F	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	89.4mcg/mL In Serum @ Unknown
751ai	44 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (prescription)	1	1						
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
752ai	44 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
753ai	44 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
754ai	44 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Morphine	2	2						
		Cocaine	3	3						
755ai	44 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (despropionyl)	2	2						
		Ethanol (beverage)	3	3						
756ai	44 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3						
757i	44 y M	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
758ai	44 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						
759i	44 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
760ph	44 y F	Acetaminophen/oxycodone	1	1	A	Ingst + Inhal	Int-A	2		
		Cocaine	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
761h	44 y F	Acetaminophen	1	1	A	Ingst	Unk	3	Acetaminophen (apap)	127 mcg/mL In Serum @ 2 h (pe)
		Ibuprofen	2	2						
		Ethanol (beverage)	3	3						
762h	44 y F	Acetaminophen	1	1	A	Ingst + Aspir	Int-U	2	Acetaminophen (apap)	74.8 mcg/mL In Blood (unspecified) @ Unknown
763h	44 y F	Acetaminophen	1	1	C	Ingst	Unk	1		
		Ethanol (beverage)	2	2						
		Medicines, Asian	3	3						
		Diphenhydramine	4	4						
764h	44 y F	Salicylate	1	1	A	Ingst	Int-S	1	Salicylate	44.2 mg/dL In Serum @ Unknown
765ai	45 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
766i	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
767i	45 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
768i	45 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methadone	2	2						
		Marijuana, N.O.S.	3	3						
769ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		<i>Mitragyna speciosa korthals</i>	2	2						
		Alprazolam	3	3						
770ai	45 y F	Oxycodone	1	1	U	Unk	Int-A	1		
		Morphine	2	2						
		Gabapentin	3	3						
771ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (despropionyl)	2	2						
772ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
773ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
774ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-M	1		
775i	45 y M	Opioid, N.O.S.	1	1	U	Unk	Unk	2		
776ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
777ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
778ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
779ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Methadone	3	3						
780ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
781ai	45 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Gabapentin	3	3						
782ai	45 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
783i	46 y F	Methadone	1	1	U	Ingst	Int-A	2		
784	46 y F	Acetaminophen	1	1	A	Ingst	Int-S	1		
		Ibuprofen	2	2						
		Hydroxyzine	3	3						
		Ziprasidone	4	4						
785pi	46 y F	Heroin	1	1	A	Unk	Int-A	2		
786ai	46 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Marijuana, N.O.S.	3	3						
787pa	46 y F	Fentanyl (non-prescription)	1	1	A	Unk	Unk	1		
788i	46 y F	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
		Ethanol (beverage)	2	2						
789ai	46 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Bromazolam	3	3						
790ai	46 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
791ai	46 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
792ai	46 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						
793ai	46 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3						
794	46 y F	Acetaminophen	1	1	C	Ingst	Int-M	2	Acetaminophen (apap)	47.2 mcg/mL In Serum @ Unknown
795ph	46 y F	Diphenhydramine	2	2	A	Ingst	Int-S	3	Acetaminophen (apap)	672 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen (ER)	1	1						
		Tetrahydrozoline	2	2						
		Ethanol (beverage)	3	3						
796ai	47 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
797ai	47 y F	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Cocaine	3	3						
798i	47 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-S	1		
		Methamphetamine	2	2						
		Bromazolam	3	3						
799i	47 y M	Fentanyl (prescription)	1	1	U	Unk	Int-A	2		
800i	47 y F	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
801ai	47 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
802ai	47 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2	U	Unk	Int-A	1		
803ai	47 y F	Fentanyl (non-prescription)	1	1						
		Methadone	2	2						
		Bromazolam	3	3						
804ai	47 y F	Oxycodone	1	1	U	Unk	Int-S	1		
		Acetaminophen	2	2						
805ai	47 y F	Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3	U	Unk	Int-A	1		
806ai	47 y F	Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3	U	Unk	Int-A	1		
807ai	47 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2	U	Unk	Int-A	1		
808ha	47 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Methadone	3	3						
809ai	47 y M	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	125 mg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	157.7 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	462 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	530 mcg/mL In Whole Blood @ Unknown
		Diphenhydramine	2	2					Diphenhydramine	4700 ng/mL In Whole Blood @ Unknown
		Ibuprofen	3	3						
		Methamphetamine	4	4					Methamphetamine	190 ng/mL In Whole Blood @ Unknown
		Methamphetamine	4	4					Amphetamine	25 ng/mL In Whole Blood @ Unknown
809ai	47 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
810ai	47 y M	Methamphetamine	2	2						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
811i	47 y M	Methamphetamine Substance (non-drug), N.O.S.	2 3	2 3						
812ha	47 y M	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
		Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	34 mcg/mL In Blood (unspecified) @ Unknown
		Hydroxyzine	2	2					Hydroxyzine	14 ng/mL In Blood (unspecified) @ Autopsy
813h	47 y F	Acetaminophen	1	1	A	Ingst	Int-U	2	Acetaminophen (apap)	10.5 mcg/mL In Serum @ Unknown
814ph	47 y M				A	Ingst	Int-S	2		
		Oxycodone	1	1						
		Guanfacine (ER)	2	2						
815ai	48 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
816ai	48 y F				U	Unk	Int-A	1		
		Oxycodone	1	1						
		Ethanol (beverage)	2	2						
		Carisoprodol	3	3						
817ai	48 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Diazepam	2	2						
818ha	48 y M	Acetaminophen	1	1	C	Ingst	Int-M	1	Acetaminophen (apap)	71 mcg/mL In Blood (unspecified) @ Unknown
819ai	48 y M	Ethanol (beverage)	2	2						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
820ai	48 y F				U	Unk	Int-S	1		
		Oxycodone	1	1						
		Ethanol (beverage)	2	2						
821ai	48 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
822ha	48 y F	Acetaminophen	1	1	A	Ingst + Inhal	Int-S	3	Acetaminophen (apap)	18 mcg/mL In Serum @ Unknown
		Hydrocarbon (fluorinated)	2	2						
		Ethanol (beverage)	3	3						
823h	48 y M	Acetaminophen	1	1	C	Ingst	Int-M	1	Acetaminophen (apap)	24 mcg/mL In Serum @ Unknown
824ai	49 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
825ai	49 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
826ai	49 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
827ai	49 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (despropionyl)	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
828i	49 y M	Methamphetamine	3	3	U	Unk	Int-A	2		
829ai	49 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
830ai	49 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (prescription)	1	1						
831ai	49 y M	Alprazolam	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
832ai	49 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
		Ethanol (beverage)	3	3						
833i	49 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
834p	49 y M	Salicylate	1	1	A	Ingst	Int-S	3		
835ha	49 y M	Colchicine	1	1	C	Unk	AR-D	2	Colchicine	8.9ng/mL In Blood (unspecified) @ 65 h (pe)
836ha	49 y F	Cocaine	2	2	A/C	Ingst	Int-S	1	Oxycodone (free)	300ng/mL In Blood (unspecified) @ Autopsy
		Oxycodone	3	1						
		Venlafaxine	1	1						
		Acetaminophen	5	2					Acetaminophen (apap)	21 mcg/mL In Blood (unspecified) @ Unknown
		Amlodipine	2	2						
		Ethanol (beverage)	6	4					Ethanol	39 mg/dL In Blood (unspecified) @ Autopsy
837h	49 y M	Hydroxyzine	4	4						
		Gabapentin	7	7						
		Methocarbamol	8	8	A	Ingst	Int-S	1	Acetaminophen (apap)	112mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
		Acetaminophen	1	1					Acetaminophen (apap)	126 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	136 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	157 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	84 mcg/mL In Blood (unspecified) @ Unknown
838h	50 y F	Cocaine	2	2	A	Ingst	Int-S	2	Acetaminophen (apap)	600 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
839ai	50 y F	Acetaminophen	1	1	U	Unk	Int-S	1		
		Dextromethorphan	2	2						
		Diphenhydramine	3	3						
840h	50 y M	Heroin	1	1	A	Ingst	Int-A	2		
		Ethanol (beverage)	2	2					Ethanol	334mg/dL In Blood (unspecified) @ Unknown
841ai	50 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
842ai	50 y F	Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2	U	Unk	Int-A	1		
843i	50 y F	Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
		Ethanol (beverage)	3	3	U	Unk	Int-A	1		
844ai	50 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2	U	Unk	Int-A	1		
845ai	50 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2	U	Unk	Int-A	1		
846ai	50 y M	Fentanyl (non-prescription)	1	1						
		Marijuana, N.O.S.	2	2	U	Unk	Int-A	1		
		Ethanol (beverage)	3	3						
847i	50 y M	Fentanyl (non-prescription)	1	1	A	Inhal	Int-A	2		
848a	50 y F	Acetaminophen	1	1	A	Ingst	Int-S	3	Acetaminophen (apap)	22.3 mcg/mL In Blood (unspecified) @ Unknown
849ai	50 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
850ai	51 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Amphetamine	3	3						
851h	51 y F	Acetaminophen Drug, N.O.S.	1	1	U	Ingst	Int-U	2		
			2	2						
852i	51 y F	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
853ai	51 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
854ai	51 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
855ai	51 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Aripiprazole	3	3						
856ai	51 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
857pa	51 y M	Fentanyl (non-prescription)	1	1	U	Inhal + Unk	Int-A	1	Fentanyl	43 ng/mL In Blood (unspecified) @ Autopsy
		Carbon monoxide	2	2					Carboxyhemoglobin	10.3 % In Blood (unspecified) @ Autopsy
858ph	51 y F	Acetaminophen/oxycodone	1	1	A/C	Ingst	Int-S	2	Acetaminophen (apap)	255.9 ng/mL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
859i	51 y M	Trazodone	2	2	U	Unk	Int-A	2		
		Fentanyl (non-prescription)	1	1						
860ai	51 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
861	51 y M	Methamphetamine	2	2	A	Ingst	Int-S	2	Acetaminophen (apap)	145 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1						
862ai	51 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
863ai	51 y F	Methamphetamine	2	2	U	Unk	Int-A	1		
		Cocaine	3	3						
		Fentanyl (non-prescription)	1	1						
864h	51 y M	Methamphetamine	2	2	A/C	Ingst	Unk	3	Acetaminophen (apap)	62.5 mcg/mL In Blood (unspecified) @ Unknown
		Cocaine	3	3						
		Acetaminophen	1	1						
865h	51 y F	Acetaminophen	1	1	A	Ingst	Int-S	3	Acetaminophen (apap)	107 mg/mL In Blood (unspecified) @ Unknown
866ha	51 y M	Drug, N.O.S.	2	2	A	Ingst	Int-S	1	Acetaminophen (apap)	150 mcg/mL In Blood (unspecified) @ Autopsy
		Acetaminophen	1	1						
		Acetaminophen	1	1						
867h	51 y M	Ethanol (beverage)	2	2	A	Ingst	Int-S	2	Salicylate	58.4 mg/dL In Blood (unspecified) @ Unknown
		Salicylate	1	1						
868i	52 y F	Pseudoephedrine	2	2	A/C	Ingst	Int-A	2		
		Opioid, N.O.S.	1	1						
869pa	52 y F				A/C	Ingst + Par + Unk	Unk	2		
		Fentanyl (prescription)	1	1					4-anpp (despropionyl fentanyl)	0.93 ng/mL In Whole Blood @ Autopsy
		Fentanyl (prescription)	1	1					Fentanyl	10 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (prescription)	1	1					Norfentanyl	5.4 ng/mL In Whole Blood @ Autopsy
		Ethanol (beverage)	2	2					Ethanol	28 mg/dL In Whole Blood @ Autopsy
		Amphetamine/dextroamphetamine	3	3					Amphetamine	13 ng/mL In Whole Blood @ Autopsy
		Amphetamine/dextroamphetamine	3	3					Bupropion	45 ng/mL In Whole Blood @ Autopsy
		Bupropion (ER)	4	4						
870i	52 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
871ai	52 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analog (acetyl)	2	2						
872ai	52 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
873a	52 y M	Fentanyl (non-prescription)	1	1	C	Ingst	Int-U	1	Acetaminophen (apap)	46.8 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
		Ethanol (beverage)	2	2						
874pha	52 y F	Ethanol (beverage)	3	3	U	Unk	Unk	1	Ethanol	13 mg/dL In Blood (unspecified) @ Unknown
		Buprenorphine	1	1						
		Buprenorphine	1	1						
		Citalopram	2	2						
		Venlafaxine	3	3						
		Venlafaxine	3	3						
		Butalbital	4	4						
		Butalbital	4	4						
		Clonazepam	5	5						
875ph	52 y F	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
		Benzodiazepine, N.O.S.	2	2						
		Cocaine	3	3						
		Amphetamine, N.O.S.	4	4						
876ai	52 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Loperamide	3	3						
877ai	52 y F	Acetaminophen/hydrocodone	1	1	U	Unk	Int-S	1		
		Lorazepam	2	2						
878	52 y M	Colchicine	1	1	C	Ingst	AR-D	2		
879h	52 y M	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	48 mcg/mL In Blood (unspecified) @ Unknown
880ai	52 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3						
881hi	52 y F	Acetaminophen	1	1	U	Ingst	Unk	1	Acetaminophen (apap)	17.4 mcg/mL In Blood (unspecified) @ Unknown
882ph	52 y M	Acetaminophen	1	1	U	Unk	Unk	2	Acetaminophen (apap)	85 mcg/mL In Blood (unspecified) @ Unknown
883h	53 y F	Acetaminophen	1	1	U	Ingst	Int-S	2	Acetaminophen (apap)	64 mcg/mL In Blood (unspecified) @ Unknown
884ai	53 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
885ai	53 y F	Fentanyl analogs (para-fluoro)	2	2						
		Morphine	1	1	U	Unk	Int-A	1		
		Oxycodone	2	2						
886ai	53 y M	Hydromorphone	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
887i	53 y M	Marijuana, N.O.S.	2	2	A/C	Unk	Int-A	2		
888i	53 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
889i	53 y M	Ethanol (beverage)	2	2	A	Unk	Int-A	2		
890ai	53 y M	Fentanyl (prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Hydroxyzine	2	2						
891h	53 y M	Gabapentin	3	3						
		Acetaminophen	1	1	U	Ingst	Unk	2	Acetaminophen (apap)	112 mcg/mL In Blood (unspecified) @ Unknown
892ai	53 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
893ai	53 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						
894ha	53 y F	Acetaminophen	1	1	U	Ingst	Int-S	1	Acetaminophen (apap)	210 mcg/mL In Serum @ 1.75 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	300 mcg/mL In Blood (unspecified) @ 1.5 h (pe)
		Lamotrigine	2	2					Lamotrigine	16.2 mcg/mL In Serum @ 3 h (pe)
		Levetiracetam	3	2					Levetiracetam	50.2 mcg/mL In Serum @ 3 h (pe)
		Topiramate	4	2					Topiramate	25.4 mcg/mL In Serum @ 3 h (pe)
895h	53 y F	Oxycodone	5	3	U	Ingst	Unk	2		
		Acetaminophen	1	1					Acetaminophen (apap)	14 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	44 mcg/mL In Blood (unspecified) @ Unknown
896ph	53 y F	Fentanyl (non-prescription)	1	1	U	Ingst	Int-S	2		
897h	53 y M	Gabapentin	2	2	U	Ingst	Int-S	2		
898ai	54 y F	Ibuprofen	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
899i	54 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
900h	54 y F	Methamphetamine	2	2	U	Ingst	Unk	3		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
901ai	54 y M	Acetaminophen	1	1					Acetaminophen (apap)	37 mg/L In Serum @ Unknown
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
902ai	54 y F	Ethanol (beverage)	2	2	U	Unk	Int-S	1		
		Acetaminophen/hydrocodone	1	1						
		Aripiprazole	2	2						
903ai	54 y F	Trazodone	3	3	U	Unk	Int-A	1		
		Heroin	1	1						
		Fentanyl analogs (despropionyl)	2	2						
904ai	54 y M	Hydromorphone	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
905	54 y F				A	Ingst	Int-S	2		
		Oxycodone	1	1						
		Diazepam	2	2						
		Salicylate	3	3					Salicylate	75 mg/dL In Blood (unspecified) @ Unknown
906ai	54 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
907i	54 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
908ai	54 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
909ai	54 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Oxycodone	2	2						
		Lorazepam	3	3						
910ai	54 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
911ai	54 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Hydroxyzine	3	3						
912i	54 y M				U	Par	Int-A	2		
		Fentanyl (non-prescription)	1	1						
913ai	54 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
914ai	54 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Alprazolam	3	3						
915ai	54 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
916h	54 y F				C	Ingst	Unk	2		
		Acetaminophen	1	1					Ethanol	0.032 mg/dL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Acetaminophen	1	1					Acetaminophen (apap)	70.1 mcg/mL In Blood (unspecified) @ Unknown
917ai	54 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
918ai	54 y F	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
919ai	54 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
920i	54 y F	Codeine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
921ai	54 y M	Cocaine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
922ai	54 y M	Diazepam	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
923ai	54 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
924ai	54 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
925	54 y M	Cocaine	3	3	C	Ingst	Unt-U	3		
926a	55 y F	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	37.6 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1					Iron	183 mcg/dL In Blood (unspecified) @ Unknown
		Iron, N.O.S.	2	2						
927i	55 y F				U	Unk	Int-A	2		
928h	55 y M	Opioid, N.O.S.	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	121 mcg/mL In Serum @ 3 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	245 mcg/mL In Serum @ 2 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	26 mcg/mL In Serum @ 5 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	369 mcg/mL In Serum @ 4 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	383 mcg/mL In Serum @ 8 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	72 mcg/mL In Serum @ 4 d (pe)
929i	55 y M				U	Unk	Int-A	2		
930ai	55 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
931ph	55 y M	Methamphetamine	2	2	U	Ingst	Unk	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Acetaminophen	1	1					Acetaminophen (apap)	133 mcg/mL In Blood (unspecified) @ 2 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	140 mcg/mL In Blood (unspecified) @ Unknown
932i	55 y M	Heroin	1	1	U	Unk	Int-A	2		
933pha	55 y M	Fentanyl (prescription)	1	1	A	Unk	Int-A	1	Fentanyl	0.47 ng/mL In Blood (unspecified) @ Unknown
		Cocaine	2	2					Benzoylcognine	73 ng/mL In Blood (unspecified) @ Unknown
934i	55 y F	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
935ha	55 y M	Acetaminophen	1	1	A	Ingst	Int-M	2		
936ai	55 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
937a	55 y F	Acetaminophen	1	1	A	Ingst	Unk	1	Acetaminophen (apap)	131.4 mcg/mL In Blood (unspecified) @ Unknown
938ai	55 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
939i	55 y F	Opioid, N.O.S.	1	1	A	Unk	Unk	2		
940i	55 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
941ai	55 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Cocaine	3	3						
942ai	55 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
943i	55 y F	Fentanyl (prescription)	1	1	A	Inhal	Int-A	2		
944h	55 y M	Acetaminophen	1	1	A/C	Ingst	Unt-G	3	Acetaminophen (apap)	25.8 mcg/mL In Blood (unspecified) @ Unknown
945ph	55 y M	Heroin	1	1	A	Unk	Int-A	3		
946ha	55 y M	Acetaminophen	1	1	C	Ingst	Unk	2	Acetaminophen (apap)	85 mcg/mL In Blood (unspecified) @ Unknown
		Ethylene glycol	2	2					Ethylene glycol	12.9 mg/dL In Blood (unspecified) @ Unknown
947pha	55 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1	4-anpp (despropionyl fentanyl)	0.46 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	1	1					Norfentanyl	0.94 ng/mL In Blood (unspecified) @ Autopsy

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Fentanyl (non-prescription)	1	1					Fentanyl	9.1 ng/mL In Blood (unspecified) @ Autopsy
		Stimulant/street drug, N.O.S.	2	2					Midazolam	47 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	3	3						
		Benzodiazepine, N.O.S.	4	4						
948h	55 y F	Acetaminophen/oxycodone	1	1	C	Ingst	Int-A	2	Acetaminophen (apap)	106.69 mcg/mL In Serum @ Unknown
949ai	56 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
950ai	56 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
951ai	56 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
952ai	56 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Oxycodone	2	2						
		Methamphetamine	3	3						
953ai	56 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
954ai	56 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
955h	56 y M	Marijuana, N.O.S.	3	3	A	Ingst	Unk	1	Acetaminophen (apap)	188 mcg/mL In Serum @ Unknown
956ai	56 y M	Acetaminophen	1	1	U	Unk	Int-A	1		
957ai	56 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
958ha	56 y F	Fentanyl (non-prescription)	1	1	A/C	Ingst	Int-S	1		
		Methamphetamine	2	2						
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	27.6 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen/hydrocodone	1	1					Hydrocodone	375 ng/mL In Blood (unspecified) @ Unknown
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	52.3 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen/oxycodone	2	2						
		Alprazolam	3	3					Alprazolam	65.2 ng/mL In Blood (unspecified) @ Unknown
		Fluoxetine	4	4					Fluoxetine	504 ng/mL In Blood (unspecified) @ Unknown
		Fluoxetine	4	4					Norfluoxetine	532 ng/mL In Blood (unspecified) @ Unknown
		Fluoxetine	4	4						
959h	56 y M	Acetaminophen	1	1	C	Ingst	Int-A	2	Acetaminophen (apap)	303 mcg/mL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Acetaminophen	1	1					Acetaminophen (apap)	79 mcg/mL In Blood (unspecified) @ 31 h (pe)
960ai	56 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Cocaine	2	2						
961h	56 y M	Marijuana, N.O.S.	3	3	C	Ingst	Int-M	3		
		Salicylate	1	1					Salicylate	34.8 mg/dL In Serum @ Unknown
		Ethanol (beverage)	2	2					Ethanol	212 mg/dL In Serum @ Unknown
962h	57 y F	Acetaminophen	1	1	U	Unk	Unk	3	Lithium	0.76 mEq/L In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	50 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Ethanol	64 mg/dL In Serum @ Unknown
		Acetaminophen	1	1					Salicylate	8.4 mg/dL In Serum @ Unknown
		Acetaminophen	1	1					Salicylate	9.6 mg/dL In Serum @ Unknown
		Salicylate	2	2						
963ai	57 y M	Ethanol (beverage)	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
964ai	57 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
965ai	57 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
966ai	57 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
967ai	57 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
968h	57 y M	Xylazine	2	2	C	Ingst	Unt-T	3		
		Acetaminophen	1	1						
		Acetaminophen/ chlorpheniramine/ dextromethorphan	2	2						
		Acetaminophen/ oxycodone	3	3						
969i	57 y F				U	Unk	Int-A	2		
970ai	57 y F	Opioid, N.O.S.	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
971ai	57 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
972ai	57 y F	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Paroxetine	3	3						
973ai	57 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Morphine	2	2						
		Gabapentin	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
974ai	57 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Quetiapine	2	2						
		Trazodone	3	3						
975h	57 y F	Acetaminophen/hydrocodone	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	25.1 mcg/mL In Blood (unspecified) @ 19 h (pe)
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	28.2 mcg/mL In Blood (unspecified) @ 15 h (pe)
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	32.1 mcg/mL In Blood (unspecified) @ 30 h (pe)
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	33.7 mcg/mL In Blood (unspecified) @ 27 h (pe)
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	41.6 mcg/mL In Blood (unspecified) @ 34 h (pe)
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	41.8 mcg/mL In Blood (unspecified) @ 38 h (pe)
976i	57 y M	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
977i	57 y M	Opioid, N.O.S.	1	1	A/C	Unk	Int-A	2		
978h	57 y M	Salicylate	1	1	A	Ingst	Int-S	2	Salicylate	94.1 mg/dL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2					Ethanol	57.2 mg/dL In Blood (unspecified) @ Unknown
979h	57 y M	Acetaminophen	1	1	C	Ingst	Unt-T	1	Acetaminophen (apap)	92 mcg/mL In Serum @ Unknown
		Ethanol (beverage)	2	2					Ethanol	11 mg/dL In Serum @ Unknown
980h	57 y M	Acetaminophen/hydrocodone	1	1	A/C	Ingst	Int-S	1	Acetaminophen (apap)	69.1 mcg/mL In Blood (unspecified) @ Unknown
		Tramadol	2	2						
		Ethanol (beverage)	3	3					Ethanol	270 mg/dL In Blood (unspecified) @ Unknown
981h	57 y F	Acetaminophen	1	1	U	Ingst	Unk	2		
982h	58 y M	Acetaminophen	1	1	A	Ingst	Int-S	3	Acetaminophen (apap)	17 mcg/mL In Serum @ 21 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	23 mcg/mL In Serum @ 15.5 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	45 mcg/mL In Serum @ 1 h (pe)
		Metformin	2	2						
		Ibuprofen	3	3						
983ai	58 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
		Ethanol (beverage)	3	3						
984ai	58 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
985pha	58 y F	Phencyclidine	3	3	U	Ingst	Int-S	1		
		Acetaminophen/oxycodone	1	1					Acetaminophen (apap)	12 mcg/mL In Serum @ 24 h (pe)
		Acetaminophen/oxycodone	1	1					Acetaminophen (apap)	75 mcg/mL In Serum @ 1 h (pe)
986	58 y F	Acetaminophen	1	1	U	Ingst	Int-S	2	Acetaminophen (apap)	242 mcg/mL In Blood (unspecified) @ Unknown
987h	58 y M	Acetaminophen	1	1	C	Ingst	Int-M	2		
		Ibuprofen	2	2						
988ai	58 y F				U	Unk	Int-A	1		
		Hydromorphone	1	1						
		Diphenhydramine	2	2						
989i	58 y F	Gabapentin	3	3	A	Inhal	Int-A	2		
		Opioid, N.O.S.	1	1						
990ai	58 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
991ai	58 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
992ha	58 y F				A	Ingst	Unt-G	1		
		Acetaminophen	1	1					Acetaminophen (apap)	78 mcg/mL In Serum @ Unknown
993h	58 y F	Acetaminophen	1	1	A	Ingst	Int-M	1		
994ph	58 y F	Acetaminophen	1	1	A	Ingst	Int-U	2	Acetaminophen (apap)	130 mcg/mL In Blood (unspecified) @ Unknown
		Salicylate	2	2						
		Ethanol (beverage)	3	3					Ethanol	44 mg/dL In Blood (unspecified) @ Unknown
995h	58 y M	Acetaminophen	1	1	A/C	Ingst	Unt-U	1	Acetaminophen (apap)	9 mcg/mL In Serum @ Unknown
996ha	58 y F	Acetaminophen	3	1		Ingst	Int-A	1	Acetaminophen (apap)	40 mcg/mL In Serum @ Unknown
		Acetaminophen	3	1					Acetaminophen (apap)	50 mg/L In Blood (unspecified) @ Unknown
		Methamphetamine	1	2					Methamphetamine	1980 ng/mL In Blood (unspecified) @ Unknown
		Methamphetamine	1	2					Amphetamine	99 ng/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	3						
997ai	59 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
998pha	59 y F	Ethanol (beverage)	3	3	U	Unk	Unk	2		
		Fentanyl (non-prescription)	1	1					Fentanyl	14 ng/mL In Blood (unspecified) @ Unknown
999ai	59 y F	Cocaine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
1000ai	59 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1001h	59 y M	Fentanyl (prescription)	1	1	U	Ingst	Int-M	3		
		Ethanol (beverage)	2	2						
		Acetaminophen/caffeine/ acetylsalicylate	1	1						
1002ha	59 y M	Oxycodone	2	2	U	Ingst	Int-S	1	Acetaminophen (apap)	141 mcg/mL In Blood (unspecified) @ 5 m (pe)
		Hydroxyzine	3	3						
		Ibuprofen	4	4						
		Acetaminophen	1	1						
1003ai	59 y M				U	Unk	Int-A	1		
1004ai	59 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
1005ai	59 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
1006h	59 y M	Fentanyl (non-prescription)	1	1	U	Ingst + Unk	Unk	3	Salicylate	29.1 mg/dL In Blood (unspecified) @ Unknown
		Ketamine	2	2						
		Marijuana, N.O.S.	3	3						
		Salicylate	1	1						
1007i	59 y M	Salicylate	1	1	U	Unk	Int-A	1	Salicylate	33.1 mg/dL In Blood (unspecified) @ Unknown
		Acetaminophen	2	2						
1008ai	59 y M	Drug, N.O.S.	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
		Methamphetamine	3	3						
1009ai	59 y M	Fentanyl (non-prescription)	1	1	A	Unk	Int-A	1		
1010ai	59 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Methadone	3	3						
1011i	59 y M	Fentanyl (non-prescription)	1	1	U	Inhal	Int-A	2		
		Methamphetamine	2	2						
		Cocaine	3	3						
1012a	59 y F	Fentanyl (prescription)	1	1	A	Ingst	Unk	3	Acetaminophen (apap)	125.6 mcg/mL In Blood (unspecified) @ Unknown
		Cocaine	2	2						
1013h	59 y M	Acetaminophen	1	1	A/C	Ingst + Unk	Int-S	3		
		Ibuprofen	1	1						
1014ai	59 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
1015h	59 y F	Fentanyl (non-prescription)	1	1	A	Unk	Unt-U	1		
		Methamphetamine	2	2						
		Oxycodone	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Salicylate	1	1					Salicylate	80 mg/dL In Blood (unspecified) @ Unknown
1016p	59 y F	Drug, N.O.S.	2	2	A	Ingst	Int-S	2		
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	431 mcg/mL In Serum @ Unknown
1017h	59 y F	Acetaminophen	1	1	U	Ingst	Int-S	3	Acetaminophen (apap)	97 mcg/mL In Serum @ Unknown
1018h	60 y M	Colchicine	1	1	A/C	Ingst	Int-S	2		
		Ethanol (beverage)	2	2					Ethanol	148.5 mg/dL In Blood (unspecified) @ Unknown
1019h	60 y F	Acetaminophen	3	3	C	Ingst	Int-M	1		
		Acetaminophen	1	1					Acetaminophen (apap)	24 mcg/mL In Serum @ 1.5 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	34 mcg/mL In Serum @ 1 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	93 mcg/mL In Serum @ Unknown
1020ai	60 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
1021h	60 y M	Ethanol (beverage)	2	2	U	Ingst	Unk	3		
		Acetaminophen	1	1					Acetaminophen (apap)	32.9 mcg/mL In Blood (unspecified) @ Unknown
1022	60 y M	Apixaban	2	2	A	Ingst	Int-S	1		
		Acetaminophen/diphenhydramine	1	1					Acetaminophen (apap)	47.2 mcg/mL In Blood (unspecified) @ 15 h (pe)
1023ai	60 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
1024pha	60 y F	Methamphetamine	2	2	A	Ingst + Unk	Int-S	1		
		Methadone	1	1					Methadone	390 ng/mL In Blood (unspecified) @ Unknown
		Methadone	1	1					Eddp (2-ethylidene-1,5-dimethyl-3,3-diphenyl pyrrolidine)	95 ng/mL In Blood (unspecified) @ Unknown
		Cocaine	2	2					Benzoyllecognine	320 ng/mL In Blood (unspecified) @ Unknown
1025ai	60 y M	Morphine	1	1	U	Unk	Unt-G	1		
		<i>Mitragyna speciosa korthals</i>	2	2						
1026ai	60 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
1027ai	60 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
1028ai	60 y M	Methamphetamine	2	2	U	Unk	Int-S	1		
		Fentanyl (non-prescription)	1	1						
1029ai	60 y F	Paroxetine	2	2	A	Unk	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1030ai	60 y M	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2	U	Unk	Int-A	1		
1031ai	60 y M	Fentanyl (non-prescription)	1	1						
		Oxycodone	2	2						
		Methamphetamine	3	3	U	Unk	Int-A	1		
1032h	60 y F	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Marijuana, N.O.S.	3	3	U	Ingst	Int-S	2	Acetaminophen (apap)	177 mcg/mL In Blood (unspecified) @ Unknown
1033h	60 y M	Acetaminophen	1	1	A	Ingst	Int-M	1	Acetaminophen (apap)	101.5 mcg/mL In Blood (unspecified) @ Unknown
1034ai	61 y F				U	Unk	Int-A	1		
1035ai	61 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
1036ai	61 y M	Methadone	2	2						
		Cocaine	3	3						
					U	Unk	Int-A	1		
1037ai	61 y F	Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2	U	Unk	Int-A	1		
1038ph	61 y F	Opioid, N.O.S.	1	1						
		Lorazepam	2	2	A	Ingst	Int-S	2	Acetaminophen (apap)	46 mcg/mL In Blood (unspecified) @ Unknown
1039ai	61 y M	Acetaminophen/oxycodone	1	1						
					U	Unk	Int-S	1		
1040ai	61 y M	Morphine	1	1						
		Diphenhydramine	2	2						
		Ethanol (beverage)	3	3	U	Unk	Int-A	1		
1041ph	61 y F	Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Cocaine	3	3	A	Ingst	Int-S	2		
1042h	61 y F	Morphine	1	1						
		Fluoxetine	2	2					Salicylate	21.1 mg/dL In Blood (unspecified) @ Unknown
		Salicylate	3	3						
1043p	61 y F	Acetaminophen	2	1	C	Ingst	Int-U	1	Acetaminophen (apap)	19.1 mcg/mL In Blood (unspecified) @ Unknown
1044ai	61 y F	Diphenhydramine	1	2	C	Ingst	Int-U	2		
		Methadone	1	1	U	Unk	Int-A	1		
1045ai	61 y M	Fentanyl (non-prescription)	1	1						
		Fentanyl analog (acetyl)	2	2						
		Methamphetamine	3	3	U	Unk	Int-A	1		
1046ai	61 y M	Fentanyl (non-prescription)	1	1						
					U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1047ai	61 y M	Cocaine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Substance (non-drug), N.O.S.	3	3						
1048h	61 y M				A/C	Ingst + Inhal + Unk	Int-A	2		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Amphetamine (hallucinogenic), 3,4-methylenedioxy methamphetamine (MDMA)	3	3						
1049h	61 y M	Acetaminophen	1	1	C	Ingst	Int-M	1	Acetaminophen (apap)	26 mcg/mL In Blood (unspecified) @ Unknown
1050ph	61 y F				A	Ingst	Int-U	3		
		Oxycodone	1	1						
		Gabapentin	2	2						
1051	61 y M	Acetaminophen	1	1	A/C	Ingst	Int-M	3	Acetaminophen (apap)	14 mcg/mL In Serum @ Unknown
1052ha	61 y F	Acetaminophen	1	1	U	Unk	Int-S	2	Acetaminophen (apap)	164 mcg/mL In Serum @ Unknown
1053i	62 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Amphetamine	3	3						
1054h	62 y F	Acetaminophen/diphenhydramine	1	1	U	Ingst	Int-S	3	Acetaminophen (apap)	24 mcg/mL In Serum @ Unknown
		Salicylate	2	2					Salicylate	4 mg/dL In Serum @ Unknown
		Ethanol (beverage)	3	3					Ethanol	326 mg/dL In Serum @ Unknown
1055ai	62 y F				U	Unk	Int-A	1		
		Oxycodone	1	1						
		Hydrocodone, N.O.S.	2	2						
		Codeine	3	3						
1056i	62 y F	Opioid, N.O.S.	1	1	A/C	Inhal	Int-A	2		
1057ha	62 y F	Acetaminophen	1	1	U	Ingst	Unk	2	Acetaminophen (apap)	150.2 mcg/mL In Whole Blood @ Unknown
1058ai	62 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Fentanyl analogs (para-fluoro)	2	2						
1059h	62 y F	Acetaminophen	1	1	A	Ingst	Unk	2	Acetaminophen (apap)	296 mcg/mL In Blood (unspecified) @ Unknown
		Salicylate	2	2					Salicylate	15 mg/dL In Blood (unspecified) @ Unknown
1060i	62 y M				A/C	Inhal	Int-A	2		
		Heroin	1	1						
1061ai	62 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
1062phi	62 y F				A	Unk	Int-A	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1063h	62 y F	Fentanyl (non-prescription)	1	1	C	Ingst	Int-M	2	Acetaminophen (apap)	116 mcg/mL In Serum @ 5 m (pe)
		Cocaine	2	2						
		Acetaminophen	1	1						
1064h	62 y M	Acetaminophen	1	1	U	Ingst	Unt-T	3	Acetaminophen (apap)	76 mcg/mL In Serum @ 2 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	109 mcg/mL In Serum @ 20 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	131.2 mcg/mL In Serum @ 14 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	216.4 mcg/mL In Serum @ Unknown
1065ai	63 y M	Oxycodone	1	1	U	Unk	Unk	2		
		Lorazepam	2	2						
		Diphenhydramine	3	3						
1066h	63 y F	Acetaminophen	1	1	U	Ingst	Unk	3	Acetaminophen (apap)	24 mcg/mL In Blood (unspecified) @ Unknown
		Diltiazem	2	2						
		Drug, N.O.S.	3	3						
1067i	63 y F	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
1068ai	63 y M	Fentanyl (non-prescription)	1	1						
		Diazepam	2	2						
1069h	63 y F	Acetaminophen	1	1	A	Ingst	Unk	2	Acetaminophen (apap)	5 mcg/mL In Serum @ 57 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	60 mcg/mL In Serum @ 32 h (pe)
1070i	63 y M	Fentanyl (prescription)	1	1	A/C	Ingst	Int-M	3	Salicylate	180 mg/L In Blood (unspecified) @ Autopsy
1071a	63 y F	Salicylate/caffeine	1	1						
		Salicylate/caffeine	1	1						
		Salicylate/caffeine	1	1	U	Unk	Int-A	1	Salicylate	19.4 mg/dL In Blood (unspecified) @ Unknown
		Salicylate/caffeine	1	1						
		Salicylate/caffeine	1	1						
1072ai	63 y M	Fentanyl (non-prescription)	1	1	A	Unk	Int-A	2		
1073i	63 y M	Opioid, N.O.S.	1	1						
		Cocaine	2	2						
1074ai	63 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Bupropion	3	3						
1075h	63 y F	Acetaminophen	1	1	A/C	Ingst	Unt-M	1	Acetaminophen (apap)	259 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen/ hydrocodone	2	2						
1076h	63 y F				U	Ingst	Int-U	3		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Acetaminophen	1	1					Acetaminophen (apap)	44 mcg/mL In Serum @ Unknown
1077i	63 y M				A/C	Unk	Int-A	2		
1078ai	63 y M	Heroin	1	1	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Xylazine	2	2						
1079i	63 y F	Alprazolam	3	3	U	Unk	Int-A	2		
		Opioid, N.O.S.	1	1						
1080ai	63 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Substance (non-drug), N.O.S.	2	2						
1081ai	63 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
1082ai	63 y M	Methamphetamine	2	2	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Cocaine	3	3						
1083h	63 y F				U	Ingst	Int-M	2		
		Acetaminophen Drug, N.O.S.	1	1						
			2	2						
1084ph	63 y F				U	Ingst	Int-A	2		
		Methadone	1	1						
1085ai	64 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
		Diazepam	3	3						
1086ai	64 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Buprenorphine	3	3						
1087	64 y F				A	Ingst	Int-S	1		
		Salicylate	1	1						
1088ai	64 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
1089h	64 y M				C	Ingst	Unt-T	3		
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	37.6 mcg/mL In Serum @ Unknown
		Acetaminophen	2	2					Acetaminophen (apap)	37.6 mcg/mL In Serum @ Unknown
1090h	64 y M				A	Ingst	Int-S	2	Acetaminophen (apap)	225 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
		Drug, N.O.S.	2	2						
1091ai	64 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
1092ai	64 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
1093ai	64 y M				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Gabapentin	2	2						
		Alprazolam	3	3						
1094pha	64 y F				A/C	Unk	Int-S	1	Diphenhydramine	2100 ng/mL In Blood (unspecified) @ Unknown
		Acetaminophen/diphenhydramine	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1095ai	64 y M	Acetaminophen/diphenhydramine	1	1	U	Unk	Int-A	1	Acetaminophen (apap)	256 mcg/mL In Plasma @ Unknown
		Gabapentin	2	2					Gabapentin	50 mcg/mL In Blood (unspecified) @ Unknown
		Fentanyl (non-prescription)	1	1						
		Ethanol (beverage)	2	2						
1096pha	64 y F	Tadalafil	3	3	A/C	Ingst	Int-S	1		
		Acetaminophen/hydrocodone	1	1					Dihydrocodeine	27.4 ng/mL In Blood (unspecified) @ Autopsy
		Acetaminophen/hydrocodone	1	1					Hydrocodone	87 ng/mL In Blood (unspecified) @ Autopsy
		Tramadol	2	2					Tramadol	1622 ng/mL In Blood (unspecified) @ Autopsy
		Tramadol	2	2					O-desmethyltramadol	382 ng/mL In Blood (unspecified) @ Autopsy
		Zolpidem	3	3					Zolpidem	562 ng/mL In Blood (unspecified) @ Autopsy
		Promethazine	4	4					Promethazine	29.7 ng/mL In Blood (unspecified) @ Autopsy
1097h	64 y M	Acetaminophen	1	1	U	Ingst	Int-U	3	Acetaminophen (apap)	11.9 mcg/mL In Serum @ 56 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	28.9 mcg/mL In Serum @ 17 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	35.9 mcg/mL In Serum @ 5 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	57.6 mcg/mL In Serum @ 2 h (pe)
1098ai	64 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
1099h	64 y F	Oxycodone	1	1	U	Ingst	Unk	3		
1100h	64 y M	Acetaminophen	1	1	A/C	Ingst	Int-S	1	Acetaminophen (apap)	33.8 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	71.4 mcg/mL In Blood (unspecified) @ Unknown
1101ha	64 y F	Acetaminophen	1	1	U	Ingst	Unk	3		
1102ai	65 y M	Salicylate	2	2	U	Unk	Unk	1		
		Morphine	1	1						
1103h	65 y F	Codeine	2	2	A	Ingst	Int-S	3		
		Acetaminophen	1	1					Acetaminophen (apap)	66 mcg/mL In Serum @ Unknown
1104ai	65 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Heroin	1	1						
1105	65 y F	Ethanol (beverage)	2	2	A	Ingst	Int-S	1		
		Acetaminophen	1	1					Acetaminophen (apap)	296 mcg/mL In Serum @ 4 h (pe)
		Tizanidine	2	2						
		Desvenlafaxine	3	3						
		Gabapentin	4	4						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1106ai	65 y M	Lorazepam	5	5	U	Unk	Int-A	1		
		Topiramate	6	6						
		Ubrogepant	7	7						
1107ph	65 y M	Fentanyl (non-prescription)	1	1	A/C	Ingst	Int-S	2		
		Methamphetamine	2	2						
		Oxycodone	3	3						
1108	65 y M	Methadone	1	1	C	Ingst	Int-M	3		
		Sodium bicarbonate	2	2						
		Metformin	3	3						
1109ai	65 y F	Acetaminophen	1	1	U	Unk	Unk	1		
1110ai	65 y M	Oxycodone	1	1	U	Unk	Int-A	1		
		Diphenhydramine	2	2						
		Hydroxyzine	3	3						
1111h	65 y F	Fentanyl (non-prescription)	1	1	A	Ingst	Int-S	3	Acetaminophen (apap)	154 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
1112h	65 y M				U	Unk	Unk	3	Acetaminophen (apap)	61.5 mcg/mL In Serum @ 30 m (pe)
		Acetaminophen	1	1						
1113h	65 y F				U	Ingst	Int-U	2	Acetaminophen (apap)	12.1 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
1114ph	65 y M	Drug, N.O.S.	2	2	C	Ingst	Int-S	2	Acetaminophen (apap)	134 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1						
		Acetaminophen	1	1						
1115h	65 y F				C	Ingst	Unk	2	Acetaminophen (apap)	107 mcg/mL In Serum @ Unknown
		Acetaminophen/oxycodone	1	1						
1116h	65 y F	Ibuprofen	2	2	C	Ingst	Int-M	1	Acetaminophen (apap)	41 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	3	3						
		Acetaminophen	1	1						
1117	65 y M				A	Ingst	Int-S	2	Salicylate	114 mg/dL In Blood (unspecified) @ Unknown
		Salicylate	1	1						
1118ai	66 y M	Morphine	1	1	U	Unk	Int-A	1		
1119h	66 y F				U	Unk	Unk	2	Acetaminophen (apap)	163 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1						
1120ai	66 y F				U	Unk	Unk	1		
		Oxycodone	1	1						
		Carisoprodol	2	2						
1121h	66 y M	Amlodipine	3	3	A/C	Ingst	Int-M	1	Acetaminophen (apap)	75 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1						
		Acetaminophen	1	1						
1122	66 y F				C	Ingst	AR-D	3		
		Acetaminophen	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1123ha	66 y M	Tramadol	1	1	A/C	Ingst	Int-A	2	Tramadol	1011 ng/mL In Blood (unspecified) @ Autopsy
		Tramadol	1	1					O-desmethyltramadol	173 ng/mL In Urine (quantitative only) @ Autopsy
		Tramadol	1	1					O-desmethyltramadol	211 ng/mL In Blood (unspecified) @ Autopsy
		Tramadol	1	1					Tramadol	897 ng/mL In Urine (quantitative only) @ Autopsy
		Acetaminophen/hydrocodone	2	2					Dihydrocodeine	10.5 ng/mL In Blood (unspecified) @ Autopsy
		Acetaminophen/hydrocodone	2	2					Hydrocodone	18.2 ng/mL In Urine (quantitative only) @ Autopsy
		Acetaminophen/hydrocodone	2	2					Acetaminophen (apap)	23.6 mcg/mL In Blood (unspecified) @ Autopsy
		Acetaminophen/hydrocodone	2	2					Hydrocodone	42.1 ng/mL In Blood (unspecified) @ Autopsy
		Acetaminophen/hydrocodone	2	2					Dihydrocodeine	7.8 ng/mL In Urine (quantitative only) @ Autopsy
		Cyclobenzaprine	3	3					Cyclobenzaprine	20.9 ng/mL In Urine (quantitative only) @ Autopsy
		Gabapentin	4	4					Gabapentin	0 mcg/mL In Urine (quantitative only) @ Autopsy
		Gabapentin	4	4					Gabapentin	15.6 mcg/mL In Blood (unspecified) @ Autopsy
		Gabapentin	4	4					Gabapentin	6.4 mcg/mL In Vitreous @ Autopsy
1124ha	66 y F	Acetaminophen	1	1	C	Ingst	Unt-T	1	Acetaminophen (apap)	75 mcg/mL In Blood (unspecified) @ Unknown
1125ai	66 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
1126i	66 y M	Opioid, N.O.S.	1	1	A	Unk	Int-A	2		
1127h	66 y M	Acetaminophen	1	1	U	Ingst	Unk	1	Acetaminophen (apap)	378 mcg/mL In Serum @ Unknown
1128ai	67 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
		Ethanol (beverage)	3	3						
1129ai	67 y F	Fentanyl (non-prescription)	1	1	U	Unk	Int-U	1		
		Methamphetamine	2	2						
1130h	67 y M	Acetaminophen	1	1	C	Ingst	Int-M	3		
1131h	67 y M	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	314 mcg/mL In Blood (unspecified) @ Unknown
		Benzodiazepine, N.O.S.	2	2						
		Metoprolol (ER)	3	3						
1132ai	67 y F	Methadone	1	1	U	Unk	Int-A	1		
1133ai	67 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1134h	67 y F	Fentanyl (non-prescription)	1	1						
		Dextromethorphan	2	2						
		Acetaminophen	1	1	C	Ingst	Unt-M	2	Acetaminophen (apap)	44 mcg/mL In Blood (unspecified) @ Unknown
1135ai	67 y M	Ethanol (non-beverage)	2	2						
		Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Fentanyl analogs (para-fluoro)	2	2						
1136ai	67 y M	Methamphetamine	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
1137ai	67 y M	Tadalafil	3	3	U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Oxycodone	2	2						
1138h	67 y F	Tramadol	3	3						
		Acetaminophen	1	1	U	Ingst	Int-M	1	Acetaminophen (apap)	132 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen/oxycodone	2	2						
1139h	67 y M	Acetaminophen/codeine	1	1	A/C	Ingst	Int-M	3	Acetaminophen (apap)	23 mcg/mL In Blood (unspecified) @ Unknown
1140h	67 y M				A/C	Unk	Unk	2		
		Tramadol	1	1						
		Tizanidine	2	2						
		Escitalopram	3	3						
		Olmesartan	4	4						
		Acetaminophen	5	5					Acetaminophen (apap)	40 mcg/mL In Blood (unspecified) @ Unknown
1141h	67 y F	Acetaminophen	1	1	C	Ingst	Int-M	2	Acetaminophen (apap)	0 mcg/mL In Serum @ 43 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	20 mcg/mL In Serum @ 30 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	31 mcg/mL In Serum @ 24 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	42 mcg/mL In Serum @ 19 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	89.8 mcg/mL In Serum @ 1 m (pe)
1142h	67 y F	Acetaminophen	1	1	A	Ingst + Unk	Unk	1	Acetaminophen (apap)	508.5 mcg/mL In Blood (unspecified) @ 36 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	624 mcg/mL In Blood (unspecified) @ 5 m (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	659 mcg/mL In Blood (unspecified) @ 48 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	666.7 mcg/mL In Blood (unspecified) @ 3 h (pe)
1143h	67 y F	Opioid, N.O.S.	2	2	C	Ingst	Int-M	3		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Acetaminophen	1	1					Acetaminophen (apap)	155 mcg/mL In Blood (unspecified) @ 0s (pa)
1144ai	68 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
1145ai	68 y F	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Morphine	1	1						
1146h	68 y M	Ethanol (beverage)	2	2						
		Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	178 mcg/mL In Serum @ Unknown
1147h	68 y M	Drug, N.O.S.	2	2	A	Ingst	Int-S	2		
1148	68 y F	Colchicine	1	1	A	Ingst	Int-S	1		
1149h	68 y F	Acetaminophen	1	1	C	Ingst + Unk	Unt-T	1		
		Acetaminophen	1	1					Ethanol	16 mg/dL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	242 mcg/mL In Serum @ Unknown
1150ai	68 y F	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Morphine	1	1						
		Codeine	2	2						
1151h	68 y F	Hydromorphone	3	3	C	Ingst	Int-M	2		
1152	68 y M	Colchicine	1	1	A	Ingst	Int-S	1		
		Salicylate	1	1					Salicylate	103 mg/dL In Blood (unspecified) @ Unknown
		Salicylate	1	1					Salicylate	65 mg/dL In Blood (unspecified) @ Unknown
		Salicylate	1	1					Salicylate	75 mg/dL In Blood (unspecified) @ Unknown
1153	68 y M	Acetaminophen	1	1	A	Ingst	Int-M	2		
1154h	68 y F	Acetaminophen	1	1	A/C	Ingst	Int-M	3		
		N-acetylcysteine	2	2						
1155	69 y F	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	16.9 mcg/mL In Blood (unspecified) @ Unknown
1156ai	69 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
1157h	69 y F	Methamphetamine	2	2	A	Ingst	Int-U	2		
		Acetaminophen	1	1					Acetaminophen (apap)	54 mcg/mL In Serum @ Unknown
1158h	69 y F	Metformin	2	2	U	Ingst	Int-M	2		
		Acetaminophen/hydrocodone	1	1						
1159h	69 y M	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	73.8 mcg/mL In Serum @ Unknown
1160a	69 y F	Acetaminophen	1	1	U	Ingst	Unk	1	Acetaminophen (apap)	220 mcg/mL In Blood (unspecified) @ Unknown
1161	69 y F	Acetaminophen/butalbital/cafeine	1	1	U	Ingst + Unk	Unk	2	Acetaminophen (apap)	120 mcg/mL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Salicylate	2	2					Salicylate	6.8mg/dL In Blood (unspecified) @ Unknown
1162h	69 y F	Acetaminophen	1	1	A/C	Ingst	Int-M	3	Acetaminophen (apap)	7.3mcg/mL In Serum @ Unknown
1163h	69 y M	Acetaminophen	1	1	A/C	Ingst	Int-M	1	Acetaminophen (apap)	56.5 mcg/mL In Blood (unspecified) @ Unknown
1164h	69 y M	Acetaminophen	1	1	A/C	Ingst	Unk	2	Acetaminophen (apap)	144 mg/L In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	200 mg/L In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	25 mg/L In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	53 mg/L In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	556 mg/L In Serum @ Unknown
1165ph	69 y F	Sevelamer	2	2	A/C	Ingst	Int-U	2		
		Fentanyl (prescription) (TD)	1	1						
1166	70 y F	Acetaminophen/oxycodone	1	1	A	Ingst	Int-S	2		
1167h	70 y F	Acetaminophen/hydrocodone	1	1	A/C	Ingst	Int-U	3		
1168ai	70 y F	Oxycodone	1	1	U	Unk	Int-A	2		
		Ethanol (beverage)	2	2						
1169ai	70 y M	Tramadol	1	1	U	Unk	Unk	1		
		Hydromorphone	2	2						
		Amitriptyline	3	3						
1170ha	70 y F	Acetaminophen	1	1	A/C	Ingst	Unt-G	2	Acetaminophen (apap)	116mcg/mL In Serum @ Unknown
1171ha	70 y M	Acetaminophen	1	1	A	Ingst	Unt-G	1	Acetaminophen (apap)	70 mcg/mL In Serum @ Unknown
1172h	70 y F	Acetaminophen/hydrocodone	1	1	U	Ingst	Int-S	2		
		Alprazolam	2	2						
		Cyclobenzaprine	3	3						
		Ethanol (beverage)	4	4						
1173ai	71 y M	Oxycodone	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Amlodipine	3	3						
1174	71 y F	Acetaminophen/oxycodone	1	1	A	Ingst	Int-A	2		
		Diazepam	2	2						
		Methocarbamol	3	3						
1175ai	71 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	1		
		Xylazine	2	2						
		Cocaine	3	3						
1176h	71 y F	Acetaminophen	1	1	C	Ingst	Int-M	1	Acetaminophen (apap)	105 mcg/mL In Serum @ 7h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	159 mcg/mL In Serum @ 1 m (pe)
1177h	71 y F				U	Ingst	Int-U	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1178h	71 y F	Salicylate	1	1					Acetaminophen (apap)	190 mcg/mL In Serum @ Unknown
		Salicylate	1	1					Salicylate	52 mg/dL In Serum @ Unknown
		Acetaminophen	2	2	C	Ingst	Int-M	1		
		Salicylate	1	1					Salicylate	98.4 mg/dL In Blood (unspecified) @ Unknown
1179h	72 y M				A/C	Ingst	AR-D	3		
1180ai	72 y M	Acetaminophen	1	1						
		Ethanol (beverage)	2	2						
1181ha	72 y F				U	Unk	Int-A	1		
		Fentanyl (non-prescription)	1	1						
		Methamphetamine	2	2						
		Cocaine	3	3						
1182	72 y M				C	Ingst	Int-U	3		
		Ibuprofen	2	2						
		Naproxen	1	1	A	Ingst	Int-S	2		
		Acetaminophen/hydrocodone	1	1						
1184h	72 y F				A/C	Ingst	Int-U	2		
1185h	72 y F	Acetaminophen	1	1	C	Ingst	Int-M	2		
		Acetaminophen	1	1					Acetaminophen (apap)	39.4 mcg/mL In Blood (unspecified) @ Unknown
1186ai	73 y M				U	Unk	Int-A	1		
1187	73 y M	Fentanyl (non-prescription)	1	1						
		Acetaminophen	1	1	U	Ingst	Int-S	2	Acetaminophen (apap)	378 mcg/mL In Blood (unspecified) @ Unknown
1188h	73 y F				A	Ingst	Int-T	3		
1189ai	73 y M				A	Unk	Int-A	1		
		Acetaminophen	1	1					Acetaminophen (apap)	42 mcg/mL In Blood (unspecified) @ Unknown
1190ha	73 y F	Fentanyl (non-prescription)	1	1						
		Acetaminophen	2	2						
		Gabapentin	3	3						
		Hydromorphone	1	1	U	Ingst + Unk	Int-U	2	Hydromorphone	27 ng/mL In Blood (unspecified) @ Autopsy
1191h	73 y M	Amitriptyline	2	2					Amitriptyline	100 ng/mL In Blood (unspecified) @ Autopsy
		Methadone	3	3						
		Tramadol	4	4					Tramadol	180 ng/mL In Blood (unspecified) @ Autopsy
		Acetaminophen	1	1	C	Ingst	Int-M	1	Acetaminophen (apap)	134 mcg/mL In Blood (unspecified) @ 0 s (pa)
1191h	73 y M	Acetaminophen	1	1					Acetaminophen (apap)	15.6 mcg/mL In Blood (unspecified) @ 75.7 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	19.2 mcg/mL In Blood (unspecified) @ 65.9 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	25.3 mcg/mL In Blood (unspecified) @ 56.9 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	25.3 mcg/mL In Blood (unspecified) @ 56.9 h (pe)

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1192h	73 y F	Acetaminophen	1	1					Acetaminophen (apap)	28.5 mcg/mL In Blood (unspecified) @ 49 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	35.4 mcg/mL In Blood (unspecified) @ 40.6 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	63.8 mcg/mL In Blood (unspecified) @ 18.4 h (pe)
		Ethanol (beverage)	2	2	A/C	Ingst	Unk	2		
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	117 mg/dL In Blood (unspecified) @ Unknown
1193h	73 y F	Acetaminophen	1	1	U	Unk	Unt-U	1	Acetaminophen (apap)	48 mg/mL In Blood (unspecified) @ Unknown
1194h	74 y M	Acetaminophen	1	1	C	Ingst + Unk	Unt-T	1	Acetaminophen (apap)	409 mcg/mL In Blood (unspecified) @ 1 h (pe)
1195h	74 y F	Salicylate	2	2					Salicylate	4.5 mg/dL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1	U	Ingst	Unk	3	Acetaminophen (apap)	38 mcg/mL In Blood (unspecified) @ Unknown
1196h	74 y F	Acetaminophen	1	1	A/C	Ingst	Int-M	3	Acetaminophen (apap)	56.5 mcg/mL In Serum @ Unknown
1197ha	74 y M	Acetaminophen/oxycodone	1	1	A	Ingst	Int-U	1	Acetaminophen (apap)	128 mg/dL In Blood (unspecified) @ Unknown
1198pha	75 y M	Oxycodone	1	1	U	Unk	Unt-T	2	Oxycodone	0.29 mg/L In Blood (unspecified) @ Autopsy
1199h	75 y F	Amitriptyline	2	2						
		Gabapentin	3	3						
		Albuterol	4	4						
1200h	75 y F	Tramadol	1	1	A/C	Ingst	Int-S	2		
1201h	75 y F	Acetaminophen	1	1	C	Ingst	Unt-T	3	Acetaminophen (apap)	36 mg/mL In Serum @ Unknown
		Acetaminophen	1	1	A	Unk	Unk	3	Acetaminophen (apap)	15.5 mg/L In Serum @ Unknown
1202a	75 y F	Acetaminophen	1	1	A	Ingst + Unk	Int-S	1	Acetaminophen (apap)	565.1 mcg/mL In Blood (unspecified) @ Unknown
1203h	75 y M	Acetaminophen/hydrocodone	1	1	A/C	Ingst	Int-S	2		
1204ha	75 y F	Acetaminophen	1	1	C	Ingst	Int-M	3	Acetaminophen (apap)	93 mcg/mL In Blood (unspecified) @ Unknown
1205h	76 y M	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	29.7 mcg/mL In Serum @ Unknown
1206	76 y F	Acetaminophen	1	1	U	Ingst	Unk	2	Acetaminophen (apap)	52.2 mcg/mL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1207h	77 y M				A	Ingst	Int-S	2		
1208h	77 y M	Salicylate	1	1	U	Ingst	Int-S	2		
		Acetaminophen	1	1					Acetaminophen (apap)	183.6 mcg/mL In Blood (unspecified) @ Unknown
1209ha	78 y F	Oxycodone	1	1	A/C	Ingst	Unk	1	Oxycodone	0.312 mg/L In Serum @ Unknown
1210ha	78 y M	Tramadol	1	1	A/C	Ingst	Int-S	1	O-desmethyiltramadol	1.2 mg/L In Blood (unspecified) @ Autopsy
		Tramadol	1	1					Tramadol	5.8 mg/L In Blood (unspecified) @ Autopsy
		Escitalopram	2	2						
1211h	78 y M	Acetaminophen	1	1	A	Ingst	Unt-G	3	Acetaminophen (apap)	130 mcg/mL In Blood (unspecified) @ Unknown
1212h	78 y M	Salicylate	1	1	C	Ingst	Int-U	3	Salicylate	42.2 mg/dL In Blood (unspecified) @ Unknown
		Caffeine/salicylamide/salicylate	2	2						
1213h	78 y F	Oxycodone	1	1	A	Ingst	Int-U	1		
		Acetaminophen	2	2					Acetaminophen (apap)	130 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (non-beverage)	3	3						
1214	79 y F	Acetaminophen/hydrocodone	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	212 mcg/mL In Blood (unspecified) @ Unknown
1215h	79 y M	Salicylate	1	1	C	Ingst	Unt-M	2	Salicylate	64.9 mg/dL In Blood (unspecified) @ Unknown
1216h	79 y F	Acetaminophen	1	1	A	Ingst	Unk	3	Acetaminophen (apap)	101 mcg/mL In Blood (unspecified) @ Unknown
1217ai	79 y M	Fentanyl (non-prescription)	1	1	U	Unk	Int-A	2		
		Amlodipine	2	2						
1218h	79 y F	Acetaminophen	1	1	C	Ingst	Unt-T	2		
1219h	79 y F	Acetaminophen	1	1	U	Ingst	Unt-T	2	Acetaminophen (apap)	244 mcg/mL In Serum @ Unknown
1220h	80 y F	Acetaminophen	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	300 mcg/mL In Serum @ Unknown
1221ha	80 y F	Acetaminophen	1	1	U	Unk	Unk	3	Acetaminophen (apap)	70.4 mcg/mL In Serum @ Unknown
1222h	80 y F	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	149 mcg/mL In Blood (unspecified) @ 36 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	27.2 mcg/mL In Blood (unspecified) @ 88 h (pe)

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1223	80 y M	Acetaminophen	1	1	A	Ingst	Int-S	3	Acetaminophen (apap)	34.6 mcg/mL In Blood (unspecified) @ 76 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	47.3 mcg/mL In Blood (unspecified) @ 63 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	83 mcg/mL In Blood (unspecified) @ 47 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	21.3 mcg/mL In Serum @ Unknown
1224ai	81 y F	Oxycodone	1	1	U	Unk	Int-S	1		
		Alprazolam	2	2						
		Temazepam	3	3						
1225ph	81 y M	Acetaminophen/hydrocodone	1	1	A/C	Ingst	Int-U	2	Acetaminophen (apap)	11.8 mcg/mL In Serum @ Unknown
1226h	82 y F	Acetaminophen	1	1					Acetaminophen (apap)	1131 mcg/mL In Serum @ Unknown
		Naproxen	2	2						
		Ibuprofen	3	3	C	Ingst	Int-S	2		
		Acetaminophen/hydrocodone	4	4						
1227	82 y F	Acetaminophen/caffeine/salicylate	1	1					Salicylate	32 mg/dL In Serum @ Unknown
		Acetaminophen/caffeine/salicylate	1	1					Acetaminophen (apap)	70 mcg/mL In Serum @ Unknown
1228h	83 y F	Acetaminophen	2	2	C	Ingst	Unk	1		
		Acetaminophen	1	1					Acetaminophen (apap)	141 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	16.6 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	2.6 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	52.8 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	6.9 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	78 mcg/mL In Serum @ Unknown
1229	83 y F	Acetaminophen	1	1	C	Ingst	Unk	1	Acetaminophen (apap)	246 mcg/mL In Serum @ Unknown
1230h	84 y F	Meloxicam	1	1						
		Gabapentin	2	2	A/C	Ingst	Int-U	2		
		Estradiol	3	3						
		Escitalopram	4	4						
		Lamotrigine (ER)	5	5						
1231p	84 y F	Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	407 mcg/mL In Serum @ Unknown
1232h	84 y F	Acetaminophen	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	150.8 mcg/mL In Serum @ Unknown
		Acetaminophen	1	1					Acetaminophen (apap)	67.4 mcg/mL In Serum @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1233h	84 y F	Acetaminophen	1	1					Acetaminophen (apap)	90 mcg/mL In Serum @ Unknown
		Atypical antipsychotic, N.O.S.	2	2						
		Beta blocker, N.O.S.	3	3						
		Salicylate	1	1	U	Ingst + Unk	Unk	1	Salicylate	42.8 mg/dL In Serum @ Unknown
1234h	84 y M				A/C	Ingst	Int-S	1		
		Oxycodone	1	1					Oxymorphone (free)	1.7 ng/mL In Blood (unspecified) @ 1 h (pe)
		Oxycodone	1	1					Oxycodone (free)	5.3 ng/mL In Blood (unspecified) @ 1 h (pe)
		Fentanyl (prescription)	2	2					Norfentanyl	41 ng/mL In Blood (unspecified) @ 1 h (pe)
1235h	85 y F	Fentanyl (prescription)	2	2					Fentanyl	73 ng/mL In Blood (unspecified) @ 1 h (pe)
					A	Ingst	Unt-U	2		
		Salicylate	1	1						
					A	Ingst	Int-S	1		
1236ha	85 y F	Acetaminophen/oxycodone	1	1					Oxymorphone (free)	11 ng/mL In Blood (unspecified) @ Unknown
		Acetaminophen/oxycodone	1	1					Acetaminophen (apap)	168 mcg/mL In Plasma @ Unknown
		Acetaminophen/oxycodone	1	1					Oxycodone (free)	550 ng/mL In Blood (unspecified) @ Unknown
1237h	85 y F				C	Ingst	AR-D	3		
		Acetaminophen	1	1					Acetaminophen (apap)	18 mcg/mL In Blood (unspecified) @ Unknown
1238	86 y F				C	Ingst	Unk	2		
		Acetaminophen	1	1					Acetaminophen (apap)	55 mcg/mL In Blood (unspecified) @ Unknown
1239	87 y F				U	Ingst	Unt-T	2		
		Acetaminophen	1	1					Acetaminophen (apap)	72 mg/mL In Serum @ Unknown
1240ha	87 y F				A	Ingst	Int-S	1		
		Acetaminophen	1	1					Acetaminophen (apap)	188 mcg/mL In Serum @ 7 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	420 mcg/mL In Serum @ 1 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	446 mcg/mL In Blood (unspecified) @ Autopsy
1241h	87 y M	Tramadol	2	2					Tramadol	1097 ng/mL In Blood (unspecified) @ Autopsy
		Tramadol	2	2					O-desmethyltramadol	152 ng/mL In Blood (unspecified) @ Autopsy
					A	Ingst	Unt-M	3		
		Acetaminophen	1	1					Acetaminophen (apap)	43.5 mcg/mL In Serum @ 16 h (pe)
1242h	89 y M				U	Ingst	Int-S	2		
		Salicylate	1	1						
1243ha	89 y F	Acetaminophen/hydrocodone	2	2					Acetaminophen (apap)	5.2 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	1	1	U	Ingst	Unk	1	Acetaminophen (apap)	143 mcg/mL In Serum @ 22 h (pe)

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1244h	90 y F	Acetaminophen	1	1	A/C	Ingst	Int-S	1	Acetaminophen (apap)	168mcg/mL In Serum @ 13 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	208 mcg/mL In Serum @ 9 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	298 mcg/mL In Serum @ 2 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	340 mg/L In Blood (unspecified) @ 3 h (pe)
		Duloxetine (ER)	2	2					Duloxetine	0.25 mg/L In Blood (unspecified) @ 3 h (pe)
1245h	90 y M	Tramadol	1	1	U	Ingst	Int-S	1		
		Acetaminophen	1	1					Acetaminophen (apap)	134.7 mcg/mL In Blood (unspecified) @ 51 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	26.7 mcg/mL In Blood (unspecified) @ 75 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	284.6 mcg/mL In Blood (unspecified) @ 39 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	376 mcg/mL In Blood (unspecified) @ 24 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	66.4 mcg/mL In Blood (unspecified) @ 63 h (pe)
1246ha	91 y M	Morphine	1	1	A	Ingst + Unk	Int-S	3	Morphine (free)	390 ng/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	2	2					Delta-9-thc	0.9ng/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	2	2					Delta-9-carboxy-thc	10 ng/mL In Blood (unspecified) @ Unknown
1247	91 y M	Oxycodone	1	1	A	Ingst	Int-S	2		
		Tramadol	2	2						
		Morphine	3	3						
1248h	91 y F	Acetaminophen/ diphenhydramine	1	1	C	Ingst	Int-S	2	Acetaminophen (apap)	140 mcg/mL In Blood (unspecified) @ 0 s (pa)
		Acetaminophen/ diphenhydramine	1	1					Acetaminophen (apap)	19 mcg/mL In Blood (unspecified) @ 68 h (pe)
		Acetaminophen/ diphenhydramine	1	1					Acetaminophen (apap)	26 mcg/mL In Blood (unspecified) @ 44 h (pe)
		Acetaminophen/ diphenhydramine	1	1					Acetaminophen (apap)	26 mcg/mL In Blood (unspecified) @ 52.7 h (pe)
		Acetaminophen/ diphenhydramine	1	1					Acetaminophen (apap)	68 mcg/mL In Blood (unspecified) @ 21.5 m (pe)
		Acetaminophen/ diphenhydramine	1	1					Acetaminophen (apap)	95 mcg/mL In Blood (unspecified) @ 9 h (pe)
1249h	91 y F	Lisinopril	2	2	A	Ingst	Int-S	1		
		Hydrochlorothiazide	3	3						
		Acetaminophen/ codeine	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1250h	92 y F	Fentanyl (prescription) (TD)	1	1	C	Derm	Unt-G	2		
1251ha	92 y M	Apixaban	2	2	A/C	Ingst	Unt-G	1		
		Acetaminophen	1	1					Acetaminophen (apap)	299 mcg/mL In Serum @ Unknown
		Ethanol (beverage)	2	2					Ethanol	0.25 % (wt/Vol) In Serum @ Unknown
		Ethanol (beverage)	2	2					Ethanol	0.328 % (wt/Vol) In Vitreous @ Autopsy
1252ph	93 y M	Acetaminophen/hydrocodone	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	263 mcg/mL In Blood (unspecified) @ 4 h (pe)
		Acetaminophen/hydrocodone	1	1					Acetaminophen (apap)	285 mcg/mL In Blood (unspecified) @ Unknown
1253ai	96 y F	Hydrocodone, N.O.S.	1	1	U	Unk	Int-S	1		
		Diphenhydramine	2	2						
		Ethanol (beverage)	3	3						
1254pi	11 m M	Fentanyl (non-prescription)	1	1	A	Ingst	Oth-M	1		
1255ph	19 m M	Methadone	1	1	A	Ingst	Unk	2		
1256p	21 m F	Fentanyl (non-prescription)	1	1	A	Unk	Unt-G	1		
1257i	30+ y F	Opioid, N.O.S.	1	1	U	Unk	Int-A	2		
1258h	50+ y F	Acetaminophen	1	1	U	Ingst	Int-S	1	Acetaminophen (apap)	137 mcg/mL In Serum @ 2 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	173 mcg/mL In Serum @ 1 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	30.6 mcg/mL In Serum @ 4 d (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	363 mcg/mL In Serum @ 4.5 h (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	500 mcg/mL In Serum @ 15 m (pe)
		Acetaminophen	1	1					Acetaminophen (apap)	66.9 mcg/mL In Serum @ 3 d (pe)
1259ph	Unknown adult (≥ 20 yrs) M	Diphenhydramine	2	2	U	Ingst	Unk	3		
		Acetaminophen	1	1					Acetaminophen (apap)	31 mcg/mL In Blood (unspecified) @ Unknown
1260pi	Unknown adult (≥ 20 yrs) F	Fentanyl (non-prescription)	3	1	A	Par	Int-A	2		
		Bromazolam	2	2						
		Drug, N.O.S.	1	3						
		Xylazine	4	4						
		Caffeine	5	5						
		Quinine	6	6						
1261i	Unknown adult (≥ 20 yrs) M				U	Unk	Int-A	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1262i	Unknown adult (≥ 20 yrs) M	Opioid, N.O.S.	1	1	U	Unk	Unk	2		
1263ai	Unknown age U	Opioid, N.O.S.	1	1	U	Unk	Unt-G	1		
1264ai	Unknown age F	Fentanyl (non-prescription) Methamphetamine	1 2	1 2	U	Unk	Unt-G	1		
1265ai	Unknown age M	Fentanyl (non-prescription) Methamphetamine	1 2	1 2	U	Unk	Unt-G	1		
		Fentanyl (non-prescription) Methamphetamine	1 2	1 2						
See Also case 24, 26, 44, 96, 98, 207, 225, 299, 1276, 1277, 1282, 1292, 1294, 1303, 1305, 1312, 1333, 1342, 1344, 1359, 1363, 1365, 1366, 1372, 1376, 1378, 1380, 1381, 1385, 1389, 1392, 1398, 1404, 1407, 1412, 1417, 1418, 1427, 1434, 1446, 1448, 1449, 1451, 1452, 1465, 1467, 1474, 1481, 1485, 1508, 1516, 1519, 1521, 1529, 1530, 1541, 1564, 1567, 1570, 1573, 1578, 1585, 1588, 1596, 1622, 1627, 1629, 1634, 1644, 1660, 1671, 1675, 1688, 1690, 1694, 1699, 1717, 1721, 1727, 1737, 1738, 1739, 1742, 1758, 1764, 1767, 1776, 1800, 1804, 1805, 1807, 1809, 1812, 1816, 1821, 1828, 1842, 1843, 1855, 1857, 1862, 1865, 1868, 1869, 1877, 1892, 1895, 1906, 1908, 1915, 1947, 1950, 1956, 1963, 1976, 1980, 1983, 2008, 2014, 2033, 2046, 2050, 2080, 2085, 2096, 2100, 2120, 2130, 2136, 2145, 2153, 2175, 2191, 2195, 2213, 2222, 2223, 2227, 2228, 2232, 2235, 2248, 2249, 2263, 2266, 2267, 2268										
Anesthetics										
[1266ph]	19 y M	Lidocaine	1	1	A	Ingst	Int-S	1	Lidocaine	3.7 mg/L In Blood (unspecified) @ Unknown
1267ai	30 y M	Lidocaine	1	1	U	Unk	Int-S	1		
[1268h]	33 y F	Marijuana, N.O.S.	2	2	A/C	Inhal	Int-A	3		
1269ph	41 y F	Nitrous oxide	1	1	U	Ingst + Inhal	Int-A	2		
		Nitrous oxide Drug, N.O.S.	1 2	1 2						
1270ai	48 y M	Ketamine Substance (non-drug), N.O.S.	1 2	1 2	U	Unk	Int-M	1		
1271h	63 y F	Bupivacaine	1	1	A	Par	Unt-T	3		
1272ai	65 y F	Ketamine Cyclobenzaprine Trazodone	1 2 3	1 2 3	U	Unk	Int-A	1		
[1273ha]	73 y M	Mepivacaine	1	1	A	Par	AR-D	2		
1274ph	97 y M	Lidocaine	1	1	A	Par	AR-D	2		
See Also case 381, 1005, 1327, 1897, 2224										
Anticholinergic Drugs										
1275	81 y F	Scopolamine (TD)	1	1	A	Derm	Unt-G	3		
See Also case 604, 1290, 1569, 1694, 1827, 2055										
Anticoagulants										
1276h	45 y F	Apixiban	1	1	A/C	Ingst	Int-S	2		
		Acetaminophen	2	2						
1277a	59 y M	Apixiban	1	1	A	Ingst	Int-S	1		
		Ibuprofen	2	2						
		Lisinopril	3	3						
		Sertraline	4	4						
1278	73 y F	Apixiban	1	1	A	Ingst	Int-S	3		
		Duloxetine (ER)	2	2						
1279h	94 y M	Apixiban	1	1	A/C	Ingst	Int-S	1		
		Ethanol (beverage)	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
See Also case 1021, 1250, 1500, 1507, 1516, 1520, 1529, 1578, 1603, 1604, 1610, 1638, 1640, 1661, 1662, 1674, 1676, 1709, 1736, 1741										
Anticonvulsants										
[1280pha]	16 y F	Lamotrigine	1	1	A	Ingst	Int-S	1	Lamotrigine	3.5 mcg/mL In Blood (unspecified) @ Unknown
1281h	19 y F	Lamotrigine	1	1	A	Ingst	Int-S	2		
1282ha	20 y F	Topiramate	1	1	A/C	Ingst	Int-S	1	Topiramate	42000 ng/mL In Blood (unspecified) @ Unknown
		Lamotrigine	2	2					Lamotrigine	100 mcg/mL In Blood (unspecified) @ Unknown
		Ibuprofen Drug, N.O.S.	3	3						
			4	4						
1283h	20 y F	Lamotrigine	1	1	A	Ingst	Int-S	1	Lamotrigine	26 mcg/mL In Blood (unspecified) @ 1 h (pe)
		Quetiapine	2	2					Quetiapine	460 ng/mL In Blood (unspecified) @ 1 h (pe)
		Bupropion	3	3					Bupropion	139 ng/mL In Blood (unspecified) @ 20 h (pe)
		Bupropion	3	3					Hydroxybupropion	338 ng/mL In Blood (unspecified) @ 20 h (pe)
1284ph	21 y F	Lacosamide	1	1	A/C	Ingst	Int-S	2		
		Clobazam	2	2						
		Fluoxetine	3	3						
		Folic acid	4	4						
		Nystatin	5	5						
1285p	29 y F	Lamotrigine	1	1	A/C	Ingst	Int-S	1	Lamotrigine	40 mcg/mL In Serum @ Unknown
		Brexiprazole	2	2						
		Ethanol (beverage)	3	3					Ethanol	134 mg/dL In Serum @ Unknown
1286h	34 y M	Oxcarbazepine	1	1	U	Ingst	Int-S	2		
1287h	41 y M	Lamotrigine (ER)	1	1	A	Ingst	Int-S	3		
		Clonidine (TD)	2	2						
		Paroxetine	3	3						
		Sertraline	4	4						
1288pha	43 y F	Valproic acid Drug, N.O.S.	1	1	U	Ingst	Int-S	2		
			2	2						
		Cocaine	3	3						
		Benzodiazepine, N.O.S.	4	4						
1289ha	47 y F	Gabapentin	1	1	U	Ingst	Int-S	1		
		Fluoxetine	2	2						
		Clonazepam	3	3						
		Pseudoephedrine	4	4						
1290phai	48 y M	Valproic acid (ER)	1	1	A/C	Ingst	Unt-G	1	Valproic acid	268 mcg/mL In Blood (unspecified) @ Unknown
		Valproic acid (ER)	1	1					Valproic acid	550 mcg/mL In Blood (unspecified) @ Autopsy
		Haloperidol	2	2						
		Benzonatate	3	3						
		Benzotropine	4	4						
1291p	55 y M	Citalopram	1	1	A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1292	55 y F	Valproic acid (ER)	3	1	A/C	Ingst	Int-S	2		
		Sertraline	2	2						
		Fluoxetine	4	3						
		Lamotrigine	4	1						
		Salicylate	2	2						
		Clonazepam	3	3						
		Amlodipine	1	4						
1293h	56 y F	Sertraline	5	5	A	Ingst	Int-S	3		
		Quetiapine	6	6						
1294pha	58 y M	Fenofibrate	7	7	A/C	Ingst + Unk	Int-A	1	Gabapentin	18 mcg/mL In Blood (unspecified) @ Unknown
		Gabapentin	1	1						
		Gabapentin	1	1						
		Fentanyl (non-prescription)	2	2						
		Fentanyl (non-prescription)	2	2						
		Fentanyl (non-prescription)	2	2						
		Cyclobenzaprine	3	3						
1295	59 y M				A	Ingst	AR-O	3		
1296h	62 y F	Phenytoin	1	1	A/C	Ingst	Int-S	1	Lamotrigine	1.6 mcg/mL In Blood (unspecified) @ Unknown
		Lamotrigine	1	1						
		Lamotrigine	1	1						
		Lamotrigine	1	1						
		Lamotrigine	1	1						
		Lamotrigine	1	1						
		Gabapentin	2	2						
1297a	67 y M	Gabapentin	3	1	A/C	Ingst	Int-S	1	Gabapentin	24 mcg/mL In Blood (unspecified) @ 7 h (pe)
		Duloxetine (ER)	2	2						
1298h	71 y F	Baclofen	1	3	A	Ingst	Int-S	2		
		Gabapentin	1	1						
		Quetiapine	2	2						
1299h	73 y M	Baclofen	3	3	C	Ingst	AR-D	3	Phenytoin	46.1 mcg/mL In Serum @ Unknown
		Phenytoin	1	1						
1300h	74 y M				A/C	Ingst	Int-S	2		
		Lamotrigine	1	1						
		Quetiapine	2	2						
		Mirtazapine	3	3						
1301ha	80 y F	Gabapentin	4	4	C	Ingst	AR-D	3	Carbamazepine	12.7 mcg/mL In Serum @ Unknown
		Carbamazepine	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1302ph	Unknown adult (≥ 20 yrs) M				A	Unk	Int-U	2		
See Also case 6, 9, 422, 461, 562, 648, 728, 770, 781, 836, 890, 894, 896, 973, 988, 1050, 1093, 1094, 1105, 1123, 1189, 1198, 1230, 1308, 1309, 1318, 1320, 1323, 1341, 1347, 1349, 1356, 1359, 1360, 1362, 1378, 1389, 1393, 1407, 1424, 1440, 1446, 1484, 1513, 1515, 1528, 1529, 1530, 1532, 1546, 1554, 1569, 1576, 1585, 1589, 1596, 1605, 1627, 1660, 1670, 1694, 1728, 1756, 1758, 1768, 1772, 1802, 1807, 1810, 1811, 1821, 1829, 1835, 1848, 1851, 1853, 1859, 1932, 1933, 2096, 2112, 2186, 2211										
Antidepressants										
1303h	12 y F				A	Ingst	Int-S	2		
		Gabapentin	1	1						
		Drug, N.O.S.	2	2						
									Salicylate	19.1 mg/dL In Blood (unspecified) @ Unknown
		Azithromycin	3	3						
		Penicillin	4	4						
1304ph	13 y F				A/C	Ingst	Int-S	1		
		Bupropion	1	1						
		Quetiapine	2	2						
1305h	13 y F				U	Ingst	Int-S	2		
		Bupropion (ER)	1	1						
		Carvedilol	2	2						
		Fluoxetine	3	3						
		Montelukast	4	4						
		Ibuprofen	5	5						
1306ph	14 y F				A	Ingst	Int-S	2		
		Bupropion	1	1						
		Pseudoephedrine	2	2						
1307ai	14 y F				U	Unk	Int-S	1		
		Bupropion	1	1						
		Atomoxetine	2	2						
		Olanzapine	3	3						
1308h	15 y F				A	Ingst	Int-S	2		
		Bupropion	1	1						
		Dicyclomine	2	2						
		Gabapentin	3	3						
1309	15 y F				U	Ingst	Int-S	2		
		Bupropion (ER)	1	1						
		Oxcarbazepine	2	2						
1310hi	15 y F				A	Ingst	Int-S	2		
		Bupropion	1	1						
1311pa	17 y F				U	Ingst	Int-S	1		
		Bupropion (ER)	1	1					Bupropion	19000 ng/mL In Blood (unspecified) @ Autopsy
									Hydroxybupropion	2500 ng/mL In Blood (unspecified) @ Autopsy
		Bupropion (ER)	1	1						
1312h	17 y M				A	Ingst	Int-S	2		
		Bupropion	1	1						
		Acetaminophen	2	2						
1313ph	18 y F				U	Ingst	Int-S	2		
		Bupropion	1	1						
1314h	18 y F				A/C	Ingst	Int-S	1		
		Lithium (ER)	1	1					Lithium	2.1 mEq/L In Serum @ 1 m (pe)
		Lithium (ER)	1	1					Lithium	3.1 mEq/L In Serum @ 4 h (pe)
		Lithium (ER)	1	1					Lithium	4.4 mEq/L In Serum @ 64 h (pe)
		Lithium (ER)	1	1					Lithium	5.6 mEq/L In Serum @ 10 h (pe)
		Lithium (ER)	1	1					Lithium	6.8 mEq/L In Serum @ 46 h (pe)
		Lithium (ER)	1	1					Lithium	6.8 mEq/L In Serum @ 52 h (pe)
		Lithium (ER)	1	1					Lithium	7.4 mEq/L In Serum @ 28 h (pe)
		Lithium (ER)	1	1					Lithium	7.6 mEq/L In Serum @ 39 h (pe)
		Lithium (ER)	1	1					Lithium	7.7 mEq/L In Serum @ 34 h (pe)
		Lithium (ER)	1	1					Lithium	8 mEq/L In Serum @ 16 h (pe)
		Lithium (ER)	1	1					Lithium	9.6 mEq/L In Serum @ 22 h (pe)

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1315h	19 y M	Bupropion (ER)	2	2	A	Ingst	Int-S	1	Bupropion	1000 ng/mL In Blood (unspecified) @ 4 h (pe)
		Trazodone	3	3						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
		Bupropion	1	1						
1316ph	19 y F	Bupropion	1	1	A	Ingst	Int-S	2	Bupropion	1100 ng/mL In Blood (unspecified) @ 5 m (pe)
		Hydroxyzine	2	2						
1317	19 y F	Bupropion (ER)	1	1	A/C	Ingst	Int-S	1	Bupropion	507 ng/mL In Blood (unspecified) @ 102 h (pe)
		Buspirone	2	2						
1318pha	19 y F	Bupropion	1	1	A/C	Ingst	Int-S	1	Bupropion	1100 ng/mL In Blood (unspecified) @ 5 m (pe)
		Bupropion	1	1						
		Fluoxetine	2	2						
		Fluoxetine	2	2						
		Gabapentin	3	3						
		Marijuana, N.O.S.	4	4						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1319	20 y F	Amitriptyline	1	1	A/C	Ingst	Int-S	2		
		Drug, N.O.S.	2	2						
1320h	22 y M	Bupropion (ER)	1	1	A/C	Ingst	Int-S	3		
		Hydroxyzine	2	2						
		Lamotrigine	3	3						
		Aripiprazole	4	4						
1321p	24 y F	Amitriptyline	1	1	A/C	Ingst	Int-S	2		
		Amphetamine/ dextroam- phetamine	2	2						
1322ph	24 y M	Dextromethorphan/ bupropion (ER)	1	1	A/C	Ingst	Int-S	2		
1323ha	24 y F	Bupropion	1	1	A/C	Ingst	Int-S	1	Bupropion	1600 ng/mL In Blood (unspecified) @ 15 m (pe)
		Bupropion	1	1					Hydroxybupropion	2600 ng/mL In Blood (unspecified) @ 15 m (pe)
		Lithium	2	2					Lithium	2.6 mEq/L In Blood (unspecified) @ Autopsy
		Mirtazapine	3	3					Mirtazapine	410 ng/mL In Blood (unspecified) @ 15 m (pe)
		Sertraline	4	4					Desmethylsertraline	1000 ng/mL In Blood (unspecified) @ 15 m (pe)
		Sertraline	4	4					Sertraline	300 ng/mL In Blood (unspecified) @ 15 m (pe)
		Lamotrigine	5	5					Lamotrigine	11 mcg/mL In Blood (unspecified) @ 15 m (pe)
1324h	25 y M	Bupropion	1	1	U	Unk	Unk	2		
		Methamphetamine	2	2						
		Risperidone	3	3						
1325	25 y F	Bupropion	1	1	A	Ingst	Int-S	2		
		Trazodone	2	2						
1326h	27 y M	Venlafaxine	1	1	A/C	Ingst	Int-S	2		
		Ethanol (beverage)	2	2						
1327h	27 y F	Isocarboxazid	1	1	A/C	Ingst	Int-S	2		
		Esketamine	2	2						
		Trazodone	3	3						
		Buspirone	4	4						
1328ph	28 y F	Paroxetine	1	1	A	Ingst	Int-S	2		
		Olanzapine	2	2						
		Hydroxyzine	3	3						
1329h	29 y F	Amitriptyline	1	1	A/C	Ingst	Int-S	2		
[1330ha]	30 y F	Bupropion	1	1	A	Ingst	Int-S	1	Bupropion	2500 ng/mL In Blood (unspecified) @ Autopsy
1331ha	30 y F	Bupropion	1	1	A/C	Ingst	Int-S	2		
		Amitriptyline	2	2						
1332h	30 y F	Bupropion (ER)	1	1	A/C	Ingst	Int-S	2		
		Quetiapine	2	2						
		Buspirone	3	3						
		Ethanol (beverage)	4	4						
1333ph	31 y F	Bupropion	1	1	U	Ingst	Int-S	2		
		Cyclic antidepressant, N.O.S.	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1334h	31 y F	Ibuprofen	3	3						
		Venlafaxine	1	1	A	Ingst	Int-S	1		
		Ethanol (beverage)	2	2					Ethanol	239 mg/dL In Blood (unspecified) @ 1 h (pe)
1335h	32 y F	Doxepin	1	1	A/C	Ingst	Int-S	1		
[1336ha]	32 y F	Amitriptyline	1	1	A/C	Ingst	Int-S	1	Amitriptyline	0.17 mg/L In Blood (unspecified) @ Autopsy
		Amitriptyline	1	1					Nortriptyline	0.2 mg/L In Blood (unspecified) @ Autopsy
1337h	33 y F	Bupropion	1	1	U	Ingst	Int-S	2		
1338ha	33 y F	Bupropion	1	1	A/C	Ingst	Int-S	2		
		Ethanol (beverage)	2	2					Ethanol	164 mg/dL In Serum @ Unknown
1339h	33 y F	Venlafaxine	1	1	A/C	Ingst	Int-S	2		
1340ph	34 y F	Doxepin	1	1	A/C	Unk	Int-S	2		
1341h	34 y F	Bupropion	1	1	A/C	Ingst	Unk	3		
		Trazodone	2	2						
		Lamotrigine	3	3						
1342ph	34 y M	Sertraline	1	1	A	Ingst	Int-S	3		
		Metoprolol (ER)	2	2						
		Doxepin	3	3						
		Buprenorphine/naloxone (sublingual film)	4	4						
		Cyclobenzaprine	5	5						
		Alprazolam	6	6						
1343h	35 y M	Bupropion	1	1	A	Ingst	Int-U	2		
		Drug, N.O.S.	2	2						
1344	35 y M	Bupropion	1	1	U	Ingst	Int-S	1		
		Salicylate	2	2					Salicylate	60 mg/dL In Blood (unspecified) @ Unknown
1345h	35 y F	Bupropion	1	1	A	Ingst	Int-S	2		
		Alcohol, N.O.S.	2	2						
1346h	36 y F	Amitriptyline	1	1	U	Ingst	Int-S	2		
1347ha	37 y M	Trazodone	1	1	A/C	Ingst	Int-S	3	Trazodone	26 mcg/mL In Blood (unspecified) @ Autopsy
		Pregabalin	2	2						
1348pa	37 y M	Trazodone	1	1	A	Ingst	Int-S	1		
1349a	38 y M	Amitriptyline	1	1	A	Ingst + Unk	Int-S	1	Amitriptyline	1600 ng/mL In Blood (unspecified) @ Unknown
		Amitriptyline	1	1					Nortriptyline	550 ng/mL In Blood (unspecified) @ Unknown
		Acetaminophen/dextromethorphan/doxylamine	3	2						
		Levetiracetam	2	2					Levetiracetam	320 mcg/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	4	4					Delta-9-carboxy-thc	69 ng/mL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Marijuana, N.O.S.	4	4					Delta-9-thc	9.2 ng/mL In Blood (unspecified) @ Unknown
1350ha	38 y M	Escitalopram	3	1	U	Ingst + Unk	Unk	2		
		Aripiprazole	4	2						
		Methamphetamine	2	2						
1351ha	39 y M	Drug, N.O.S.	1	3	A/C	Ingst	Int-S	1		
		Venlafaxine	1	1					Venlafaxine	13000 ng/mL In Blood (unspecified) @ Unknown
		Venlafaxine	1	1					O-desmethyl-venlafaxine	690 ng/mL In Blood (unspecified) @ Unknown
		Bupropion	2	2					Bupropion	190 ng/mL In Blood (unspecified) @ Unknown
		Bupropion	2	2					Hydroxybupropion	830 ng/mL In Blood (unspecified) @ Unknown
		Atenolol	3	3					Atenolol	49000 ng/mL In Blood (unspecified) @ Unknown
1352ha	39 y F	Bupropion	1	1	A/C	Ingst	Int-S	1	Bupropion	40 mg/L In Blood (unspecified) @ Unknown
[1353pha]	40 y M	Tianeptine	1	1	U	Ingst	Int-U	1	Tianeptine	600 ng/mL In Blood (unspecified) @ Unknown
		Caffeine	3	3						
		Diphenhydramine	2	3					Diphenhydramine	170 ng/mL In Blood (unspecified) @ Unknown
1354ha	40 y M	Bupropion (ER)	1	1	A	Ingst	Int-S	1		
		Escitalopram	2	2						
1355h	41 y M	Lithium	1	1	A	Ingst	Int-S	2	Lithium	0.6 mmol/L In Serum @ 72 h (pe)
		Lithium	1	1					Lithium	1.3 mmol/L In Serum @ 48 h (pe)
		Lithium	1	1					Lithium	1.3 mmol/L In Serum @ 54 h (pe)
		Lithium	1	1					Lithium	1.5 mmol/L In Serum @ 36 h (pe)
		Lithium	1	1					Lithium	1.6 mmol/L In Serum @ 65 h (pe)
		Lithium	1	1					Lithium	2.7 mmol/L In Serum @ 42 h (pe)
		Lithium	1	1					Lithium	2.7 mmol/L In Serum @ 51 h (pe)
		Lithium	1	1					Lithium	5 mmol/L In Serum @ 5 h (pe)
		Lithium	1	1					Lithium	6 mmol/L In Serum @ 24 h (pe)
		Lithium	1	1					Lithium	6 mmol/L In Serum @ 9 h (pe)
1356ha	41 y M	Bupropion (ER)	1	1	A/C	Ingst + Par + Unk	Int-S	1	Bupropion	120 ng/mL In Blood (unspecified) @ Autopsy
		Lamotrigine	3	2					Cyclobenzaprine	24.7 ng/mL In Blood (unspecified) @ Autopsy

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1357h	42 y M	Lamotrigine	3	2					Lamotrigine	36 mcg/mL In Blood (unspecified) @ Autopsy
		Methamphetamine	2	2					Methamphetamine	516 ng/mL In Blood (unspecified) @ Autopsy
		Methamphetamine	2	2					Gabapentin	7 mcg/mL In Blood (unspecified) @ Autopsy
		Hydroxyzine	4	3					Hydroxyzine	3705 ng/mL In Blood (unspecified) @ Autopsy
		Trazodone	5	4					Trazodone	3.1 mcg/mL In Blood (unspecified) @ Autopsy
		Trazodone	5	4					Mcpp (meta-chlorophenylpiperazine)	483 ng/mL In Blood (unspecified) @ Autopsy
1357h	42 y M	Cyclobenzaprine	6	6						
		Gabapentin	7	7						
1358h	44 y F	Escitalopram	1	1	A/C	Ingst + Aspir	Int-S	2		
		Trazodone	2	2	A	Ingst	Int-S	2		
1359p	46 y F	Bupropion (ER)	1	1						
		Alprazolam	2	2	A/C	Ingst	Int-S	2		
1360p	46 y M	Bupropion	1	1						
		Brexiprazole	2	2						
		Asenapine	3	3						
		Atomoxetine	4	4						
		Trazodone	5	5						
		Tizanidine	6	6						
		Quetiapine	7	7						
		Hydroxyzine	8	8						
		Gabapentin	9	9						
		Topiramate	10	10						
		Meloxicam	11	11						
1361h	46 y F	Bupropion	1	1	A/C	Ingst	Int-S	2		
		Propranolol	2	2						
		Lamotrigine	3	3						
		Olanzapine	4	4						
		Hydroxyzine	5	5						
1362	48 y F	Lithium	1	1	A/C	Ingst	Unt-U	3	Lithium	3.8 mEq/L In Blood (unspecified) @ Unknown
1363	50 y F	Bupropion	1	1	U	Ingst	Int-S	2		
		Gabapentin	2	2						
1364h	50 y F	Venlafaxine (ER)	1	1	A/C	Unk	Int-S	3		
		Acetaminophen/hydrocodone	2	2					Acetaminophen (apap)	39.2 mcg/mL In Blood (unspecified) @ Unknown
1364h	50 y F	Ondansetron	3	3						
		Tizanidine	4	4						
		Alprazolam	5	5						
1365h	51 y F	Bupropion	1	1	A/C	Ingst	Int-S	2		
		Vortioxetine	2	2						
		Ethanol (beverage)	3	3					Ethanol	377 mg/dL In Blood (unspecified) @ Unknown
1366	51 y M	Venlafaxine (ER)	1	1	A	Ingst	Int-S	2		
		Propranolol	2	2						
		Oxycodone (ER)	3	3						
		Meloxicam	4	4						
1366	51 y M	Bupropion	1	1	A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1367pha	52 y F	Acetaminophen/hydrocodone	2	2					Acetaminophen (apap)	115.4 mcg/mL In Blood (unspecified) @ Unknown
		Fluoxetine	3	3						
		Ibuprofen	4	4						
		Trazodone	1	1	A/C	Ingst	Int-S	2	Trazodone	18.7 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2						
		Bupropion	3	3					Bupropion	410 ng/mL In Blood (unspecified) @ Unknown
		Sertraline	4	4					Sertraline	694 ng/mL In Blood (unspecified) @ Unknown
1368h	53 y M	Sertraline	4	4					Desmethylsertraline	767 ng/mL In Blood (unspecified) @ Unknown
		Chlorophenhyll-piperazine	5	5						
1369pha	53 y M	Amitriptyline	1	1	A	Ingst	Int-S	2		
1370hi	53 y M	Bupropion (ER)	1	1	U	Ingst	Int-S	1	Bupropion	4200 ng/mL In Blood (unspecified) @ Autopsy
1371ph	53 y M	Lithium	1	1	A/C	Ingst	Unk	2		
1372pha	54 y M				U	Ingst + Inhal	Unk	2		
		Amitriptyline	1	1						
		Ethanol (beverage)	2	2						
		Methamphetamine	3	3						
					A/C	Ingst	Int-S	2		
1373a	54 y M	Amitriptyline	5	1					Buprenorphine	11 ng/mL In Blood (unspecified) @ Autopsy
		Buprenorphine	1	1						
		Lorazepam	2	2						
		Benzodiazepine, N.O.S.	3	3						
		Ethanol (beverage)	4	4					Ethanol	208 mg/dL In Blood (unspecified) @ Autopsy
					A/C	Ingst	Int-S	1	Nortriptyline	240 ng/mL In Whole Blood @ Unknown
		Amitriptyline	1	1					Amitriptyline	2400 ng/mL In Whole Blood @ Unknown
		Cocaine	2	2					Benzoyllecognine	1900 ng/mL In Whole Blood @ Unknown
		Cocaine	2	2					Ecgonine methyl ester	280 ng/mL In Whole Blood @ Unknown
		Marijuana, N.O.S.	3	3					Delta-9-carboxy-thc	17 ng/mL In Whole Blood @ Unknown
1374h	55 y F	Tianeptine	1	1	C	Ingst	Int-A	2		
1375ph	55 y F				U	Ingst	Int-S	2		
		Bupropion	1	1						
1376	55 y F	Methocarbamol	2	2						
					U	Ingst	Int-S	2		
		Amitriptyline	1	1						
		Opioid, N.O.S.	2	2						
		Diazepam	3	3						
1377h	56 y M	Drug, N.O.S.	4	4						
					U	Ingst	Int-S	3		
1378h	56 y M	Venlafaxine	1	1						
		Duloxetine (ER)	2	2						
		Lithium	1	1	U	Ingst	Unk	1	Lithium	0.9 mEq/L In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1379h	56 y F	Lithium	1	1	A/C	Ingst	Int-S	2	Lithium	3 mEq/L In Blood (unspecified) @ Unknown
		Lithium	1	1					Lithium	7 mEq/L In Blood (unspecified) @ Unknown
		Acetaminophen	2	2					Acetaminophen (apap)	5.6mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	2	2					Acetaminophen (apap)	71.5 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen/ diphenhydramine	3	3						
		Temazepam	4	4						
		Diazepam	5	5						
		Sildenafil	6	6						
		Oxcarbazepine	7	7						
		Lurasidone	8	8						
		Lamotrigine	9	9						
		Chlorpromazine	10	10						
		Levothyroxine	11	11						
		Melatonin	12	12						
1380pha	57 y F	Venlafaxine	1	1	A/C	Ingst	Unk	1		
		Bupropion (ER)	3	1					Hydroxybupropion	100 ng/mL In Serum @ Unknown
		Bupropion (ER)	3	1					Bupropion	61 ng/mL In Serum @ Unknown
		Ethanol (beverage)	1	2					Ethanol	194 mg/dL In Serum @ Unknown
		Acetaminophen	2	3					Acetaminophen (apap)	11.6 mcg/mL In Serum @ Unknown
1381ha	57 y M				A/C	Ingst	Int-S	2		
1382a	58 y M	Bupropion	1	1						
		Ethanol (beverage)	2	2						
		Naltrexone	3	3						
		Opioid, N.O.S.	4	4						
1383ph	59 y F	Trazodone	1	1	A/C	Ingst	Int-S	2		
1383ph	59 y F	Cyclic antidepressant, N.O.S.	1	1	A	Ingst	Int-S	3		
1384ph	59 y F	Cyclic antidepressant, N.O.S.	1	1	A	Ingst	Int-S	2		
1385ha	60 y F				A	Ingst	Int-S	2		
		Amitriptyline	2	1						
		Diazepam	1	1					Nordiazepam	150 ng/mL In Blood (unspecified) @ Unknown
		Diazepam	1	1					Oxazepam	22 ng/mL In Blood (unspecified) @ Unknown
		Diazepam	1	1					Diazepam	56 ng/mL In Blood (unspecified) @ Unknown
		Hydrocodone, N.O.S.	3	3						
		Marijuana	4	4					Delta-9-carboxy-thc	9.1 ng/mL In Blood (unspecified) @ Unknown
		Bupropion	1	1					Bupropion	34000 ng/mL In Blood (unspecified) @ Unknown
		Bupropion	1	1					Hydroxybupropion	6400 ng/mL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1387ha	61 y F	Venlafaxine	2	2					O-desmethyl-venlafaxine	1200 ng/mL In Blood (unspecified) @ Unknown
		Venlafaxine	2	2					Venlafaxine	24000 ng/mL In Blood (unspecified) @ Unknown
		<i>Mitragyna speciosa korthals</i>	3	3					Mitragynine	44 ng/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	4	4					Ethanol	230 mg/dL In Blood (unspecified) @ Unknown
		Paroxetine	5	5					Paroxetine	280 ng/mL In Blood (unspecified) @ Unknown
		Citalopram	6	6					Citalopram	100 ng/mL In Blood (unspecified) @ Unknown
		Doxylamine	7	7					Doxylamine	760 ng/mL In Blood (unspecified) @ Unknown
		Dextromethorphan	8	8					Dextro / levo methorphan	61 ng/mL In Blood (unspecified) @ Unknown
		Dextromethorphan	8	8					Dextrorphan / levorphanol	81 ng/mL In Blood (unspecified) @ Unknown
		Pseudoephedrine	9	9					Pseudoephedrine	1100 ng/mL In Blood (unspecified) @ Unknown
		Pseudoephedrine	9	9					Norpseudoephedrine	17 ng/mL In Blood (unspecified) @ Unknown
		Mirtazapine	1	1	A	Ingst	Int-S	3	Mirtazapine	860 ng/mL In Blood (unspecified) @ Autopsy
1388h	61 y F	Aripiprazole	2	2					Aripiprazole	100 ng/mL In Blood (unspecified) @ Autopsy
		Paroxetine	3	3					Paroxetine	17 ng/mL In Blood (unspecified) @ Autopsy
1389h	62 y M	Duloxetine (ER)	1	1	A/C	Ingst	Unk	3		
1389h	62 y M	Nortriptyline	1	1	A	Ingst + Aspir	Int-S	2		
		Clonazepam	2	2						
		Bupropion	3	3						
		Duloxetine (ER)	4	4						
		Tamsulosin	5	5						
		Gabapentin	6	6						
		Dextromethorphan	7	7						
		Pregabalin	8	8						
		Acetaminophen	9	9						
		Atypical antipsychotic, N.O.S.	10	10						
1390i	62 y F	Bupropion	1	1	A	Ingst	Int-S	2		
1391ha	63 y F	Ethanol (beverage)	2	2						
		Bupropion	1	1	A/C	Ingst	Int-S	1	Bupropion	12000 ng/mL In Blood (unspecified) @ 4 d (pe)
		Bupropion	1	1					Hydroxybupropion	7400 ng/mL In Blood (unspecified) @ 4 d (pe)
		Clonidine	3	2					Clonidine	61 ng/mL In Blood (unspecified) @ 4 d (pe)
		Escitalopram	2	3					Citalopram	6900 ng/mL In Blood (unspecified) @ 4 d (pe)

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1392h	63 y F	Amitriptyline	1	1	A/C	Ingst	Int-S	2		
		Fluoxetine	3	2						
		Rizatriptan	4	3						
		Propranolol	2	4						
		Hydroxyzine	5	5						
1393ai	64 y F	Amitriptyline	1	1	U	Unk	Int-A	1		
		Diphenhydramine	2	2						
		Gabapentin	3	3						
1394h	64 y F	Duloxetine (ER)	1	1	U	Ingst	Int-S	2		
		Phentermine	2	2						
		Levocetirizine	3	3						
		Pantoprazole	4	4						
1395h	65 y F	Bupropion (ER)	1	1	A/C	Ingst	Unk	3		
		Metformin	2	2						
1396	65 y M	Lithium	1	1	A	Ingst	Int-S	1		
		Trazodone	2	2						
		Quetiapine	3	3						
		Hydroxyzine	4	4						
1397h	69 y M	Doxepin	1	1	A	Ingst	Int-S	2		
1398ph	69 y F	Bupropion	2	1	A/C	Ingst	Int-S	2		
		Insulin (glargine) (ER)	1	2						
		Trazodone	3	3						
		Tramadol	4	4						
1399h	69 y F	Lithium (ER)	1	1	A/C	Ingst	Int-S	2	Lithium	2.8 mEq/L In Serum @ Unknown
		Amitriptyline	2	2						
		Clonazepam	3	3						
1400ha	70 y M	Tianeptine	1	1	U	Ingst	AR-D	3		
1401h	71 y F	Trazodone	1	1	A/C	Ingst	Int-S	2		
		Clonazepam	2	2						
1402ha	72 y M	Escitalopram	1	1	A/C	Ingst	Int-S	3		
		Bupropion	2	2						
1403ai	72 y F	Citalopram	1	1	U	Unk	Int-S	1		
		Tizanidine	2	2						
		Zolpidem	3	3						
1404ai	72 y M	Bupropion	1	1	U	Unk	Int-A	1		
		Morphine	2	2						
		Hydrocodone, N.O.S.	3	3						
1405h	75 y F	Lithium	1	1	C	Ingst	AR-D	3	Lithium	2.4 mEq/L In Blood (unspecified) @ Unknown
1406ha	75 y F	Lithium	1	1	C	Ingst	AR-D	3	Lithium	1.98 mEq/L In Blood (unspecified) @ Unknown
1407	76 y F	Sertraline	1	1	A/C	Ingst	Int-S	2		
		Lamotrigine	2	2						
		Amitriptyline	3	3						
		Tramadol	4	4						
1408ph	79 y M	Amitriptyline	1	1	A/C	Ingst	Int-S	1	Tca (totall tricyclic concentration)	155 mg/mL In Serum @ Unknown
1409h	82 y M	Bupropion	1	1	A/C	Ingst	Int-S	1		
		Metoprolol	2	1						
		Famotidine	3	3						
1410ph	83 y M	Amitriptyline	1	1	A/C	Ingst	Unt-T	2		
		Citalopram	2	2						
1411h	84 y F				C	Ingst	AR-D	3		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1412i	60+ y M	Lithium	1	1					Lithium	1.6 mEq/L In Blood (unspecified) @ Unknown
		Paroxetine	1	1	A/C	Ingst + Par	Unk	2	Paroxetine	500 mg/L In Blood (unspecified) @ Unknown
		Buprenorphine	2	2					Buprenorphine	12.7 ng/mL In Blood (unspecified) @ Unknown
		Buprenorphine	2	2					Norbuprenorphine	2.4 ng/mL In Blood (unspecified) @ Unknown
See Also case 4, 18, 36, 53, 67, 80, 89, 146, 158, 203, 294, 334, 335, 388, 422, 442, 475, 604, 609, 610, 638, 678, 836, 858, 869, 874, 902, 958, 972, 974, 1028, 1041, 1074, 1105, 1140, 1169, 1190, 1198, 1210, 1230, 1243, 1272, 1277, 1278, 1283, 1284, 1287, 1289, 1291, 1292, 1297, 1300, 1426, 1442, 1443, 1446, 1447, 1473, 1479, 1484, 1487, 1488, 1490, 1495, 1500, 1501, 1506, 1507, 1511, 1512, 1521, 1528, 1529, 1530, 1532, 1543, 1547, 1552, 1554, 1555, 1556, 1558, 1561, 1565, 1572, 1573, 1575, 1580, 1589, 1598, 1602, 1605, 1608, 1627, 1629, 1631, 1638, 1653, 1662, 1667, 1669, 1670, 1680, 1695, 1698, 1701, 1704, 1705, 1706, 1747, 1749, 1751, 1759, 1764, 1768, 1779, 1789, 1802, 1803, 1804, 1806, 1810, 1811, 1819, 1821, 1827, 1833, 1835, 1841, 1848, 1851, 1853, 1860, 1865, 1871, 1904, 1905, 1944, 1947, 1966, 1998, 2006, 2055, 2089, 2128, 2136, 2234, 2241, 2246, 2259										
Antihistamines										
1413	1 y M	Diphenhydramine	1	1	A	Ingst	Oth-M	3	Diphenhydramine	390 ng/mL In Blood (unspecified) @ Unknown
[1414pha]	3 y M	Diphenhydramine	1	1	A	Unk	Oth-M	1	Diphenhydramine	5000 ng/mL In Serum @ Unknown
		Diphenhydramine	1	1					Diphenhydramine	5707 ng/mL In Whole Blood @ Autopsy
1415h	16 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1		
1416pa	17 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1	Diphenhydramine	29000 ng/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2					Ethanol	26 mg/dL In Plasma @ Unknown
1417pha	19 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1	Diphenhydramine	18000 ng/mL In Blood (unspecified) @ Unknown
		Acetaminophen	2	2					Acetaminophen (apap)	714 mcg/mL In Blood (unspecified) @ Unknown
1418pha	20 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1	Norfentanyl	1.1 ng/mL In Blood (unspecified) @ Unknown
		Diphenhydramine	1	1					Diphenhydramine	1774 ng/mL In Blood (unspecified) @ Unknown
		Diphenhydramine	1	1					Fentanyl	47 ng/mL In Blood (unspecified) @ Unknown
		Fentanyl (non-prescription)	2	2						
1419i	20 y F	Diphenhydramine	1	1	U	Unk	Int-S	1		
1420h	20 y M	Diphenhydramine	1	1	A	Ingst	Unk	2		
1421ha	20 y F	Diphenhydramine	1	1	C	Ingst	Int-S	1		
1422	20 y M	Diphenhydramine	1	1	A	Ingst	Int-U	2		
1423ph	23 y F	Diphenhydramine	1	1	A	Ingst	Int-S	2		
1424pha	23 y M	Diphenhydramine	1	1	A	Ingst	Int-S	1	Diphenhydramine	1800 mcg/mL In Gastric (stomach content) @ Autopsy
		Diphenhydramine	1	1					Diphenhydramine	38 mcg/mL In Bile @ Autopsy

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1425pha	24 y F	Acetaminophen/ dextromethorphan/doxylamine	2	2	A	Ingst	Int-S	1	Dextro / levo methorphan	1200 ng/mL In Gastric (stomach content) @ Autopsy
		Acetaminophen/ dextromethorphan/doxylamine	2	2					Doxylamine	14 mcg/mL In Gastric (stomach content) @ Autopsy
		Acetaminophen/ dextromethorphan/doxylamine	2	2					Doxylamine	14000 ng/mL In Gastric (stomach content) @ Autopsy
		Acetaminophen/ dextromethorphan/doxylamine	2	2					Doxylamine	250 ng/mL In Bile @ Autopsy
		Acetaminophen/ dextromethorphan/doxylamine	2	2					Levetiracetam	28 mcg/mL In Bile @ Autopsy
		Acetaminophen/ dextromethorphan/doxylamine	2	2					Dextro / levo methorphan	320 ng/mL In Bile @ Autopsy
		Acetaminophen/ dextromethorphan/doxylamine	2	2					Dextrorphan / levorphanol	390 ng/mL In Bile @ Autopsy
		Acetaminophen/ dextromethorphan/doxylamine	2	2					Acetaminophen (apap)	60 mcg/dL In Blood (unspecified) @ Unknown
		Acetaminophen/ dextromethorphan/doxylamine	2	2					Acetaminophen (apap)	60 mcg/dL In Serum @ Unknown
		Ethanol (beverage)	3	3					Ethanol	21 mg/dL In Serum @ Unknown
		Ethanol (beverage)	3	3					Ethanol	45 mg/dL In Gastric (stomach content) @ Autopsy
		Levetiracetam	4	4						
		Isopropanol	5	5						
		Diphenhydramine	1	1						
1426pha	24 y F	Benzodiazepine, N.O.S.	2	2	A	Ingst	Int-S	1		
		Lurasidone	3	3						
		Aripiprazole	4	4						
		Ethanol (beverage)	5	5					Ethanol	51 mg/dL In Serum @ 45 m (pe)
1427h	25 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1	Diphenhydramine	7900 ng/mL In Blood (unspecified) @ Unknown
		Diphenhydramine	2	2						
		Fluoxetine	3	3					Norfluoxetine	370 ng/mL In Blood (unspecified) @ Unknown
		Fluoxetine	3	3					Fluoxetine	730 ng/mL In Blood (unspecified) @ Unknown
1428i	25 y F	Diphenhydramine	2	1	A	Ingst	Int-S	2		
		Ibuprofen	1	2					Ibuprofen	820 mcg/mL In Serum @ Unknown
1429pha	26 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1		
		Atomoxetine	2	1						
		Diphenhydramine	1	1					Diphenhydramine	15589 ng/mL In Blood (unspecified) @ Autopsy
1430ph	28 y F	Diphenhydramine	1	1	A	Ingst	Int-S	2		
1431ai	31 y F	Diphenhydramine	1	1	U	Unk	Unk	1		
		Cyclobenzaprine	2	2						
		Ethanol (beverage)	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1432ai	35 y M	Diphenhydramine	1	1	U	Unk	Int-S	1		
		Marijuana, N.O.S.	2	2						
1433ha	37 y M	Diphenhydramine	1	1	U	Ingst	Int-S	2	Diphenhydramine	5091 ng/mL In Blood (unspecified) @ Unknown
		Metoprolol	2	2					Metoprolol	2700 ng/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	3	3					Ethanol	47 mg/dL In Blood (unspecified) @ Unknown
		Cyclobenzaprine	4	4					Cyclobenzaprine	18 ng/mL In Blood (unspecified) @ Unknown
1434ha	43 y F	Diphenhydramine	1	1	A/C	Ingst	Int-S	1	Diphenhydramine	4300 ng/mL In Blood (unspecified) @ Autopsy
		Acetaminophen	2	2					Acetaminophen (apap)	41.2 mcg/mL In Serum @ Unknown
		Oxycodone (ER)	3	4					Hydrocodone	130 ng/mL In Blood (unspecified) @ Autopsy
		Acetaminophen/hydrocodone	5	5						
		Phentermine	4	5					Phentermine	34 ng/mL In Blood (unspecified) @ Autopsy
		Tramadol	6	6					Tramadol	270 ng/mL In Blood (unspecified) @ Autopsy
1435h	43 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1		
		Ethanol (beverage)	2	2					Ethanol	65 mg/dL In Blood (unspecified) @ Unknown
1436ai	44 y M	Diphenhydramine	1	1	U	Unk	Int-S	1		
1437ai	45 y F	Promethazine	1	1	U	Unk	Int-S	1		
		Hydroxyzine	2	2						
1438ph	47 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1		
1439ai	50 y F	Diphenhydramine	1	1	U	Unk	Int-S	1		
		Hydroxyzine	2	2						
1440ai	51 y F	Diphenhydramine	1	1	U	Unk	Unk	1		
		Gabapentin	2	2						
1441ph	51 y M	Diphenhydramine	1	1	A	Ingst	Int-S	1		
		Ethanol (beverage)	2	2					Ethanol	0.296 g/dL In Blood (unspecified) @ Unknown
1442ph	55 y F	Hydroxyzine	1	1	A	Ingst	Int-S	2		
		Fluoxetine	2	2						
		Sertraline	3	3						
1443h	60 y F	Hydroxyzine	1	1	A/C	Ingst	Int-S	2		
		Aripiprazole	2	2						
		Amlodipine	3	3						
		Escitalopram	4	4						
1444a	66 y F	Diphenhydramine	1	1	A	Ingst	Int-S	1	Diphenhydramine	9700 ng/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2					Ethanol	232 mg/dL In Blood (unspecified) @ Unknown
1445ai	74 y F	Diphenhydramine	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1446ha	75 y F	Ethanol (beverage)	2	2	A/C	Ingst	Unk	1	Acetaminophen (apap)	15.7 mcg/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	3	3						
		Diphenhydramine	1	1						
		Acetaminophen	2	2						
		Bupropion	3	3						
1447h	83 y F	Citalopram	4	4	U	Ingst	Int-S	2		
		Trazodone	5	5						
		Gabapentin	6	6						
		Hydroxyzine	1	1						
		Mirtazapine	2	2						
See Also case 29, 81, 123, 203, 283, 323, 347, 409, 432, 462, 475, 477, 554, 558, 694, 722, 724, 730, 742, 746, 763, 784, 794, 808, 812, 836, 839, 890, 911, 988, 1001, 1039, 1042, 1065, 1096, 1109, 1253, 1258, 1316, 1320, 1328, 1353, 1356, 1359, 1360, 1392, 1393, 1394, 1396, 1409, 1467, 1500, 1511, 1526, 1529, 1530, 1538, 1541, 1547, 1548, 1554, 1561, 1568, 1589, 1602, 1614, 1630, 1652, 1708, 1732, 1737, 1753, 1759, 1779, 1800, 1802, 1816, 1825, 1841, 1860, 1881, 1899, 1904, 1928, 2043, 2059, 2102, 2108, 2114, 2148										
Antimicrobials										
1448h	41 y M	Linezolid	1	1	C	Ingst + Par	Unt-T	3		
1449h	53 y M	Fentanyl (prescription)	2	2	U	Ingst	AR-D	3		
		Plant (Ashwagandha)	3	3						
		Linezolid	1	1						
1450h	59 y M	Methadone	2	2	A/C	Par	AR-D	1	Vancomycin	50 mcg/mL In Serum @ Unknown
		Vancomycin	1	1						
1451pha	62 y F	Cefepime	2	2	A/C	Ingst	Int-S	1		
		Hydroxychloroquine	1	1						
		Diltiazem	2	2						
1452h	83 y M	Acetaminophen	3	3	A	Ingst	Unk	3		
		Fluconazole	1	1						
[1453h]	88 y F	Acetaminophen	2	2	A	Par	Unt-T	2		
		Acyclovir	1	1						
See Also case 728, 1260, 1284, 1303, 1467, 1528, 1725, 1726, 1804										
Antineoplastics										
1454h	37 y M	Fluorouracil	1	1	U	Par	AR-D	3		
[1455h]	66 y F	Oxaliplatin	2	2	A	Par	AR-D	1	Methotrexate	0.91 µmol/L In Blood (unspecified) @ 330 h (pe)
		Methotrexate	1	1						
		Methotrexate	1	1						
		Methotrexate	1	1						
		Methotrexate	1	1						
		Methotrexate	1	1						
		Methotrexate	1	1						
		Methotrexate	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Methotrexate	1	1					Methotrexate	2.16 µmol/L In Blood (unspecified) @ 181 h (pe)
		Methotrexate	1	1					Methotrexate	2.55 µmol/L In Blood (unspecified) @ 141 h (pe)
		Methotrexate	1	1					Methotrexate	2.88 µmol/L In Blood (unspecified) @ 156 h (pe)
		Methotrexate	1	1					Methotrexate	26 µmol/L In Blood (unspecified) @ 40 h (pe)
		Methotrexate	1	1					Methotrexate	34 µmol/L In Blood (unspecified) @ 24 h (pe)
		Methotrexate	1	1					Methotrexate	5.8 µmol/L In Blood (unspecified) @ 108 h (pe)
		Methotrexate	1	1					Methotrexate	8.2 µmol/L In Blood (unspecified) @ 64 h (pe)
1456h	68 y M	Arsenic trioxide	1	1	A	Ingst	Int-S	2		
1457	68 y F	Methotrexate	1	1	C	Ingst	Unt-T	2		
1458ha	74 y F	Methotrexate	1	1	A	Ingst	AR-D	3		
1459h	78 y M	Methotrexate	1	1	A/C	Ingst	AR-D	1	Methotrexate	0.07 µmol/L In Serum @ 10 d (pe)
		Methotrexate	1	1					Methotrexate	0.09 µmol/L In Serum @ 9 d (pe)
1460h	79 y F	Methotrexate	1	1	A/C	Ingst	AR-D	2		
Asthma Therapies										
[1461]	61 y M	Theophylline	1	1	C	Ingst	Unt-U	1	Theophylline	150 mcg/mL In Blood (unspecified) @ Unknown
		Theophylline	1	1					Theophylline	150 mcg/mL In Serum @ Unknown
1462p	74 y M	Theophylline	1	1	U	Ingst	Unk	2	Theophylline	41.3 mcg/mL In Blood (unspecified) @ Unknown
See Also case 414, 1198, 1305, 1577, 1670, 1768										
Cardiovascular Drugs										
[1463ph]	2 y F	Flecainide	1	1	A	Ingst	Unt-G	1		
		Atenolol	2	2						
1464	12 y F	Amlodipine	1	1	A	Ingst	Int-S	2		
1465ha	15 y M	Propafenone (ER)	1	1	A	Ingst	Int-S	1	Propafenone	4.1 mcg/mL In Blood (unspecified) @ Autopsy
		Acetaminophen	2	2					Acetaminophen (apap)	113.1 mcg/dL In Blood (unspecified) @ 15 m (pe)
		Acetaminophen	2	2					Acetaminophen (apap)	89 mcg/mL In Blood (unspecified) @ Autopsy
1466h	17 y F	Metoprolol	1	1	A	Ingst	Int-S	2		
		Ranolazine (ER)	2	2						
		Vitamin D	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1467a	17 y F	Pantoprazole	4	4	A	Ingst	Int-S	1	Acetaminophen (apap)	186 mcg/mL In Blood (unspecified) @ Unknown
		Carvedilol	1	1						
		Glipizide	2	2						
		Acetaminophen	3	3						
		Furosemide	4	4	A	Ingst	Int-S	1		
		Atorvastatin	5	5						
		Losartan	6	6						
		Famotidine	7	7						
		Sulfamethoxazole/trimethoprim	8	8	A	Ingst	Int-S	1		
		Ibuprofen	9	9						
1468ha	17 y F	Amlodipine	1	1						
		Metoprolol	2	2						
1469ai	17 y F	Amlodipine	1	1	U	Unk	Int-S	1		
		Metoprolol	2	2						
1470h	20 y F	Guanfacine	1	1	A/C	Ingst	Int-S	2		
1471h	23 y M	Verapamil	1	1	A	Ingst	Int-S	2		
		Metoprolol	2	2						
1472phai	23 y F	Propranolol	1	1	A/C	Ingst	Int-S	1	Propranolol	170 ng/mL In Blood (unspecified) @ Autopsy
1473ai	25 y F	Propranolol	1	1	U	Unk	Int-S	1		
		Citalopram	2	2						
		Bupropion	3	3						
1474a	25 y F	Propranolol	1	1	A	Ingst + Unk	Int-S	1	Propranolol	15000 ng/mL In Blood (unspecified) @ Unknown
		Tramadol	2	2					Tramadol	130 ng/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	3	3					Delta-9-carboxy-thc	50 ng/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	3	3					Delta-9-thc	7.9 ng/mL In Blood (unspecified) @ Unknown
1475ph	25 y M	Propranolol	1	1	A/C	Ingst	Int-S	2		
[1476ha]	26 y F	Flecainide	1	1	A	Ingst	Int-U	1	Flecainide	5 mg/L In Blood (unspecified) @ Autopsy
1477h	27 y F	Amlodipine	2	1	A	Ingst	Int-S	2		
		Verapamil	1	2						
1478h	28 y M	Sildenafil	1	1	A	Ingst	Int-S	2	Ethanol	58 mg/dL In Blood (unspecified) @ Unknown
		Amlodipine	2	2						
		Ethanol (beverage)	3	3						
1479ph	28 y F	Propranolol	1	1	A	Ingst	Int-S	2		
		Lithium	2	2						
		Escitalopram	3	3						
1480	29 y M	Flecainide	1	1	A/C	Ingst	Int-S	2		
1481h	30 y F	Propranolol	1	1	A/C	Ingst	Int-S	2		
		Clonidine	2	2						
		Lorazepam	3	3						
		Tramadol	4	4						
1482h	31 y F				A	Ingst	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1483ph	32 y F	Amlodipine	1	1	A	Ingst	Int-S	2		
1484h	33 y F	Metoprolol	1	1	A/C	Ingst	Int-S	3		
		Propranolol	1	1						
		Carbamazepine	2	2					Carbamazepine-10,11-epoxide	1.6mcg/mL In Blood (unspecified) @ Unknown
		Carbamazepine	2	2					Carbamazepine	10.8mcg/mL In Blood (unspecified) @ Unknown
		Carbamazepine	2	2					Carbamazepine	13.2mcg/mL In Blood (unspecified) @ Unknown
		Carbamazepine	2	2					Carbamazepine	13.4mcg/mL In Blood (unspecified) @ Unknown
		Carbamazepine	2	2					Carbamazepine	8.6mcg/mL In Blood (unspecified) @ Unknown
		Carbamazepine	2	2					Carbamazepine	8.8mcg/mL In Blood (unspecified) @ Unknown
		Carbamazepine	2	2					Carbamazepine	9.4mcg/mL In Blood (unspecified) @ Unknown
1485ph	34 y F	Fluoxetine	3	3	U	Ingst + Unk	Int-U	3		
		Metoprolol	4	1						
		Fentanyl (non-prescription)	1	2						
		Amphetamine	2	3						
1486h	34 y M	Xylazine	3	4	A/C	Ingst	Int-S	2		
		Amlodipine	1	1						
		Hydrochlorothiazide	2	2						
		Ethanol (beverage)	3	3					Ethanol	91 mg/dL In Serum @ 2 h (pe)
1487h	35 y F	Atenolol	1	1	A	Ingst	Int-S	1		
1488h	35 y M	Citalopram	2	2	A/C	Ingst	Int-S	2		
		Amlodipine	1	1						
		Citalopram	2	2						
		Ethanol (beverage)	3	3						
1489	37 y M	Amlodipine	1	1	A/C	Ingst	Int-S	2		
1490ph	37 y M	Metoprolol	1	1	A/C	Ingst	Int-S	2		
		Escitalopram	2	2						
		Amphetamine/dextro-amphetamine	3	3						
		Clonazepam	4	4						
		Lorazepam	5	5						
		Ethanol (beverage)	6	6						
1491h	38 y M	Calcium antagonist, N.O.S.	1	1	A/C	Ingst	Int-S	2		
		Angiotensin receptor blocker, N.O.S.	2	2						
1492	38 y M	Digoxin	1	1	A/C	Ingst	AR-D	3	Digoxin	4.7ng/mL In Serum @ Unknown
1493	40 y F	Amiodarone	1	1	A	Ingst + Aspir	Int-S	2		
		Potassium	2	2						
		Magnesium	3	3						
		Calcium	4	4						
		Clonazepam	5	5						
		Gemfibrozil	6	6						
1494h	41 y F	Amlodipine	1	1	A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1495h	42 y F	Ethanol (beverage)	2	2						
		Amlodipine	1	1	U	Ingst	Int-S	2		
		Venlafaxine	2	2						
		Metoprolol	3	3						
		Ethanol (beverage)	4	4						
1496h	42 y F	Amlodipine	1	1	A/C	Ingst	Unt-T	2		
1497	42 y F	Amlodipine	1	1	U	Ingst	Int-S	2		
		Pantoprazole (ER)	2	2						
		Alprazolam	3	3						
1498a	43 y M	Verapamil (ER)	1	1	A/C	Ingst	Int-S	1		
1499ha	44 y F	Amlodipine	2	1	A/C	Ingst	Int-S	2		
		Beta blocker, N.O.S.	1	1						
		Ethanol (beverage)	3	3						
1500ha	44 y M	Amlodipine	1	1	A/C	Ingst	Int-S	1		
		Tizanidine	2	2						
		Warfarin	3	3						
		Trazodone	4	4						
		Hydroxyzine	5	5						
1501h	44 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Amitriptyline	2	2						
		Drug, N.O.S.	3	3						
1502ph	44 y F	Verapamil	1	1	U	Ingst	Int-S	2		
1503h	44 y M	Verapamil	1	1	A/C	Ingst	Int-S	1		
1504ph	44 y M	Beta blocker, N.O.S.	1	1	U	Ingst	Unk	2	Metformin	1 mcg/mL In Blood (unspecified) @ Unknown
		Beta blocker, N.O.S.	1	1					Ethanol	112 mg/dL In Blood (unspecified) @ Unknown
		Ethylene glycol (antifreeze)	2	2						
		Methanol	3	3						
		Metformin	4	4						
		Ethanol (beverage)	5	5						
1505h	45 y F	Amlodipine	1	1	A	Ingst	Int-U	2		
1506h	45 y F	Nifedipine (ER)	1	1	A	Ingst	Int-S	2		
		Duloxetine (ER)	2	2						
1507	46 y F	Flecainide	1	1	A/C	Ingst	Int-S	1		
		Venlafaxine (ER)	2	2						
		Propranolol	3	3						
		Apixiban	4	4						
		Substance (non-drug), N.O.S.	5	5						
1508ha	46 y M	Diltiazem	2	1	A/C	Ingst	Int-M	3		
		Methamphetamine	1	1					Amphetamine	11 ng/mL In Serum @ 2 h (pe)
		Methamphetamine	1	1					Methamphetamine	50 ng/mL In Serum @ 2 h (pe)
		Salicylate	3	2					Salicylate	39.6 mg/dL In Serum @ 2 h (pe)
		Salicylate	3	2					Salicylate	40 mg/dL In Serum @ 5 h (pe)
1509h	46 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Losartan	2	2						
		Atorvastatin	3	3						
1510ha	46 y F	Clonidine	1	1	A/C	Ingst	Int-S	1	Clonidine	40 ng/mL In Serum @ Unknown
		Nifedipine	2	2						
		Carvedilol	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1511a	46 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2	Amlodipine	10 ng/mL In Serum @ Autopsy
		Fluoxetine	2	2					Norfluoxetine	110 ng/mL In Serum @ Autopsy
		Fluoxetine	2	2					Fluoxetine	220 ng/mL In Serum @ Autopsy
		Hydroxyzine	3	3					Hydroxyzine	220 ng/mL In Serum @ Autopsy
1512ha	46 y M	Rosuvastatin	4	4	A/C	Ingst	Int-S	3		
		Amlodipine	1	1						
		Fluoxetine	2	2						
		Ethanol (beverage)	3	3					Ethanol	251 mg/dL In Serum @ 2 h (pe)
1513ha	47 y F	Amlodipine	1	1	A	Ingst	Int-S	1	Amlodipine	72 ng/mL In Blood (unspecified) @ Unknown
		Lisinopril	2	2						
		Quetiapine	3	3					Quetiapine	1800 ng/mL In Blood (unspecified) @ Unknown
		Zolpidem	4	4					Zolpidem	220 ng/mL In Blood (unspecified) @ Unknown
		Gabapentin	5	5					Gabapentin	13 mcg/mL In Blood (unspecified) @ Unknown
		Risperidone	6	6					Risperidone	80 ng/mL In Blood (unspecified) @ Unknown
1514ha	47 y F	Propranolol	1	1	A/C	Ingst	Int-S	1	Propranolol	610 ng/mL In Blood (unspecified) @ Unknown
		Lorazepam	2	2					Lorazepam	0.038 mg/L In Blood (unspecified) @ Unknown
1515a	47 y F	Amlodipine	1	1	A	Ingst	Int-S	1		
		Losartan	2	2						
		Gabapentin	3	3					Gabapentin	14 mcg/mL In Blood (unspecified) @ Unknown
		Cyclobenzaprine	4	4					Cyclobenzaprine	2000 ng/mL In Blood (unspecified) @ Unknown
1516h	48 y F	Metoprolol	1	1	A/C	Ingst	Int-S	1		
		Amlodipine	2	2						
		Salicylate	3	3						
		Losartan	4	4						
		Clopidogrel	5	5						
		Atorvastatin	6	6						
1517ha	48 y M	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Carvedilol	2	2						
		Losartan	3	3						
		Furosemide	4	4						
1518h	48 y F	Beta blocker, N.O.S.	1	1	A	Ingst	Int-S	2		
1519	48 y F	Amlodipine	1	1	A	Ingst	Int-S	2		
		Acetaminophen	2	2					Acetaminophen (apap)	14.6 mcg/mL In Blood (unspecified) @ Unknown
1520	48 y M	Amlodipine	1	1	A	Ingst	Int-S	2		
		Furosemide	2	2						
		Clonidine	3	3						
		Clopidogrel	4	4						
1521ha	48 y F	Amlodipine	1	1	A/C	Ingst	Int-S	1	Amlodipine	350 ng/mL In Serum @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1522ai	49 y F	Bupropion	2	2					Bupropion	379 ng/mL In Serum @ Unknown
		Acetaminophen	3	3					Acetaminophen (apap)	67.5 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	4	4					Ethanol	96 mg/dL In Whole Blood @ Unknown
		Duloxetine (ER)	5	5					Duloxetine	272 ng/mL In Serum @ Unknown
		Methocarbamol	6	6						
1523	49 y F	Clonidine	1	1	U	Unk	Int-S	3		
1524h	49 y M	Diltiazem	1	1	A	Ingst	Int-S	2		
		Lisinopril	2	2						
1525h	50 y M	Amlodipine	1	1	A	Ingst + Unk	Int-S	2		
		Methamphetamine	2	2						
1526h	50 y F	Carvedilol	1	1	A/C	Ingst	Int-S	1		
		Amlodipine	2	2						
1527a	50 y M	Verapamil	2	1	U	Ingst	Unk	2		
		Drug, N.O.S.	3	2						
		Diphenhydramine	1	3						
		Ethanol (beverage)	4	4					Ethanol	24 mg/dL In Blood (unspecified) @ Unknown
1528h	51 y M	Amlodipine	1	1	A/C	Ingst	Int-S	1	Amlodipine	410 ng/mL In Blood (unspecified) @ Unknown
		Metoprolol (ER)	2	2						
		Ethanol (beverage)	3	3					Ethanol	330 mg/dL In Whole Blood @ Unknown
1529h	52 y F				A	Ingst	Int-S	1		
		Amlodipine	1	1						
		Bupropion	2	2						
		Quetiapine	3	3						
		Valproic acid	4	4						
		Doxepin	5	5						
		Bictegravir/emtricitabine/tenofovir	6	6						
1530ha	52 y M	Diltiazem	1	1	A/C	Ingst	Int-S	2		
		Metoprolol	2	2						
		Bupropion (ER)	3	3						
		Acetaminophen	4	4						
		Apixiban	5	5						
		Metolazone	6	6						
		Fluoxetine	7	7						
		Gabapentin	8	8						
		Hydroxyzine	9	9						
		Zolpidem	10	10						
		Lamotrigine	11	11						
		Diphenhydramine	12	12						
		Levothyroxine	13	13						
1530ha	52 y M	Amlodipine	1	1	A/C	Ingst	Int-S	1	Amlodipine	240 ng/mL In Blood (unspecified) @ Unknown
		Clonidine	2	2					Clonidine	40 ng/mL In Blood (unspecified) @ Unknown
		Gabapentin	3	3					Gabapentin	4.3 mcg/mL In Blood (unspecified) @ Unknown
		Venlafaxine	4	4					O-desmethyl-venlafaxine	28 ng/mL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1531h	52 y M	Venlafaxine	4	4					Venlafaxine	430 ng/mL In Blood (unspecified) @ Unknown
		Hydroxyzine	5	5					Hydroxyzine	4100 ng/mL In Blood (unspecified) @ Unknown
		Acetaminophen	6	6					Acetaminophen (apap)	35 mcg/mL In Blood (unspecified) @ Unknown
		Insulin, N.O.S.	7	7						
1532h	52 y M	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Lisinopril	2	2						
		Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Escitalopram	2	2					Escitalopram	241 ng/mL In Blood (unspecified) @ Unknown
1533h	53 y M	Gabapentin	3	3					Gabapentin	14 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	4	4					Ethanol	245 mg/dL In Blood (unspecified) @ Unknown
		Amlodipine	1	1	A	Ingst	Int-S	1	Amlodipine	960 ng/mL In Serum @ Unknown
1534h	53 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2		
1535h	53 y M	Amlodipine	1	1	A	Ingst	Int-S	2		
		Clonidine	2	2						
		Drug, N.O.S.	3	3						
1536ha	53 y F	Amlodipine	1	1	U	Ingst	Int-S	1	Amlodipine	530 ng/mL In Serum @ 1 h (pe)
		Metoprolol	2	2					Metoprolol	140 ng/mL In Serum @ 1 h (pe)
		Ethanol (beverage)	3	3						
1537	53 y F	Carvedilol	1	1	A	Ingst	Int-S	2		
1538p	53 y M	Clonidine	1	1	U	Ingst	Int-S	2		
		Diphenhydramine	2	2						
		Lisinopril	3	3						
		Hydrochlorothiazide	4	4						
1539ha	53 y F	Ethanol (beverage)	5	5						
		Amlodipine	1	1	A/C	Ingst	Int-S	1	Amlodipine	1.7 mg/L In Blood (unspecified) @ Autopsy
		Labetalol	2	2					Labetalol	12 mg/L In Blood (unspecified) @ Autopsy
1540	54 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2		
1541ha	54 y F	Amlodipine	1	1	A/C	Ingst + Unk	Int-S	1	Amlodipine	13 ng/mL In Blood (unspecified) @ Unknown
		Diphenhydramine	2	2					Diphenhydramine	1300 ng/mL In Blood (unspecified) @ Unknown
		Oxycodone	3	3					Oxycodone (free)	22 ng/mL In Blood (unspecified) @ Unknown
1542h	54 y F	Diltiazem	1	1	U	Ingst	Int-S	2		
		Triazolam	2	2						
1543ph	54 y M	Flecainide	1	1	A	Ingst	Int-S	1		
		Metoprolol	2	2						
		Fluoxetine	3	3						
		Quetiapine	4	4						
1544h	55 y M	Amlodipine	1	1	A/C	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1545h	55 y M	Bisoprolol	2	2	A/C	Ingst	Int-S	2		
		Ethanol (beverage)	3	3						
		Tetrahydrozoline	4	4						
1546	55 y F	Amlodipine	1	1	A	Ingst	Int-U	2		
		Carvedilol	2	2						
		Alprazolam	3	3						
		Atorvastatin	4	4						
1547h	55 y M	Verapamil	1	1	A/C	Ingst	Int-S	2		
		Propranolol	2	2						
		Gabapentin	3	3						
1548a	55 y F	Amlodipine	1	1	U	Ingst	Int-S	2	Hydroxyzine	2600 ng/mL In Whole Blood @ Autopsy
		Carvedilol	2	2						
		Escitalopram	3	3						
		Hydroxyzine	4	4						
		Ethanol (beverage)	5	5						
1549h	56 y F	Propranolol	1	1	A	Ingst	Int-S	2		
		Metoprolol	2	2						
		Drug, N.O.S.	3	3						
1550	56 y F	Amlodipine	1	1	A	Ingst	Int-S	3		
1551	56 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Ethanol (beverage)	2	2						
1552h	56 y F				U	Ingst	Int-S	2		240 mg/dL In Blood (unspecified) @ Unknown
		Calcium antagonist, N.O.S.	1	1						
		Antihyperlipidemic, N.O.S.	2	2						
		Valsartan and hydrochlorothiazide	3	3						
		Metoprolol (ER)	4	4						
		Furosemide	5	5						
		Alprazolam	6	6						
1553h	56 y M	Escitalopram	7	7	A/C	Ingst	Int-S	2		
		Verapamil (ER)	1	1						
		Hydrochlorothiazide	2	2						
		Lisinopril	3	3						
1554ha	56 y F	Atorvastatin	4	4	A/C	Ingst	Int-S	1	Amlodipine	470 ng/mL In Blood (unspecified) @ Unknown
		Amlodipine	1	1						
		Diltiazem	2	2						
		Hydroxyzine	3	3					Hydroxyzine	130 ng/mL In Blood (unspecified) @ Unknown
		Duloxetine (ER)	4	4						
1555a	57 y M	Trazodone	5	5	A	Ingst	Int-S	1	Trazodone	1 mcg/mL In Blood (unspecified) @ Unknown
		Gabapentin	6	6					Gabapentin	53 mcg/mL In Blood (unspecified) @ Unknown
		Diltiazem (ER)	1	1						
		Clonidine	2	2					Clonidine	47 ng/mL In Blood (unspecified) @ Autopsy
		Duloxetine (ER)	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1556h	57 y F	Hydrochlorothiazide/ lisinopril	4	4	A	Ingst	Int-S	3	Quetiapine	1300 ng/mL In Blood (unspecified) @ Autopsy
		Hydralazine	5	5						
		Quetiapine	6	6						
		Amlodipine	1	1						
		Metoprolol	2	2						
1557hi	57 y F	Escitalopram	3	3	A/C	Ingst	Int-S	1		
		Ethanol (beverage)	4	4						
		Omeprazole	5	5						
		Amlodipine	1	1						
		Hydrochlorothiazide/ irbesartan	2	2						
1558h	57 y M				A/C	Ingst	Int-S	2		
		Amlodipine	1	1						
1559h	57 y F	Duloxetine (ER)	2	2	A/C	Ingst	Int-S	2		
		Amlodipine	1	1						
1560ph	58 y F	Atorvastatin	2	2	A	Ingst	Int-S	2		
		Ethanol (beverage)	3	3						
		Propranolol	1	1						
		Lisinopril	2	2						
		Losartan	3	3						
1561ha	58 y F	Ethanol (beverage)	4	4	A/C	Ingst	Int-S	2	Ethanol	214 mg/dL In Serum @ Unknown
		Amlodipine	1	1						
		Atenolol	2	2						
		Paroxetine	3	3						
		Doxylamine	4	4						
1562	58 y F	Dextromethorphan	5	5	U	Unk	Unk	2	Dextro / levo methorphan	82.4 ng/mL In Blood (unspecified) @ Unknown
1563ha	58 y M	Metoprolol	1	1	A/C	Ingst	Int-S	1		
		Amlodipine	1	1						
		Metoprolol (ER)	2	2						
		Hydralazine	3	3						
		Empagliflozin	4	4						
1564ha	58 y M	Amlodipine	1	1	A	Ingst + Unk	Int-S	2	Amlodipine	650 ng/mL In Blood (unspecified) @ Autopsy
1565	58 y M	Methamphetamine	2	2	A	Ingst	Int-S	3		
		Fentanyl (non-prescription)	3	3						
1566h	59 y F	Nebivolol	1	1	U	Ingst	Int-S	2		
		Mirtazapine	2	2						
1567ai	59 y F	Metoprolol	1	1	U	Unk	Int-S	1		
		Metformin	2	2						
1568h	59 y F	Clonidine	1	1	A/C	Ingst	Int-S	1		
		Acetaminophen	2	2						
		Marijuana, N.O.S.	3	3						
1569	59 y M	Verapamil	1	1	A/C	Ingst	Int-S	2		
		Diphenhydramine	2	2						
1570ph	60 y F	Amlodipine	1	1	U	Ingst	Int-S	2	Carbamazepine	37.2 mcg/mL In Blood (unspecified) @ Unknown
		Carbamazepine	2	2						
1570ph	60 y F	Haloperidol	3	3	U	Ingst	Int-S	2		
		Benzotropine	4	4						
		Metoprolol	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1571	60 y M	Tramadol	2	2	A/C	Ingst	Unt-T	2		
		Zolpidem	3	3						
1572h	60 y M	Amlodipine	1	1	A	Ingst	Int-S	2		
1573h	60 y F	Diltiazem	1	1	A/C	Ingst	Int-S	2	Ethanol	201 mg/dL In Blood (unspecified) @ Unknown
		Quetiapine	2	2						
		Trazodone	3	3						
		Ropinirole	4	4						
		Lisinopril	5	5						
		Cholecalciferol	6	6						
		Amlodipine	1	1						
1574	60 y M	Escitalopram	2	2	A/C	Ingst	Int-S	2		
		Meloxicam	3	3						
		Drug, N.O.S.	4	4						
		Ethanol (beverage)	5	5						
		Diltiazem	1	1						
1575h	60 y F				A	Ingst	Int-S	2		
1576ha	60 y F	Metoprolol	1	1	A	Ingst	Int-S	1	Amlodipine	840 ng/mL In Blood (unspecified) @ Unknown
		Amitriptyline	2	2						
		Hydrochlorothiazide/ lisinopril	3	3						
		Cariprazine	4	4						
		Amlodipine	1	1						
		Gabapentin	2	2					Gabapentin	1.4 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	3	3						
1577h	61 y F				A/C	Ingst	Int-S	2		
1578h	61 y M	Carvedilol	1	1	A/C	Ingst	Int-S	2		
		Montelukast	2	2						
		Amlodipine	1	1						
		Salicylate	2	2						
		Clopidogrel	3	3						
1579h	61 y M	Vitamin D	4	4	A/C	Ingst	Int-S	3		
		Amlodipine	1	1						
		Alcohol, N.O.S.	2	2						
		Ethanol (beverage)	3	3						
1580h	61 y F				A/C	Ingst	Int-S	2		
1581h	62 y M	Amlodipine	1	1	C	Ingst	Unt-U	2	Digoxin	2.4 ng/mL In Blood (unspecified) @ Unknown
		Brexpiprazole	2	2						
		Duloxetine (ER)	3	3						
		Lorazepam	4	4						
		Levothyroxine	5	5						
		Atorvastatin	6	6						
1582	62 y F	Digoxin	1	1	A/C	Ingst	Int-S	1		
		Amlodipine	1	1						
		Losartan	3	2						
		Chlorthalidone	4	3						
		Tizanidine	2	4						
[1583pha]	63 y F	Propranolol	1	1	A	Ingst	Int-S	1	Propranolol	1800 ng/mL In Serum @ Unknown
1584h	63 y M				A/C	Ingst	Unk	1		
1585ha	63 y F	Digoxin	1	1	A	Ingst	Int-S	1	Digoxin	2.7 ng/mL In Blood (unspecified) @ Unknown
		Amlodipine	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Tramadol	2	2					O-desmethyiltramadol	1700 ng/mL In Blood (unspecified) @ Unknown
		Tramadol	2	2					Tramadol	9000 ng/mL In Blood (unspecified) @ Unknown
		Hydrocodone, N.O.S.	3	3					Hydrocodone (free)	140 ng/mL In Blood (unspecified) @ Unknown
		Oxycodone	4	4					Oxycodone (free)	66 ng/mL In Blood (unspecified) @ Unknown
		Gabapentin	5	5					Gabapentin	66 mcg/mL In Blood (unspecified) @ Unknown
1586h	63 y F				U	Ingst	Int-S	2		
1587h	63 y M	Amlodipine	1	1	A	Ingst	AR-D	3		
1588h	63 y F	Diltiazem	1	1	A	Ingst	Int-S	1		
		Amlodipine	1	1						
		Acetaminophen	2	2					Acetaminophen (apap)	304 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	2	2					Acetaminophen (apap)	370 mcg/mL In Blood (unspecified) @ Unknown
1589h	64 y F				A/C	Ingst	Int-S	1		
		Carvedilol	1	1						
		Bupropion	2	2						
		Hydroxyzine	3	3						
		Buspirone	4	4						
		Lamotrigine (ER)	5	5						
		Glimepiride	6	6						
1590	64 y F				A	Ingst	Int-S	2		
1591hi	64 y M	Amlodipine	1	1	A	Ingst	Int-S	2		
1592h	64 y F	Amlodipine	1	1	A	Ingst	Int-S	1		
1593a	64 y M				A/C	Ingst + Inhal + Par	Int-S	1		
		Metoprolol (ER)	1	1					Metoprolol	1900 ng/mL In Blood (unspecified) @ Unknown
		Lisinopril	2	2						
		Insulin, N.O.S.	3	3						
		Cocaine	4	4					Benzoyllecognine	3100 ng/mL In Blood (unspecified) @ Unknown
		Cocaine	4	4					Ecgonine methyl ester	620 ng/mL In Blood (unspecified) @ Unknown
		Empagliflozin	5	5						
		Metformin	6	6					Metformin	180 mcg/mL In Blood (unspecified) @ Unknown
		Amlodipine	7	7					Amlodipine	170 ng/mL In Blood (unspecified) @ Unknown
1594h	64 y M				C	Ingst	AR-D	3		
		Digoxin	1	1					Digoxin	3.8 ng/mL In Blood (unspecified) @ Unknown
[1595ha]	64 y M				A/C	Ingst	Int-S	1		
1596ha	64 y M	Diltiazem	1	1	A	Ingst	Int-S	1		
		Amlodipine	1	1						
		Propranolol	2	2					Propranolol	4500 ng/mL In Blood (unspecified) @ Autopsy

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Cyclobenzaprine	3	3					Cyclobenzaprine	27 ng/mL In Blood (unspecified) @ Autopsy
		Gabapentin	4	4					Gabapentin	30 mcg/mL In Blood (unspecified) @ Autopsy
		Primidone	5	5					Primidone	33 mcg/mL In Blood (unspecified) @ Autopsy
		Acetaminophen/oxycodone	6	6					Oxycodone (free)	160 ng/mL In Blood (unspecified) @ Autopsy
		Acetaminophen/oxycodone	6	6					Acetaminophen (apap)	8 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	7	7					Ethanol	154 mg/dL In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	7	7					Ethanol	186 mg/dL In Blood (unspecified) @ Unknown
1597h	64 y M	Diltiazem	1	1	A/C	Ingst	Int-S	2		
1598h	64 y F	Ranolazine (ER)	1	1	A/C	Ingst	Int-S	2		
		Metoprolol (ER)	2	2						
		Buspirone	3	3						
		Duloxetine (ER)	4	4						
1599h	64 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Lisinopril	2	2						
1600ha	65 y F	Diltiazem (ER)	1	1	A/C	Ingst	Int-S	1	Diltiazem	1489 ng/mL In Blood (unspecified) @ Unknown
		Flecainide	2	2					Flecainide	271 ng/mL In Blood (unspecified) @ Unknown
1601	65 y M	Amlodipine	1	1	A	Ingst	Int-S	1		
1602h	65 y F	Propranolol	1	1	A	Ingst	Int-S	1		
		Trazodone	2	2						
		Diphenhydramine	3	3						
		Benzodiazepine, N.O.S.	4	4						
		Lorazepam	5	5						
1603h	65 y F	Amlodipine	1	1	A/C	Ingst	Int-S	1		
		Atenolol	2	2						
		Furosemide	3	3						
		Apixiban	4	4						
1604h	65 y F	Metoprolol	1	1	A/C	Ingst	Int-S	2		
		Carvedilol	2	2						
		Baclofen	3	3						
		Lisinopril	4	4						
		Losartan	5	5						
		Clopidogrel	6	6						
1605h	65 y M	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Metformin	2	2						
		Glipizide	3	3						
		Venlafaxine	4	4						
		Oxcarbazepine	5	5						
		Finasteride	6	6						
1606h	65 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Clonazepam	2	2						
1607ha	65 y M	Amlodipine	1	1	A/C	Ingst	Int-S	1	Amlodipine	180 ng/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2					Ethanol	69 mg/dL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1608h	66 y F	Risperidone	3	3	A/C	Ingst	Int-S	2	9-hydroxyrisperidone	140 ng/mL In Blood (unspecified) @ Unknown
		Risperidone	3	3					Risperidone	200 ng/mL In Blood (unspecified) @ Unknown
		Amlodipine	1	1						
		Mirtazapine	2	2						
1609h	66 y M	Aripiprazole	3	3	A	Par	Unt-T	3		
1610h	67 y M	Amiodarone	1	1	A/C	Ingst	Int-S	2		
1611	67 y M	Flecainide	1	1	A	Ingst	Int-S	2		
		Diltiazem	2	2						
		Rivaroxaban	3	3						
		Nifedipine (ER)	1	1						
		Carvedilol	2	2						
		Amiodarone	3	3						
		Finasteride	4	4						
		Losartan	5	5						
1612ha	67 y F	Ferrous sulfate	6	6	A/C	Ingst	Int-S	1		
		Vitamin (cyanocobalamin/ folate)	7	7						
		Amlodipine	1	1					Amlodipine	370 ng/mL In Blood (unspecified) @ Unknown
1613hi	67 y M	Propranolol	2	2	A	Ingst	Int-S	2	Propranolol	240 ng/mL In Blood (unspecified) @ Unknown
		Amlodipine	1	1						
		Losartan	2	2						
1614	67 y F	Lisinopril	3	3	A/C	Ingst	Int-S	2		
1615h	67 y F	Amlodipine	1	1	C	Unk	Unk	2		
		Hydroxyzine	2	2						
		Hydrochlorothiazide	3	3						
1616h	67 y F	Digoxin	1	1	A	Ingst	Int-S	2	Digoxin	7.4 ng/mL In Serum @ Unknown
1617h	68 y F	Metoprolol (ER)	1	1	A/C	Ingst	Int-S	2		
1618h	68 y F	Verapamil	1	1	A/C	Ingst	Int-S	1		
1619ai	68 y F	Amlodipine	1	1	U	Unk	Int-S	1		
1620ha	68 y M	Propranolol	1	1	C	Ingst	Int-S	1		
		Metoprolol	2	2						
		Ethanol (beverage)	3	3						
		Amlodipine	1	1						
		Baclofen	2	2					Baclofen	1.7 mcg/mL In Blood (unspecified) @ Unknown
		Clonazepam	3	3					7-aminoclonazepam	120 ng/mL In Blood (unspecified) @ Unknown
		Clonazepam	3	3					Clonazepam	58 ng/mL In Blood (unspecified) @ Unknown
		Marijuana	4	4					Delta-9-thc	2.6 ng/mL In Blood (unspecified) @ Unknown
		Marijuana	4	4					Delta-9-carboxy-thc	9.5 ng/mL In Blood (unspecified) @ Unknown
1621h	68 y F	Amlodipine	1	1	A	Ingst	Int-S	1		
1622ha	68 y F	Amlodipine	1	1	A	Ingst	Int-S	2		
		Tramadol	2	2						
		Quetiapine	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1623h	68 y M	Amlodipine Clonidine Clonazepam Carvedilol Ethanol (beverage)	1 2 3 4 5	1 2 3 4 5	A/C	Ingst + Unk	Int-S	2		
									Ethanol	0.003 mg/dL In Blood (unspecified) @ Unknown
1624h	69 y M	Metoprolol (ER)	1	1	C	Ingst	Unt-T	2		
1625ph	69 y F	Diltiazem (ER) Lisinopril	1 2	1 2	U	Ingst	Int-S	2		
1626	69 y F	Verapamil	1	1	A/C	Ingst	Int-S	2		
1627h	69 y F	Propranolol Tizanidine Buprenorphine Gabapentin Trazodone	1 2 3 4 5	1 2 3 4 5	A	Ingst	Int-S	3		
1628h	69 y F	Amlodipine Sodium hydroxide Metoprolol	1 3 2	1 1 2	A	Ingst	Int-S	1		
1629ha	70 y M	Verapamil Acetaminophen/ hydrocodone Trazodone	1 2 3	1 2 3	A/C	Ingst	Int-S	1		
1630h	70 y M	Amlodipine Carvedilol Cetirizine Atorvastatin Ethanol (beverage)	1 2 3 4 5	1 2 3 4 5	A/C	Ingst + Unk	Int-S	2		
									Ethanol	150 mg/dL In Blood (unspecified) @ Unknown
1631ph	70 y M	Amlodipine Trazodone Carvedilol	1 2 3	1 2 3	A/C	Ingst	Int-S	3		
1632p	70 y F	Propranolol Carvedilol Benzodiazepine, N.O.S.	1 2 3	1 2 3	A	Ingst	Int-S	2		
1633ha	70 y F	Amlodipine Hydrochlorothiazide	1 2	1 2	A/C	Ingst	Int-S	2		
1634h	70 y M	Metoprolol (ER) Nebivolol Ethanol (beverage) Acetaminophen Acetaminophen	1 2 3 4 4	1 2 3 4 4	A/C	Ingst	Int-S	2		
									Ethanol	155 mg/dL In Serum @ Unknown
									Acetaminophen (apap)	10 mcg/mL In Serum @ Unknown
									Acetaminophen (apap)	16.9 mcg/mL In Serum @ Unknown
1635h	70 y M	Carvedilol	1	1	A	Ingst	Unt-T	2		
1636h	70 y F	Diltiazem Atorvastatin	1 2	1 2	A/C	Ingst	Int-S	2		
1637h	70 y M	Digoxin	1	1	C	Ingst	AR-D	3	Digoxin	4.7 ng/mL In Serum @ Unknown
1638h	71 y M	Diltiazem (ER) Citalopram Galantamine Lisinopril Apixaban Bisacodyl	1 2 3 4 5 6	1 2 3 4 5 6	A/C	Ingst	Unt-G	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1639h	71 y F	Magnesium	7	7	C	Ingst + Par	Unt-T	2		
		Vitamin D	8	8						
		Metoprolol	1	1						
		Propranolol	2	2						
		Verapamil	3	3						
1640h	71 y M	Diltiazem	4	4	A/C	Ingst	Int-S	2		
		Diltiazem	1	1						
		Isosorbide mononitrate (ER)	2	2						
		Apixaban	3	3						
		Zolpidem	4	4						
1641h	71 y F	Losartan	5	5	A	Ingst	Int-S	3		
		Amlodipine	1	1						
		Alprazolam	2	2						
1642h	71 y M	Amlodipine	1	1	A	Ingst	Int-S	2		
		Carvedilol	2	2						
		Lisinopril	3	3						
1643h	71 y M	Amlodipine	1	1	A/C	Ingst	Int-S	1		
1644pha	71 y M	Amlodipine	1	1	A/C	Ingst	Int-S	1	Amlodipine	85 mcg/L In Blood (unspecified) @ Unknown
		Amlodipine	1	1						
[1645h]	71 y M	Ibuprofen	2	2	A	Ingst	Unt-T	1		
		Acetaminophen	3	3						
		Amlodipine	1	1						
1646a	71 y F	Amlodipine	1	1	A/C	Ingst	Int-S	1	Amlodipine	1000 ng/mL In Blood (unspecified) @ Unknown
		Amlodipine	1	1						
1647h	71 y F	Amlodipine	1	1	A	Ingst	Int-S	2		
1648h	71 y M	Nifedipine	1	1	A/C	Ingst	Int-S	2		
		Carvedilol	2	2						
		Ethanol (beverage)	3	3						
1649pha	71 y F	Amlodipine	1	1	U	Ingst	Int-S	1	Amlodipine	1500 ng/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2						
		Ethanol (beverage)	2	2						
1650h	72 y M	Digoxin	1	1	A/C	Ingst	AR-D	3	Digoxin	6.6 ng/mL In Blood (unspecified) @ 2 d (pe)
1651pa	72 y F	Metoprolol	1	1	A	Ingst	Int-S	2		
1652	72 y M	Calcium antagonist, N.O.S.	1	1	A	Ingst	Int-S	2		
		Alpha-adrenergic blocker	2	2						
		Cetirizine	3	3						
1653h	72 y F	Propranolol	1	1	A/C	Ingst	Int-S	2		
		Buspirone	2	2						
		Venlafaxine	3	3						
		Clonazepam	4	4						
1654h	72 y M	Carvedilol	1	1	A/C	Ingst	Int-S	2		
		Amlodipine	2	2						
		Losartan	3	3						
		Drug, N.O.S.	4	4						
1655h	72 y M	Isosorbide mononitrate (ER)	2	1	A/C	Ingst	Int-S	2		
		Ranolazine (ER)	3	2						
		Buspirone	4	3						

(Continued)

Table 21. Continued.

										Blood					
Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Concentration @ Time					
1656h	73 y F	Metoprolol	1	4	A	Ingst	Unk	2							
		Trandolapril/verapamil (ER)	1	1											
1657ph	74 y F				A	Ingst	Int-S	2							
		Amlodipine	1	1											
		Labetalol	2	2											
		Hydralazine	3	3											
		Valsartan	4	4											
1658	74 y M				A/C	Ingst	Int-S	2							
		Amlodipine	1	1											
		Memantine	2	2											
		Lisinopril	3	3											
1659h	75 y F				A/C	Ingst	Int-S	2							
		Amlodipine	1	1											
		Metoprolol	2	2											
1660ai	75 y M				U	Unk	Int-S	1							
		Diltiazem	1	1											
		Acetaminophen	2	2											
		Levetiracetam	3	3											
1661h	75 y M				A/C	Ingst	Int-S	2							
		Diltiazem	1	1											
		Cilostazol	2	2											
		Losartan	3	3											
		Ethanol (beverage)	4	4											
1662ha	76 y F				A/C	Ingst	Int-S	1							
		Digoxin	1	1					Digoxin	7.9ng/mL In Blood (unspecified) @ Unknown					
		Diltiazem	2	2					Diltiazem	130ng/mL In Blood (unspecified) @ Unknown					
		Diltiazem	2	2					Diltiazem	2500 ng/mL In Blood (unspecified) @ Autopsy					
		Levothyroxine	3	3											
		Beta blocker, N.O.S.	4	4											
		Quetiapine	5	5											
		Rivaroxaban	6	6											
		Sertraline	7	7											
		Sertraline	7	7					Sertraline	110ng/mL In Blood (unspecified) @ Unknown					
		Sertraline	7	7	Sertraline	870ng/mL In Blood (unspecified) @ Autopsy									
1663ha	76 y M				A	Ingst	Int-S	2							
		Calcium antagonist, N.O.S.	1	1											
		Beta blocker, N.O.S.	2	2											
		Hydralazine	3	3											
1664h	76 y F				A/C	Ingst	Int-S	2							
		Amlodipine	1	1											
		Quetiapine	2	2											
1665h	76 y F				A/C	Ingst	Int-S	2							
		Amlodipine	1	1											
		Lisinopril	2	2											
		Insulin, N.O.S.	3	3											
1666h	76 y M				A/C	Ingst	Unt-T	2							
		Digoxin	1	1					Digoxin	0.9ng/mL In Blood (unspecified) @ Unknown					
		Metoprolol (ER)	2	2											
1667ha	77 y F	Amlodipine	1	1	U	Ingst	Int-S	1						Amlodipine	390ng/mL In Blood (unspecified) @ Unknown
		Nortriptyline	2	2	Nortriptyline	550ng/mL In Blood (unspecified) @ Unknown									
1668h	77 y M				A/C	Ingst	Int-S	2							
		Ethanol (beverage)	3	3											
		Diltiazem	1	1											
		Amlodipine	2	2											
1669ha	77 y F	Amlodipine	1	1							A/C	Ingst	Int-S	1	

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1670h	77 y F	Risperidone	2	2	A/C	Ingst	Int-S	2	Risperidone	200 ng/mL In Blood (unspecified) @ Unknown
		Mirtazapine	3	3					Mirtazapine	330 ng/mL In Blood (unspecified) @ Unknown
		Carvedilol	1	1						
		Metoprolol	2	2						
		Trazodone	3	3						
		Duloxetine (ER)	4	4						
		Alprazolam	5	5						
		Gabapentin	6	6						
		Hydralazine	7	7						
		Ethanol (beverage)	8	8						
		Escitalopram	9	9						
		Memantine	10	10						
		Atorvastatin	11	11						
		Montelukast	12	12						
		Levothyroxine	13	13						
		Magnesium oxide (cyanocobalamin)	14	14						
1671h	77 y F				A/C	Ingst	Int-S	2		
		Metoprolol	1	1						
		Temazepam	2	2						
1672h	77 y M	Acetaminophen/hydrocodone	3	3	A	Ingst	Int-S	2	Acetaminophen (apap)	34 mcg/mL In Blood (unspecified) @ Unknown
		Digoxin	1	1					Digoxin	12.9 ng/mL In Blood (unspecified) @ Unknown
		Amlodipine	2	2						
1673h	78 y M	Flecainide	1	1	C	Ingst	AR-D	2		
1674	78 y F				A/C	Ingst	Int-S	2		
		Verapamil	1	1						
		Metoprolol	2	2						
1675h	78 y F	Apixaban	3	3	A/C	Ingst	Int-S	1		
		Amlodipine/benazepril	1	1						
		Acetaminophen	2	2					Acetaminophen (apap)	24.9 mcg/mL In Serum @ 5.5 h (pe)
1676h	78 y M	Acetaminophen	2	2	A/C	Ingst + Par	Int-S	1	Acetaminophen (apap)	60.2 mcg/mL In Serum @ 15 m (pe)
		Lorazepam	3	3						
		Amlodipine	1	1						
1677h	78 y F	Metoprolol	2	2	A	Ingst	Int-S	2		
		Metformin	3	3						
		Lisinopril	4	4						
1678h	78 y F	Glipizide	5	5	A	Ingst	AR-D	3		
		Insulin (glargine) (ER)	6	6						
		Hydrochlorothiazide	7	7						
1679h	79 y F	Clopidogrel	8	8	A/C	Par	AR-D	3		
		Digoxin	1	1					Digoxin	2.1 ng/mL In Blood (unspecified) @ Unknown
		Digoxin	1	1					Digoxin	7.17 ng/mL In Serum @ 9 h (pe)
1680h	79 y F	Digoxin	1	1	A/C	Ingst	Int-S	3		
		Amlodipine	1	1						
		Alprazolam	2	2						
1681h	79 y M	Lorazepam	3	3	C	Ingst	Unk	3		
		Mirtazapine	4	4						
		Digoxin	1	1					Digoxin	3.8 ng/mL In Serum @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1682h	79 y F	Amlodarone	2	2						
		Digoxin	1	1	A/C	Ingst + Par	AR-D	3	Digoxin	4.5 ng/mL In Serum @ 6 h (pe)
1683h	80 y F				A/C	Ingst	Unt-T	1		
		Amlodipine	1	1						
		Metoprolol (ER)	2	2						
		Losartan	3	3						
1684h	80 y F				A	Ingst	Int-S	2		
		Amlodipine	1	1						
		Clonazepam	2	2						
1685	81 y F				A/C	Ingst	Unt-T	2		
		Amlodipine	1	1						
1686	81 y F				A/C	Ingst	Unt-T	3		
		Amlodipine	1	1						
1687h	81 y F				A	Ingst	Int-S	3		
		Propranolol	1	1						
		Losartan	2	2						
		Clonidine	3	3						
1688h	81 y M				A/C	Ingst	Int-S	2		
		Metoprolol	1	1						
		Salicylate	2	2					Salicylate	21.7 mg/dL In Serum @ Unknown
		Insulin (lispro)	3	3						
		Insulin (glargine) (ER)	4	4						
		Furosemide	5	5						
		Losartan	6	6						
		Atorvastatin	7	7						
1689h	81 y F				U	Ingst	Int-S	2		
		Atenolol	1	1						
		Amlodipine	2	2						
1690ha	82 y M				A/C	Ingst	Int-S	2		
		Carvedilol	1	1						
		Amlodipine	2	2						
		Codeine	3	3						
1691	82 y M				A	Ingst	Int-S	2		
		Carvedilol	1	1						
1692h	82 y M				A/C	Ingst	Unt-T	2		
		Amlodipine	1	1						
1693hi	82 y F				A/C	Ingst	Unt-T	2		
		Amlodipine	1	1						
1694h	83 y M				A/C	Ingst	Int-S	2		
		Amlodipine	1	1						
		Acetaminophen	2	2					Acetaminophen (apap)	119 mcg/mL In Blood (unspecified) @ Unknown
		Salicylate	3	3					Salicylate	33 mg/dL In Serum @ Unknown
		Salicylate	3	3					Salicylate	37 mg/dL In Serum @ Unknown
		Furosemide	4	4						
		Gabapentin	5	5						
		Iron, N.O.S.	6	6						
		Metformin	7	7						
		Antacid (proton pump inhibitor, N.O.S.)	8	8						
		Drug, anticholinergic	9	9						
1695h	83 y F				A/C	Ingst	Unt-U	3		
		Amlodipine	1	1						
		Carvedilol	2	2						
		Mirtazapine	3	3						
		Hydrochlorothiazide/triamterene	4	4						
		Atorvastatin	5	5						
1696	83 y F				A	Ingst	Int-S	1		
		Amlodipine	1	1						
		Metoprolol	2	2						
1697h	83 y M				A/C	Ingst	Unt-T	2		
		Nifedipine (ER)	1	1						
		Amlodipine	2	2						
		Metoprolol	3	3						
		Diltiazem	4	4						
		Losartan	5	5						
1698ha	83 y F				A/C	Ingst	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1699p	83 y M	Amlodipine	1	1	A	Ingst	Int-S	2	Amlodipine	220 ng/mL In Blood (unspecified) @ Unknown
		Glipizide	2	2					Duloxetine	146 ng/mL In Blood (unspecified) @ Unknown
		Duloxetine (ER)	3	3						
		Metformin	4	3						
		Pravastatin	5	4					Haloperidol	12.3 ng/mL In Blood (unspecified) @ Unknown
		Haloperidol	6	6						
1700h	84 y F	Beta blocker, N.O.S.	1	1	A/C	Ingst	Unt-T	3	Acetaminophen (apap)	235.8 mcg/mL In Serum @ Unknown
		Amiodarone	2	2						
		Acetaminophen	3	3						
1701ha	84 y F	Beta blocker, N.O.S.	2	1	A/C	Ingst	Int-S	2		
		Calcium antagonist, N.O.S.	1	2						
		Clonidine	3	3						
1702	84 y F	Amlodipine	1	1	A/C	Ingst	Unt-T	2		
		Baclofen	2	2						
		Duloxetine (ER)	3	3						
		Mirtazapine	4	4						
		Losartan	5	5						
		Pantoprazole	6	6						
		Atorvastatin	7	7						
		Melatonin	8	8						
1703	85 y F	Diltiazem (ER)	1	1	C	Ingst	AR-D	1		
		Metoprolol	2	2						
1704h	85 y F	Verapamil (ER)	1	1	A/C	Ingst	Int-S	2		
		Amlodipine	1	1						
		Trazodone	2	2						
		Tamsulosin	3	3						
		Allopurinol	4	4						
1705h	85 y F	Amlodipine	1	1	A	Ingst	Int-S	2		
		Fluoxetine	2	2						
		Pravastatin	3	3						
1706	86 y M	Amlodipine	1	1	A/C	Ingst	Int-S	2		
		Atenolol	2	2						
		Lisinopril	3	3						
		Pantoprazole (ER)	4	4						
		Mirtazapine	5	5						
		Prednisone	6	6						
1707h	86 y F	Amlodipine	1	1	A/C	Ingst	Unt-G	2		
1708h	86 y F	Amlodipine	1	1	A/C	Ingst	Int-S	1		
		Diltiazem	1	1						
		Hydrochlorothiazide	2	2						
		Famotidine	3	4						
1709h	87 y M	Losartan	4	4	A/C	Ingst	Int-S	2		
		Carvedilol	1	1						
		Nitroglycerin	2	2						
1710h	88 y F	Apixaban	3	3	C	Ingst	AR-D	3	Digoxin	5.4 ng/mL In Blood (unspecified) @ Unknown
		Digoxin	1	1						
1711h	88 y M	Nateglinide	2	2	A	Ingst	Int-S	1		
		Metformin	3	3						
		Linagliptin	4	4						
		Amlodipine	1	1						
1712h	88 y F	Melatonin	2	2	A/C	Ingst	Int-S	3		
		Verapamil	1	1						
1713h	88 y M				A	Par	Unt-T	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1714h	89 y F	Digoxin	1	1	U	Ingst + Unk	Int-S	2	Digoxin	1.1 mcg/mL In Blood (unspecified) @ Unknown
		Digoxin	1	1					Digoxin	4.3 mcg/mL In Blood (unspecified) @ Unknown
		Digoxin	1	1					Digoxin	5 mcg/mL In Blood (unspecified) @ Unknown
		Nebivolol Amlodipine Ethanol (beverage)	1 2 3	1 2 3					Ethanol	80 mg/dL In Blood (unspecified) @ Unknown
1715h	89 y M	Digoxin	1	1	C	Ingst	AR-D	3	Digoxin	5.5 ng/mL In Serum @ 21 h (pe)
1716h	90 y F	Diltiazem	1	1	A	Par	AR-D	3		
1717h	90 y M	Carvedilol Amiodarone Ibuprofen	1 2 3	1 2 3	A/C	Ingst	Int-S	2		
1718h	90 y F	Metoprolol	1	1	A/C	Ingst	Int-U	3		
1719	90 y F	Metoprolol Diltiazem	1 2	1 2	A/C	Ingst	Unt-T	2		
1720ph	90 y F	Propranolol	1	1	U	Ingst	Unk	2		
1721h	91 y F	Losartan Naproxen	1 2	1 2	A	Ingst	Int-S	2		
1722	94 y M	Metoprolol Quetiapine Omeprazole (ER)	1 2 3	1 2 3	A/C	Ingst	Int-S	2		
1723	96 y F	Digoxin	1	1	A/C	Ingst	Unt-T	3		
1724h	97 y F	Amlodipine	1	1	A/C	Ingst	Int-S	2		
See Also case 36, 80, 152, 294, 347, 626, 648, 814, 836, 1066, 1095, 1120, 1131, 1136, 1140, 1173, 1217, 1232, 1248, 1277, 1287, 1292, 1305, 1342, 1351, 1360, 1365, 1378, 1389, 1391, 1392, 1409, 1433, 1443, 1451, 1736, 1741, 1747, 1749, 1763, 1764, 1766, 1769, 1772, 1776, 1782, 1789, 1799, 1800, 1805, 1811, 1821, 1827, 1874, 1899, 1956, 1992, 2088, 2100, 2170, 2246, 2253										
Cold and Cough Preparations										
1725ph	19 y F	Benzonatate Metronidazole Doxycycline	1 2 3	1 2 3	A	Ingst	Int-S	2		
1726p	24 y F	Benzonatate Nitrofurantoin Prednisone	1 2 3	1 2 3	A	Ingst	Int-S	2		
1727h	25 y F	Acetaminophen/ decongestant/ dextromethorphan	1	1	A	Ingst + Par	Int-S	1	Acetaminophen (apap)	119 mcg/mL In Blood (unspecified) @ 20 h (pe)
		Acetaminophen/ decongestant/ dextromethorphan	1	1					Acetaminophen (apap)	66 mcg/mL In Blood (unspecified) @ 43 h (pe)
		Acetaminophen/ decongestant/ dextromethorphan	1	1					Acetaminophen (apap)	75 mcg/mL In Blood (unspecified) @ 25 h (pe)
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3						
		Salicylate	4	4					Salicylate	4.1 mg/dL In Blood (unspecified) @ 20 h (pe)
1728h	32 y F				A	Ingst	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1729h	33 y F	Acetaminophen/ dextromethorphan/ phenylephrine	1	1					Acetaminophen (apap)	67 mcg/mL In Blood (unspecified) @ Unknown
		Topiramate	2	2						
		Acetaminophen/ dextromethorphan/ doxylamine	1	1	C	Ingst	Int-M	3	Acetaminophen (apap)	42 mcg/mL In Blood (unspecified) @ Unknown
1730h	49 y F	Acetaminophen/ antihistamine/ dextromethorphan	1	1	A	Ingst	Int-S	1	Acetaminophen (apap)	577 mcg/mL In Serum @ 19h (pe)
1731p	53 y F	Benzodiazepine, N.O.S.	2	2	A	Ingst	Int-S	2		
1732h	81 y F	Benzonatate	1	1	A	Ingst	Int-S	2		
[1733pha]	12 m F	Acetaminophen/ dextromethorphan/ doxylamine	1	1						
		Diphenhydramine	2	2	U	Unk	Unt-G	1		
		Benzonatate	1	1					4-butylaminobenzoic acid	0mcg/mL In Blood (unspecified) @ Autopsy
		Benzonatate	1	1					Benzonatate	7.3mg/L In Urine (quantitative only) @ 2h (pe)
See Also case 6, 136, 346, 412, 462, 839, 867, 951, 968, 1133, 1289, 1290, 1306, 1349, 1386, 1389, 1424, 1561, 1762, 1894, 2059										
Diagnostic Agents										
1734i	45 y F	Gadolinium	1	1	A	Oth	Unt-T	3		
Dietary Supplements/Herbals/ Homeopathic										
[1735h]	45 y F	Dietary supplement, non-botanical, N.O.S.	1	1	C	Ingst	AR-O	2		
See Also case 79, 334, 763, 1378, 1448, 1701, 1711										
Diuretics										
1736h	54 y F	Thiazide	1	1	A/C	Ingst	Unk	2		
		Beta blocker, N.O.S.	2	2						
		Antiplatelet drug, N.O.S.	3	3						
See Also case 1248, 1467, 1486, 1517, 1520, 1529, 1538, 1552, 1553, 1582, 1603, 1614, 1633, 1676, 1688, 1694, 1695, 1708, 1741, 1763, 1772, 1808										
Electrolytes and Minerals										
1737h	13 y F	Ferrous sulfate	1	1	A	Ingst	Int-S	1	Iron	6752 mcg/dL In Blood (unspecified) @ Unknown
		Hydroxyzine	2	2						
		Diphenhydramine/ ibuprofen	3	3						
		Ethanol (beverage)	4	4					Ethanol	30.8mg/dL In Blood (unspecified) @ Unknown
1738ha	15 y F	Iron, N.O.S.	1	1	A	Ingst	Int-S	1	Iron	1013 mcg/dL In Blood (unspecified) @ Unknown
1739ha	18 y F	Ibuprofen	2	2	A	Ingst	Int-S	1		
		Iron	1	1					Iron	368 mcg/dL In Blood (unspecified) @ 1 d (pe)
		Iron	1	1					Iron	378 mcg/dL In Blood (unspecified) @ Unknown

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
[1740h]	28 y M	Iron	1	1					Iron	65 mcg/dL In Blood (unspecified) @ 4 d (pe)
		Acetaminophen	2	2					Acetaminophen (apap)	18 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	2	2					Acetaminophen (apap)	5.8 mcg/mL In Blood (unspecified) @ 1 d (pe)
		Salicylate	3	3					Salicylate	11 mg/dL In Blood (unspecified) @ Unknown
		Salicylate	3	3					Salicylate	8.4 mg/dL In Blood (unspecified) @ 1 d (pe)
		Selenium	1	1	A/C	Ingst	Int-S	1	Selenium	194.5 mcg/L In Blood (unspecified) @ 5 d (pe)
		Selenium	1	1					Selenium	203 mcg/L In Blood (unspecified) @ 7 d (pe)
1741ha	70 y M	Selenium	1	1					Selenium	203.4 mcg/L In Urine (quantitative only) @ 5 d (pe)
		Selenium	1	1					Selenium	7280 mcg/L In Blood (unspecified) @ 0 d (pe)
		Ethanol (beverage)	2	2	A/C	Ingst	Int-S	2		
		Potassium Metoprolol	3 1	1 2					Metoprolol	250 ng/mL In Blood (unspecified) @ Unknown
		Apixaban	2	3					Apixaban	2500 ng/mL In Blood (unspecified) @ Unknown
1742h	74 y M	Torsemide	4	4					Torsemide	8200 ng/mL In Blood (unspecified) @ Unknown
		Tamsulosin	5	5	A	Ingst	Int-S	1		
		Sodium bicarbonate	1	1						
		Acetaminophen/diphenhydramine/salicylate	2	2					Acetaminophen (apap)	100 mcg/mL In Serum @ 1.5 h (pe)
		Acetaminophen/diphenhydramine/salicylate	2	2					Acetaminophen (apap)	12 mcg/mL In Serum @ 26 h (pe)
[1743h]	77 y F	Acetaminophen/diphenhydramine/salicylate	2	2					Acetaminophen (apap)	90 mcg/mL In Serum @ 4.5 h (pe)
		Magnesium	1	1	A	Ingst	Int-M	2	Magnesium	11.3 mg/dL In Blood (unspecified) @ Unknown
		See Also case 279, 326, 356, 445, 469, 926, 1107, 1493, 1611, 1638, 1670, 1694, 1782, 1785								
Eye/Ear/Nose/Throat Preparations										
1744	42 y M	Propylhexedrine	1	1	A	Ingst	Int-U	2		
See Also case 795, 1544										
Gastrointestinal Preparations										
[1745ph]	28 y M	Loperamide	1	1	A	Ingst	Unk	2		
1746ph	43 y M				A	Ingst	Int-A	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1747h	62 y M	Loperamide	1	1	A	Ingst	Int-S	2		
		Oxybutynin	1	1						
		Sertraline	2	2						
		Glipizide	3	3						
		Metformin	4	4						
		Lisinopril	5	5						
		Omeprazole (ER)	6	6						
1748a	62 y M	Dicyclomine	1	1	A	Ingst + Unk	Int-S	3		
		Cocaine	2	2						
1749ph	78 y F	Dicyclomine	1	1	A/C	Ingst	Int-S	2		
		Fluoxetine	2	2						
		Venlafaxine	3	3						
		Baclofen	4	4						
		Alprazolam	5	5						
		Sotalol	6	6						
		Amlodipine	7	7						
See Also case 121, 414, 470, 490, 876, 1308, 1363, 1394, 1466, 1493, 1497, 1556, 1638, 1694, 1701, 1706, 1722, 1772, 1785, 1804, 1827, 2205										
Hormones and Hormone Antagonists										
1750	16 y F	Metformin	1	1	A	Ingst	Int-S	2		
1751h	34 y F	Metformin	1	1	C	Ingst + Par	AR-D	2		
		Lithium	2	2						
		Trazodone	3	3						
		Risperidone	4	4						
		Semaglutide	5	5						
1752h	37 y M	Insulin (lispro)	1	1	A	Par	Int-S	1		
1753pha	39 y M	Insulin, N.O.S.	1	1	A/C	Ingst + Par	Int-S	2		
		Diphenhydramine	2	2						
1754h	39 y F	Metformin	1	1	A	Ingst	Int-S	3		
1755	40 y M	Insulin, N.O.S.	1	1	A	Derm + Unk	Int-S	2		
1756h	41 y F	Metformin	1	1	A/C	Ingst	Int-S	1		
		Gabapentin	2	2						
1757ha	42 y M	Insulin (lispro)	1	1	A	Ingst	Int-S	2		
		Insulin (glargine) (ER)	2	2						
1758h	42 y F	Insulin, N.O.S.	1	1	A	Ingst + Par	Int-S	2		
		Gabapentin	2	2						
		Tramadol	3	3						
1759ha	42 y F	Metformin	1	1	A	Ingst + Unk	Int-S	1	Metformin	260 mcg/mL In Blood (unspecified) @ Autopsy
		Fluoxetine	2	2					Norfluoxetine	270 ng/mL In Blood (unspecified) @ Autopsy
		Fluoxetine	2	2					Fluoxetine	7100 ng/mL In Blood (unspecified) @ Autopsy
		Hydroxyzine	4	3					Hydroxyzine	660 ng/mL In Blood (unspecified) @ Autopsy
		Risperidone	3	4					Risperidone	150 ng/mL In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	5	5					Ethanol	239 mg/dL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	6	6					Delta-9-carboxy-thc	39 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	6	6					Delta-9-thc	7.9 ng/mL In Blood (unspecified) @ Autopsy
1760h	44 y M	Insulin (detemir) (ER)	1	1	A/C	Par	Int-S	3		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1761h	45 y F	Insulin (aspart)	2	2						
1762h	46 y F	Metformin	1	1	A/C	Ingst	Int-S	2		
		Metformin	1	1	A/C	Ingst	Int-M	2		
		Acetaminophen/ dextromethorphan/ doxylamine/ phenylephrine	2	2					Acetaminophen (apap)	24 mcg/mL In Blood (unspecified) @ Unknown
1763ha	47 y M	Metformin	1	1	A	Ingst + Unk	Int-S	1	Metformin	73 mcg/mL In Blood (unspecified) @ Autopsy
		Clonidine	2	2					Clonidine	19 ng/mL In Blood (unspecified) @ Autopsy
		Chlorthalidone	3	3					Chlorthalidone	2800 ng/mL In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	4	4					Ethanol	258 mg/dL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	5	5					Delta-9-thc	0.75 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	5	5					Delta-9-carboxy-thc	18 ng/mL In Blood (unspecified) @ Autopsy
1764h	47 y M	Metformin	1	1	C	Ingst	Int-S	2		
		Metoprolol (ER)	2	2						
		Trazodone	3	3						
		Buspirone	4	4						
		Lorazepam	5	5						
		Sertraline	6	6						
		Celecoxib	7	7						
1765ha	48 y M	Metformin	1	1	A/C	Ingst + Par	Int-S	1		
		Ethanol (beverage)	2	2					Ethanol	133 mg/dL In Serum @ Unknown
1766ph	50 y M	Insulin, N.O.S.	3	3	A/C	Ingst	Int-S	1		
		Metformin	1	1						
		Metoprolol	2	2						
1767h	51 y F	Drug, N.O.S.	1	1	A	Ingst	Int-S	1		
		Metformin/sitagliptin	2	1						
		Ibuprofen	3	3						
		Acetaminophen	4	4						
1768	52 y M	Metformin	1	1	A	Ingst	Int-S	2		
		Tizanidine	2	2						
		Memantine	3	3						
		Budesonide/formoterol	4	4						
		Carbamazepine	5	5						
		Doxepin	6	6						
		Gabapentin	7	7						
		Pregabalin	8	8						
1769h	52 y M	Metformin/sitagliptin	1	1	A/C	Ingst	Int-S	2		
		Thioxanthene	2	2						
		Empagliflozin	3	3						
		Levothyroxine	4	4						
		Atorvastatin	5	5						
		Temazepam	6	6						
		Fenofibrate	7	7						
1770h	53 y M	Metformin	1	1	U	Unk	Unk	2		
[1771a]	53 y M	Metformin	1	1	A/C	Ingst	Int-S	1	Metformin	67 mcg/mL In Serum @ Autopsy
1772ha	54 y M	Metformin	1	1	A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1773ha	54 y F	Metoprolol	2	2					Metoprolol	150 ng/mL In Blood (unspecified) @ Unknown
		Amlodipine	3	3						
		Glipizide	4	4						
		Quetiapine	5	5						
		Atorvastatin	6	6						
		Isosorbide mononitrate	7	7						
		Gabapentin	8	8					Gabapentin	3.2 mcg/mL In Blood (unspecified) @ Unknown
		Furosemide	9	9						
		Angiotensin converting enzyme inhibitor, N.O.S.	10	10						
		Omeprazole (ER)	11	11						
1774ha	57 y F	Insulin (regular)	1	1						
		Marijuana, N.O.S.	2	2						
1775h	57 y M	Metformin	1	1	A	Ingst	Int-S	2		
1776ha	57 y F	Insulin (lispro)	1	1						
		Insulin (glargine) (ER)	2	2						
		Insulin, N.O.S.	1	1	U	Ingst + Par	Int-S	2		
		Propranolol	2	2					Propranolol	26 ng/mL In Blood (unspecified) @ Unknown
1777h	59 y M	Clonidine	3	3						
		Meloxicam	4	4						
1778h	59 y F	Metformin	1	1	C	Ingst	AR-D	3		
1779ai	60 y M	Metformin	1	1	U	Ingst	AR-D	3		
		Metformin	1	1	U	Unk	Int-S	1		
1780h	62 y F	Insulin, N.O.S.	1	1						
		Diphenhydramine	2	2						
		Duloxetine (ER)	3	3						
1781h	62 y F	Semaglutide	1	1	A	Par	Unt-G	3		
		Metformin	1	1	A/C	Ingst	Int-S	1	Metformin	76 mcg/mL In Serum @ Unknown
		Metformin	1	1						
1782h	64 y M	Metformin (ER)	1	1	A	Ingst	Unt-T	3		
1783h	64 y F	Magnesium oxide	2	2						
		Carvedilol	3	3						
		Glimepiride	1	1	A/C	Ingst	Int-S	2		
1784h	64 y M	Metformin	1	1	A/C	Ingst	AR-D	2		
1785	66 y F	Metformin	1	1	A/C	Ingst	Int-S	2		
		Omeprazole	2	2						
		Calcium	3	3						
1786h	66 y M	Metformin (ER)	1	1	A	Ingst	Int-S	1		
1787h	70 y F	Metformin	1	1	A/C	Ingst	Unk	3		
1788h	70 y M	Metformin	1	1	C	Ingst	AR-D	3		
1789h	70 y M	Metformin	1	1	A	Ingst	Int-U	1		
		Bupropion	2	2						
		Lisinopril	3	3						
1790h	73 y M	Metformin	1	1	C	Ingst	Unt-T	2		
1791ha	74 y F	Metformin	1	1	A	Ingst	Int-S	1		
1792h	75 y M	Metformin	1	1	A	Ingst	Int-S	1		
1793h	77 y F	Metformin	1	1	A/C	Ingst	AR-D	3		
1794ph	79 y M	Metformin	1	1	A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1795h	79 y M	Metformin	1	1	A	Ingst	AR-D	3		
1796h	79 y M	Semaglutide	1	1	C	Par	Unt-T	3		
1797h	80 y F	Metformin	1	1	U	Ingst	Unk	3		
1798h	82 y F	Metformin (ER)	1	1	A	Ingst	Int-S	2		
1799h	88 y M	Metformin	1	1	U	Ingst	Int-S	1		
		Fenofibrate	2	2						
		Pramipexole	3	3						
1800h	89 y F	Metformin	1	1	A/C	Ingst	Int-S	1		
		Amlodipine	2	2						
		Tramadol	3	3						
		Famotidine	4	4						
		Vitamin (ascorbic acid/copper/vitamin A)	5	5						
See Also case 152, 244, 353, 414, 982, 1107, 1157, 1230, 1378, 1395, 1398, 1467, 1504, 1529, 1530, 1563, 1566, 1580, 1589, 1593, 1605, 1662, 1665, 1670, 1676, 1688, 1694, 1698, 1706, 1710, 1726, 1747, 1827, 1879										
Miscellaneous Drugs										
1801ph	29 y F	Cyproheptadine/pyridoxine/lysine	1	1	C	Ingst	Int-M	3		
1802ph	64 y F	Ropinirole	1	1	A/C	Ingst	Int-U	2		
		Gabapentin	2	2						
		Hydroxyzine	3	3						
		Venlafaxine	4	4						
		Ziprasidone	5	5						
1803	90 y M	Memantine	1	1	A/C	Ingst	Int-M	3		
		Trazodone	2	2						
See Also case 193, 637, 1154, 1164, 1307, 1359, 1429, 1572, 1605, 1611, 1638, 1658, 1670, 1704, 1768, 1799, 1827, 1836, 2183										
Muscle Relaxants										
1804h	50 y F	Baclofen	1	1	A	Ingst	Int-S	2		
		Bupropion	2	2						
		Acetaminophen/hydrocodone	3	3						
		Nitrofurantoin	4	4						
		Oxybutynin	5	5						
		Duloxetine (ER)	6	6						
1805h	53 y M	Baclofen	1	1	A/C	Ingst	Unk	3		
		Acetaminophen	2	2						
		Clonidine	3	3						
1806ai	55 y F	Cyclobenzaprine	1	1	U	Unk	Int-A	1		
		Methamphetamine	2	2						
		Duloxetine (ER)	3	3						
1807p	55 y F	Baclofen	1	1	A/C	Ingst	Int-S	2		
		Pregabalin	2	2						
		Lorazepam	3	3						
		Oxycodone	4	4						
		Zolpidem	5	5						
1808ph	55 y F	Tizanidine	1	1	A/C	Ingst	Int-S	2		
		Benzodiazepine, N.O.S.	2	2						
		Thiazide	3	3						
1809pha	56 y F	Cyclobenzaprine	1	1	U	Ingst + Unk	Int-S	2		
		Oxycodone	2	2					Oxycodone	0.13 mcg/mL In Unknown @ Unknown
1810h	58 y F	Baclofen	1	1	A	Ingst	Int-S	2		
		Gabapentin	2	2						
		Duloxetine (ER)	3	3						
1811	59 y M				A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time				
1812ha	59 y F	Methocarbamol	1	1	C	Ingst	Int-M	3						
		Pregabalin	2	2										
		Amlodipine	3	3										
		Bupropion (ER)	4	4										
1813h	63 y F	Methocarbamol	1	1	A/C	Ingst	Int-S	3						
		Oxycodone	2	2										
1814p	63 y M	Baclofen	1	1	A/C	Ingst	Int-S	2						
1815	63 y F	Cyclobenzaprine	1	1	A/C	Ingst	Int-S	2						
1816ha	66 y F	Carisoprodol	1	1	A	Ingst	Unk	1						
		Metaxalone	1	1					Metaxalone	15 mcg/mL In Blood (unspecified) @ Autopsy				
		Tramadol	2	2					O-desmethyiltramadol	117 ng/mL In Blood (unspecified) @ Autopsy				
		Tramadol	2	2					Tramadol	618 ng/mL In Blood (unspecified) @ Autopsy				
[1817ha]	67 y M	Diphenhydramine	3	3					Diphenhydramine	0.4 mcg/mL In Blood (unspecified) @ Autopsy				
		Fentanyl (prescription)	4	4					Fentanyl	6.7 ng/mL In Blood (unspecified) @ Autopsy				
		Baclofen	1	1					A/C	Ingst	Int-S	1	Baclofen	6.8 mcg/mL In Blood (unspecified) @ 0 m (pe)
		Zolpidem	2	2					Zolpidem	71 ng/mL In Blood (unspecified) @ 0 m (pe)				
1818h	71 y M	Marijuana	3	3	A	Ingst	AR-D	3						
		Baclofen	1	1							Baclofen	0.098 mcg/mL In Blood (unspecified) @ 7 h (pe)		
1819ph	72 y F				A/C	Ingst	Int-S	2						
		Tizanidine	1	1										
		Zolpidem	2	2										
1820h	77 y M	Citalopram	3	3	A/C	Ingst	Unt-T	3						
		Tizanidine	1	1										
1821h	87 y F				A/C	Ingst	Int-S	2						
		Baclofen	1	1										
		Pregabalin	2	2										
		Amlodipine	3	3										
		Escitalopram	4	4										
Sumatriptan 5 5														
See Also case 216, 487, 694, 816, 836, 1105, 1120, 1123, 1140, 1172, 1174, 1272, 1294, 1297, 1298, 1342, 1356, 1359, 1363, 1375, 1403, 1431, 1433, 1500, 1515, 1521, 1582, 1596, 1604, 1620, 1627, 1701, 1749, 1768, 1827, 1841, 1849, 2006, 2112, 2139, 2162, 2177, 2233														
Sedative/Hypnotics/ Antipsychotics														
[1822ha]	11 y F				C	Par	AR-D	3						
1823ha	19 y F	Propofol	1	1	A	Ingst	Unk	3						
		Olanzapine	1	1										
1824pha	19 y M				A	Ingst	Int-S	1						
		Midazolam	1	1					Alpha-hydroxy-midazolam (=1-hydroxy-midazolam)	37246 ng/mL In Urine (quantitative only) @ 2 d (pe)				
		Temazepam	2	2					Temazepam	194 ng/mL In Urine (quantitative only) @ 2 d (pe)				
		Temazepam	2	2					Oxazepam	257 ng/mL In Urine (quantitative only) @ 2 d (pe)				
1825pi	25 y F				A	Ingst	Int-S	2						
		Ziprasidone	1	1										
		Hydroxyzine	2	2										

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1826pa	29 y M	Benzodiazepine, N.O.S.	1	1	A	Ingst	Int-S	2		
1827hi	30 y M	Haloperidol	1	1	A/C	Ingst	Int-S	3		
		Doxepin	5	2						
		Linaclotide	2	2						
		Benzotropine	6	3						
		Atomoxetine	4	4						
		Cyclobenzaprine	3	5						
		Prazosin	7	7						
		Levothyroxine	8	8						
1828ha	31 y M	Quetiapine	1	1	A/C	Ingst	Int-S	2		
		Olanzapine	2	2						
		Ibuprofen	3	3						
1829h	31 y F	Haloperidol	1	1	A	Ingst	Int-S	2		
		Oxcarbazepine	2	2						
		Risperidone	3	3						
1830ai	35 y F	Bromazolam	1	1	U	Unk	Int-A	1		
1831ha	39 y M	Quetiapine	1	1	A/C	Ingst	Int-S	1	Quetiapine	8100 ng/mL In Blood (unspecified) @ Unknown
1832ai	40 y M	Alprazolam	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
1833i	41 y M	Quetiapine	1	1	U	Unk	Int-S	1		
		Citalopram	2	2						
		Ethanol (beverage)	3	3						
1834p	42 y F	Alprazolam	1	1	A/C	Ingst	Int-S	2		
		Olanzapine	2	2						
		Ethanol (beverage)	3	3					Ethanol	143 mcg/dL In Blood (unspecified) @ Unknown
1835ai	44 y F	Quetiapine	1	1	U	Unk	Unk	1		
		Trazodone	2	2						
		Gabapentin	3	3						
1836p	44 y M	Propofol	1	1	A	Ingst	Int-S	2		
		Etomidate	2	2						
		Rocuronium	3	3						
1837h	44 y F	Alprazolam	1	1	A	Ingst	Int-S	3		
1838h	45 y M	Quetiapine	1	1	A/C	Ingst	Int-U	2		
		Methamphetamine	2	2						
1839h	45 y F	Quetiapine	1	1	A/C	Ingst + Aspir	Int-S	3		
1840h	47 y F	Quetiapine	1	1	A	Ingst	Int-S	3		
1841h	48 y F	Quetiapine	1	1	A/C	Ingst	Int-S	2		
		Clonazepam	1	1						
		Quetiapine	2	2						
		Hydroxyzine	3	3						
		Venlafaxine	4	4						
		Eszopiclone	5	5						
		Tizanidine	6	6						
		Amphetamine/dextro-amphetamine	7	7						
1842h	49 y F	Lorazepam	1	1	A/C	Ingst	Int-S	3	Salicylate	25.8 mg/dL In Serum @ 18 h (pe)
1843ph	49 y F	Salicylate	2	2	U	Ingst + Unk	Int-S	2		
		Alprazolam	1	1						
		Acetaminophen/hydrocodone	2	2					Acetaminophen (apap)	129 mcg/mL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	3	3						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1844h	51 y F	Acetaminophen	4	4	A	Ingst	Int-S	3		
1845ai	51 y M	Quetiapine	1	1	U	Unk	Int-A	1		
		Temazepam	1	1						
		Quetiapine	2	2						
		Ethanol (beverage)	3	3						
1846p	52 y M	Alprazolam	1	1	A	Ingst	Int-S	2		
		Zolpidem	2	2						
1847i	54 y F	Quetiapine	1	1	U	Unk	Unk	1		
		Alprazolam	2	2						
		Zolpidem	3	3						
1848ph	55 y F	Quetiapine	1	1	A/C	Ingst	Int-S	2		
		Olanzapine	2	2						
		Sertraline	3	3						
		Pregabalin	4	4						
		Eszopiclone	5	5						
1849p	55 y M	Clonazepam	2	1	A	Ingst	Int-S	2		
		Cyclobenzaprine	1	2						
1850p	55 y M	Quetiapine	1	1	A/C	Ingst	Int-S	2		
1851h	57 y F	Clonazepam	1	1	A/C	Ingst	Int-S	3		
		Valproic acid	2	2						
		Venlafaxine	3	3						
1852p	58 y F	Zolpidem	1	1	A/C	Ingst	Int-S	2		
		Ethanol (beverage)	2	2					Ethanol	356 mg/dL In Blood (unspecified) @ Unknown
1853pha	58 y F	Quetiapine	1	1	A/C	Ingst	Int-S	1	Quetiapine	8700 ng/mL In Blood (unspecified) @ 12 h (pe)
		Gabapentin	2	2					Gabapentin	7.4 mcg/mL In Blood (unspecified) @ 12 h (pe)
		Citalopram	3	3					Citalopram	200 ng/mL In Blood (unspecified) @ 12 h (pe)
1854	59 y M	Alprazolam	1	1	A/C	Ingst	Int-S	2		
1855p	60 y F	Alprazolam	1	1	A	Ingst	Int-S	2		
		Acetaminophen	2	2					Acetaminophen (apap)	57 mcg/mL In Serum @ Unknown
1856ai	61 y F	Alprazolam	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
1857pha	61 y M	Alprazolam	1	1	A	Ingst	Int-S	3	Alprazolam	23 ng/mL In Blood (unspecified) @ Autopsy
		Buprenorphine/naloxone (sublingual film)	2	2					Buprenorphine	0.96 ng/mL In Blood (unspecified) @ Autopsy
		Buprenorphine/naloxone (sublingual film)	2	2					Norbuprenorphine	2.1 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana	3	3					Delta-9-thc	0.72 ng/mL In Blood (unspecified) @ Autopsy
1858h	61 y M	Quetiapine	1	1	A/C	Ingst	Int-S	3		
1859pha	61 y M	Alprazolam	1	1	C	Ingst	Int-U	1	Alprazolam	220 ng/mL In Blood (unspecified) @ Unknown
		Gabapentin	2	2					Gabapentin	33 mcg/mL In Blood (unspecified) @ Unknown
1860ha	62 y F				A/C	Ingst	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1861h	63 y F	Quetiapine	1	1	A/C	Ingst	Int-S	2		
		Temazepam	2	2						
		Venlafaxine	3	3						
		Hydroxyzine	4	4						
		Oxazepam	5	5						
1862h	64 y F	Alprazolam	1	1	U	Ingst	Int-S	2		
1863h	67 y F	Quetiapine	1	1	A	Ingst	Int-S	2		
		Acetaminophen	2	2						
1864h	67 y F	Chlorpromazine	1	1	A/C	Ingst	Int-S	2		
1865h	69 y F	Quetiapine	1	1	A/C	Ingst	Int-S	2		
		Clonazepam	1	1						
		Cyclic antidepressant, N.O.S.	2	2						
1866h	69 y M	Nabumetone	3	3	A	Ingst	Int-U	1		
		Benzodiazepine, N.O.S.	1	1						
1867h	70 y F				A/C	Ingst	Int-S	2		
1868p	70 y F	Lorazepam	1	1	A/C	Ingst	Unk	2		
		Zolpidem	2	2						
		Benzodiazepine, N.O.S.	1	1						
1869h	70 y F	Opioid, N.O.S.	2	2	A	Ingst	Int-S	2		
		Diazepam	1	1						
1870ha	71 y F	Methadone	2	2	A/C	Ingst	Int-S	1	Quetiapine	16000 ng/mL In Blood (unspecified) @ Autopsy
		Quetiapine (ER)	1	1						
		Clonazepam	2	2					7-aminoclonazepam	26 ng/mL In Blood (unspecified) @ Autopsy
		Clonazepam	2	2					Clonazepam	6.1 ng/mL In Blood (unspecified) @ Autopsy
1871h	73 y F				C	Ingst	AR-D	3		
1872a	73 y M	Aripiprazole	1	1	A	Ingst	Int-S	3	Diazepam	150 ng/mL In Blood (unspecified) @ Autopsy
		Bupropion	2	2						
		Duloxetine (ER)	3	3						
		Buspirone	4	4						
		Diazepam	1	1						
		Diazepam	1	1					Nordiazepam	190 ng/mL In Blood (unspecified) @ Unknown
		Diazepam	1	1					Nordiazepam	270 ng/mL In Blood (unspecified) @ Autopsy
		Diazepam	1	1					Diazepam	390 ng/mL In Blood (unspecified) @ Unknown
1873h	73 y F				A/C	Ingst	Int-S	3		
1874ai	78 y M	Lorazepam	1	1	U	Unk	Int-S	1		
		Pentobarbital	1	1						
		Amlodipine	2	2						
1875	80 y F	Marijuana, N.O.S.	3	3	A	Ingst	Int-S	2		
1876h	82 y M	Zolpidem	1	1	U	Ingst	Int-S	3	Ethanol	4 mg/dL In Blood (unspecified) @ Unknown
		Eszopiclone	1	1						
		Ethanol (beverage)	2	2						
1877h	85 y M				A	Ingst	Int-S	1		
		Chloral hydrate	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Acetaminophen	2	2					Acetaminophen (apap)	49.3 mcg/mL In Blood (unspecified) @ Unknown
1878ai	86 y F				U	Unk	Int-S	1		
		Alprazolam	1	1						
1879	86 y F				A/C	Ingst + Par	Int-S	2		
		Alprazolam	1	1						
		Insulin, N.O.S.	2	2						
1880h	88 y F				A	Ingst	Int-S	2		
		Doxylamine	1	1						
1881a	91 y M				A/C	Ingst	Int-S	1		
		Clonazepam	1	1					7-aminoclonazepam	13 ng/mL In Blood (unspecified) @ Autopsy
		Clonazepam	1	1					Clonazepam	33 ng/mL In Blood (unspecified) @ Autopsy
		Hydroxyzine	2	2					Hydroxyzine	670 ng/mL In Blood (unspecified) @ Autopsy
1882	93 y M				A	Ingst	Int-S	2		
		Temazepam	1	1						
See Also case 6, 9, 26, 36, 57, 80, 89, 94, 125, 126, 146, 152, 203, 299, 348, 363, 368, 374, 377, 387, 390, 403, 406, 411, 418, 429, 440, 449, 467, 506, 507, 515, 533, 536, 543, 550, 551, 558, 581, 583, 591, 596, 599, 600, 604, 609, 615, 630, 632, 643, 646, 648, 661, 667, 685, 703, 722, 730, 735, 743, 769, 784, 789, 797, 802, 817, 830, 855, 874, 875, 877, 902, 905, 909, 914, 921, 947, 958, 974, 1037, 1065, 1068, 1078, 1085, 1093, 1096, 1105, 1131, 1172, 1174, 1224, 1232, 1260, 1283, 1284, 1285, 1288, 1289, 1290, 1292, 1298, 1300, 1304, 1307, 1317, 1320, 1324, 1327, 1328, 1332, 1342, 1350, 1358, 1359, 1360, 1363, 1372, 1376, 1378, 1385, 1387, 1389, 1396, 1399, 1401, 1403, 1425, 1443, 1481, 1490, 1493, 1497, 1513, 1514, 1528, 1529, 1542, 1543, 1545, 1552, 1555, 1569, 1570, 1572, 1575, 1580, 1589, 1598, 1602, 1606, 1607, 1608, 1620, 1622, 1623, 1632, 1640, 1641, 1653, 1655, 1662, 1664, 1669, 1670, 1671, 1675, 1680, 1684, 1698, 1722, 1730, 1749, 1751, 1759, 1764, 1769, 1772, 1802, 1807, 1808, 1817, 1819, 1896, 1899, 1905, 1922, 1966, 2050, 2085, 2091, 2162, 2186, 2205, 2211, 2233, 2235, 2241, 2253, 2259, 2263										
Stimulants and Street Drugs										
1883i	14 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
[1884ha]	14 y F				A	Ingst	Unk	1		
		Methamphetamine	1	1					Methamphetamine	1300 ng/mL In Blood (unspecified) @ Autopsy
		Methamphetamine	1	1					Amphetamine	150 ng/mL In Blood (unspecified) @ Autopsy
1885pa	17 y M				U	Unk	Unk	2		
		Cocaine	1	1						
		Drug, N.O.S.	2	2					Fentanyl	0.006 mg/L In Blood (unspecified) @ Unknown
		Drug, N.O.S.	2	2					Benzoylcognine	1.7 mg/L In Blood (unspecified) @ Unknown
1886ph	17 y F				A	Unk	Int-A	2		
		Methamphetamine	1	1						
1887ai	19 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
1888ha	19 y M				A	Ingst + Unk	Int-S	3		
		Lysergic acid diethylamide (LSD)	1	1					Lsd (lysergic acid diethylamide)	0.2 ng/mL In Blood (unspecified) @ Autopsy
		Mushroom (psilocybin/psilocin)	2	2						
		Marijuana, N.O.S.	3	3					Delta-9-thc	17 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	3	3					Delta-9-carboxy-thc	500 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	3	3					11-oh-thc (11-hydroxy-delta-9-tetrahydrocannabinol)	6 ng/mL In Blood (unspecified) @ Autopsy
1889ai	19 y M				U	Unk	Int-A	1		
		Lysergic acid diethylamide (LSD)	1	1						
		Mushroom (psilocybin/psilocin)	2	2						
1890pa	20 y M				U	Unk	Int-A	2		
		Cocaine	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1891ha	21 y M	Ethanol (beverage)	2	2					Ethanol	43 mg/dL In Blood (unspecified) @ Unknown
		Methamphetamine	1	1	U	Unk	Unk	1	Amphetamine	275 ng/mL In Blood (unspecified) @ Unknown
		Methamphetamine	1	1					Methamphetamine	4493 mg/mL In Blood (unspecified) @ Unknown
1892h	21 y M				A/C	Unk	Int-A	2		
		Methamphetamine	1	1						
		Fentanyl (non-prescription)	2	2						
1893ai	22 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
1894ph	22 y M				A	Ingst	Int-A	2		
		Cocaine	1	1						
		Amphetamine/dextro-amphetamine (ER)	2	2						
		Chlorpheniramine/dextromethorphan	3	3						
1895ai	23 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Protonitazene	2	2						
		Marijuana, N.O.S.	3	3						
1896pha	23 y M				A/C	Ingst + Unk	Int-M	1		
		Cocaine	1	1					Cocaine	15000 ng/mL In Other @ Autopsy
		Cocaine	1	1					Benzoyllecognine	3500 ng/mL In Other @ Autopsy
		Alprazolam	2	2					Alprazolam	18 ng/mL In Other @ Autopsy
		Marijuana	3	3					Delta-9-thc	25 ng/mL In Other @ Autopsy
		Marijuana	3	3					11-oh-thc (11-hydroxy-delta-9-tetrahydrocannabinol)	7.3 ng/mL In Other @ Autopsy
		Marijuana	3	3					Delta-9-carboxy-thc	76 ng/mL In Other @ Autopsy
1897ai	24 y M				U	Inhal + Unk	Int-A	1		
		Methamphetamine	1	1						
		Nitrous oxide	2	2						
1898	24 y M				A	Unk	Int-S	2		
		Methamphetamine	1	1						
1899h	24 y F				A/C	Ingst	Int-S	2		
		Methamphetamine	1	1						
		Prazosin	2	2						
		Olanzapine	3	3						
		Hydroxyzine	4	4						
		Foreign body, N.O.S.	5	5						
1900ph	25 y M				A	Ingst	Int-U	2		
		Stimulant/street drug, N.O.S.	1	1						
1901ai	26 y F				U	Unk	Int-A	1		
		Cocaine	1	1						
		Ethanol (beverage)	2	2						
1902ai	26 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
1903h	27 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Methamphetamine	2	2						
		Amphetamine (hallucinogenic), 3,4-methylene-dioxymethamphetamine (MDMA)	3	3						
1904ai	27 y M				U	Unk	Int-S	1		
		Amphetamine	1	1						
		Escitalopram	2	2						
		Hydroxyzine	3	3						
1905i	28 y M				U	Unk	Int-S	1		
		Amphetamine	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time		
1906ai	28 y M	Nortriptyline	2	2	U	Unk	Int-A	1				
		Clonazepam	3	3								
		Methamphetamine	1	1								
[1907pha]	28 y M	Fentanyl (non-prescription)	2	2	A	Ingst	Int-U	1	Amphetamine	180 ng/mL In Blood (unspecified) @ Unknown		
		Methamphetamine	1	1							Methamphetamine	27000 ng/mL In Blood (unspecified) @ Unknown
		Methamphetamine	1	1								
1908ha	29 y F	Methamphetamine	1	1	C	Ingst + Inhal	Unt-M	3	Amphetamine	22 ng/mL In Blood (unspecified) @ Unknown		
		Methamphetamine	1	1							Methamphetamine	380 ng/mL In Blood (unspecified) @ Unknown
		Acetaminophen	2	2								
1909ai	29 y M	Methamphetamine	1	1	U	Unk	Int-A	1				
1910p	29 y M	Cocaine	1	1	A	Ingst + Unk	Unt-G	2	Ethanol	11 mg/dL In Blood (unspecified) @ Unknown		
		Amphetamine	2	2								
		Ethanol (beverage)	3	3								
1911ai	29 y M	Methamphetamine	1	1	U	Unk	Int-A	1				
		Marijuana, N.O.S.	2	2								
		Substance (non-drug), N.O.S.	3	3								
1912ai	30 y M	Cocaine	1	1	U	Unk	Int-A	1				
		Ethanol (beverage)	2	2								
1913ai	30 y M	Methamphetamine	1	1	U	Unk	Int-A	1				
		Substance (non-drug), N.O.S.	2	2								
1914h	30 y M	Cocaine	1	1	A	Ingst + Unk	Unk	3	Ethanol	15 mg/dL In Plasma @ Unknown		
		Ethanol (beverage)	2	2								
1915pha	31 y M	Cocaine	1	1	A	Dem+Unk+Rec	Unk	1	Cocaine	2 mg/L In Blood (unspecified) @ Unknown		
		Codeine	2	2					Codeine	0.017 mg/L In Blood (unspecified) @ Unknown		
		Oxycodone	3	3					Oxycodone	0.09 mg/L In Blood (unspecified) @ Unknown		
		Oxymorphone	4	4					Oxymorphone	0.024 mg/L In Blood (unspecified) @ Unknown		
1916ai	31 y M	Methamphetamine	1	1	U	Unk	Int-A	1				
1917ai	31 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1				
		Methamphetamine	1	1								
1918h	31 y M	Marijuana, N.O.S.	2	2	A	Unk	Unk	3				
1919h	31 y M	Cocaine	1	1	U	Unk	Unk	2				
		Methamphetamine	1	1								
1920h	31 y M	Drug, N.O.S.	2	2	A	Unk	Int-S	2				
		Cocaine	1	1								
1921ph	31 y M	Narcotic antagonist (long acting), N.O.S.	2	2	A	Inhal	Int-A	3				
		Cocaine	1	1								

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1922h	31 y M	Alprazolam	2	1	U	Ingst	Int-S	2		
		Amphetamine	1	1						
1923ai	32 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
1924ai	32 y F	Cocaine	1	1	U	Unk	Int-A	1		
1925ai	32 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
1926	32 y F	Methamphetamine	1	1	A	Unk	Int-S	1	Methamphetamine	8.6 mg/L In Blood (unspecified) @ Unknown
		Cocaine	2	2						
1927a	32 y F	Cocaine	1	1	A	Unk	Int-A	2		
1928ai	33 y F	<i>Mitragyna speciosa korthals</i>	1	1	U	Unk	Int-A	1		
		Diphenhydramine	2	2						
		Hydroxyzine	3	3						
1929a	33 y M	Cocaine	1	1	A	Ingst	Int-A	1		
		Marijuana, N.O.S.	2	2						
		Methamphetamine	3	3						
1930h	33 y M	Methamphetamine	1	1	A	Unk	Int-S	1		
1931ai	33 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
1932ha	33 y M	Methamphetamine	1	1	C	Unk	Int-A	1	Methamphetamine	150 ng/mL In Blood (unspecified) @ Autopsy
		Methamphetamine	1	1					Amphetamine	29 ng/mL In Blood (unspecified) @ Autopsy
		Gabapentin	2	2					Gabapentin	4.7 mcg/mL In Blood (unspecified) @ Autopsy
		Marijuana	3	3					Delta-9-carboxy-thc	10 ng/mL In Blood (unspecified) @ Autopsy
1933ai	33 y M	Methamphetamine	1	1	U	Unk	Int-M	1		
		Gabapentin	2	2						
1934h	33 y M	Amphetamine	1	1	A	Unk	Int-S	2		
		Amphetamine (hallucinogenic), 3,4-methylenedioxy methamphetamine (MDMA)	2	2						
		Ethanol (beverage)	3	3						
1935ai	34 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
1936	34 y M	Methamphetamine	1	1	A	Ingst	Int-A	1		
1937h	34 y F	Cocaine	1	1	A	Unk	Int-U	2		
1938ai	34 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
1939pha	34 y M	Cocaine	1	1	A	Ingst	Int-A	1		
		Ethanol (beverage)	2	2						
1940ai	35 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
		Ethanol (beverage)	3	3						
1941pha	35 y M	Cocaine	1	1	A	Unk	Unk	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1942h	35 y M	<i>Mitragyna speciosa korthals</i>	2	2						
		Drug, N.O.S.	3	3						
		Methamphetamine	1	1	A	Ingst	Int-M	1		
1943h	36 y M	Body stuffer	2	2	A	Ingst	Int-M	2		
1944pha	36 y F	Cocaine	1	1	U	Ingst	Int-A	3	Benzoyllecognine	1 mg/L In Serum @ Unknown
		Cocaine	1	1						
1945ai	36 y M	Bupropion	2	2	U	Unk	Int-A	1		
		Methamphetamine	1	1						
1946h	36 y F	Cocaine	2	2						
					A/C	Ingst	Int-U	3		
1947ai	36 y F	<i>Mitragyna speciosa korthals</i>	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
1948pha	36 y F	Tramadol	2	2						
		Fluoxetine	3	3						
					A	Ingst	Int-U	1		
		Cocaine	2	1					Ethanol	258 mg/dL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	4	2						
		Ethylene glycol	3	3					Ethylene glycol	13.9 mg/dL In Blood (unspecified) @ Unknown
		Ethylene glycol	3	3						
1949ai	37 y F	Xylazine	1	4	U	Unk	Int-A	1	Xylazine	2.6 ng/mL In Blood (unspecified) @ Unknown
1950ha	37 y M	Cocaine	1	1						
		Cocaethylene	2	2						
		Marijuana, N.O.S.	3	3						
					A/C	Ingst + Inhal + Unk	Int-A	1		
		Cocaine	1	1					Benzoyllecognine	840 ng/mL In Serum @ Unknown
		Buprenorphine	2	2						
		Buprenorphine	2	2					Buprenorphine	3.7 ng/mL In Serum @ Unknown
1951ai	37 y F				U	Unk	Int-A	1		
1952	37 y M	Methamphetamine	1	1						
		Cocaine	1	1	A	Ingst	Int-A	1	Cocaine	8500 ng/mL In Blood (unspecified) @ Autopsy
1953ai	37 y F				U	Unk	Int-A	1		
1954ai	37 y F	Cocaine	1	1	U	Unk	Int-A	1		
1955ai	37 y F	Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
1956ai	37 y M				U	Unk	Int-A	1		
1957ai	37 y M	Methamphetamine	1	1						
		Buprenorphine	2	2						
		Sildenafil	3	3						
1958ai	37 y F				U	Unk	Int-A	1		
1959ai	37 y M	Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
1960	37 y M	Substance (non-drug), N.O.S.	1	1						
			2	2						
1961p	38 y F	Cocaine	1	1	U	Unk	Int-A	1		
1961p	38 y F	Methamphetamine	1	1	A	Ingst	Int-A	2		
		Street drug, N.O.S.	1	1	A	Ingst	Int-S	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
1962h	38 y M	Cocaine	1	1	A	Unk	Int-A	1		
		Drug, N.O.S.	2	2						
1963ai	38 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methadone	2	2						
1964ph	38 y M	Methamphetamine	1	1	A	Ingst	Int-A	2		
1965ai	38 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
1966pai	38 y F	Methamphetamine	1	1	A	Ingst + Inhal	Int-A	1		
		Mushroom (processed, muscimol/ibotenic acid/hallucinogenic tryptamines/caffeine)	2	2						
		Mushroom (processed, muscimol/ibotenic acid/caffeine)	3	3						
		Trazodone	4	4						
		Lorazepam	5	5						
		Ethanol (beverage)	6	6						
1967ha	38 y M	Methamphetamine	1	1	U	Unk	Int-A	3	Amphetamine	0.079 mg/L In Blood (unspecified) @ 4 h (pe)
		Methamphetamine	1	1					Methamphetamine	0.86 mg/L In Blood (unspecified) @ 4 h (pe)
1968ai	39 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
1969ai	39 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
1970ai	39 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
1971ai	39 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Gamma-hydroxybutyric acid	2	2						
		Marijuana, N.O.S.	3	3						
1972ai	39 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
1973ai	39 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
1974	39 y M	Cocaine	1	1	A	Unk	Unk	2		
1975ai	40 y M	Methamphetamine	1	1	U	Unk	Int-A	3		
		Marijuana, N.O.S.	2	2						
1976ph	40 y M	Methamphetamine	1	1	A	Ingst	Int-A	2		
		Fentanyl (non-prescription)	2	2						
1977ai	40 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
1978ai	40 y M	Methamphetamine	1	1	U	Unk	Int-S	1		
1979ai	40 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
1980ph	41 y M	Cocaine	1	1	U	Unk	Int-A	2		
		Fentanyl (prescription)	2	2						
1981ha	41 y M				A	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		<i>Mitragyna speciosa korthals</i>	1	1					Mitragynine	6.3 mg/L In Blood (unspecified) @ Autopsy
1982ai	41 y M	Drug, N.O.S.	2	2	U	Unk	Int-A	1		
1983ha	41 y F	Methamphetamine	1	1	U	Unk	Int-A	1	Methamphetamine	10000 ng/mL In Blood (unspecified) @ Unknown
		Methamphetamine	1	1					Amphetamine	81 ng/mL In Blood (unspecified) @ Unknown
		Methamphetamine	1	1						
1984h	41 y M	Ibuprofen	2	2	U	Unk	Unk	2		
1985ai	41 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
1986ai	41 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
1987ai	41 y F	Marijuana, N.O.S.	2	2	U	Unk	Int-A	1		
1988ai	41 y M	Methamphetamine	1	1	U	Unk	Int-A	2		
1989i	41 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
1990h	41 y M				A	Ingst + Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2					Ethanol	210 mg/dL In Serum @ Unknown
1991ai	42 y F				U	Unk	Int-A	1		
1992ai	42 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Amlodipine	2	2						
1993ai	42 y M				U	Unk	Int-A	1		
1994ai	42 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
1995ai	43 y F				A	Unk	Int-A	1		
		Methamphetamine	1	1						
[1996ha]	43 y M				A/C	Unk	Int-A	1		
		Methamphetamine	1	1					Amphetamine	62 ng/mL In Blood (unspecified) @ Unknown
		Methamphetamine	1	1					Methamphetamine	780 ng/mL In Blood (unspecified) @ Unknown
1997ai	43 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
1998ai	43 y F				U	Unk	Int-S	1		
		Cocaine	1	1						
		Mirtazapine	2	2						
		Ethanol (beverage)	3	3						
1999ha	43 y M				U	Unk	Int-S	1		
		Cocaine	1	1					Ecgonine methyl ester	360 ng/mL In Serum @ Unknown
		Cocaine	1	1					Benzoylcognine	510 ng/mL In Serum @ Unknown
2000pha	44 y M				A	Ingst	Int-M	3		
		Cocaine	2	1						
		Methamphetamine	1	1						
2001ai	44 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2002ai	44 y F				U	Unk	Int-A	1		
		Cocaine	1	1						
2003ai	44 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2004pi	44 y M				A	Ingst	Int-A	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2005ph	44 y M	Mushroom (processed, N.O.S.)	1	1						
					U	Ingst	Int-S	2		
2006ph	44 y F	Cocaine	1	1	A/C	Ingst	Int-S	2		
		Lisdexamfetamine (ER)	1	1						
		Tizanidine	2	2						
		Bupropion (ER)	3	3						
2007ai	45 y F				U	Unk	Int-A	1		
2008i	45 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Methanol	2	2						
		Tramadol	3	3						
2009	45 y M				A	Derm	Int-A	3		
		Methamphetamine	1	1						
		Hydrochloric acid	2	2						
		Cocaine	3	3						
2010ai	45 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2011ai	45 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2012ai	45 y F				U	Unk	Int-A	1		
		Cocaine	1	1						
2013ai	45 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2014h	45 y M				A	Inhal + Unk	Int-A	2		
		Cocaine	1	1						
		Fentanyl (prescription)	2	2						
2015h	45 y F				A/C	Unk	Int-A	2		
		Methamphetamine	1	1						
2016h	45 y F				A	Ingst	Int-A	2		
		Amphetamine (hallucinogenic), 3,4-methylenedioxyamphetamine (MDMA)	1	1						
2017ai	46 y M				U	Unk	Int-A	1		
2018ai	46 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2019ha	46 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2020ai	47 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2021ai	47 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2022ai	47 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2023ai	47 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2024ai	47 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2025ai	47 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
2026ai	47 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2027ai	47 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2028ai	48 y M	Ethanol (beverage)	3	3						
		Methamphetamine	1	1	U	Unk	Int-A	1		
2029ai	48 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2030i	48 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2031ai	49 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2032ai	49 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
2033h	49 y F	Cocaine	1	1		Unk	Int-A	2		
		Fentanyl (prescription)	2	2						
2034ai	49 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
2035a	49 y M	Cocaine	1	1	A	Ingst	Int-M	1		
2036h	49 y M	Methamphetamine	1	1	A	Ingst	Int-A	3	Methamphetamine	670 ng/mL In Blood (unspecified) @ 5 m (pe)
		Methamphetamine	1	1					Amphetamine	85 ng/mL In Blood (unspecified) @ 5 m (pe)
		Mushroom (unknown)	2	1						
2037ai	50 y F	Cocaine	1	1	U	Unk	Int-A	1		
2038ai	50 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2039ai	50 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
2040ai	50 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2041ai	50 y M	Cocaine	1	1	U	Unk	Int-A	1		
2042ai	50 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2043ai	50 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Diphenhydramine	2	2						
		Hydroxyzine	3	3						
2044ph	50 y M	Cocaine	1	1	A	Unk	Unk	2		
2045i	50 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
2046	50 y M	<i>Mitragyna speciosa korthals</i>	2	1	C	Ingst	AR-D	3		
		Nonsteroidal antiinflammatory drug, N.O.S.	1	2						
2047ai	51 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2048ai	51 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Marijuana, N.O.S.	3	3						
2049ai	51 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2050h	51 y F	Amphetamine/dextro-amphetamine	1	1	U	Ingst	Int-S	2		
		Acetaminophen	2	2					Acetaminophen (apap)	303 mcg/mL In Blood (unspecified) @ Unknown
		Benzodiazepine, N.O.S.	3	3						
2051ai	51 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Marijuana, N.O.S.	3	3						
2052ai	51 y M				U	Unk	Int-A	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2053ai	51 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
2054ai	51 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Substance (non-drug), N.O.S.	2	2						
		Ethanol (beverage)	3	3						
2055ai	51 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Benzotropine	2	2						
		Mirtazapine	3	3						
2056ai	52 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
2057ai	52 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2058ai	52 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2059i	52 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Diphenhydramine	2	2						
		Pseudoephedrine	3	3						
2060ai	52 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
2061i	52 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2062ai	52 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2063ai	52 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Cocaine	2	2						
		Substance (non-drug), N.O.S.	3	3						
2064h	52 y M				A/C	Ingst + Unk	Unk	2		
		Amphetamine/dextro amphetamine	1	1						
2065ph	52 y M				U	Unk	Int-A	2		
		Cocaine	1	1						
		Stimulant/street drug, N.O.S.	2	2						
		Marijuana, N.O.S.	3	3						
2066ai	53 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2067ai	53 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2068ai	53 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2069ai	53 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Ethanol (beverage)	2	2						
2070ai	54 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2071ai	54 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Ethanol (beverage)	2	2						
		Marijuana, N.O.S.	3	3						
2072ai	54 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2073ai	54 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2074ai	54 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2075h	54 y M				U	Unk	Int-A	2		
		Cocaine	1	1						
		Ethanol (beverage)	2	2						
2076ai	54 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2077ai	54 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2078ph	54 y M				U	Unk	Unk	2		
		Amphetamine	1	1						
2079ai	55 y F				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2080ai	55 y F	Methamphetamine	1	1						
		Methamphetamine	1	1	U	Unk	Int-A	1		
		Morphine	2	2						
2081ai	55 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2082ai	55 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
2083h	55 y M				A	Par	Int-A	2		
		Methamphetamine	1	1						
2084h	55 y M				U	Unk	Unt-E	2		
		Stimulant/street drug, N.O.S	1	1						
2085ai	55 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Morphine	2	2						
		Chlordiazepoxide	3	3						
2086ai	55 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2087ai	56 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2088ai	56 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Ethanol (beverage)	2	2						
		Amlodipine	3	3						
2089ai	56 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Trazodone	2	2						
		Citalopram	3	3						
2090ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2091ai	56 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Alprazolam	2	2						
		Ethanol (beverage)	3	3						
2092ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2093ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2094ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2095ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2096ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Buprenorphine	2	2						
		Gabapentin	3	3						
2097ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2098ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2099ai	56 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2100ha	56 y M				A/C	Ingst	Int-S	2		
		Methamphetamine	1	1					Amphetamine	11 ng/mL In Whole Blood @ Autopsy
		Methamphetamine	1	1					Methamphetamine	150 ng/mL In Whole Blood @ Autopsy
		Metoprolol	3	1						
		Morphine	2	2					Morphine (free)	23 ng/mL In Whole Blood @ Autopsy
2101ai	57 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2102ai	57 y F				U	Unk	Int-A	1		
		Cocaine	1	1						
		Diphenhydramine	2	2						
2103ai	57 y M				U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2104ai	57 y M	Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2105ai	57 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2106ai	57 y M	Methamphetamine	1	1	U	Unk	Int-A	2		
2107ai	57 y M	Cocaine	1	1	U	Unk	Int-A	1		
2108ai	57 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2109ai	57 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Diphenhydramine	2	2						
2110ai	57 y M	Marijuana, N.O.S.	3	3	U	Unk	Int-A	1		
2111p	57 y M	Methamphetamine	1	1	A	Ingst	Int-A	2		
2112pha	57 y M	Marijuana, N.O.S.	1	1	A	Ingst + Unk	Int-A	1		
		Cocaine	1	1						
		Methamphetamine	2	2						
		Ethanol (beverage)	3	3					Ethanol	98 mg/dL In Blood (unspecified) @ Unknown
2113ai	58 y F	Gabapentin	4	4						
		Methocarbamol	5	5						
2114ai	58 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2115ai	58 y F	Methamphetamine	1	1						
		Diphenhydramine	2	2						
2116ai	58 y M	Cocaine	1	1	U	Unk	Int-A	1		
2117ai	58 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Cocaine	2	2						
2118ai	58 y F	Methamphetamine	1	1	U	Unk	Int-A	2		
2119ai	58 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2120ai	58 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2121ha	58 y M	Methamphetamine	1	1	A	Ingst + Inhal	Int-A	1		
		Morphine	2	2						
		Methamphetamine	1	1					Methamphetamine	2945 ng/mL In Blood (unspecified) @ Autopsy
		Methamphetamine	1	1					Amphetamine	90.1 ng/mL In Blood (unspecified) @ Autopsy
		Ethanol (beverage)	2	2					Ethanol	14 mg/dL In Blood (unspecified) @ Unknown
2122ai	58 y F	Marijuana	3	3	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2123a	58 y F				A	Ingst	Oth-W	3		
2124ai	59 y F	Cocaine	1	1	U	Unk	Int-A	1		
2125ai	59 y F	Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
2126i	59 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2127ai	59 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2128ai	59 y M	Methamphetamine	1	1	U	Unk	Int-S	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2129ai	59 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Citalopram	2	2						
		Mirtazapine	3	3						
2130ai	59 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Cocaehtylene	2	2						
		Ethanol (beverage)	3	3						
2131i	59 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Acetaminophen	3	3						
2132ai	60 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
2133pa	2 m M	Methamphetamine	1	1	U	Unk	Unt-U	2	Methamphetamine	0.014 mg/L In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	2	2						
2134ai	60 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2135ai	60 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
2136ai	60 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Acetaminophen	2	2						
		Duloxetine (ER)	3	3						
2137ai	60 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
2138ai	60 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
2139ai	60 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Cyclobenzaprine	2	2						
		Substance (non-drug), N.O.S.	3	3						
2140ai	60 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2141ai	60 y M	Ethanol (beverage)	2	2	U	Unk	Int-A	1		
		Methamphetamine	1	1						
2142pi	60 y M	Methamphetamine	1	1	A	Inhal	Int-M	2		
2143ai	61 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
2144ai	61 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
2145ai	61 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Oxycodone	2	2						
2146ai	61 y M	Oxymorphone	3	3	U	Unk	Int-A	2		
		Methamphetamine	1	1						
2147ai	61 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2148ai	61 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Diphenhydramine	2	2						
2149pai	61 y M	Methamphetamine	1	1	A/C	Unk	Int-A	1	Methamphetamine	1300 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	2	2					Amphetamine	310 ng/mL In Blood (unspecified) @ Autopsy
2150ai	62 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2151ai	62 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
2152ai	62 y F	Methamphetamine	1	1	U	Unk	Int-A	3		
2153ai	62 y M	Methamphetamine	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Methamphetamine	1	1						
		Fentanyl (non-prescription)	2	2						
2154ai	63 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2155ai	63 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2156ai	64 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2157ai	64 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2158i	64 y M				U	Inhal+Par	Int-A	2		
		Cocaine	1	1						
2159ph	64 y M				U	Unk	Int-U	2		
		Stimulant/street drug, N.O.S.	1	1						
		Ethanol (beverage)	2	2					Ethanol	110mg/dL In Serum @ Unknown
2160ai	64 y M				U	Unk	Int-A	2		
		Methamphetamine	1	1						
2161ai	64 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2162ai	64 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Cyclobenzaprine	2	2						
		Aripiprazole	3	3						
2163ai	64 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Substance (non-drug), N.O.S.	2	2						
		Marijuana, N.O.S.	3	3						
2164ai	64 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
2165ai	65 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
2166ai	65 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2167ai	65 y M				A	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2168ai	65 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2169ai	65 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2170ai	65 y M				U	Unk	Int-A	1		
		Cocaine	1	1						
		Sildenafil	2	2						
		Ethanol (beverage)	3	3						
2171ai	65 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2172ai	65 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2173ai	65 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2174pha	65 y M				A	Inhal	Int-A	3		
		Cocaine	1	1						
2175ai	66 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Salicylate	2	2						
2176ai	66 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
2177ai	66 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Cyclobenzaprine	2	2						
2178ai	66 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Ethanol (beverage)	2	2						
2179ai	66 y M				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2180ai	66 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
2181ai	67 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2182i	67 y M	Methamphetamine	1	1	U	Unk	Int-A	3		
		Marijuana, N.O.S.	2	2						
2183ai	67 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethyl chloride	2	2						
		Donepezil	3	3						
2184ai	67 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2185ai	67 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Ethanol (beverage)	2	2						
		Marijuana, N.O.S.	3	3						
2186ai	67 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Quetiapine	2	2						
		Gabapentin	3	3						
2187h	67 y F	Methamphetamine	1	1	U	Ingst + Oth	Int-U	2		
		Substance (non-drug), N.O.S.	2	2						
2188ai	68 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2189ai	68 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2190ai	68 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2191ai	68 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Oxycodone	2	2						
		Ethanol (beverage)	3	3						
2192i	68 y M	Methamphetamine	1	1	U	Unk	Int-M	1		
2193ai	68 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
2194ai	68 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2195ai	68 y M	Cocaine	1	1	U	Unk	Int-A	1		
		Acetaminophen	2	2						
2196ai	68 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Marijuana, N.O.S.	2	2						
2197h	68 y F	Methamphetamine	1	1	A	Ingst	Int-A	2		
2198ai	69 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2199ai	69 y M	Methamphetamine	1	1	U	Unk	Int-S	1		
		Substance (non-drug), N.O.S.	2	2						
2200ai	69 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
2201ai	69 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
2202ai	70 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
2203ai	70 y M	Amphetamine	1	1	U	Unk	Int-A	1		
2204ai	71 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
2205h	71 y F	Methylphenidate	1	1	A/C	Ingst	Int-S	2		
		Atropine/diphenoxylate	2	2						
		Zolpidem	3	3						
2206ai	72 y M	Methamphetamine	1	1	U	Unk	Int-A	1		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2207ai	72 y M	Ethanol (beverage)	2	2						
2208ai	73 y F	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1	U	Unk	Int-A	1		
		Substance (non-drug), N.O.S.	2	2						
2209ai	74 y M				U	Unk	Int-A	1		
2210ai	75 y M	Methamphetamine	1	1	U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Substance (non-drug), N.O.S.	2	2						
2211ai	76 y F				U	Unk	Int-A	1		
		Methamphetamine	1	1						
		Aripiprazole	2	2						
		Gabapentin	3	3						
2212h	Unknown adult (≥ 20 yrs) M				A	Unk	Int-A	2		
		Cocaine	1	1						
		Methamphetamine	2	1						
		Drug, N.O.S.	3	2						
2213ph	Unknown adult (≥ 20 yrs) U				A	Par	Int-A	2		
		Methamphetamine	1	1						
		Fentanyl (non-prescription)	2	2						
2214ai	Unknown age M				U	Unk	Unt-G	1		
		Methamphetamine	1	1						
2215ai	Unknown age F				U	Unk	Unt-G	1		
		Methamphetamine	1	1						
2216ai	Unknown age F				U	Unk	Unt-G	1		
		Methamphetamine	1	1						
2217ai	Unknown age M				U	Unk	Unt-G	1		
		Methamphetamine	1	1						
2218ai	Unknown age F				U	Unk	Unt-G	1		
		Methamphetamine	1	1						
2219ai	Unknown age M				U	Unk	Unt-G	1		
		Methamphetamine	1	1						
		Marijuana, N.O.S.	2	2						
See Also case 1, 6, 31, 41, 48, 52, 67, 203, 221, 223, 244, 284, 299, 325, 331, 336, 341, 342, 344, 345, 351, 352, 354, 357, 359, 360, 363, 364, 366, 369, 370, 371, 373, 374, 375, 376, 377, 379, 381, 382, 384, 386, 387, 390, 391, 393, 394, 396, 398, 399, 401, 402, 403, 404, 405, 407, 410, 412, 416, 420, 421, 423, 424, 426, 427, 428, 429, 435, 437, 439, 444, 449, 450, 451, 453, 454, 459, 463, 464, 465, 471, 474, 477, 478, 479, 480, 483, 485, 486, 487, 491, 492, 493, 494, 495, 496, 497, 499, 500, 502, 503, 506, 511, 514, 515, 516, 517, 518, 519, 520, 521, 523, 527, 528, 530, 531, 532, 533, 534, 535, 537, 541, 542, 543, 544, 547, 549, 550, 552, 553, 555, 556, 561, 562, 564, 571, 572, 573, 576, 578, 579, 580, 581, 584, 585, 586, 587, 588, 589, 590, 592, 593, 594, 597, 599, 600, 605, 607, 608, 614, 616, 617, 618, 619, 624, 625, 627, 631, 634, 635, 636, 637, 640, 641, 646, 651, 652, 653, 655, 660, 661, 662, 664, 667, 668, 669, 673, 676, 679, 680, 681, 682, 687, 688, 690, 691, 692, 696, 697, 698, 700, 703, 704, 705, 706, 707, 709, 711, 712, 714, 715, 717, 719, 723, 725, 731, 733, 734, 736, 737, 738, 740, 741, 744, 747, 748, 751, 752, 753, 754, 756, 758, 760, 765, 766, 768, 769, 773, 777, 778, 779, 781, 782, 786, 789, 790, 791, 792, 793, 796, 797, 801, 804, 805, 806, 807, 808, 809, 810, 819, 821, 824, 825, 826, 827, 829, 832, 835, 837, 842, 843, 844, 845, 846, 849, 850, 853, 854, 855, 860, 862, 863, 869, 875, 876, 880, 886, 892, 893, 898, 899, 906, 908, 910, 911, 914, 915, 917, 919, 920, 921, 923, 924, 930, 933, 936, 940, 941, 942, 947, 949, 951, 952, 954, 957, 960, 964, 970, 971, 972, 983, 984, 990, 991, 996, 997, 998, 1005, 1007, 1009, 1010, 1011, 1013, 1014, 1023, 1024, 1025, 1026, 1027, 1029, 1030, 1031, 1035, 1040, 1044, 1046, 1047, 1048, 1053, 1061, 1062, 1073, 1081, 1082, 1086, 1088, 1091, 1092, 1106, 1125, 1128, 1129, 1135, 1136, 1156, 1175, 1180, 1246, 1260, 1263, 1264, 1265, 1267, 1288, 1318, 1321, 1324, 1349, 1350, 1353, 1356, 1367, 1371, 1373, 1385, 1386, 1394, 1432, 1434, 1445, 1474, 1485, 1490, 1508, 1524, 1564, 1567, 1593, 1620, 1727, 1748, 1759, 1763, 1773, 1806, 1817, 1838, 1841, 1857, 1874, 2224, 2227, 2231, 2235, 2241, 2263										
Unknown Drug										
2220p	3 y M				A	Unk	Unt-G	2		
		Drug, N.O.S.	1	1						
2221p	20 y F				A	Ingst	Int-S	3		
		Drug, N.O.S.	1	1						
2222ph	21 y M				U	Par + Unk	Int-S	2		
		Drug, N.O.S.	1	1						
		Fentanyl (non-prescription)	2	2						
2223i	22 y M				A	Unk	Int-A	2		
		Drug, N.O.S.	1	1						
		Opioid, N.O.S.	2	2						
2224pha	23 y F				A	Unk	Unk	3		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2225p	23 y F	Drug, N.O.S.	1	1	A	Unk	Int-A	2	Ketamine	1843 ng/mL In Blood (unspecified) @ Unknown
		Ketamine	2	2						
		Ketamine	2	2					Norketamine	801 ng/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	3	3					Delta-9-carboxy-thc	129 ng/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	3	3					Delta-9-thc	18 ng/mL In Blood (unspecified) @ Unknown
		Marijuana, N.O.S.	3	3					11-oh-thc (11-hydroxy-delta-9-tetrahydrocannabinol)	6.7 ng/mL In Blood (unspecified) @ Unknown
		Drug, N.O.S.	1	1					Ethanol	35 mg/dL In Blood (unspecified) @ Unknown
		Ethanol (beverage)	2	2						
2226i	27 y M				U	Unk	Int-A	2		
2227ph	27 y M	Drug, N.O.S.	1	1	A	Ingst	AR-F	3		
		Codeine	2	2						
		Ethanol (beverage)	3	3						
		Marijuana	4	4						
2228ph	28 y M				U	Unk	Int-S	2		
		Drug, N.O.S.	1	1						
		Acetaminophen/diphenhydramine	2	2						
		Ethanol (beverage)	3	3						
2229ph	29 y F	Fentanyl (non-prescription)	4	4	A	Ingst	Int-S	2	Ethanol	108 mg/dL In Serum @ Unknown
		Drug, N.O.S.	1	1						
		Ethanol (beverage)	2	2						
2230h	30 y M				U	Unk	Int-U	2		
2231	30 y M	Drug, N.O.S.	1	1	U	Ingst	Unk	2		
		Cocaine	2	2						
2232ph	31 y F	Drug, N.O.S.	1	1	U	Unk	Int-A	2		
		Fentanyl (non-prescription)	2	2						
2233a	32 y F				U	Ingst + Unk	Unk	2	Lorazepam	11 ng/mL In Blood (unspecified) @ Autopsy
		Drug, N.O.S.	1	1					Norfentanyl	2.1 ng/mL In Blood (unspecified) @ Autopsy
		Drug, N.O.S.	1	1					Fentanyl	2.3 ng/mL In Blood (unspecified) @ Autopsy
		Cyclobenzaprine	2	2					Cyclobenzaprine	130 ng/mL In Blood (unspecified) @ Autopsy
		Zolpidem	3	3					Zolpidem	40 ng/mL In Blood (unspecified) @ Autopsy
		Clonazepam	4	4					7-aminoclonazepam	8.3 ng/mL In Blood (unspecified) @ Autopsy
		Alprazolam	5	5					Alprazolam	12 ng/mL In Blood (unspecified) @ Autopsy
2234pha	32 y M	Drug, N.O.S.	1	1	A/C	Ingst	Int-S	2	Trazodone	2200 ng/mL In Blood (unspecified) @ Autopsy
		Trazodone	2	2						
2235pha	34 y M				U	Unk	Unk	1		
		Drug, N.O.S.	1	1						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
		Cocaine	2	2					Cocaine	1000 ng/mL In Urine (quantitative only) @ Autopsy
		Cocaine	2	2					Benzoylecognine	1300 ng/mL In Blood (unspecified) @ Autopsy
		Cocaine	2	2					Benzoylecognine	16000 ng/mL In Urine (quantitative only) @ Autopsy
		Fentanyl (non-prescription)	3	3					Acetyl fentanyl	0.56 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	3	3					Norfentanyl	2.4 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	3	3					Fentanyl	31 ng/mL In Blood (unspecified) @ Autopsy
		Fentanyl (non-prescription)	3	3					4-anpp (despropionyl fentanyl)	9.3 ng/mL In Blood (unspecified) @ Autopsy
		Buprenorphine	4	4					Buprenorphine	71 ng/mL In Urine (quantitative only) @ Autopsy
		Clonazepam	5	5					7-aminoclonazepam	26 ng/mL In Blood (unspecified) @ Autopsy
		Clonazepam	5	5					Clonazepam	6.7 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	6	6					11-oh-thc (11-hydroxy-delta-9-tetrahydrocannabinol)	2.1 ng/mL In Blood (unspecified) @ Autopsy
		Marijuana, N.O.S.	6	6					Delta-9-thc	3.9 ng/mL In Blood (unspecified) @ Autopsy
		Amphetamine/dextroamphetamine	7	7					Amphetamine	10000 ng/mL In Blood (unspecified) @ Autopsy
		Amphetamine/dextroamphetamine	7	7					Amphetamine	10000 ng/mL In Urine (quantitative only) @ Autopsy
		Amphetamine/dextroamphetamine	7	7					Amphetamine	130 ng/mL In Blood (unspecified) @ Autopsy
2236i	34 y M	Drug, N.O.S.	1	1	U	Unk	Int-A	2		
2237ph	35 y M	Drug, N.O.S.	1	1	U	Unk	Unk	2		
2238	37 y F	Drug, N.O.S.	1	1	A	Ingst	Int-M	2		
2239ph	37 y M	Drug, N.O.S.	1	1	A	Unk	Int-A	2		
2240i	38 y M	Drug, N.O.S.	1	1	A	Unk	Int-U	2		
2241ph	38 y F	Drug, N.O.S.	1	1	U	Ingst + Unk	Int-S	2		
		Lurasidone	2	2						
		Sertraline	3	3						
		Zolpidem	4	4						
		Lorazepam	5	5						
		Buspirone	6	6						
		Amphetamine/dextroamphetamine	7	7						
2242h	38 y M	Ethanol (beverage)	8	8	U	Unk	Int-S	2		
2243ph	40 y M	Drug, N.O.S.	1	1	A/C	Unk	Unk	2		
2244ph	40 y M	Drug, N.O.S.	1	1	A	Unk	Unk	2		
2245	40 y F	Drug, N.O.S.	1	1	A	Ingst	Int-A	2		

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2246h	40 y M	Drug, N.O.S.	1	1						
		Drug, N.O.S.	3	1	C	Ingst + Unk	Int-S	2		
		Bupropion	2	2						
		Metoprolol	1	3						
		Clonidine	4	4						
2247ai	42 y F	Drug, N.O.S.	1	1	U	Unk	Int-A	1		
2248p	43 y F	Drug, N.O.S.	1	1	U	Unk	Int-A	2		
		Fentanyl (non-prescription)	2	2						
2249h	44 y F	Drug, N.O.S.	1	1	A	Ingst	Int-S	2		
		Acetaminophen	2	2					Acetaminophen (apap)	65 mcg/mL In Serum @ Unknown
		Acetaminophen	2	2					Acetaminophen (apap)	85 mg/L In Serum @ Unknown
2250i	45 y M	Drug, N.O.S.	1	1	A	Unk	Int-A	2		
2251ph	45 y M	Drug, N.O.S.	1	1	U	Unk	Unk	2		
2252ph	50 y M	Drug, N.O.S.	1	1	A	Oth	Int-A	2		
		Drug, N.O.S.	1	1					Ethanol	40 mg/dL In Serum @ Unknown
		Ethanol (beverage)	2	2						
2253h	52 y F	Drug, N.O.S.	1	1	U	Ingst	Int-S	2	Acetaminophen (apap)	83 mcg/mL In Serum @ Unknown
		Amiodarone	2	2						
		Digoxin	3	3						
		Phenobarbital	4	4						
2254i	53 y M	Drug, N.O.S.	1	1	U	Unk	Unk	2		
2255i	53 y M	Drug, N.O.S.	1	1	A	Unk	Int-A	2		
2256i	54 y M	Drug, N.O.S.	1	1	A	Unk	Unk	2		
2257pha	55 y F	Drug, N.O.S.	1	1	A/C	Ingst	Unk	2		
2258	55 y F	Drug, N.O.S.	1	1	A	Ingst	Int-S	2		
2259pha	56 y F	Drug, N.O.S.	1	1	U	Ingst	Int-S	1		
		Quetiapine	2	2					Quetiapine	2.8 mg/L In Blood (unspecified) @ Autopsy
		Clonazepam	3	3					Clonazepam	0.11 mg/L In Blood (unspecified) @ Autopsy
		Clonazepam	3	3					Aminoclonazepam	0.16 mg/L In Blood (unspecified) @ Autopsy
		Venlafaxine	4	4					O-desmethyl-venlafaxine	1.8 mg/L In Blood (unspecified) @ Autopsy
2260ha	58 y M	Drug, N.O.S.	1	1	A	Ingst	Unk	3		
2261ph	58 y M	Drug, N.O.S.	1	1	C	Ingst + Unk	Int-U	2		
		Ethanol (beverage)	2	2					Ethanol	317 mg/dL In Blood (unspecified) @ Unknown
2262ai	60 y F	Drug, N.O.S.	1	1	U	Unk	Int-A	1		
2263pha	60 y F	Drug, N.O.S.	3	1	U	Ingst	Int-A	1		
		Fentanyl (non-prescription)	2	2						
		Zolpidem	4	2						
		Quetiapine	5	3						
		Cocaine	1	4						

(Continued)

Table 21. Continued.

Annual Report ID	Age	Substances	Substance Rank	Cause Rank	Chronicity	Route	Reason	RCF	Analyte	Blood Concentration @ Time
2264i	61 y F	Drug, N.O.S.	1	1	A	Unk	Int-U	2		
2265h	64 y F	Drug, N.O.S.	1	1	A	Unk	Unk	2		
		Ethanol (beverage)	2	2					Ethanol	94 mg/dL In Serum @ Unknown
2266h	67 y F	Drug, N.O.S.	1	1	U	Unk	Unk	2		
		Acetaminophen	2	2					Acetaminophen (apap)	21 mcg/mL In Serum @ 2 h (pe)
2267	69 y F	Drug, N.O.S.	1	1	A	Ingst	Int-S	2		
		Oxycodone	2	2						
2268	82 y M	Drug, N.O.S.	1	1	A	Ingst	Int-S	2	Acetaminophen (apap)	678 mcg/mL In Blood (unspecified) @ Unknown
		Acetaminophen	2	2						
2269ph	88 y F	Drug, N.O.S.	1	1	A/C	Ingst	Int-S	3		
		Ethanol (beverage)	2	2					Ethanol	80 mg/dL In Blood (unspecified) @ Unknown
2270h	20+ y M	Drug, N.O.S.	1	1	U	Unk	Unk	2		
2271	30+ y F	Drug, N.O.S.	1	1	A	Ingst	Int-A	3		

See Also case 7, 39, 42, 51, 54, 80, 85, 363, 457, 461, 524, 721, 723, 851, 865, 1006, 1015, 1066, 1083, 1090, 1113, 1146, 1260, 1269, 1282, 1288, 1302, 1319, 1343, 1350, 1376, 1501, 1526, 1535, 1549, 1573, 1654, 1767, 1871, 1885, 1919, 1941, 1942, 1962, 1981, 2212

Listing of 2,622 (1,491 Direct + 1,131 Indirect) fatalities classified as Relative Contribution to Fatality category 1-Undoubtedly responsible, 2-Probably responsible, or 3-Contributory.

Annual Report ID: Bracketed [case number]=Narrative provided for this case in [Appendix C](#) i=death, indirect report; identified through other sources (news feeds, medical examiner data, or other) about which no inquiry to the PC was made, **p**=prehospital cardiac and/or respiratory arrest, **h**=hospital records reviewed, **a**=autopsy report reviewed.

Age Gender: **y** years, **m** months, **d** days, **F** female, **M** male, **F-Pregnant** pregnant, **U** unknown

Substance: **EC** enteric coated, **ER** extended release, **TD** transdermal

Cause Rank: Designates and orders the substance(s) deemed to have made the greatest contribution to the death. Cause rank permits 2 or more substances to be judged indistinguishable in terms of contribution to the death. Additional agents implicated in the death are listed below the primary agent(s) in the order of their contribution to the fatality.

Chronicity: **C**=chronic exposure, **A**=acute exposure, **A/C**=acute on chronic, **U**=unknown

Route: **Aspir**=Aspiration (with ingestion), **B-S**=Bite/sting, **Derm**=Dermal, **Ingst**=Ingestion, **Inhal**=Inhalation/nasal, **Oc**=Ocular, **Ot**=Otic, **Oth**=Other, **Par**=Parenteral, **Rec**=Rectal, **Unk**=Unknown, **Vag**=Vaginal

Reason: **AR-D**=Adverse reaction – Drug, **AR-F**=AR – Food, **AR-O**=AR – Other, **Int-A**=Intentional – Abuse, **Int-M**=Int – Misuse, **Int-S**=Int – Suspected Suicide, **Int-U**=Int – Unknown, **Oth-C**=Other – Contamination/tampering, **Oth-M**=Oth – Malicious, **Oth-W**=Oth – Withdrawal, **Unk**=Unknown reason, **Unt-B**=Unintentional – Bite/sting, **Unt-E**=Unt – Environmental, **Unt-F**=Unt – Food poisoning, **Unt-G**=Unt – General, **Unt-M**=Unt – Misuse, **Unt-O**=Unt – Occupational, **Unt-T**=Unt – Therapeutic error, **Unt-U**=Unt – Unknown

RCF (Relative Contribution to Fatality): 1=Undoubtedly responsible, 2=Probably responsible, 3=Contributory. Provided by the RPC for Indirect cases and the America's Poison Centers Fatality Review Team for the direct (non-Indirect) cases.

Appendix B

Table 22(A) & Table 22(B). Demographic profile of single substance exposure cases by generic category
Nonpharmaceuticals (Table 22(A))

Table 22A. Demographic profile of single substance nonpharmaceuticals exposure cases by generic category.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Adhesives/Glues																			
Miscellaneous Adhesives/Glues																			
Cyanoacrylates (superglues, etc)	4,581	4,491	1,727	327	370	1,628	11	386	42	4,332	89	18	37	1,250	574	840	170	4	0
Epoxy	1,238	1,186	369	125	47	512	4	110	19	1,133	26	2	21	265	145	234	72	2	0
Non-toxic adhesives/glues (white glue, paper glue, etc)	859	789	400	249	59	64	4	10	3	699	86	2	0	49	80	51	0	0	0
Toluene/xylene (adhesives only)	180	176	44	9	6	88	0	28	1	157	9	4	5	39	25	37	12	1	0
Unknown types of adhesive, glue, cement or paste	3,404	3,191	1,208	353	173	1,097	22	313	25	2,968	124	29	58	516	459	443	99	2	0
Category total:	10,262	9,833	3,748	1,063	655	3,389	41	847	90	9,289	334	55	121	2,119	1,283	1,605	353	9	0
Alcohols																			
Miscellaneous alcohols																			
Ethanol (beverages)	52,827	8,967	2,533	303	1,091	4,261	11	588	180	3,513	4,356	578	298	3,598	894	1,473	1,370	363	49
Ethanol (non-beverage, non-rubbing)	1,361	816	383	32	40	306	1	49	5	685	99	7	7	161	143	87	33	14	3
Higher alcohols (butanol, amyl alcohol, propanols, etc)	127	72	25	7	6	28	0	5	1	66	5	0	0	20	10	21	6	0	0
Isopropanol (excluding rubbing alcohols and cleaning agents)	3,768	3,165	999	118	206	1,593	5	222	22	2,321	758	26	24	967	503	599	315	60	0
Methanol (excluding automotive products and cleaning agents)	705	566	67	15	26	375	0	76	7	476	46	14	5	227	101	119	32	14	9
Other types of alcohol	105	82	34	2	1	37	1	7	0	71	8	1	0	19	20	12	6	0	0
Unknown types of alcohol	852	245	39	3	23	146	1	24	9	63	131	21	14	148	19	44	49	23	4
Rubbing alcohols																			
Rubbing alcohols: ethanol with methyl salicylate	9	9	4	0	0	5	0	0	0	8	1	0	0	4	3	1	1	0	0
Rubbing alcohols: ethanol without methyl salicylate	284	256	112	8	10	105	0	20	1	209	40	2	3	58	37	39	12	2	0
Rubbing alcohols: isopropanol with methyl salicylate	131	119	68	3	5	36	2	3	2	103	13	1	1	35	29	19	8	1	0
Rubbing alcohols: isopropanol without methyl salicylate	7,649	6,657	2,713	262	367	2,868	10	394	43	5,272	1,244	37	44	1,706	1,037	1,155	448	49	0
Rubbing alcohols: unknown	338	275	64	8	20	156	1	24	2	171	93	4	2	115	19	54	39	7	0
Category total:	68,156	21,229	7,041	761	1,795	9,916	32	1,412	272	12,958	6,794	691	398	7,058	2,815	3,623	2,319	533	65
Arts/crafts/office supplies																			
Miscellaneous arts/crafts/office supplies																			
Artist paints (non-water color)	3,408	3,279	2,257	334	170	420	8	78	12	3,132	114	6	16	139	417	195	20	1	0
Artist paints (water color#00029; Chalks	1,054	1,022	777	140	37	56	4	8	0	969	45	1	3	18	137	24	0	0	0
Clays	543	526	449	41	16	12	3	2	3	510	13	1	1	35	67	27	1	0	0
Crayons	2,257	2,163	1,661	273	62	140	13	13	1	2,089	53	4	8	103	237	108	6	0	0
Glazes	1,325	1,240	848	250	62	66	10	4	0	1,180	50	4	2	37	107	30	1	0	0
Office supplies: miscellaneous	130	122	30	24	12	44	1	9	2	104	14	1	2	10	15	2	0	0	0
Other types of arts/crafts/writing products	64	61	37	8	1	13	0	2	0	56	2	2	1	5	6	5	1	0	0
	6,376	5,979	3,943	895	261	628	34	198	20	5,646	254	33	29	322	825	303	20	2	0
Pencils	1,219	1,153	338	604	132	45	13	15	6	928	182	21	5	49	138	73	3	0	0
Pens or inks	7,831	7,556	4,382	2,011	606	375	50	113	19	6,925	525	29	40	303	990	249	16	1	0
Typewriter correction fluids	241	233	123	43	27	36	0	3	1	196	29	4	0	18	49	14	1	0	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Unknown types of arts/crafts/ writing products	90	87	56	16	7	4	2	2	0	79	6	0	0	5	15	6	1	0	0
Category Total:	24,538	23,421	14,901	4,639	1,393	1,839	138	447	64	21,814	1,287	106	107	1,044	3,003	1,048	72	4	0
Automotive products																			
Automotive products: brake fluids	783	742	145	14	40	459	2	74	8	701	24	12	1	255	139	201	35	3	0
Automotive products: ethylene glycol (including antifreeze)	7,079	6,626	422	150	543	4,803	11	628	69	5,801	616	127	28	2,276	1,169	1,085	366	106	15
Automotive products: glycol and methanol mixtures	88	84	18	4	5	44	0	12	1	75	7	1	0	22	26	14	2	2	0
Automotive products: hydrocarbons (transmission fluids, power steering fluids, etc)	1,688	1,570	384	54	127	846	1	150	8	1,488	60	9	3	521	237	497	71	8	1
Automotive products: methanol (dry gas, windshield washing solutions, etc)	1,679	1,594	164	61	147	1,015	0	192	15	1,459	88	14	13	380	330	293	55	9	3
Automotive products: other glycols	233	223	56	22	15	108	0	14	8	210	6	4	1	42	34	33	5	0	0
Miscellaneous automotive/aircraft/boat products																			
Automotive/aircraft/boat products: non-toxic	15	14	7	1	0	4	0	2	0	14	0	0	0	2	4	2	0	0	0
Automotive/aircraft/boat products: other	1,213	1,157	314	32	64	656	2	81	8	1,112	23	11	9	352	187	309	64	4	0
Automotive/aircraft/boat products: unknown	124	113	21	2	6	60	0	23	1	101	6	3	3	48	12	34	8	0	0
Category total:	12,902	12,123	1,531	340	947	7,995	16	1,176	118	10,961	830	181	58	3,898	2,138	2,468	606	132	19
Batteries																			
Disc batteries																			
Disc batteries: alkaline (mno2)	336	326	204	43	10	57	0	12	0	302	13	8	1	228	180	21	4	0	0
Disc batteries: lithium	975	927	120	68	102	500	9	115	13	848	55	6	11	324	209	194	45	4	0
Disc batteries: mercuric oxide	2	2	1	0	0	1	0	0	0	2	0	0	0	1	0	0	0	0	0
Disc batteries: nickel cadmium	4	4	1	1	0	1	0	1	0	4	0	0	0	0	0	0	0	0	0
Disc batteries: other	21	20	11	4	1	2	1	1	0	19	1	0	0	11	5	1	0	0	0
Disc batteries: silver oxide	22	22	16	3	0	3	0	0	0	22	0	0	0	14	12	0	0	0	0
Disc batteries: unknown	1,892	1,836	1,215	252	83	240	8	33	5	1,688	102	17	2	1,360	768	130	36	10	0
Disc batteries: zinc-air	221	205	41	7	1	143	1	11	1	191	7	1	2	176	147	14	2	0	0
Miscellaneous batteries																			
Automotive/aircraft/boat batteries	1,119	1,099	76	34	63	742	4	166	14	1,069	14	2	8	321	125	298	77	2	0
Other types of battery	240	231	63	16	25	101	0	23	3	208	13	0	6	68	51	40	4	1	0
Penlight/flashlight/dry cell batteries	6,979	6,788	4,383	605	350	1,102	23	287	38	6,225	462	44	16	1,571	2,149	580	73	2	0
Unknown types of battery	62	55	19	7	4	22	0	3	0	53	2	0	0	16	20	10	0	0	0
Category total:	11,873	11,515	6,150	1,040	639	2,914	46	652	74	10,631	669	78	46	4,090	3,666	1,288	241	19	0
Bites and envenomations																			
Aquatic																			
Fish stings	443	439	16	41	56	298	3	21	4	435	1	0	1	215	7	134	65	2	0
Jellyfish and other coelenterate stings	296	294	46	62	40	129	2	14	1	294	0	0	0	77	1	94	28	3	0
Other or unknown marine animal bites and/or envenomations	750	728	439	51	29	178	5	24	2	695	16	11	4	115	92	67	24	0	0
Exotic snakes																			
Exotic snake: unknown if poisonous	5	5	0	0	2	3	0	0	0	5	0	0	0	3	0	1	1	0	0
Exotic snakes: non-poisonous	19	19	5	3	2	9	0	0	0	19	0	0	0	18	0	14	1	0	0
Exotic snakes: poisonous	57	54	0	3	3	47	0	1	0	53	1	0	0	46	2	16	20	4	0
Insects																			
Ant or fire ant bites	363	338	105	29	20	148	1	28	7	319	3	15	1	56	12	65	16	0	0

(Continued)

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age						Reason			Treated in		Outcome							
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death		
Miscellaneous bites and envenomations	2,753	2,648	400	232	119	1,524	7	323	43	2,644	0	2	2	433	35	772	185	13	4		
	794	788	218	97	51	355	4	54	9	780	1	0	7	164	22	237	36	0	0		
	236	235	73	14	7	121	1	16	3	232	0	2	1	37	16	78	9	0	0		
	90	79	20	9	5	38	2	4	1	79	0	0	0	12	2	26	2	1	0		
	3,056	2,958	667	336	134	1,464	6	295	56	2,675	10	246	14	585	110	677	167	1	0		
	7,866	7,854	1,025	772	538	4,971	4	486	58	7,851	0	2	1	1,219	104	3,965	506	28	0		
	601	583	124	51	32	281	3	83	9	583	0	0	0	138	24	99	9	0	0		
	Mammals																				
	Bat bites	1,103	1,086	125	101	83	614	18	119	26	1,076	2	0	0	629	264	132	1	0	0	
	Cat bites	998	988	89	94	100	605	6	84	10	988	0	0	0	684	28	404	23	1	0	
Miscellaneous snake bites and envenomations	3,013	2,998	468	582	326	1,423	23	137	39	2,997	1	0	0	2,348	33	1,445	121	10	0		
	16	16	2	0	2	11	1	0	0	16	0	0	0	10	2	4	0	0	0		
	17	17	3	0	1	8	1	3	1	16	0	0	0	9	0	6	0	0	0		
	Other mammal bites	867	851	89	87	63	445	10	119	833	1	5	2	432	86	180	17	0	0		
	Raccoon bites	180	176	10	14	29	101	0	19	3	170	3	1	0	117	70	0	0	0		
	Rodent or lagomorph bites (squirrels, rats, mice, gerbils, hamsters, rabbits, etc)	927	882	188	144	78	371	10	80	11	826	7	45	0	308	59	183	10	1	0	
	Skunk bites	18	17	0	0	0	9	0	7	1	17	0	0	0	4	1	3	0	0	0	
	Miscellaneous bites and envenomations																				
	Other or unknown animal bites	292	286	54	31	29	136	3	27	6	274	6	3	2	110	19	70	32	0	0	
	Other or unknown reptile bites	259	257	69	52	18	97	5	13	3	239	16	1	1	93	20	74	15	0	1	
Miscellaneous snake bites and envenomations	1,659	1,615	435	123	78	782	3	167	27	1,581	6	20	5	317	59	280	86	4	1		
	spider bite and/or envenomation																				
	Miscellaneous snake bites and envenomations																				
	Unknown or known non-poisonous snake bites	682	671	49	119	90	393	1	16	3	669	1	0	0	490	46	366	51	1	0	
	Unknown types of snake envenomation	1,974	1,943	127	263	218	1,284	2	36	13	1,941	0	0	0	1,789	55	914	587	33	2	
	Snakes																				
	Copperhead envenomations	2,672	2,604	95	192	234	2,052	1	20	10	2,597	6	0	0	2,513	19	769	1,550	61	0	
	Coral envenomations	84	82	3	3	8	65	0	2	1	82	0	0	0	78	3	40	23	0	0	
	Cottonmouth envenomations	285	280	5	19	43	210	0	2	1	278	0	0	1	269	11	95	140	9	0	
	Rattlesnake envenomations	850	809	34	40	51	675	0	5	4	807	1	0	0	800	13	229	429	55	0	
Spiders	1,212	1,174	67	82	114	898	0	9	4	1,166	4	0	3	1,134	24	423	561	37	0	0	
	Unknown crotalid envenomations																				
	Black widow spider bites and/or envenomations	1,616	1,601	129	89	102	1,189	1	81	10	1,598	2	1	0	711	44	546	334	13	0	
	Brown recluse spider bites and/or envenomations	827	812	72	63	70	515	1	80	11	809	0	0	1	383	10	176	198	29	0	
	Other necrotizing spider bites and/or envenomations	44	44	10	5	2	20	0	7	0	44	0	0	0	8	2	15	0	0	0	
	Other spider bites and/or envenomations	2,262	2,242	327	164	147	1,361	2	212	29	2,223	7	6	0	577	42	593	168	4	0	
	Tarantula bites and/or envenomations	30	30	4	1	2	20	0	3	0	30	0	0	0	8	2	8	1	0	0	
	Category total:	39,216	38,503	5,592	3,968	2,926	22,850	126	2,597	444	37,941	95	360	46	16,939	1,286	13,270	5,416	310	8	
	Building and construction products																				
	Insulation																				
Asbestos		372	322	44	16	15	177	3	59	8	313	3	2	4	56	47	22	2	0	0	
Fiberglass		797	758	251	82	58	279	4	72	12	699	25	10	14	127	89	128	21	0	0	
Other types of insulation		283	270	102	12	9	115	4	26	2	261	3	1	3	71	37	49	9	0	0	
Unknown types of insulation		334	313	188	29	14	67	0	12	3	294	8	2	7	59	39	40	11	0	0	
Urea or formaldehyde insulations	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	

Continued

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Miscellaneous building and construction products	1,726	1,664	959	111	41	424	12	106	11	1,610	30	5	13	292	143	13	1	0	
	1,034	978	297	29	44	531	2	68	7	944	11	6	14	148	227	132	4	2	
	2,501	2,273	975	158	100	803	7	179	51	2,140	78	7	33	323	403	106	6	0	
	131	126	31	1	16	66	0	11	1	120	3	1	2	22	30	10	1	0	
	56	55	8	1	0	39	0	7	0	55	0	0	0	10	10	5	0	0	
	7,235	6,760	2,856	439	297	2,501	32	540	95	6,437	161	34	90	1,007	1,052	309	12	2	
	Chemicals																		
	Acids																		
	1,711	1,414	49	30	141	1,016	1	162	15	1,331	34	14	12	627	106	504	156	13	3
	618	507	17	0	21	424	0	41	4	494	8	1	2	379	70	173	96	9	0
5,030	4,266	694	276	283	2,518	26	411	58	3,955	179	53	55	1,542	435	1,233	407	18	1	
132	97	6	3	3	75	0	8	2	93	0	2	0	59	3	41	14	1	0	
Miscellaneous chemicals	1,718	1,486	456	68	134	688	2	119	19	1,324	89	22	20	379	171	394	60	4	1
	3,505	3,026	420	90	195	1,924	6	369	22	2,851	75	22	59	1,461	250	1,026	458	52	0
	2,083	1,531	146	65	109	947	0	200	64	1,416	66	22	16	579	134	432	154	17	0
	5,175	4,918	1,108	321	533	2,493	4	422	37	4,664	123	55	56	561	765	381	41	0	0
	21	15	1	3	1	8	0	2	0	12	2	0	1	8	1	3	2	1	0
	178	115	2	0	2	95	0	13	3	76	12	20	0	77	19	22	5	5	1
	4	0	0	0	0	3	0	1	0	4	0	0	0	0	1	0	0	0	0
	631	535	80	22	30	357	3	34	9	363	120	17	2	238	101	65	44	49	9
	640	557	22	14	54	381	2	76	8	514	18	9	11	245	53	169	37	1	0
	262	220	47	3	5	144	0	20	1	208	4	3	4	103	32	71	25	2	0
Cleaning substances (household)	201	161	14	4	11	115	0	14	3	157	1	0	2	73	17	60	15	0	0
	1,076	961	215	225	126	332	11	44	8	702	238	5	6	323	163	140	60	16	19
	19,985	15,622	5,199	1,605	789	6,515	54	1,313	147	14,203	699	170	453	3,186	2,078	2,911	606	58	6
	1	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0
	596	488	168	30	24	220	2	39	5	425	26	3	26	132	68	94	18	4	0
	220	202	10	6	11	148	0	23	4	194	2	1	4	97	15	59	24	0	0
	38	27	8	3	1	12	0	3	0	18	3	4	1	14	6	4	5	0	0
	592	557	121	20	14	318	3	72	9	536	12	0	7	140	155	129	24	5	0
	7,070	6,546	1,060	358	315	3,644	34	964	171	4,836	236	899	270	2,100	710	1,455	488	61	3
	51,487	43,256	9,843	3,146	2,802	22,378	148	4,350	589	38,377	1,947	1,322	1,007	12,323	5,353	9,366	2,759	316	43
Cleaning substances (household)																			
Automatic dishwasher detergents																			
Automatic dishwasher detergents: granules (unit dose)	1,939	1,912	1,756	16	11	104	2	21	2	1,896	1	4	1	126	316	315	15	0	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Automatic dishwasher detergents: granules (various containers)	1,209	1,191	916	14	23	186	0	48	4	1,174	4	7	3	98	162	175	21	0	0
Automatic dishwasher detergents: granules with liquids (unit dose)	5,748	5,697	5,111	73	39	366	13	87	8	5,647	13	17	10	439	914	1,112	56	0	0
Automatic dishwasher detergents: liquids (unit dose)	442	441	357	9	5	63	1	6	0	436	0	0	0	69	47	91	12	1	0
Automatic dishwasher detergents: liquids (various containers)	588	559	319	15	15	181	0	27	2	549	2	4	4	123	78	122	33	2	0
Automatic dishwasher detergents: tablets	1,959	1,939	1,672	25	20	182	3	36	1	1,922	5	6	2	173	305	422	21	0	0
Automatic dishwasher rinse agents	596	570	396	7	16	125	0	25	1	557	5	3	2	64	76	108	12	0	0
Other or unknown types of automatic dishwasher detergent	2,559	2,515	2,064	44	23	310	6	61	7	2,495	1	11	4	246	310	455	32	4	1
Bleaches																			
Bleaches: borates	176	135	42	3	8	66	0	15	1	128	5	1	1	27	17	27	3	1	0
Bleaches: hypochlorite (liquid and dry)	35,312	27,052	8,272	965	1,858	13,174	76	2,470	237	23,966	2,173	433	281	7,729	3,623	7,245	1,080	47	1
Bleaches: non-hypochlorite	163	129	30	5	6	73	0	15	0	105	12	1	9	34	13	39	5	1	0
Bleaches: other or unknown (household)	1,524	1,071	265	40	101	570	1	86	8	895	126	21	18	389	126	286	45	1	0
Cleaners																			
Anionic or nonionic cleansers	1,048	942	606	34	15	246	1	39	1	892	38	8	2	119	262	139	11	1	0
Other or unknown types of household cleanser	2,325	2,038	1,037	53	101	695	5	140	7	1,895	88	30	11	429	335	347	57	5	0
Disinfectants																			
Disinfectants: hypochlorite (non-bleach products)	1,482	1,253	457	47	66	550	1	120	12	1,179	45	11	9	317	161	309	65	3	0
Disinfectants: other or unknown	7,393	6,467	2,926	594	386	2,014	29	476	42	5,925	369	68	62	839	989	1,191	115	5	0
Disinfectants: phenol	351	323	148	25	18	110	1	20	1	288	27	5	1	63	53	52	12	0	0
Disinfectants: pine oil	2,761	2,152	877	123	90	874	9	167	12	1,994	96	29	17	394	321	419	48	5	0
Drain cleaners																			
Drain cleaners: acids	218	166	9	2	4	123	1	23	4	155	5	1	3	56	9	46	19	1	0
Drain cleaners: alkalis	2,326	1,930	233	46	61	1,321	4	247	18	1,813	77	14	19	659	195	557	206	33	0
Drain cleaners: hydrochloric acid	24	18	2	0	2	14	0	0	0	16	2	0	0	9	1	4	4	1	0
Drain cleaners: other or unknown	679	573	255	14	9	248	1	39	7	546	16	3	3	124	95	94	25	3	1
Drain cleaners: sulfuric acid	656	574	36	159	17	277	0	79	6	558	9	0	5	154	23	277	66	6	0
Fabric softeners/antistatic agents																			
Fabric softener/antistatic agent: other or unknown	12	11	8	1	0	1	0	1	0	10	1	0	0	1	2	0	0	0	0
Fabric softeners/antistatic agents: aerosol or spray	59	58	39	2	2	11	1	3	0	56	0	0	2	6	11	8	2	0	0
Fabric softeners/antistatic agents: dry or powder (unit dose)	2	2	2	0	0	0	0	0	0	2	0	0	0	0	0	1	0	0	0
Fabric softeners/antistatic agents: dry or powder (various containers)	2	2	2	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
Fabric softeners/antistatic agents: liquid (unit dose)	10	8	5	0	1	2	0	0	0	8	0	0	0	0	0	2	0	0	0
Fabric softeners/antistatic agents: liquid (various containers)	705	616	339	28	30	170	0	48	1	582	23	3	4	85	97	79	7	1	1
Fabric softeners/antistatic agents: solid or sheet	340	320	254	14	9	30	0	11	2	304	8	4	1	11	47	14	2	0	0
Glass cleaners																			
Glass cleaners: ammonia containing	463	398	270	11	23	81	0	11	2	375	20	0	2	32	88	49	4	0	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Glass cleaners: anionics or nonionics	172	150	101	3	7	32	0	7	0	139	8	2	1	22	29	18	2	0	0
Glass cleaners: isopropanol	791	673	397	41	32	162	2	34	5	619	42	5	5	93	110	80	8	0	0
Glass cleaners: other or unknown types of household	2,486	2,105	1,264	111	167	437	5	104	17	1,890	173	19	7	312	377	304	30	3	0
Hand dishwashing																			
Anionic or nonionic hand dishwashing detergents	7,943	6,563	2,907	416	248	2,375	18	552	47	6,203	127	168	39	613	635	1,278	81	4	1
Other or unknown types of household hand dishwashing detergent	1,058	913	398	60	37	339	3	75	1	854	18	40	0	81	87	165	9	0	0
Laundry additives																			
Enzyme and/or microbiological laundry additives	57	49	25	1	2	16	0	5	0	47	1	0	1	13	8	10	1	1	0
Laundry bluing and/or brightening agents (without detergent)	54	48	17	1	0	25	0	4	1	48	0	0	0	5	9	5	1	1	0
Laundry detergent boosters	1,031	961	703	64	32	137	2	19	4	919	29	4	5	82	182	78	7	0	0
Other or unknown laundry additives or miscellaneous products	1,541	1,437	1,011	68	43	244	3	61	7	1,372	46	11	4	160	235	190	25	1	0
Water softeners	39	35	16	3	0	13	0	3	0	33	0	2	0	5	8	5	1	0	0
Laundry detergents																			
Laundry detergents: granules (unit dose)	308	293	186	13	21	59	0	12	2	280	11	1	1	60	60	53	10	1	0
Laundry detergents: granules (various containers)	2,472	2,292	1,368	82	93	564	10	171	4	2,169	88	12	14	344	307	392	52	2	0
Laundry detergents: granules with liquids (unit dose)	174	172	142	5	4	15	0	5	1	169	3	0	0	44	22	52	5	0	0
Laundry detergents: liquids (unit dose)	10,835	10,683	8,944	576	236	720	25	160	22	10,428	194	8	20	3,651	1,447	4,055	484	13	1
Laundry detergents: liquids (various containers)	5,026	4,701	2,712	170	208	1,359	9	228	15	4,431	203	30	17	834	585	1,060	96	7	1
Laundry detergents: other or unknown types of household laundry detergent and/or fabric cleaner	286	259	134	15	9	69	7	25	0	241	9	4	1	39	37	46	5	1	0
Laundry detergents: soaps	224	206	127	9	6	46	0	17	1	196	4	2	4	28	28	27	2	1	0
Laundry prewash/stain removers																			
Laundry prewash/stain removers: aerosol or spray solvent based	67	64	50	2	4	7	0	1	0	63	0	0	1	10	6	9	2	0	0
Laundry prewash/stain removers: aerosol or spray surfactant based	466	428	241	6	12	135	0	32	2	402	8	3	5	52	43	84	14	1	0
Laundry prewash/stain removers: dry solvent based	2	2	2	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
Laundry prewash/stain removers: dry surfactant based	122	114	81	4	5	22	1	1	0	112	0	0	1	13	17	17	4	0	0
Laundry prewash/stain removers: liquid solvent based	188	180	119	2	5	46	0	6	2	176	3	0	1	20	36	24	0	1	0
Laundry prewash/stain removers: liquid surfactant based	1,531	1,435	1,154	36	27	181	1	33	3	1,391	28	8	2	155	202	275	26	0	0
Laundry prewash/stain removers: other or unknown	1,108	991	712	25	21	200	3	28	2	960	19	7	1	108	180	197	10	0	0
Laundry prewash/stain removers: other or unknown solvent based	29	27	16	0	1	7	1	2	0	27	0	0	0	1	5	0	0	0	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Laundry prewash/stain removers: other or unknown surfactant based	29	27	23	1	0	3	0	0	0	27	0	0	0	1	4	1	1	0	0
Miscellaneous cleaners																			
Miscellaneous cleaning agents: acids	3,161	2,748	1,207	75	98	1,123	1	228	16	2,611	61	36	30	366	463	475	65	4	1
Miscellaneous cleaning agents: alkalis	7,979	6,778	3,058	179	279	2,704	14	513	31	6,397	205	68	90	1,538	1,032	1,658	289	21	0
Miscellaneous cleaning agents: anionics or nonionics	4,741	4,188	2,279	223	189	1,232	11	233	21	3,944	141	44	38	516	611	699	80	3	0
Miscellaneous cleaning agents: cationics	2,207	2,033	828	104	118	825	8	119	31	1,874	105	22	23	448	296	503	67	5	1
Miscellaneous cleaning agents: ethanol (excluding automotive products)	212	194	109	13	20	40	0	12	0	175	10	7	1	17	33	34	0	0	0
Miscellaneous cleaning agents: glycols (excluding automotive products)	293	268	123	27	12	80	0	24	2	243	18	5	1	34	40	57	3	0	0
Miscellaneous cleaning agents: isopropanol (excluding automotive products and glass)	546	506	188	71	27	178	5	31	6	439	56	7	3	93	77	93	14	1	0
Miscellaneous cleaning agents: methanol (excluding automotive products)	30	27	4	0	2	16	0	5	0	26	1	0	0	10	4	9	0	0	0
Miscellaneous cleaning agents: other or unknown household cleaning agents	5,785	5,168	2,793	276	218	1,515	23	306	37	4,828	183	89	43	784	956	925	106	8	1
Miscellaneous cleaning agents: phenol (excluding disinfectants)	15	14	2	1	0	9	0	2	0	12	1	0	0	8	3	6	0	0	0
Miscellaneous cleaning substances (household)																			
Ammonia cleaners (all purpose)	646	451	93	13	33	254	2	48	8	405	25	6	10	102	48	126	22	1	0
Carpet, upholstery, leather, or vinyl cleaners	2,632	2,409	1,439	91	72	661	4	130	12	2,319	40	14	24	306	359	426	39	0	0
Hydrofluoric acid or bifluoride wheel cleaners	57	55	6	0	3	44	0	2	0	53	1	1	0	37	6	17	19	0	0
Starches, fabric finishes, or sizing	76	72	49	4	1	15	0	1	2	70	2	0	0	3	10	12	0	0	0
Oven cleaners																			
Oven cleaners: acids	14	13	2	0	3	6	1	1	0	13	0	0	0	7	2	5	2	0	0
Oven cleaners: alkalis	2,084	1,999	325	74	155	1,156	22	251	16	1,900	24	43	24	596	233	540	158	8	0
Oven cleaners: detergent types	14	11	5	2	0	2	0	2	0	9	1	0	1	0	2	0	0	0	0
Oven cleaners: other or unknown	276	263	32	10	22	167	0	29	3	238	2	15	6	69	42	52	22	1	0
Rust removers																			
Rust removers: acids other than hydrofluoric acid types	482	404	71	15	17	254	0	46	1	380	13	1	9	121	65	119	22	3	0
Rust removers: alkalis	4	4	0	0	0	4	0	0	0	4	0	0	0	2	0	2	0	0	0
Rust removers: hydrofluoric acid	108	100	10	1	4	75	0	7	3	98	2	0	0	48	17	37	8	0	1
Rust removers: other or unknown	142	120	12	2	4	83	0	18	1	115	2	0	3	28	15	23	10	0	0
Spot removers/dry cleaning agents																			
Spot removers/dry cleaning agents: anionics or nonionics	102	84	59	1	2	19	0	2	1	80	1	1	2	14	17	13	2	0	0
Spot removers/dry cleaning agents: glycols	18	17	8	0	0	8	0	1	0	14	2	1	0	4	0	4	1	0	0

(Continued)



Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Spot removers/dry cleaning agents: isopropanol	9	8	3	0	0	4	0	1	0	8	0	0	0	1	1	1	0	0	
	7	6	3	0	0	3	0	0	0	5	0	0	1	0	3	1	0	0	
	644	604	350	16	20	162	1	51	4	581	13	3	6	81	117	73	8	2	0
Spot removers/dry cleaning agents: other hydrocarbon and/or non-halogenated containing	166	149	92	5	2	45	0	5	0	144	1	2	2	40	30	38	4	0	0
Spot removers/dry cleaning agents: other or unknown																			
Toilet bowl cleaners																			
Toilet bowl cleaners: acids	2,772	2,150	1,128	47	76	730	2	154	13	2,049	80	5	14	450	417	460	72	9	0
Toilet bowl cleaners: alkalis	2,594	2,120	1,514	54	59	394	5	86	8	2,035	64	6	12	302	469	328	23	5	0
Toilet bowl cleaners: other or unknown	3,759	3,311	2,645	87	43	425	13	95	3	3,234	50	8	6	352	656	290	23	3	0
Wall/floor/tile cleaners																			
Wall/floor/tile/all-purpose cleaning agents: acids	1,057	781	444	25	21	238	1	45	7	740	27	2	9	97	130	140	19	1	0
Wall/floor/tile/all-purpose cleaning agents: alkalis	7,571	6,468	3,172	228	293	2,350	7	379	39	6,083	233	77	51	1,314	1,068	1,376	199	9	3
Wall/floor/tile/all-purpose cleaning agents: anionics or nonionics	9,003	7,544	4,016	296	331	2,476	13	375	37	7,074	320	76	30	1,526	1,267	1,287	144	6	0
Wall/floor/tile/all-purpose cleaning agents: cationics	2,422	2,062	1,315	75	76	505	5	77	9	1,933	83	20	19	302	294	468	38	2	1
Wall/floor/tile/all-purpose cleaning agents: ethanol	193	167	130	6	3	24	0	4	0	159	3	3	2	11	22	42	3	0	0
Wall/floor/tile/all-purpose cleaning agents: glycols	785	636	355	31	22	181	0	46	1	610	15	4	5	106	105	104	13	0	0
Wall/floor/tile/all-purpose cleaning agents: isopropanol	727	588	353	29	25	146	5	29	1	552	27	6	2	87	100	80	11	1	0
Wall/floor/tile/all-purpose cleaning agents: other or unknown	2,008	1,765	1,122	63	53	426	7	79	15	1,676	45	20	12	224	306	304	27	2	0
Category total:	176,451	152,155	80,899	6,227	6,482	47,959	395	9,343	850	142,698	6,042	1,602	1,085	29,605	22,651	33,237	4,368	253	15
Cosmetics/personal care products																			
Dental care products																			
False teeth cleaning agents	2,273	2,253	205	42	45	1,695	1	248	17	2,184	34	6	25	141	321	176	10	2	0
Other dental care products (excluding fluoride supplements)	1,230	1,190	462	76	37	491	1	100	23	1,100	14	0	68	99	146	102	19	0	0
Toothpastes (with fluoride)	9,466	9,186	7,530	458	203	802	15	161	17	8,922	163	9	60	223	1,388	526	17	1	0
Toothpastes (without fluoride)	1,523	1,437	1,182	51	21	142	2	36	3	1,398	18	2	15	28	137	64	0	0	0
Hair care products																			
Curl activators	31	28	21	3	0	3	0	1	0	27	0	1	0	1	2	2	0	0	0
Hair coloring agents (excluding peroxides)	1,568	1,503	561	45	95	637	1	142	22	1,343	33	4	116	314	175	296	62	1	0
Hair oils	807	781	597	24	24	110	2	22	2	764	9	2	4	97	111	85	15	1	0
Hair relaxers (with other alkalines)	55	54	26	4	3	16	0	5	0	49	0	0	5	28	11	20	6	0	0
Hair relaxers (with other non-alkalines)	17	17	8	0	0	8	0	1	0	15	1	0	1	1	0	5	0	0	0
Hair relaxers (with sodium hydroxide)	85	83	43	4	4	26	1	5	0	72	1	1	9	36	8	27	5	2	0
Hair rinses, conditioners, relaxers	1,761	1,630	1,260	85	51	184	6	40	4	1,556	53	6	8	127	241	152	11	0	0
Hair sprays	725	650	358	30	40	169	2	43	8	595	41	4	8	90	94	91	13	1	0
Other hair care products (excluding peroxides)	2,700	2,573	1,693	170	126	468	10	100	6	2,434	46	9	62	368	387	421	46	2	0
Permanent wave solutions	80	79	25	4	8	35	0	7	0	74	0	0	5	32	6	28	4	0	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Shampoos	4,399	4,112	2,526	298	209	899	13	147	20	3,817	202	11	57	450	423	708	54	3	0
Hand sanitizers																			
Hand sanitizers: ethanol based	19,097	18,390	11,653	2,288	1,053	2,998	28	331	39	15,509	2,452	284	24	2,266	3,486	1,625	448	61	0
Hand sanitizers: isopropanol based	227	204	112	19	13	49	1	10	0	165	35	2	0	38	47	24	11	0	0
Hand sanitizers: non-alcohol based	434	423	339	25	17	30	3	9	0	394	25	2	2	18	41	23	3	1	0
Hand sanitizers: unknown	2,027	1,785	803	303	149	450	9	61	10	1,331	380	49	8	406	296	260	70	13	2
Miscellaneous cosmetics/ personal care products																			
Baby oils	702	676	585	10	9	57	1	13	1	656	9	2	6	70	109	66	5	1	0
Bath oils and/or bubble baths	3,174	3,091	2,551	326	52	121	8	28	5	3,011	51	4	14	173	477	261	11	1	0
Creams, lotions, and make-up	15,004	14,352	11,148	536	345	1,835	28	423	37	13,826	208	32	241	606	1,770	761	65	2	0
Deodorants	5,943	5,773	4,454	325	398	480	16	80	20	5,366	314	30	32	316	703	488	28	2	0
Depilatories	445	431	129	23	32	201	1	35	10	323	22	4	81	129	40	109	39	1	0
Douches	34	33	18	0	2	11	0	2	0	33	0	0	0	4	5	1	0	0	0
Eye products	799	764	514	33	30	155	2	26	4	727	11	2	24	84	101	70	16	1	0
Lipsticks and lip balms (with camphor)	592	575	478	26	17	39	2	11	2	560	10	0	5	16	123	24	0	0	0
Lipsticks and lip balms (without camphor)	3,330	3,198	2,690	199	65	173	5	54	12	3,011	87	4	84	94	371	161	9	0	0
Perfumes, colognes, and aftershaves	5,926	5,661	3,796	531	470	694	13	131	26	5,101	397	104	23	568	889	1,008	59	1	0
Peroxides	5,566	4,890	1,207	244	339	2,557	9	488	46	4,512	215	25	103	1,012	550	1,169	164	10	1
Powders made of material other than talc	902	853	634	53	31	100	3	28	4	800	35	3	11	73	106	116	4	1	0
Powders made of talc	674	643	433	45	25	112	3	24	1	615	20	4	4	72	92	81	9	1	0
Soaps (bar, hand or complexion)	9,834	9,191	5,347	688	472	2,214	37	389	44	8,539	447	54	102	768	995	1,319	98	5	1
Suntan and/or sunscreen products	4,513	4,409	3,559	264	85	399	14	74	14	4,298	43	14	46	185	469	386	19	0	0
Mouthwashes																			
Mouthwashes: ethanol containing	4,039	3,584	608	340	317	1,951	4	350	14	2,864	627	15	49	589	419	359	148	25	3
Mouthwashes: fluoride containing	3,929	3,870	2,188	919	81	561	5	107	9	3,775	64	2	19	70	465	137	5	0	0
Mouthwashes: non ethanol containing	2,398	2,297	554	234	155	1,135	1	202	16	2,165	99	3	21	116	282	125	11	1	0
Mouthwashes: unknown	164	119	18	11	6	65	1	17	1	95	22	1	1	28	7	20	5	1	0
Nail products																			
Acrylic nail adhesives	2,133	2,111	790	394	303	536	9	66	13	2,048	44	5	10	818	225	565	139	1	0
Acrylic nail primers	191	182	130	9	13	28	0	2	0	178	2	0	2	62	34	49	6	0	0
Acrylic nail removers	18	17	6	3	0	7	0	1	0	15	2	0	0	2	4	2	0	0	0
Miscellaneous nail products	831	796	499	41	31	180	5	37	3	758	16	3	14	140	116	140	18	0	0
Nail polish removers (acetone containing)	1,272	1,220	693	66	101	297	3	57	3	1,133	65	11	5	194	192	218	22	1	0
Nail polishes	3,172	3,033	2,457	176	87	240	6	62	5	2,944	53	9	24	275	442	410	23	0	0
Other nail polish removers	409	392	267	22	17	76	0	8	2	372	18	1	1	53	83	55	8	1	0
Unknown nail polish removers	1,744	1,625	910	113	110	411	6	72	3	1,512	88	13	8	242	261	240	16	2	0
Category total:	126,243	120,164	76,067	9,560	5,691	23,847	277	4,256	466	110,986	6,476	737	1,407	11,522	16,650	12,975	1,721	146	7
Deodorizers																			
Air freshener																			
Air fresheners: aerosols	1,710	1,646	1,049	107	65	320	7	91	7	1,561	43	23	14	139	257	215	21	0	0
Air fresheners: liquids	7,943	7,832	6,611	216	110	691	10	176	18	7,656	81	32	25	544	1,267	1,101	37	1	0
Air fresheners: solids	968	935	771	34	16	83	7	20	4	910	9	4	5	76	117	69	3	1	0
Air fresheners: unknown form	1,079	1,052	758	61	24	149	0	57	3	1,020	16	10	5	107	149	117	13	0	0
Miscellaneous deodorizers																			
Diaper pail deodorizers (excluding moth repellants)	17	17	16	0	0	1	0	0	0	17	0	0	0	2	3	2	0	0	0

(Continued)



Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in Health care facility		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	None	Minor	Moderate	Major	Death	
Other types of deodorizer (not for personal use)	5,722	5,489	3,616	285	163	1,158	13	231	23	5,239	159	37	37	652	881	771	48	3	0
Toilet bowl deodorizers	195	192	136	17	4	27	1	7	0	180	10	2	0	27	35	19	4	0	0
Unknown types of deodorizer (not for personal use)	47	45	26	4	1	9	0	5	0	42	1	0	2	7	4	6	0	0	0
Category total:	17,681	17,208	12,983	724	383	2,438	38	587	55	16,625	319	108	88	1,554	2,713	2,300	126	5	0
Dyes																			
Miscellaneous dyes																			
Dyes: fabrics	353	337	228	29	14	57	2	5	2	318	7	7	4	27	53	12	1	0	0
Dyes: foods (including easter egg)	433	395	261	68	17	31	6	12	0	355	22	2	14	14	68	29	9	0	0
Dyes: leathers	49	47	33	3	1	5	0	5	0	46	1	0	0	3	11	3	0	0	0
Dyes: other	479	446	213	77	68	73	2	12	1	407	23	2	13	44	84	29	4	0	0
Dyes: unknown	53	52	29	9	1	9	0	2	2	47	2	0	3	5	7	4	1	0	0
Category total:	1,367	1,277	764	186	101	175	10	36	5	1,173	55	11	34	93	223	77	15	0	0
Essential oils																			
Miscellaneous essential oil																			
Cinnamon oil	528	448	273	32	25	88	2	25	3	376	31	10	29	51	27	122	20	0	0
Clove oil	487	428	214	12	7	165	1	28	1	373	20	7	25	63	64	88	6	1	0
Eucalyptus oil	1,269	1,150	631	44	34	379	4	56	2	1,088	34	5	21	195	250	175	21	2	0
Miscellaneous essential oils	13,064	12,403	8,689	424	269	2,336	42	587	56	11,919	196	61	197	945	2,064	1,709	92	4	0
Pennyroyal oil	37	37	3	11	0	17	0	5	1	25	11	0	1	5	12	4	1	0	0
Tea tree oil	2,281	2,117	956	75	79	853	2	145	7	1,964	84	15	42	214	395	192	25	0	0
Category total:	17,666	16,583	10,766	598	414	3,838	51	846	70	15,745	376	98	315	1,473	2,812	2,290	165	7	0
Fertilizers																			
Miscellaneous fertilizers																			
Household plant foods (generally for indoor plants)	1,445	1,376	671	98	54	443	3	102	5	1,346	15	5	6	78	234	67	6	0	0
Other types of fertilizer	1,481	1,361	755	115	25	366	7	79	14	1,310	29	8	11	97	197	103	11	0	0
Outdoor fertilizers	1,620	1,535	888	100	35	425	5	74	8	1,481	28	9	14	131	290	115	22	0	0
Plant hormones	98	89	41	3	1	35	0	9	0	87	2	0	0	6	20	5	1	0	0
Unknown types of fertilizer	103	85	49	3	3	24	0	5	1	82	1	1	0	11	11	9	3	0	0
Category total:	4,747	4,446	2,404	319	118	1,293	15	269	28	4,306	75	23	31	323	752	299	43	0	0
Fire extinguishers																			
Miscellaneous fire extinguisher																			
Miscellaneous fire extinguishers	2,242	2,118	173	240	234	1,126	43	250	52	1,945	70	59	32	700	280	568	144	4	0
Category total:	2,242	2,118	173	240	234	1,126	43	250	52	1,945	70	59	32	700	280	568	144	4	0
Foreign bodies/toys/miscellaneous toys/miscellaneous																			
Miscellaneous foreign bodies/toys/miscellaneous																			
Ashes	281	259	188	5	8	46	0	9	3	242	5	6	2	13	42	10	0	0	0
Bubble blowing solutions	2,150	2,119	1,806	192	36	60	5	18	2	2,070	34	4	5	137	186	374	9	1	0
Charcoals	544	428	268	21	19	99	1	20	0	376	24	11	15	34	71	33	1	0	0
Christmas ornaments	207	204	145	16	6	30	0	6	1	201	1	0	2	23	41	10	2	0	0
Coins	3,623	3,557	2,477	964	48	48	10	8	2	3,437	97	8	4	1,349	744	384	49	3	0
Descendants	14,988	14,823	9,854	2,410	466	1,523	78	411	81	14,013	588	143	24	828	1,889	180	6	0	0
Expanding water beads	11,016	10,952	8,629	1,935	203	133	30	17	5	10,541	333	35	7	2,373	3,617	426	30	0	0
Feces/Urine	5,632	4,793	3,222	227	144	853	17	293	37	4,561	45	125	27	241	574	154	9	0	0
Glass	1,903	1,842	421	104	90	872	5	318	32	1,734	15	77	13	159	296	99	9	1	0
Glow products	7,262	7,239	4,587	2,167	201	166	49	50	19	7,057	141	5	16	294	652	1,361	24	1	0
Incense (punk)	280	274	170	8	8	65	3	18	2	264	2	4	3	24	31	18	3	1	0
Other types of foreign body, toy, or miscellaneous substance	31,276	29,748	19,764	4,477	1,058	3,113	159	1,028	149	28,186	987	272	181	3,326	5,002	1,388	144	10	0
Oxygen absorbers	1,087	1,079	326	315	88	268	10	67	5	964	85	21	5	59	233	43	0	0	0
Soil	1,664	1,664	1,053	64	29	230	7	61	8	1,401	20	2	24	102	193	103	6	0	0
Toys	12,634	12,484	9,655	2,152	255	273	53	83	13	12,144	262	30	22	1,100	1,983	622	30	0	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Unknown types of foreign body, toy, or miscellaneous substance	3,958	3,900	3,076	533	110	128	14	33	6	3,797	74	8	11	327	698	176	3	1	1
Thermometers																			
Thermometers: mercury	781	778	111	65	28	365	2	193	14	764	5	5	4	50	155	11	1	0	0
Thermometers: other	378	374	92	56	11	126	8	71	10	364	3	7	0	25	67	20	1	0	0
Thermometers: unknown	63	62	15	2	2	26	0	16	1	60	2	0	0	6	4	1	0	0	0
Category total:	99,727	96,367	65,859	15,713	2,810	8,424	451	2,720	390	92,176	2,723	763	365	10,470	16,478	5,413	327	18	1
Fumes/gases/vapors																			
Miscellaneous fumes/gases/vapors																			
Carbon dioxide	792	757	59	89	75	415	2	100	17	701	30	8	12	170	96	177	42	1	0
Carbon monoxide	16,512	15,097	1,581	1,084	1,066	8,717	138	2,138	373	14,462	378	46	47	6,750	2,766	3,401	1,638	387	79
Chloramine gas	3,468	3,271	90	41	184	2,126	15	579	236	3,184	71	3	11	475	327	744	158	6	1
Chlorine gas	6,763	6,374	301	276	332	4,056	34	1,194	181	6,116	168	8	65	1,444	721	1,498	491	16	1
Chlorine gas (when household acid is mixed with hypochlorite)	4,880	4,676	188	118	247	3,300	23	734	66	4,544	124	0	6	606	779	1,164	254	6	0
Hydrogen sulfide (sewer gas)	874	769	40	28	28	531	5	123	14	756	4	4	3	373	97	240	100	17	4
Methane and natural gas	6,034	5,721	917	400	324	2,945	48	968	119	5,660	15	12	22	1,050	1,365	1,068	158	8	1
Other types of fume, gas or vapor	2,884	2,628	259	138	158	1,606	26	376	65	2,338	208	18	40	679	336	535	194	22	4
Polymer fume fever	11	11	1	0	1	6	0	3	0	11	0	0	0	1	0	2	0	0	0
Simple asphyxiants	2,783	2,539	219	226	229	1,518	13	294	40	2,288	204	12	18	896	347	645	225	24	2
Unknown types of fume, gas or vapor	2,522	2,455	76	92	118	1,342	22	635	170	2,172	15	219	10	653	189	428	116	5	0
Category total:	47,523	44,298	3,731	2,492	2,762	26,562	326	7,144	1,281	42,232	1,217	330	234	13,097	7,023	9,902	3,376	492	92
Heavy metals																			
Miscellaneous heavy metals																			
Aluminum	778	704	331	64	34	209	0	59	7	659	17	15	5	72	86	38	15	1	0
Arsenic (excluding pesticides)	1,210	1,003	115	73	43	638	7	110	17	517	7	253	35	464	131	100	45	6	0
Barium, soluble salts	75	38	2	1	4	24	0	7	0	22	3	4	7	13	0	6	4	0	0
Cadmium	95	63	9	0	1	45	0	7	1	41	1	4	0	34	15	4	1	1	0
Copper	749	643	92	68	95	321	1	64	2	537	36	17	33	185	61	156	39	2	1
Fireplace flame colors	13	13	6	4	1	2	0	0	0	13	0	0	0	6	5	3	1	0	0
Gold	4	2	0	0	0	2	0	0	0	2	0	0	0	2	1	0	0	0	0
Lead	3,606	3,381	1,464	303	162	1,084	48	284	36	3,045	46	106	37	1,601	844	234	127	12	0
Manganese	49	49	6	1	6	32	0	4	0	34	1	2	3	23	4	9	4	1	0
Mercury (other)	316	289	15	8	24	200	2	31	9	198	7	22	47	110	51	18	11	1	0
Mercury, elemental (excluding thermometer)	843	801	50	38	37	517	5	135	19	679	10	26	47	196	141	52	30	1	1
Metal fume fever	243	214	13	4	23	148	1	25	0	204	5	0	4	92	13	68	22	0	0
Other types of heavy metal	2,269	1,835	549	129	128	816	9	186	18	1,416	95	37	270	256	212	309	37	1	0
Thallium	79	63	2	0	0	43	0	17	1	21	2	33	0	33	1	4	1	2	1
Unknown types of heavy metal	81	76	7	4	5	42	1	17	0	51	0	7	5	34	5	9	1	0	1
Category total:	10,434	9,174	2,661	697	563	4,123	74	946	110	7,439	230	526	493	3,120	1,570	1,010	338	28	4
Hydrocarbons																			
Miscellaneous hydrocarbons																			
Benzene	77	61	1	0	2	49	0	8	1	46	3	10	1	39	8	10	10	0	0
Carbon tetrachloride	21	19	1	1	2	13	0	2	0	17	2	0	0	4	4	4	1	1	0
Diesel fuels	1,228	1,169	126	21	86	732	4	125	75	1,094	55	11	6	401	126	366	62	3	0
Freon and other propellants	6,297	373	241	297	457	4,062	29	796	94	5,460	501	19	42	1,891	813	1,481	500	57	11
Gasolines	9,716	9,279	955	464	848	5,796	17	1,106	93	8,526	525	112	48	1,981	1,081	2,684	315	29	0
Kerosenes	389	360	116	15	19	165	1	36	8	338	14	2	2	121	53	97	25	2	0
Lamp oils	664	648	440	24	16	134	3	30	1	627	12	5	0	190	135	147	58	9	0
Lighter fluids and/or naphtha	1,396	1,334	677	42	82	412	6	103	12	1,219	40	55	11	444	238	309	75	8	0

(Continued)



Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age						Reason			Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death	
Industrial cleaners	Lubricating oils and/or motor oils	3,254	2,989	1,243	145	208	1,157	9	212	15	2,819	99	41	11	620	571	70	5	0	
	Mineral seal oil	18	15	7	0	1	6	0	1	0	13	2	0	0	6	4	0	0	0	
	Mineral spirits	1,471	1,342	256	46	69	834	3	122	12	1,256	56	15	9	477	155	107	5	0	
	Other types of halogenated hydrocarbon	176	157	14	3	11	110	0	18	1	147	4	1	3	53	13	20	0	0	
	Other types of hydrocarbon	2,752	2,499	866	110	130	1,177	5	179	32	2,381	65	20	18	755	381	124	8	0	
	Toluene and/or xylene (excluding adhesives)	486	415	31	7	19	310	0	46	2	399	7	4	2	215	37	143	5	0	
	Turpentine	188	165	29	4	7	108	0	16	1	137	22	4	0	44	30	31	7	1	
	Unknown types of hydrocarbon	462	419	150	14	30	185	3	32	5	393	20	1	4	116	88	35	4	0	
	Category total:	28,595	26,923	5,285	1,137	1,987	15,250	80	2,832	352	24,872	1,427	300	157	7,356	3,737	6,879	1,457	136	12
	Miscellaneous industrial cleaners																			
Infectious and toxin-mediated diseases	Industrial cleaner: disinfectants	1,995	1,823	150	120	151	1,073	9	306	14	1,636	133	30	10	525	197	102	3	0	
	Industrial cleaner: other or unknown	1,507	1,373	243	75	114	792	6	115	28	1,260	56	39	12	481	142	106	4	0	
	Industrial cleaners: acids	1,696	1,471	257	46	92	895	4	158	19	1,364	60	25	16	496	185	111	2	0	
	Industrial cleaners: alkalis	2,536	2,333	318	58	181	1,535	3	219	19	2,213	74	27	13	1,186	202	327	22	0	
	Industrial cleaners: anionics or nonionics	484	425	135	19	26	188	0	55	2	403	18	4	0	121	41	99	13	2	0
	Industrial cleaners: cationics	1,460	1,356	252	114	96	761	4	116	13	1,166	132	29	21	426	182	398	65	3	0
	Category total:	9,678	8,781	1,355	432	660	5,244	26	969	95	8,042	473	154	72	3,235	949	2,846	724	36	0
	Infectious diseases																			
	Algae associated exposures																			
	Cyanobacteria exposure (blue green algae)	605	591	111	107	73	220	9	61	10	571	3	3	11	78	139	16	0	0	0
Botulinum toxins	Karenia brevis exposure (red algae tide)	8	7	1	2	0	4	0	0	7	0	0	0	0	3	0	1	2	0	0
	Unknown type of algae associated exposure	16	12	0	3	2	5	0	2	9	9	1	0	1	5	0	0	1	0	0
	Botulism	457	428	99	11	10	264	1	39	4	261	11	5	130	165	37	29	37	23	0
	Ichthyosarcotoxins																			
	Ciguatera poisoning	85	81	2	3	5	64	0	7	0	65	0	0	15	46	1	19	27	4	0
	Ciguatera fish poisoning	9	7	0	0	0	7	0	0	0	6	0	0	1	0	0	0	1	0	0
	Other types of seafood poisoning	128	112	15	4	8	70	1	12	2	82	2	3	24	26	12	16	11	0	0
	Palytoxin poisoning	75	75	2	1	1	59	0	9	3	64	0	2	9	33	5	27	13	3	0
	Paralytic shellfish poisoning	75	73	4	6	3	47	0	13	0	70	1	0	2	34	11	11	22	1	0
	Scombrotoxin poisoning	121	111	9	5	5	80	0	12	0	73	0	1	36	29	5	24	17	1	0
Infectious diseases	Tetrodotoxin poisoning	309	305	58	60	21	138	3	22	3	281	8	1	14	53	49	37	8	1	0
	Bacterial diseases	734	692	113	40	43	372	7	102	15	610	3	25	43	168	51	46	2	0	0
	Fungal diseases	95	92	15	1	2	60	0	14	0	88	0	1	3	10	3	7	2	1	0
	Other types of bacterial food poisoning (salmonella, shigella, vibrio, staphylococcus, streptococcus, etc)	41	36	6	4	1	16	1	7	1	34	0	1	1	2	5	4	0	0	0
	Parasitic diseases	19	17	4	0	0	11	0	2	0	13	0	1	2	6	0	0	1	0	0
	Prion diseases	3	3	0	1	0	1	0	1	0	2	0	0	0	0	0	0	1	0	0
	Unknown types of bacterial food poisoning	243	213	47	17	12	110	0	26	1	207	0	4	2	16	13	14	2	0	0
	Unknown types of suspected food poisoning	8,467	8,024	1,697	634	489	3,898	38	1,131	137	7,272	32	150	540	584	603	1,142	192	0	0
	Viral diseases	723	319	59	26	14	170	3	33	14	296	0	1	3	176	11	21	23	0	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age						Reason			Treated in		Outcome					
												Health care facility							
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	None	Minor	Moderate	Major	Death	
Category total:	12,213	11,198	2,242	925	689	5,596	63	1,493	190	10,011	61	198	837	1,436	945	1,543	426	36	0
Information calls																			
Food information calls																			
Obsolete: information calls about food products, additives or supplements	787	721	202	78	30	308	1	93	9	593	5	31	88	32	83	42	11	0	0
Obsolete: information calls about possibly spoiled foods	565	561	134	55	25	250	2	78	17	532	4	7	15	10	14	25	3	0	0
Category total:	1,352	1,282	336	133	55	558	3	171	26	1,125	9	38	103	42	97	67	14	0	0
Lacrimators																			
Miscellaneous lacrimators																			
Lacrimators: capsicum defense sprays	3,385	3,343	583	750	490	1,016	134	309	61	2,696	119	344	50	791	124	1,426	185	3	0
Lacrimators: cn (chloroacetophenone)	170	160	25	23	18	61	0	27	6	111	7	25	6	56	3	72	14	0	0
Lacrimators: cs (o-chlorobenzylidene malonitrile)	34	29	3	0	2	19	0	3	2	23	0	3	3	16	4	7	4	1	0
Lacrimators: other	50	20	1	2	0	12	0	3	2	20	0	0	0	3	1	7	1	0	0
Lacrimators: unknown	75	70	4	8	7	29	0	20	2	49	3	10	3	31	0	39	5	0	0
Category total:	3,714	3,622	616	783	517	1,137	134	362	73	2,899	129	382	62	897	132	1,551	209	4	0
Matches/fireworks/explosives																			
Miscellaneous matches/fireworks/explosives																			
Explosives	141	133	40	14	19	49	0	8	3	111	10	6	3	51	24	24	3	4	0
Fireworks	699	690	564	47	10	48	1	18	2	673	11	3	0	111	193	54	14	2	0
Matches	151	147	114	5	6	18	0	4	0	140	4	2	0	7	32	6	0	0	0
Other types of match, firework, or explosive	58	57	32	6	2	7	0	10	0	53	1	2	0	8	18	10	0	0	0
Unknown types of match, firework, or explosive	4	4	1	1	1	1	0	0	0	4	0	0	0	3	3	1	0	0	0
Category total:	1,053	1,031	751	73	38	123	1	40	5	981	26	13	3	180	270	95	27	6	0
Miscellaneous foods																			
Foods																			
Capsicum peppers	1,297	1,231	302	188	155	440	6	121	19	910	75	31	171	163	45	445	51	1	0
Food additives	4,462	3,408	1,412	217	147	1,347	7	238	40	2,688	490	19	185	481	463	566	80	11	2
Food products	24,129	22,195	8,597	1,925	1,088	7,765	140	2,337	343	18,138	421	1,115	2,370	1,402	2,378	1,725	256	8	0
Other adverse reactions to food	713	680	114	47	49	331	3	130	6	343	7	37	290	125	40	90	47	3	0
Category total:	30,601	27,514	10,425	2,377	1,439	9,883	156	2,826	408	22,079	993	1,202	3,016	2,171	2,926	2,826	434	23	2
Mushrooms, naturally occurring																			
Miscellaneous mushrooms, naturally occurring																			
Group 1 mushrooms: cyclopeptides	88	76	22	5	3	40	0	4	2	57	13	0	3	44	19	16	7	6	1
Group 1a mushrooms: orellanine	5	5	4	0	0	1	0	0	0	5	0	0	0	2	2	1	0	0	0
Group 2 mushrooms: muscimol (ibotenic acid)	166	133	14	4	18	88	0	9	0	33	91	0	8	97	14	30	50	7	0
Group 3 mushrooms: monomethylhydrazine (mmh)	53	53	3	3	3	39	0	5	0	42	1	0	10	28	8	27	4	0	0
Group 4 mushrooms: muscarine and histamine	57	52	10	3	7	28	0	4	0	44	7	0	1	36	6	24	11	1	0
Group 5 mushrooms: coprine	12	10	5	4	0	1	0	0	0	9	1	0	0	0	5	0	0	0	0
Group 6 mushrooms: hallucinogenics (psilocybin and psilocin)	2,085	1,389	148	61	489	623	0	47	21	312	1,006	6	43	1,061	74	344	532	29	0
Group 7 mushrooms: gastrointestinal irritants	403	381	142	27	36	161	0	14	1	307	62	0	11	168	69	144	46	1	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age										Reason			Treated in Health care facility		Outcome			
			≤20					≥20					Unint	Int	Other	Adv Rxn	None	Minor	Moderate	Major	Death
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age												
Mushrooms: miscellaneous, non-toxic	407	324	123	23	15	138	1	23	1	262	20	0	42	55	43	18	1	0			
Mushrooms: other potentially toxic	262	233	78	13	10	111	0	19	2	174	11	0	45	49	57	11	0	0			
Mushrooms: unknown	4,752	4,434	2,322	580	307	1,068	19	113	25	3,527	712	10	148	1,319	743	291	29	1			
Category total:	8,290	7,090	2,871	723	888	2,298	20	238	52	4,772	1,924	16	311	3,011	1,429	970	74	2			
Other/unknown nondrug substances																					
Miscellaneous other/unknown nondrug substances																					
Other non-drug substances unlikely to be drug products	17,942	16,215	5,030	1,368	869	6,385	128	2,083	352	12,696	577	1,789	393	1,838	2,187	624	123	9			
	6,788	6,513	988	303	228	3,994	19	802	179	4,944	149	734	149	466	670	242	60	4			
Category total:	24,730	22,728	6,018	1,671	1,097	10,379	147	2,885	531	17,640	726	2,523	542	2,304	2,857	866	183	13			
Paints and stripping agents																					
Miscellaneous paints and stripping agents																					
Other types of paint, varnish or lacquer	485	453	148	35	21	198	1	48	2	438	6	0	8	66	77	20	1	0			
Unknown types of paint, varnish or lacquer	3,323	3,093	1,630	204	104	877	27	230	21	2,974	61	11	31	455	272	64	5	0			
Varnishes and lacquers	460	419	76	11	21	236	3	69	3	397	6	2	14	58	91	26	2	0			
Paints																					
Anti-algae paints	22	21	2	0	0	17	0	2	0	20	0	0	1	3	6	2	0	0			
Anti-corrosion paints	29	28	4	1	2	18	0	3	0	24	1	1	2	2	5	5	0	0			
Oil-base paints	1,718	1,607	365	112	100	816	4	196	14	1,503	44	13	39	180	342	68	12	0			
Water base paints (acrylic, latex, etc)	2,644	2,544	1,651	165	94	524	6	94	10	2,455	51	7	28	310	219	32	1	0			
Wood stains	573	544	163	19	13	261	2	79	7	526	4	1	11	46	111	14	0	0			
Stripping agents																					
Methylene chloride stripping agents	189	178	22	2	11	121	1	20	1	174	2	0	1	19	58	25	1	0			
Other types of stripping agent	449	415	82	5	15	270	0	40	3	398	3	2	10	46	117	53	2	0			
Unknown types of stripping agent	75	60	6	2	4	40	0	8	0	56	2	0	2	2	20	10	0	0			
Category total:	9,967	9,362	4,149	556	385	3,378	44	789	61	8,965	180	37	147	1,187	1,318	319	24	0			
Pesticides																					
Fumigants																					
Aluminum phosphide	68	59	3	0	5	45	0	5	1	55	4	0	0	13	14	5	1	1			
Methyl bromide	24	23	7	1	2	11	0	1	1	21	0	0	1	2	4	4	0	0			
Other fumigants	43	41	4	2	3	24	0	8	0	39	0	1	1	6	15	1	0	0			
Sulfuryl fluoride	235	194	32	13	5	123	0	18	3	183	3	2	6	37	34	11	0	0			
Unknown fumigants	78	74	4	6	3	50	1	6	4	68	2	2	1	9	18	4	0	0			
Fungicides (non-medical)																					
Carbamate fungicides	56	46	10	2	2	25	0	7	0	46	0	0	0	9	14	2	0	0			
Copper compound fungicides	164	152	18	3	1	110	0	19	1	150	1	0	1	6	22	23	1	0			
Mercurial fungicides	1	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0			
Non-mercurial fungicide	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
Other types of non-medical fungicide	601	493	89	12	14	325	0	49	4	462	10	9	11	75	106	22	1	0			
Other/unknown type of non-medical fungicide	4	3	0	0	1	2	0	0	0	3	0	0	0	0	0	0	0	0			
Phthalimide fungicides	14	10	5	0	1	3	0	1	0	10	0	0	0	1	0	0	0	0			
Unknown types of non-medical fungicide	29	20	4	0	0	15	0	1	0	20	0	0	0	1	8	0	0	0			
Wood preservatives	103	94	10	1	2	64	2	15	0	90	0	2	1	11	16	4	0	0			

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome							
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death	
Herbicides (including algaecides, defoliant, desiccant, plant growth regulators)	6	4	0	0	0	4	0	0	0	4	0	0	0	1	1	2	0	0	0	
	1,246	1,079	214	43	29	631	2	150	10	1,025	15	9	25	179	208	227	26	2	0	
	593	543	116	19	10	338	0	58	2	520	14	4	3	78	133	91	16	1	1	
	2,397	2,121	316	82	58	1,337	1	301	26	1,989	40	27	37	424	299	441	58	2	2	
	1,769	1,434	328	57	43	817	6	169	14	1,390	19	5	15	239	217	247	36	2	0	
	102	89	3	0	5	77	0	4	0	78	3	4	3	51	19	23	12	3	2	
	114	79	16	3	5	49	0	6	0	74	2	2	1	21	14	16	2	0	0	
	374	312	59	20	7	174	1	49	2	286	7	10	7	73	36	64	8	1	0	
	24	15	5	4	0	6	0	0	0	14	1	0	0	3	5	2	0	0	0	
	Insecticides (including insect growth regulators, molluscicides, nematocides)	919	848	245	43	18	418	0	109	15	791	23	16	8	167	164	117	29	2	0
300		282	40	9	10	165	0	55	3	264	11	4	2	46	34	60	11	2	0	
92		78	27	4	1	33	1	10	2	76	2	0	0	9	12	7	2	0	0	
190		183	20	7	12	103	1	38	2	167	10	2	4	57	14	59	13	0	0	
109		72	23	6	1	34	0	5	3	67	0	0	4	10	4	5	0	1	0	
24		21	8	0	0	10	0	3	0	17	0	1	1	5	6	3	0	0	0	
43		40	23	0	4	10	0	3	0	36	2	0	2	0	3	7	1	0	0	
2,083		1,949	598	74	69	974	13	195	26	1,784	104	19	23	601	381	357	108	19	1	
13		13	5	0	0	6	0	2	0	12	0	1	0	3	3	3	0	0	0	
422		407	53	20	19	254	0	57	4	387	12	5	2	76	38	106	22	1	0	
Organophosphate insecticides in combination with non-carbamate insecticides	9,780	9,780	4,524	422	230	3,628	38	852	86	9,425	109	52	147	1,004	1,447	1,152	130	11	1	
	3,272	3,100	655	172	120	1,708	14	406	25	2,879	107	34	70	604	343	834	149	4	0	
	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	
	18,057	18,057	3,712	786	666	10,442	40	2,219	192	16,802	506	185	478	3,212	2,115	4,314	668	27	4	
	24	24	0	1	0	15	1	6	1	18	0	0	6	3	2	5	0	0	0	
	3,681	3,338	617	183	122	1,790	33	529	64	2,944	98	154	89	835	363	622	164	12	0	
	Miscellaneous pesticides	18	14	2	1	1	7	0	3	0	12	1	0	0	4	4	2	1	0	0
		10,236	10,121	9,093	165	61	598	31	153	20	10,024	41	25	14	649	1,396	231	13	1	0
	Repellents	576	545	150	26	21	280	2	64	2	508	10	4	22	74	64	112	12	0	0
1		1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	
2,893		2,830	1,325	316	142	828	6	192	21	2,611	44	47	114	257	350	678	47	0	0	
2,481		2,442	1,721	179	39	411	6	78	8	2,353	17	26	35	160	343	343	13	0	0	
1,268		1,240	457	75	25	464	9	195	15	1,167	28	12	21	187	234	148	20	4	0	

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age						Reason			Treated in		Outcome					
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Other types of moth repellent	1	1	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0
Paradichlorobenzene moth repellants (excluding deodorizing products)	125	123	48	10	2	42	1	18	2	118	2	0	3	15	33	28	3	0	0
Unknown types of insect repellent	186	177	75	12	6	59	2	21	2	170	2	2	3	22	12	34	6	0	0
Unknown types of moth repellent	971	952	428	51	19	295	22	129	8	901	23	7	13	134	196	103	12	1	0
Rodenticides																			
ANTU (1-naphthalenylthiourea)	6	6	0	3	3	0	0	0	0	4	2	0	0	1	1	0	0	0	0
Bromethalin rodenticides	1,818	1,748	1,142	71	27	368	9	103	28	1,651	60	15	14	450	476	68	11	0	1
Cholecalciferol rodenticides	539	512	367	20	8	92	0	20	5	481	18	11	2	146	144	16	1	0	0
Cyanide rodenticides	1	1	0	0	0	1	0	0	0	1	0	0	0	0	0	1	0	0	0
Long-acting anticoagulant rodenticides	2,923	2,833	2,014	111	39	475	5	160	29	2,720	64	19	15	738	706	76	11	3	0
Other types of rodenticide	1,081	1,038	694	40	8	236	2	48	10	970	44	8	7	258	217	63	7	2	0
Phu (m-3-pyridylmethyl-n1-p-nitrophenyl urea)	1	1	1	0	0	0	0	0	0	1	0	0	0	1	1	0	0	0	0
Sodium monofluoroacetate	3	2	0	0	0	2	0	0	0	2	0	0	0	0	0	1	0	0	0
Strychnine rodenticides	66	48	3	0	2	32	1	8	2	19	5	18	2	22	12	5	3	1	0
Unknown types of rodenticide	2,335	2,128	1,132	101	41	625	9	179	41	1,807	148	120	19	629	460	99	33	3	2
Warfarin type anticoagulant rodenticides	76	73	48	2	2	16	0	4	1	68	3	2	0	18	20	3	0	1	0
Zinc phosphide rodenticides	67	62	15	0	0	39	0	8	0	57	4	0	1	23	21	13	3	0	0
Category total:	76,404	71,977	30,510	3,179	1,914	28,691	259	6,739	685	67,845	1,621	866	1,234	11,745	10,723	11,040	1,705	108	15
Photographic products																			
Miscellaneous photographic products																			
Developers, fixing baths, stop baths	61	54	7	2	11	26	1	6	1	48	2	0	3	4	7	13	0	0	0
Other types of photographic product	424	400	273	45	10	59	1	9	3	383	14	1	0	20	57	19	7	0	0
Photographic coating fluids	55	55	45	6	0	1	1	2	0	54	1	0	0	2	6	1	0	0	0
Unknown types of photographic product	3	3	2	0	0	1	0	0	0	3	0	0	0	0	0	0	0	0	0
Category total:	543	512	327	53	21	87	3	17	4	488	17	1	3	26	70	33	7	0	0
Plants																			
Miscellaneous plants																			
Plants: amygdalin and/or cyanogenic glycosides	7,977	7,880	3,757	933	273	2,421	16	451	29	7,202	293	49	303	575	1,351	376	72	2	0
Plants: anticholinergics	805	763	374	67	39	228	5	42	8	667	67	7	13	111	134	53	41	6	0
Plants: cardiac glycosides (excluding drugs)	1,707	1,671	793	158	79	511	3	118	9	1,497	100	13	52	256	389	148	37	7	1
Plants: colchicine	23	23	10	2	0	10	0	1	0	20	3	0	0	5	8	3	1	0	0
Plants: depressants	111	97	44	17	5	29	2	0	0	74	16	0	4	9	17	12	1	0	0
Plants: gastrointestinal irritants (excluding oxalate containing plants)	8,070	7,853	5,014	878	229	1,403	30	268	31	7,321	340	9	158	697	1,156	740	98	4	0
Plants: hallucinogenics (code as street drug unless plant part involved)	277	224	44	18	55	88	0	16	3	122	78	0	22	76	30	32	25	2	1
Plants: nicotine (excluding tobacco products)	1,031	993	216	130	58	475	16	92	6	899	61	8	20	168	292	115	25	1	0
Plants: non-toxic	3,366	3,044	1,615	592	105	526	30	143	33	2,606	225	13	189	183	338	198	26	3	0
Plants: other toxic types	6,477	6,180	3,682	780	184	1,207	28	264	35	5,649	297	28	180	590	1,067	452	98	8	2
Plants: oxalates	6,931	6,834	5,041	774	181	655	23	149	11	6,427	293	12	82	555	921	1,349	70	1	0
Plants: skin irritants (excluding oxalate containing plants)	6,097	5,723	2,009	540	296	2,157	27	634	60	5,190	170	12	326	729	460	885	270	7	0
Plants: solanine	3,006	2,950	1,744	225	74	721	5	165	16	2,733	47	11	147	191	498	233	25	1	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age					Reason			Treated in		Outcome							
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death	
Polishes and waxes	Plants: stimulants	193	172	59	17	11	67	2	13	3	149	12	1	7	24	31	14	4	0	0
	Plants: toxalbumins	428	411	109	35	23	191	1	48	4	340	50	8	10	132	93	77	18	1	0
	Plants: unknown toxic types or unknown if toxic	7,845	7,411	4,635	928	219	1,229	62	306	32	6,786	362	31	210	604	1,082	599	92	4	0
	Category total:	54,344	29,146	6,094	1,831	11,918	250	2,710	280	47,682	2,414	202	1,723	4,905	7,867	5,286	903	47	4	
	Miscellaneous polishes and waxes																			
	Floor waxes, polishes, or sealers	196	174	62	3	4	80	4	20	1	163	6	0	5	28	27	31	8	0	0
	Furniture polishes	673	648	497	15	7	111	0	15	3	621	12	8	6	70	141	89	8	1	0
	Miscellaneous polishes and waxes (excluding mineral seal oils)	1,136	1,067	641	35	24	298	6	55	8	1,031	20	5	9	148	200	128	23	0	0
	Category total:	2,005	1,889	1,200	53	35	489	10	90	12	1,815	38	13	20	246	368	248	39	1	0
	Radiation																			
Miscellaneous radiation	Alpha radiation	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Beta radiation	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Gamma radiation	1	1	0	0	1	0	0	0	0	1	0	0	0	1	0	0	0	0	0
	Ionizing radiation: type unknown	102	95	2	3	3	53	1	29	4	60	0	13	12	32	10	11	5	0	0
	Radon	99	80	7	5	5	44	1	16	2	76	1	2	0	17	8	9	1	0	0
	Specific nonpharmaceutical radionuclides	105	81	1	1	8	52	0	17	2	70	3	3	3	34	11	14	1	0	0
	X-ray radiation	15	14	0	0	2	8	0	3	1	4	1	0	8	2	0	0	2	0	0
	Miscellaneous radiation	3	3	0	0	0	3	0	0	0	2	0	1	0	0	0	1	0	0	0
	Nonpharmaceutical radiation: type unknown																			
	Non-ionizing radiation																			
Sporting equipment	Extremely low-frequency radiation	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
	Infrared radiation	4	4	0	0	0	3	1	0	0	3	0	0	0	1	0	0	0	0	0
	Microwave radiation	18	15	0	0	0	8	0	7	0	13	0	1	1	4	1	0	0	0	0
	Non-ionizing radiation: type unknown	24	21	0	1	1	12	0	7	0	16	1	4	0	7	0	1	4	0	0
	Radio frequency radiation	15	15	0	2	2	8	0	2	1	13	0	2	0	6	1	1	2	0	0
	Ultraviolet radiation	23	17	1	0	1	10	0	4	1	15	1	0	1	8	0	1	3	0	0
	Visible light radiation (lasers)	12	12	1	1	0	8	0	2	0	7	1	1	2	2	2	1	0	0	0
	Category total:	424	359	13	13	22	210	3	87	11	281	8	27	27	114	33	39	18	0	0
	Miscellaneous sporting equipment																			
	Miscellaneous sporting equipment																			
Swimming pool/aquarium	Fishing baits	65	65	43	11	1	6	1	3	0	60	5	0	0	6	11	8	1	0	0
	Fishing products, miscellaneous	5	5	4	1	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0
	Golf balls (including liquid center of golf balls)	4	4	1	1	0	2	0	0	0	4	0	0	0	0	0	0	0	0	0
	Gun bluing compounds	14	13	3	0	1	8	0	1	0	13	0	0	0	7	2	5	4	0	0
	Hunting products, miscellaneous	438	430	204	40	30	124	2	26	4	385	19	18	1	133	112	62	4	1	0
	Other types of sporting equipment	27	27	14	7	3	2	0	1	0	24	2	1	0	6	6	3	0	0	0
	Category total:	553	544	269	60	35	142	3	31	4	491	26	19	1	152	131	78	9	1	0
	Swimming pool/aquarium																			
	Miscellaneous swimming pool/aquarium																			
	Miscellaneous swimming pool or aquarium product	Algicides	379	355	101	28	13	179	1	31	2	341	5	3	4	48	51	70	10	0
Aquarium products, miscellaneous		815	755	523	59	17	125	2	26	3	717	15	21	1	62	140	42	9	1	0
Bromine shock treatments		58	50	20	8	0	19	0	3	0	45	1	0	4	11	7	8	1	0	0
Chlorine shock treatments		2,647	2,489	423	253	176	1,354	12	251	20	2,391	50	17	26	689	163	799	218	17	0
Other types of swimming pool or aquarium product		2,143	2,027	456	214	124	983	8	220	22	1,938	31	7	42	492	177	730	142	6	0

(Continued)

Table 22A. Continued.

	No. of case mentions	No. of single exposures	Age						Reason			Treated in		Outcome					
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Swimming pool and aquarium test kits	137	118	50	13	4	39	0	11	1	116	1	0	1	18	16	23	2	0	0
Category total:	6,179	5,794	1,573	575	334	2,699	23	542	48	5,548	103	48	78	1,320	554	1,672	382	24	0
Tobacco/nicotine/cigarette products																			
Ecigarettes: nicotine containing																			
Ecigarettes: nicotine device flavor unknown	4,919	4,783	4,387	70	91	182	6	40	7	4,588	121	11	54	509	745	1,206	54	5	0
Ecigarettes: nicotine device with added flavors	2,164	2,139	1,923	73	41	88	1	11	2	2,011	74	1	29	181	377	575	27	2	0
Ecigarettes: nicotine device without added flavors	1,007	981	896	18	18	44	1	4	0	944	19	2	13	97	142	214	9	1	1
Ecigarettes: nicotine liquid flavor unknown	436	424	349	1	12	56	0	6	0	396	15	2	10	80	73	110	7	0	1
Ecigarettes: nicotine liquid with added flavors	158	153	124	7	3	17	0	2	0	139	5	3	1	23	44	38	1	1	0
Ecigarettes: nicotine liquid without added flavors	25	24	20	0	1	2	0	1	0	23	1	0	0	4	5	7	0	0	0
Miscellaneous tobacco products																			
Chewing tobacco	741	732	633	23	9	55	0	9	3	709	10	2	6	120	168	169	18	0	0
Cigarettes	2,053	1,944	1,660	18	36	181	8	38	3	1,846	45	15	27	252	451	292	20	0	0
Cigars	134	124	83	2	2	31	0	4	2	98	11	1	14	19	28	20	6	0	0
Dissolvable tobacco	16	16	10	0	2	3	0	1	0	14	1	0	1	7	3	1	0	0	0
Filter tips only (i.e. butts)	20	19	10	0	1	4	0	4	0	18	1	0	0	1	5	3	0	0	0
Heat not burn tobacco	3	3	2	0	0	1	0	0	0	2	1	0	0	0	2	0	0	0	0
Other types of tobacco product	2,953	2,851	1,951	69	104	627	3	85	12	2,558	183	12	88	603	763	534	73	5	0
Snuff	237	231	192	5	3	26	0	4	1	217	6	1	5	46	58	53	10	0	0
Unknown types of tobacco product	1,753	1,641	1,174	53	58	278	2	66	10	1,442	127	11	50	331	324	356	49	2	0
Category total:	16,619	16,065	13,414	339	381	1,595	21	275	40	15,005	620	61	298	2,273	3,188	3,578	274	16	2
Waterproofers/sealants																			
Miscellaneous waterproofers/sealants																			
Waterproofers/sealants: aerosols	127	121	33	5	6	62	1	12	2	110	4	0	7	23	18	23	5	0	0
Waterproofers/sealants: liquids	111	104	34	4	3	50	0	11	2	102	0	0	2	12	14	24	2	0	0
Waterproofers/sealants: solids	3	3	2	0	0	1	0	0	0	2	0	0	1	1	1	0	1	0	0
Waterproofers/sealants: unknown form	27	25	6	1	1	14	0	3	0	25	0	0	0	6	5	7	0	0	0
Category total:	268	253	75	10	10	127	1	26	4	239	4	0	10	42	38	54	8	0	0
Weapons of mass destruction																			
Miscellaneous weapons of mass destruction																			
Anthrax	6	6	0	0	0	5	0	0	1	3	0	1	0	2	2	0	1	0	0
Nerve gases	5	5	0	0	0	5	0	0	0	1	0	3	0	2	0	0	0	0	0
Other biological weapons	2	2	1	0	0	1	0	0	0	2	0	0	0	0	0	1	0	0	0
Other chemical weapons	6	4	1	0	0	1	0	1	1	3	0	1	0	2	0	0	1	0	0
Other nuclear weapons	1	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0
Other suspicious powders	127	114	13	4	5	71	0	17	4	67	11	31	1	42	11	24	9	1	0
Other suspicious substances (non-powder)	330	307	36	13	11	186	1	45	15	128	19	133	2	118	23	51	24	10	0
Suspicious powders in envelope or package	53	49	0	2	2	33	0	9	3	21	3	19	1	24	7	12	8	0	0
Category total:	530	488	51	19	18	303	1	72	24	226	33	188	4	190	43	88	43	11	0
Nonpharmaceuticals total:																			

Table 22B. Demographic profile of single substance pharmaceuticals exposure cases by generic category.

	No. of case mentions	No. of single exposures				Age			Reason				Treated in		Outcome			
		≤ 5	6-12	13-19	≥ 20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Analgesics																		
Acetaminophen alone																		
Acetaminophen alone, adult	45,857	31,162	6,344	1,597	7,928	14,186	129	963	16,187	14,182	7	402	18,253	7,760	5,285	2,757	881	73
Acetaminophen alone, pediatric	17,253	15,860	14,247	1,350	117	112	4	16	15,587	167	9	56	2,445	3,261	294	40	11	1
Acetaminophen alone, unknown if adult or pediatric	27,595	17,241	4,513	968	4,093	7,256	86	311	9,149	7,504	5	226	10,187	4,088	3,008	1,786	640	84
Acetaminophen combinations																		
Acetaminophen in combination with other drugs, adult	6,252	3,597	747	266	1,521	980	13	66	1,368	2,111	6	66	2,382	776	931	581	46	1
Acetaminophen in combination with other drugs, pediatric	529	461	403	53	2	3	0	0	451	3	0	4	32	94	15	4	1	0
Acetaminophen with codeine	1,452	654	55	11	66	470	7	45	324	265	1	54	342	141	135	70	17	2
Acetaminophen with diphenhydramine	4,941	2,870	420	98	521	1,752	15	64	1,016	1,791	1	28	1,947	522	657	598	131	4
Acetaminophen with hydrocodone	7,834	3,353	270	98	275	2,556	17	136	1,724	1,386	10	153	1,908	739	662	435	147	12
Acetaminophen with other opioids or opioid analogs	86	41	6	1	3	27	0	4	24	13	0	4	18	9	9	3	1	0
Acetaminophen with oxycodone	4,575	1,954	156	22	238	1,431	15	89	786	1,021	22	78	1,293	389	379	311	209	7
Acetaminophen with propoxyphene	15	5	0	0	1	3	0	1	2	2	0	0	3	1	1	2	0	0
Acetylsalicylic acid alone																		
Acetylsalicylic acid alone, low-dose	3,936	1,791	1,069	157	167	360	8	29	1,383	371	1	25	637	527	112	98	7	0
Acetylsalicylic acid alone, not low-dose	1,504	975	234	37	243	440	3	17	432	511	0	21	631	194	181	196	29	1
Acetylsalicylic acid alone, unknown dose	7,861	3,342	979	176	586	1,512	27	60	1,633	1,504	7	49	2,103	721	493	684	115	13
Acetylsalicylic acid combinations																		
Acetylsalicylic acid in combination with other drugs, adult	766	514	121	18	35	328	0	12	302	176	0	19	311	83	86	110	17	1
Acetylsalicylic acid in combination with other drugs, pediatric	2	2	0	1	0	1	0	0	1	1	0	0	1	0	0	0	0	0
Acetylsalicylic acid with codeine	15	7	0	0	0	7	0	0	1	5	0	1	5	0	3	2	0	0
Acetylsalicylic acid with other opioids or opioid analogs	4	2	0	0	0	1	0	1	0	1	0	0	1	0	0	0	1	0
Miscellaneous analgesics																		
Non-aspirin salicylates (excluding topicals and/or gastrointestinal drugs)	323	271	111	13	38	95	0	14	241	21	0	9	48	43	46	11	1	0
Other analgesics	314	221	72	13	26	99	0	9	140	67	0	12	85	31	49	21	3	0
Phenacetin	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Phenazopyridine	1,050	858	552	20	37	220	2	26	736	59	0	58	226	227	82	54	10	1
Salicylamide	4	3	0	0	0	2	0	1	3	0	0	0	1	0	0	0	0	0
Unknown analgesics	110	34	3	0	19	12	0	0	7	26	0	1	30	11	8	6	1	0
Nonsteroidal antiinflammatory drugs																		
Colchicine	531	361	35	5	13	293	0	15	272	50	0	35	187	59	95	30	6	3
Cyclooxygenase-2 inhibitors	1,201	573	125	26	31	354	1	35	497	62	1	13	98	111	27	12	0	0
Ibuprofen	73,703	54,824	32,878	3,244	7,771	9,819	154	905	43,053	11,051	16	492	13,778	11,833	4,277	1,231	94	2
Ibuprofen with diphenhydramine	1,477	885	130	25	150	540	4	36	443	426	0	10	468	156	157	140	12	0
Ibuprofen with hydrocodone	8	6	0	0	1	5	0	0	5	1	0	0	1	1	2	0	0	0
Indomethacin	191	109	20	3	6	74	0	6	62	35	0	12	40	21	25	4	1	0
Ketoprofen	51	27	4	0	1	21	0	1	20	7	0	0	8	3	3	1	0	0

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Naproxen Other types of nonsteroidal antiinflammatory drug Unknown types of nonsteroidal antiinflammatory drug	8,351 8,277 12	4,607 4,825 4	1,256 1,338 2	150 185 0	957 260 1	2,018 2,636 1	3 2 0	197 377 0	26 27 0	2,848 4,147 2	1,608 479 2	2 1 0	121 178 0	1,809 804 2	711 324 1	200 79 0	8 2 0	2 1 0	
	Other acetaminophen and acetylsalicylic acid combinations																		
	Acetaminophen and acetylsalicylic acid with other ingredients Acetaminophen and acetylsalicylic acid without other ingredients	5,425 127	3,402 72	764 19	187 4	952 3	1,402 39	3 0	84 7	10 0	1,641 45	1,631 20	3 0	104 2	1,846 39	786 9	487 17	15 1	0 0
Pharmaceutical and illegal opioid preparations	3,842	2,163	643	56	69	1,225	3	149	18	1,223	562	115	203	1,520	326	502	471	92	1
	Buprenorphine	52	14	2	1	9	0	2	0	11	1	0	2	6	4	3	1	0	0
	Butorphanol	670	358	53	48	35	206	0	12	4	261	72	3	17	118	70	52	17	7
Other acetaminophen and acetylsalicylic acid combinations	Codeine	15	9	3	1	4	0	1	0	8	1	0	0	5	2	1	2	0	0
	Difenoxin	1	1	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0
	Dihydrocodeine	2,855	1,470	124	18	65	1,166	0	75	22	334	954	81	40	1,193	208	274	392	7
	Fentanyl (prescription)	1,711	778	30	15	673	706	0	30	8	52	673	39	6	690	118	161	265	7
	Heroin	1,401	484	48	8	32	336	1	48	11	275	152	8	34	211	93	87	16	0
	Hydrocodone alone or in combination (excluding combination products with acetaminophen, acetylsalicylic acid or ibuprofen)	673	306	18	11	6	249	0	13	9	196	78	4	25	156	56	45	44	14
	Hydromorphone	4	2	0	0	0	2	0	0	0	0	1	0	1	2	0	1	1	0
	Levorphanol	19	9	0	2	1	4	0	2	0	2	4	0	2	5	0	2	1	2
	Meperidine	1,667	840	71	9	31	680	2	38	9	356	331	58	62	698	112	121	205	155
	Methadone	1,539	741	66	5	25	595	1	43	6	472	203	13	44	418	151	116	119	52
Morphine	5	3	1	0	0	2	0	0	0	2	0	1	0	2	1	1	0	0	
Naluphine Non-prescription fentanyl Other or unknown opioids Oxycodone alone or in combination (excluding combination products with acetaminophen or acetylsalicylic acid)	5,487 2,505 6,283	2,851 1,425 2,634	176 57 293	25 12 104	174 56 200	2,342 1,236 1,864	0 1 0	93 59 138	41 4 35	355 99 1,428	2,120 833 970	286 407 40	11 27 144	2,400 1,228 1,479	474 67 548	588 167 513	530 271 347	533 661 177	64 52 4
	Oxymorphone	23	8	0	0	1	7	0	0	4	3	0	0	4	1	1	3	0	0
	Pentazocine	9	8	0	0	1	6	0	1	1	5	0	2	7	1	1	3	1	0
	Propoxyphene	6	2	1	0	0	0	0	1	0	1	0	1	0	0	0	0	0	0
	Remifentanyl	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sufentanil Synthetic opioids, analogs and precursors (excluding pharmaceutical preparations)	2 45	1 17	0 0	0 0	0 2	1 12	0 0	0 0	0 3	1 1	0 13	0 2	0 1	1 15	0 0	0 2	0 7	1 2	0 0
	Tapentadol	118	64	2	2	3	50	0	7	23	30	3	6	48	9	12	17	6	0
	Tramadol	5,538	2,234	233	53	147	1,685	0	101	1,030	1,031	15	110	1,481	475	531	384	99	3
	Serotonin 5-ht 1b,1d receptor agonists	435	206	60	37	34	68	0	6	1	157	31	0	17	66	57	26	12	0
	Serotonin 5-ht 1b,1d receptor agonists: other or unknown agonists: sumatriptan	837	416	104	37	41	209	0	22	3	310	56	0	47	173	100	78	28	3
Category total:	267,376	171,957	68,844	9,159	27,001	61,720	126	4,368	739	111,134	54,683	1,168	3,035	73,896	37,353	22,028	12,915	4,886	356
Anesthetics																			
Inhalation anesthetics																			
Nitrous oxide	780	627	14	23	142	425	0	19	4	165	434	6	13	440	101	122	193	40	1

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in			Outcome					
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Other types of inhalation anesthetic	102	88	1	3	3	74	0	7	0	80	3	0	4	55	9	33	18	3	0
Unknown types of inhalation anesthetic	1	1	0	0	0	1	0	0	0	0	0	0	1	1	0	0	1	0	0
Local and/or topical anesthetics																			
Dibucaine	7	7	3	0	0	4	0	0	6	6	0	0	1	1	3	0	1	0	0
Lidocaine	2,907	2,594	712	119	162	1,347	2	244	8	2,118	129	9	323	557	423	352	147	27	2
Other or unknown local and/or topical anesthetic	2,700	2,477	1,116	114	106	957	3	167	14	2,106	115	14	228	500	539	269	90	37	3
Miscellaneous anesthetics																			
Ketamine and analogs	633	253	10	4	14	205	0	18	2	73	140	10	27	198	21	56	75	17	1
Other types of anesthetic	34	25	4	0	1	16	0	4	0	16	3	0	6	9	5	5	3	1	0
Unknown types of anesthetic	7	3	1	1	0	1	0	0	0	2	1	0	0	2	0	1	0	0	0
Category total:	7,171	6,075	1,861	264	428	3,030	5	459	28	4,566	825	39	603	1,763	1,101	838	528	125	7
Anticholinergic drugs																			
Miscellaneous anticholinergic drugs																			
Anticholinergic drugs (excluding cough and cold preparations, and plants)	3,518	1,726	157	69	92	1,245	5	148	10	1,324	177	16	181	467	227	196	157	20	2
Category total:	3,518	1,726	157	69	92	1,245	5	148	10	1,324	177	16	181	467	227	196	157	20	2
Anticoagulants																			
Miscellaneous anticoagulants																			
Glycoprotein IIa/IIb inhibitors	4	2	0	0	0	1	0	1	0	2	0	0	0	0	0	0	0	0	0
Heparins	243	203	27	3	4	140	0	19	10	167	11	0	23	83	43	14	19	0	0
Other antiplatelets	2,704	768	134	8	9	539	0	73	5	712	38	0	16	118	147	28	9	2	0
Other types of anticoagulant	8,629	3,863	495	24	30	3,020	2	274	18	3,607	173	2	64	593	663	91	67	11	0
Unknown types of anticoagulant	18	8	1	0	0	7	0	0	0	4	0	0	0	3	1	0	0	0	0
Warfarin (excluding rodenticides)	1,028	496	58	9	8	387	0	33	1	416	57	4	12	159	82	24	46	8	1
Category total:	12,626	5,340	715	44	51	4,094	2	400	34	4,908	283	6	115	956	936	157	141	21	1
Anticonvulsants																			
Anticonvulsants: carbamazepine and analogs																			
Carbamazepine	1,913	919	65	11	46	773	0	22	2	414	333	1	133	648	104	206	253	49	1
Oxcarbazepine	4,097	1,729	331	214	435	703	1	41	4	989	691	1	29	931	313	403	256	27	1
Anticonvulsants: gamma aminobutyric acid and analogs																			
Gabapentin	21,550	7,323	1,039	174	437	5,337	3	282	51	3,537	3,454	30	161	4,049	1,695	1,443	824	178	2
Other types of gamma aminobutyric acid anticonvulsant	3,304	1,139	176	28	50	831	0	50	4	666	404	14	37	579	208	253	139	17	0
Anticonvulsants: hydantoins																			
Fosphenytoin	9	6	1	0	0	5	0	0	0	4	0	0	2	5	0	2	1	0	0
Phenytoin	1,028	708	21	0	11	666	0	7	3	248	97	1	303	580	64	183	252	39	3
Miscellaneous anticonvulsants																			
Felbamate	55	19	5	0	3	11	0	0	0	19	0	0	0	4	6	1	0	0	0
Lamotrigine	11,434	4,317	559	187	769	2,596	3	189	14	2,699	1,419	5	161	2,122	701	939	646	114	2
Levetiracetam	6,513	3,358	960	374	366	1,559	4	88	7	2,732	549	5	45	900	744	377	107	19	1
Other types of anticonvulsant (excluding barbiturates)	2,377	1,059	187	115	110	591	2	49	5	871	148	1	28	382	190	163	111	31	0
Primidone	410	136	8	0	5	117	0	6	0	84	32	1	12	69	29	32	21	5	0
Succinimides	224	160	63	51	25	19	1	1	0	147	11	0	2	37	46	17	3	0	0
Topiramate	3,688	1,268	274	103	254	594	0	41	2	726	475	1	54	676	284	278	175	9	1
Unknown types of anticonvulsant (excluding barbiturates)	6	1	0	0	1	0	0	0	0	0	1	0	0	1	1	0	0	0	0
Valproic acid	6,480	2,446	201	136	329	1,709	1	58	12	1,065	873	6	384	1,625	476	464	507	78	0
Zonisamide	859	393	93	40	54	184	1	20	1	339	50	1	1	94	76	33	16	1	0

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in		Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Category total:	63,947	24,981	3,983	1,433	2,894	15,696	16	854	105	14,540	8,537	67	1,352	12,702	4,937	4,794	3,311	567	11
Antidepressants																			
Lithium	6,458	3,691	72	30	406	3,092	0	71	20	866	1,059	9	1,498	3,155	390	811	1,468	188	9
Miscellaneous antidepressants																			
Antidepressants: type unknown to consumer	32	8	0	0	2	5	0	0	1	1	6	0	1	7	1	2	2	1	0
Bupropion	21,036	11,337	746	314	1,547	8,179	5	503	43	7,536	3,534	13	177	6,093	2,109	1,734	2,058	611	9
Other types of antidepressant	459	270	1	2	13	243	0	8	3	12	190	46	17	245	17	66	91	43	2
Trazodone	23,208	8,681	688	270	1,317	6,146	1	211	48	2,507	5,967	7	119	6,507	1,510	2,577	2,092	152	7
Monoamine oxidase inhibitors (maoi)																			
Isocarboxazid	4	1	0	0	0	1	0	0	0	1	0	0	0	0	0	1	0	0	0
Other types of monoamine oxidase inhibitor (maoi)	48	22	0	0	0	22	0	0	0	20	0	0	2	2	4	2	3	0	0
Phenelzine	19	9	0	0	0	9	0	0	0	4	4	0	0	7	1	2	2	2	0
Selegiline	27	12	0	0	0	12	0	0	0	10	1	0	1	3	2	2	2	0	0
Tranylcypromine	32	15	1	0	3	11	0	0	0	7	4	0	4	10	0	2	6	0	0
Selective serotonin reuptake inhibitors (ssri)																			
Citalopram	3,884	1,539	290	95	336	761	1	48	8	834	660	3	29	805	336	269	237	28	0
Escitalopram	16,285	7,550	1,041	497	2,610	3,121	4	238	39	3,705	3,610	15	175	4,077	1,633	1,575	1,013	49	1
Fluoxetine	16,260	7,336	829	641	3,222	2,433	4	174	33	3,121	4,009	10	155	4,362	1,922	1,604	795	81	0
Fluvoxamine	667	226	24	8	45	134	0	15	0	148	58	0	17	73	37	24	23	2	0
Other types of selective serotonin reuptake inhibitor (ssri)	1,214	552	160	14	62	292	0	22	2	380	142	1	25	241	108	112	54	3	0
Paroxetine	1,911	713	127	24	128	391	0	39	4	389	273	7	41	344	149	130	89	7	0
Sertraline	22,744	11,420	2,270	709	3,829	4,254	7	294	57	5,752	5,245	17	328	6,288	2,328	2,669	1,495	80	1
Serotonin norepinephrine reuptake inhibitors (snri)																			
Duloxetine	6,343	2,282	434	81	379	1,287	1	86	14	1,203	927	20	114	1,220	467	496	302	26	1
Nefazodone	5	2	0	0	0	2	0	0	0	1	1	0	0	1	1	0	0	0	0
Other types of serotonin norepinephrine reuptake inhibitor (snri)	84	36	2	4	6	22	0	2	0	23	7	0	3	14	5	8	2	0	0
Venlafaxine	7,252	2,991	482	100	519	1,775	2	103	10	1,742	1,119	16	102	1,542	643	570	420	65	2
Tetracyclic antidepressants																			
Mirtazapine	5,103	1,544	181	47	218	1,038	0	54	6	584	900	7	44	1,019	317	431	232	23	1
Tricyclic antidepressants (tca)																			
Anitriptyline	3,848	1,735	178	76	254	1,177	0	43	7	641	976	4	54	1,253	229	271	510	253	9
Amoxapine	10	2	0	1	0	1	0	0	0	2	0	0	0	2	0	1	0	0	0
Clomipramine	296	135	10	2	10	106	0	6	1	77	44	0	12	66	30	19	25	2	0
Desipramine	32	8	1	0	1	6	0	0	0	6	1	0	0	4	2	0	3	0	0
Doxepin	1,326	480	34	18	35	375	0	16	2	175	275	1	17	337	61	92	130	49	3
Imipramine	139	56	13	4	4	32	0	3	0	45	9	0	2	29	11	11	6	2	0
Loxapine	66	20	1	0	1	18	0	0	0	4	16	0	0	17	6	3	7	1	0
Nortriptyline	745	349	30	12	31	259	0	17	0	220	97	5	19	167	53	62	62	17	0
Other types of tricyclic antidepressant (tca)	131	46	1	1	12	31	0	1	1	10	25	1	1	37	5	3	12	9	2
Protriptyline	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Tricyclic antidepressants (tca) formulated with a benzodiazepine	5	2	0	0	0	1	0	1	0	1	0	0	1	0	0	1	0	0	0
Tricyclic antidepressants (tca) formulated with a phenothiazine	2	1	0	0	0	1	0	0	0	0	1	0	0	1	0	0	0	1	0
Tricyclic antidepressants (tca): type unknown to consumer	14	5	0	0	1	4	0	0	0	0	4	0	1	5	0	1	2	1	0
Trimipramine	2	2	1	0	1	0	0	0	0	1	1	0	0	1	1	0	0	0	0

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age					Reason				Treated in				Outcome			
			≤ 5	6-12	13-19	≥ 20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Category total:	139,697	63,078	7,617	2,950	14,992	35,241	25	1,954	299	30,028	29,165	182	2,959	37,934	12,378	13,551	11,143	1,696	47
Antihistamines																			
Histamine h2 antagonists																			
Cimetidine and other histamine-2 blockers	9,258	6,919	5,246	373	161	959	11	147	22	6,619	201	1	80	474	1,387	169	14	1	0
Less sedating antihistamines																			
Cetirizine	20,427	15,404	10,051	2,294	1,008	1,812	8	216	15	14,481	771	12	91	1,333	2,815	517	62	2	0
Fexofenadine	3,274	2,377	709	250	232	1,036	2	139	9	2,204	138	3	24	221	370	86	19	0	0
Levocetirizine	2,213	1,628	972	302	104	202	0	42	6	1,538	67	1	16	135	306	58	9	0	0
Loratadine	13,216	10,123	5,528	1,663	743	1,917	9	243	20	9,484	522	5	77	971	1,845	327	69	4	0
Other less sedating antihistamines	175	135	92	6	5	30	0	2	0	130	4	0	0	22	32	12	1	0	0
Miscellaneous antihistamines																			
Other antihistamines alone (excluding cough and cold preparations)	1,228	689	327	124	110	111	3	11	3	509	156	0	18	230	113	93	56	7	1
Sedating antihistamines																			
Diphenhydramine alone	39,755	26,338	8,880	1,733	6,697	8,415	17	463	133	13,547	12,039	21	433	13,825	4,398	4,394	5,431	834	15
Hydroxyzine	21,502	7,934	720	361	2,459	4,139	2	199	54	2,018	5,718	3	122	5,897	1,865	2,283	1,391	94	0
Obsolete: diphenhydramine alone (prescription)	6	4	0	0	1	3	0	0	0	2	1	0	1	2	2	1	0	0	0
Obsolete: diphenhydramine alone (unknown if over the counter or prescription)	224	194	133	22	11	19	0	9	0	177	14	0	2	32	10	7	3	0	0
Other sedating antihistamines	6,337	4,044	1,922	329	354	1,276	3	142	18	3,199	655	11	140	1,190	917	510	254	28	1
Category total:	117,615	75,789	34,580	7,457	11,885	19,919	55	1,613	280	53,908	20,286	57	1,004	24,332	14,060	8,457	7,309	970	17
Antimicrobials																			
Anthelmintics																			
Diethylcarbamazine	17	17	7	0	0	8	0	1	1	16	1	0	0	1	2	0	1	0	0
Levamisole	25	20	2	0	1	14	0	3	0	16	2	0	1	6	0	3	2	1	0
Other types of anthelmintic	2,309	2,099	871	182	70	826	3	141	6	1,629	340	6	107	486	406	214	95	11	1
Piperazine	75	69	52	1	1	13	0	1	1	68	1	0	0	6	15	2	0	0	0
Unknown types of anthelmintic	5	5	2	0	1	2	0	0	0	4	1	0	0	0	0	0	0	0	0
Antibiotics																			
Systemic antibiotic preparations (oral, intravenous, intramuscular)	27,857	21,970	9,189	2,529	1,223	7,719	49	1,153	108	18,933	1,023	16	1,927	2,706	3,256	1,643	329	34	1
Topical antibiotic preparations (dermal, otc, ophthalmic, nasal)	4,367	4,155	2,556	255	111	949	16	252	16	3,977	51	4	117	114	557	185	6	0	0
Unknown types of antibiotic preparation	274	200	107	13	12	58	1	7	2	171	6	0	23	19	23	14	6	0	0
Antifungals																			
Systemic antifungal preparations (oral, intravenous, intramuscular)	1,326	971	377	42	27	440	2	79	4	875	12	1	78	91	138	53	17	3	0
Topical antifungal preparations (dermal, otc, ophthalmic, nasal)	5,682	5,328	2,997	193	96	1,666	9	344	23	5,098	62	6	142	328	690	340	44	1	0
Unknown types of antifungal preparation	8	8	0	0	0	6	0	2	0	7	0	0	1	0	0	0	0	0	0
Antiparasitics																			
Antimalarials	977	524	102	33	44	310	0	33	2	431	53	0	35	193	125	51	43	13	1
Metronidazole	1,556	842	184	17	56	479	0	100	6	681	66	1	90	130	105	93	16	1	0
Other types of antiparasitic	30	23	8	1	3	11	0	0	0	16	1	0	5	5	3	6	0	0	0
Antituberculars																			
Isoniazid	60	32	3	0	2	27	0	0	0	16	1	0	12	25	7	4	5	8	0
Other types of antitubercular	33	12	1	0	2	7	0	2	0	9	3	0	0	2	2	1	0	0	0
Rifampin	103	67	13	6	10	35	0	3	0	45	12	0	9	27	10	12	5	0	0
Unknown types of antitubercular	2	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in				Outcome				
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Antivirals	462	163	24	29	15	88	0	7	0	124	21	0	12	55	36	14	14	5	0
	1,817	1,133	52	7	33	911	1	123	6	927	95	0	104	191	178	88	20	1	0
	649	584	177	128	57	187	1	33	1	533	4	0	45	57	85	40	9	1	0
	1,999	1,403	285	25	65	901	3	119	5	1,165	112	0	113	253	214	107	59	14	2
	164	157	49	8	10	71	0	17	2	143	1	0	12	7	12	11	0	0	0
	33	25	6	0	2	17	0	0	0	23	1	0	1	3	5	2	0	0	0
Miscellaneous antimicrobials	240	206	117	13	3	66	0	7	0	195	5	2	1	29	37	15	3	0	0
	9	5	1	0	0	2	0	1	1	5	0	0	0	0	0	0	0	0	0
	50,079	40,019	17,182	3,482	1,844	14,814	85	2,428	184	35,108	1,874	36	2,835	4,734	5,906	2,898	674	93	5
	2,869	2,264	244	47	60	1,722	2	178	11	2,094	46	4	111	756	416	238	156	33	9
	2,869	2,264	244	47	60	1,722	2	178	11	2,094	46	4	111	756	416	238	156	33	9
Miscellaneous antineoplastics	3,795	3,413	1,449	660	254	902	9	125	14	2,736	408	8	236	485	497	398	188	2	0
	107	75	7	0	1	65	1	1	0	39	4	0	29	51	9	6	17	12	3
	4,253	2,511	1,290	381	138	599	3	92	8	2,370	119	1	12	273	431	53	10	0	0
	2,184	2,103	719	420	131	710	4	118	1	2,014	62	2	20	578	154	793	170	1	0
	283	230	80	33	9	230	1	7	0	209	12	0	9	33	50	26	4	1	0
	1,061	914	173	135	43	471	0	87	5	823	45	1	43	93	122	87	44	2	0
	3	2	1	0	0	1	0	0	0	2	0	0	0	0	0	0	0	0	0
Cardiovascular drugs	11,686	9,248	3,719	1,629	576	2,848	18	430	28	8,193	650	12	349	1,513	1,263	1,363	433	18	3
Angiotensin converting enzyme inhibitor	1,220	608	181	34	27	340	0	24	2	538	63	1	5	173	170	37	19	1	0
Angiotensin converting enzyme inhibitor, alone																			
Angiotensin receptor blocker																			
Antihyperlipidemic																			
Antihyperlipidemic, alone																			

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age					Reason				Treated in			Outcome					
			Age					Reason				Health care facility			Outcome					
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death	
Antihypertensives	7,066	3,833	922	1,701	772	388	2	38	10	3,086	623	0	90	1,839	1,000	637	622	27	1	
	6	4	1	0	0	3	0	0	0	4	0	0	0	2	1	0	0	0	0	
	5	2	0	0	0	2	0	0	0	0	1	0	0	2	0	0	0	0	0	
Beta blockers	174	73	11	1	3	54	0	3	1	70	3	0	0	22	20	5	5	0	0	
	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Calcium antagonist	30,519	11,783	2,468	395	585	7,880	0	428	27	9,290	2,185	6	208	4,698	3,454	953	1,132	165	20	
	262	152	13	8	3	120	0	8	0	137	9	0	3	54	59	10	5	1	1	
	160	101	23	4	1	70	0	3	0	96	5	0	0	46	36	7	6	0	0	
	15	8	0	1	1	6	0	0	0	7	1	0	0	2	1	0	1	0	0	
	3	2	0	0	0	2	0	0	0	2	0	0	0	1	0	0	0	0	0	
	60	42	9	3	1	28	0	1	0	38	4	0	0	12	15	1	6	1	0	
	17,205	6,693	1,166	212	215	4,816	4	266	14	5,704	850	0	106	3,271	2,299	522	525	119	43	
Miscellaneous cardiovascular drugs	6,554	1,727	223	45	176	1,179	0	95	9	1,067	568	2	74	774	385	292	176	12	0	
	2,843	1,393	100	17	13	1,176	0	80	7	1,247	57	0	75	720	413	127	183	40	7	
	1,246	882	37	5	5	828	0	6	1	295	30	0	508	774	87	82	392	117	30	
	11,842	5,786	1,605	1,454	1,162	1,487	5	62	11	3,700	1,895	17	100	3,806	951	1,114	1,802	192	1	
	1,911	656	101	9	32	488	0	23	3	510	123	0	19	322	190	86	63	5	0	
	913	234	36	2	3	180	0	12	1	217	11	0	4	86	54	29	20	0	0	
	699	369	170	21	4	153	0	20	1	287	68	1	8	176	137	33	31	1	1	
	6	6	0	0	0	5	0	1	0	1	0	0	5	6	1	0	0	1	0	
	563	210	26	6	10	153	0	14	1	167	33	0	6	83	48	19	24	6	2	
	2,054	1,293	400	79	58	656	3	84	13	1,020	140	4	115	470	327	170	76	6	0	
	40	12	5	0	0	3	0	2	2	9	2	1	0	6	3	1	1	0	0	
	27	11	2	1	1	7	0	0	0	10	0	0	1	3	2	0	1	0	0	
	237	186	49	16	11	98	0	10	2	167	8	1	9	68	25	49	26	0	0	
	129,177	52,014	11,500	4,591	3,591	30,114	25	2,043	150	42,063	7,991	45	1,523	20,783	13,507	4,918	5,427	710	108	
	Cold and cough preparations																			
Acetaminophen and acetylsalicylic acid with decongestant and/or antihistamine	2	1	1	0	0	0	0	0	0	1	0	0	0	1	1	0	0	0	0	
	6	3	1	1	1	0	0	0	0	2	1	0	0	1	2	0	0	0	0	

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age					Reason				Treated in			Outcome			
			Age					Reason				Health care facility			Outcome			
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	None	Minor	Moderate	Major	Death
Acetaminophen and acetylsalicylic acid with decongestant without opioids	2	1	0	0	0	1	0	0	0	0	1	0	0	1	0	1	0	0
Acetaminophen, acetylsalicylic acid, and dextromethorphan with antihistamine	6	5	2	0	1	2	0	0	0	0	4	1	0	0	1	1	0	0
Acetaminophen, acetylsalicylic acid, and dextromethorphan with decongestant	8	4	2	1	1	0	0	0	0	0	3	1	0	0	0	0	0	0
Acetaminophen, acetylsalicylic acid, and dextromethorphan with decongestant and antihistamine	1	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0
Obsolete: acetaminophen, acetylsalicylic acid, and dextromethorphan combinations with decongestant and/or antihistamine without phenylpropanolamine	2	1	1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0
Acetaminophen with decongestant and/or antihistamine																		
Acetaminophen and codeine with antihistamine	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Acetaminophen and codeine with decongestant and antihistamine	5	3	1	0	0	2	0	0	0	0	3	0	0	0	1	0	0	0
Acetaminophen and dextromethorphan with antihistamine	5,946	2,379	645	138	419	1,077	0	79	21	1,253	986	4	94	1,169	447	254	28	3
Acetaminophen and dextromethorphan with antihistamine	5,495	2,682	1,193	239	343	816	4	79	8	2,080	462	1	115	671	528	100	13	1
Acetaminophen and dextromethorphan with decongestant and antihistamine	2,553	1,518	823	169	143	349	1	31	2	1,232	217	1	56	326	281	158	60	3
Acetaminophen and other opioid with antihistamine	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Acetaminophen and other opioid with decongestant and antihistamine	4	1	0	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0
Acetaminophen with antihistamine	372	232	50	10	45	121	0	6	0	91	131	1	7	147	34	55	62	1
Acetaminophen without opioids	1,009	547	333	89	59	60	0	6	0	462	63	1	18	113	94	41	17	2
Acetaminophen with decongestant and antihistamine without opioids	722	380	165	30	45	129	1	8	2	297	55	0	22	75	59	39	14	0
Obsolete: acetaminophen and dextromethorphan combinations with decongestant and/or antihistamine without phenylpropanolamine	6	6	4	1	0	1	0	0	0	5	0	0	0	1	1	0	0	0
Acetylsalicylic acid with decongestant and/or antihistamine																		
Acetylsalicylic acid and dextromethorphan with antihistamine	6	3	2	0	0	1	0	0	0	2	1	0	0	1	0	1	0	0

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in		Outcome					
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major
Acetylsalicylic acid and dextromethorphan with decongestant and antihistamine	4	3	3	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0
Acetylsalicylic acid with antihistamine	8	7	5	0	0	1	0	1	0	6	0	0	1	2	0	1	0	0
Acetylsalicylic acid with antihistamine without opioids	60	41	27	7	1	4	0	2	0	39	0	0	2	6	6	3	1	0
Acetylsalicylic acid with decongestant and antihistamine	2	1	0	0	0	1	0	0	0	1	0	0	0	0	1	0	0	0
Acetylsalicylic acid with decongestant without opioids																		
Antihistamine and/or decongestant																		
Antihistamine and decongestant with codeine	29	11	4	3	0	4	0	0	0	10	0	0	0	1	3	1	1	0
Antihistamine and decongestant with dextromethorphan	2,787	2,248	1,363	501	110	243	1	28	2	2,102	98	1	39	307	430	170	56	3
Antihistamine and decongestant with other opioid	8	6	3	1	0	2	0	0	0	4	1	0	0	2	1	0	1	0
Antihistamine and decongestant without opioid	2,468	1,913	1,142	270	108	345	1	41	6	1,777	85	0	33	274	379	109	34	2
Antihistamine with codeine	94	70	7	5	8	47	0	2	1	37	27	1	4	32	9	12	8	2
Antihistamine with dextromethorphan	2,956	2,333	717	316	210	1,045	1	42	2	1,586	678	3	37	853	308	364	363	37
Antihistamine with other opioid	51	40	3	3	1	32	0	1	0	30	9	0	1	15	12	1	4	2
Antihistamine without opioid	1,072	672	292	21	76	272	0	10	1	370	285	0	10	356	145	98	129	20
Decongestant with codeine	18	13	1	3	0	7	0	1	1	12	0	0	1	4	5	1	0	0
Decongestant with dextromethorphan	1,695	1,283	881	243	38	106	1	10	4	1,185	60	0	30	153	233	64	27	3
Decongestant with other opioid	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
Decongestant without opioid	2,971	1,967	835	111	172	752	2	88	7	1,766	117	2	64	235	380	128	43	1
Obsolete: antihistamine and/or decongestant with dextromethorphan without phenylpropanolamine	2	2	1	0	0	1	0	0	0	2	0	0	0	0	0	1	0	0
Obsolete: antihistamine and/or decongestant without phenylpropanolamine and opioid	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
Miscellaneous cold and cough preparations																		
Acetaminophen in combination with dextromethorphan (without decongestants or antihistamines)	265	152	72	22	17	39	0	2	0	113	30	2	5	34	40	14	6	1
Cough and cold preparations (not otherwise classified)	3,353	2,453	1,800	166	120	324	7	26	10	2,095	276	7	53	413	355	152	118	12
Dextromethorphan preparations (not otherwise classified)	11,465	8,308	2,601	1,018	1,323	3,126	3	217	20	5,466	2,517	12	244	3,069	1,073	1,260	1,281	94
Dextromethorphan with expectorants	866	622	271	86	81	176	0	8	0	471	127	1	18	165	76	69	58	4
Expectorants without dextromethorphan	2,173	1,410	367	79	138	725	1	88	12	1,205	141	0	50	203	241	67	25	0
Non-Opioid antitussives excluding dextromethorphan	3,004	2,305	483	147	253	1,263	4	144	11	1,874	294	1	129	780	605	349	93	16
Obsolete: expectorants or antitussives (without narcotics or narcotic analogs)	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
Obsolete: unknown types of cough and cold preparation	8	3	1	0	0	2	0	0	0	1	2	0	0	2	1	0	1	0

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in		Outcome							
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death	
Non-acetylsalicylic acid salicylates with decongestant and/or antihistamine	4	3	0	2	0	0	1	0	0	2	0	0	0	0	0	1	0	0	0	
	Phenylpropanolamine containing preparations																			
	2	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	
	3	1	1	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0
	34	19	11	1	2	5	0	0	0	17	2	0	0	0	4	0	2	1	0	0
	3	3	2	1	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0
	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	2	1	0	1	0	0	0	0	2	0	0	0	1	0	1	0	0	0	0	
49	34	22	3	2	6	0	1	0	30	3	0	0	6	10	1	1	0	0	0	
12	8	2	1	1	4	0	0	0	6	1	0	0	3	0	1	1	0	0	0	
phenylpropanolamine without opioid																				

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age					Reason				Treated in			Outcome				
			Age					Reason				Treated in			Outcome				
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Non-acetylsalicylic acid salicylates, phenylpropanolamine, and dextromethorphan combinations with decongestant and/or antihistamine	2	2	1	0	0	1	0	0	0	2	0	0	0	0	1	0	0	0	
Other phenylpropanolamine preparations (excluding street drugs and diet aids)	237	189	85	1	2	92	0	8	1	188	0	0	1	16	54	5	0	0	
Category total:	51,867	33,900	14,234	3,691	3,722	11,185	28	929	111	25,851	6,674	38	1,034	9,448	5,818	3,870	2,762	244	8
Diagnostic agents																			
Miscellaneous diagnostic agents																			
Diagnostic tablets for glucose or ketones	3	3	0	0	0	3	0	0	0	3	0	0	0	1	0	0	0	0	0
Other types of diagnostic agent	1,631	1,546	383	55	23	872	3	185	25	1,402	23	6	108	199	199	213	31	3	1
Unknown types of diagnostic agent	6	4	0	1	1	2	0	0	0	2	0	0	2	2	2	0	0	0	0
Category total:	1,640	1,553	383	56	24	877	3	185	25	1,407	23	6	110	202	201	213	31	3	1
Dietary supplements/herbals/homeopathic																			
Amino acids																			
Creatine	298	242	164	27	14	29	0	6	2	215	13	1	12	30	49	15	4	0	0
Other amino acid dietary supplements	787	483	231	23	28	183	0	16	2	362	41	3	74	99	82	64	22	1	0
Botanical products																			
Blue cohosh	1	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0
Citrus aurantium (single ingredient)	4	3	1	0	0	2	0	0	0	2	0	0	1	0	1	0	1	0	0
Echinacea	70	44	36	4	0	4	0	0	0	39	1	0	3	2	8	3	0	0	0
Ginkgo biloba	105	74	34	2	2	30	0	6	0	61	7	0	6	9	13	10	1	0	0
Ginseng	58	38	21	0	1	13	0	2	1	28	1	0	9	9	8	5	4	0	0
Kava kava	181	98	10	5	9	70	0	3	1	33	32	1	30	52	7	31	20	3	0
Ma huang/ephedra (single ingredient)	6	4	1	0	0	3	0	0	0	1	2	0	0	3	1	1	1	0	0
Multi-botanicals with citrus aurantium	24	21	9	3	0	8	0	1	0	16	3	0	2	7	1	3	2	0	0
Multi-botanicals with ma huang	15	13	3	1	2	7	0	0	0	7	5	0	1	7	2	1	4	0	0
Multi-botanicals without ma huang or citrus aurantium	1,932	1,600	1,052	112	60	331	0	42	3	1,319	115	4	149	296	289	133	61	9	1
Other single ingredient botanicals	4,024	3,219	1,686	181	83	1,056	6	187	20	2,691	152	33	320	437	472	326	75	7	0
St. John's wort	149	117	94	5	8	8	0	2	0	107	9	0	1	13	19	7	1	0	0
Valerian	168	89	30	1	6	46	0	6	0	56	22	0	10	27	14	6	5	0	0
Yohimbe	59	41	4	0	5	31	0	1	0	15	13	0	12	29	1	12	14	4	0
Cultural medicines																			
Asian medicines	102	73	32	2	6	29	0	3	1	59	2	1	9	30	12	11	5	0	0
Ayurvedic medicines	1,065	727	430	60	77	144	4	12	0	569	82	1	72	196	151	65	27	6	0
Hispanic medicines	16	10	3	0	1	6	0	0	0	5	3	0	2	5	2	2	1	0	0
Other cultural medicines	83	67	25	4	3	31	0	4	0	40	8	1	17	28	7	7	5	1	0
Energy products																			
Energy drinks: caffeine containing (from any source including guarana, kola nut, tea, yerba mate, cocoa, etc)	1,642	1,418	840	202	163	175	5	30	3	1,116	154	7	125	229	222	213	79	4	0
Energy drinks: caffeine only (without guarana, kola nut, tea, yerba mate, cocoa, etc)	1,058	869	569	84	96	106	0	13	1	703	99	4	58	132	167	120	56	4	0
Energy drinks: ethanol and caffeine containing (from any source including guarana, kola nut, tea, yerba mate, cocoa, etc)	121	18	6	0	4	7	0	1	0	9	8	1	0	5	2	4	2	0	0

(Continued)



Table 22B. Continued.

No. of case mentions	No. of single exposures	Age				Reason				Treated in				Outcome				
		≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
2	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
Energy drinks: ethanol and caffeine only (without guarana, kola nut, tea, yerba mate, cocoa, etc)																		
1	1	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0
Energy drinks: ethanol containing without caffeine (from any source)																		
25	22	16	2	0	3	0	1	0	18	0	0	3	3	5	6	0	1	0
Energy drinks: no caffeine (from any source)																		
644	478	267	76	48	72	2	12	1	358	55	1	55	75	82	66	25	3	0
195	152	75	20	15	39	0	3	0	103	29	0	15	40	25	23	14	1	0
Energy products: other																		
223	160	64	8	16	64	0	6	2	94	26	1	36	54	20	26	16	2	0
Androgen or androgen precursor dietary supplements																		
26	15	9	0	0	3	0	3	0	13	0	0	2	0	1	0	0	0	0
48,434	42,692	34,202	4,991	1,671	1,528	71	184	45	39,955	2,334	61	189	6,172	6,916	3,169	102	4	0
2,615	2,422	1,955	277	81	92	0	14	3	2,278	110	2	21	293	413	224	7	1	0
Melatonin in combination with other substances																		
43	27	10	0	2	11	0	4	0	24	1	0	2	3	2	7	0	0	0
Phytoestrogen dietary supplements																		
Miscellaneous dietary supplements/herbals/homeopathic																		
8,337	7,694	6,785	319	74	413	21	71	11	7,430	93	3	139	538	1,172	197	21	6	0
2,560	1,989	1,223	222	73	403	6	53	9	1,706	114	6	159	264	314	153	30	3	1
Homeopathic agents																		
Unknown dietary supplements or homeopathic agents																		
1,039	520	389	35	6	82	0	5	3	480	20	3	17	40	83	30	2	0	0
588	453	329	6	11	87	3	15	2	429	12	0	9	27	72	14	4	1	0
Fatty acid supplements																		
Glucosamine (with or without chondroitin)																		
1,183	640	405	32	22	156	0	23	2	554	32	2	48	63	99	34	7	1	1
Other single ingredient non-botanical dietary supplements																		
77,883	66,535	51,011	6,705	2,587	5,273	118	729	112	60,898	3,598	136	1,608	9,218	10,735	4,988	618	62	3
Category total:																		
Diuretics																		
Miscellaneous diuretics																		
3,135	778	203	20	22	494	0	31	8	698	57	4	15	216	166	112	27	1	1
Furosemide																		
4,370	1,414	396	77	79	790	0	60	12	1,194	157	1	56	379	313	113	65	2	0
Other types of diuretic																		
4,046	1,216	355	64	49	686	1	60	1	1,074	118	1	17	261	294	76	29	2	0
Thiazide																		
52	17	5	1	2	8	0	1	0	12	3	1	0	4	3	2	1	0	0
Unknown types of diuretic																		
11,603	3,425	959	162	152	1,978	1	152	21	2,978	335	7	88	860	776	303	122	5	1
Category total:																		
Electrolytes and minerals																		
Miscellaneous electrolytes and minerals																		
10,966	9,418	7,938	624	157	565	27	95	12	9,089	234	4	62	467	1,514	191	32	1	0
Calcium and calcium salts																		
166	131	46	11	13	43	1	17	0	119	9	0	3	26	24	9	3	0	0
Chromium, trivalent																		
172	146	54	14	8	59	2	8	1	91	27	1	22	42	24	17	5	0	0
Colloidal silver																		
683	646	442	70	15	96	0	21	2	578	9	2	52	54	90	63	10	0	0
Fluoride (excluding vitamins, hydrofluoric acid & mouthwashes)																		
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Germanium and germanium salts																		
8,555	6,431	2,540	269	791	2,521	4	278	28	5,089	786	4	501	1,619	1,158	903	205	12	0
Iron and iron salts (excluding vitamins with iron)																		
8,959	6,868	3,981	502	245	1,831	15	279	15	6,033	338	18	447	812	1,104	711	110	3	2
310	246	99	26	27	88	1	5	0	157	49	0	38	67	48	44	16	5	0
Magnesium and magnesium salts																		
Multi-mineral and multi-herbal dietary supplement																		
247	185	97	17	6	56	0	7	2	143	11	1	23	16	29	22	3	0	0
214	184	86	13	8	64	0	10	3	143	11	4	25	22	30	11	3	0	0
Multi-mineral dietary supplements																		
Other types of electrolyte or mineral																		
1,995	651	140	21	32	412	0	41	5	508	108	0	24	199	128	39	47	4	0
113	81	20	1	4	47	0	7	2	60	3	2	14	25	11	9	6	1	0
Potassium and potassium salts																		
Selenium and selenium salts																		

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures		Age					Reason				Treated in		Outcome				
				≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major
Sodium and sodium salts Unknown types of electrolyte or mineral Vanadium and vanadium salts Zinc and zinc salts Category total: Eye/ear/nose/throat preparations Miscellaneous eye/ear/nose/throat preparations Topical steroids for eye/nose/throat	3,733	3,206	1,747	511	142	647	15	127	15	2,787	272	37	90	498	500	369	47	0	0
	19	18	11	3	1	3	0	0	0	15	1	0	1	2	5	1	0	0	0
	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	2,882	2,462	1,145	111	146	912	8	138	8	1,917	119	2	413	195	261	382	27	2	0
	39,020	30,673	18,346	2,193	1,595	7,344	69	1,033	93	26,729	1,977	75	1,715	4,044	4,926	2,771	514	28	2
	1,835	1,557	623	263	52	485	3	127	4	1,467	38	0	52	60	228	111	6	0	0
	1,968	1,819	457	91	140	949	3	173	6	1,550	92	12	157	213	312	157	34	3	1
	1,060	1,031	608	33	44	268	3	72	3	982	10	2	35	38	134	81	11	0	0
	26	26	9	0	2	12	0	2	1	21	5	0	0	11	6	2	1	1	0
	12	11	1	0	1	4	0	4	1	10	0	0	0	1	2	0	1	0	0
Ophthalmic preparations Contact lens products Glaucoma medications Other ophthalmic sympathomimetics Other types of ophthalmic preparation Tetrahydrozoline, ophthalmic preparations Unknown types of ophthalmic preparation	1,535	1,452	782	44	51	475	5	90	5	1,397	24	5	14	222	164	240	41	1	0
	466	59	10	5	282	377	0	43	3	377	8	1	16	41	64	39	11	1	0
	649	631	341	29	25	197	0	35	4	558	28	14	19	142	205	37	14	0	0
	2,969	2,823	1,325	109	73	1,075	8	221	12	2,609	86	27	88	204	464	101	30	4	0
	818	726	229	26	45	345	1	70	10	496	123	81	17	257	175	59	47	2	0
	57	45	11	0	2	21	0	8	3	22	6	13	3	17	4	3	5	1	0
	438	433	171	34	12	178	2	34	2	426	1	0	4	38	61	101	3	0	0
	2,418	2,394	526	97	70	1,409	5	271	16	2,349	5	1	32	365	168	808	67	0	0
	26	24	2	4	1	14	0	2	1	23	0	1	0	3	2	6	1	0	0
	Throat preparations Other types of throat preparation Throat lozenges with local anesthetics Throat lozenges without local anesthetics Category total: Gastrointestinal preparations Antacids Antacids: other types Antacids: proton pump inhibitors Antacids: salicylate-containing Antidiarrheals Antidiarrheals: diphenoxylate and atropine containing Antidiarrheals: loperamide Antidiarrheals: non-opioid containing (excluding salicyl containing) Antidiarrheals: paregoric containing	854	824	132	62	89	450	1	83	7	764	38	2	18	45	110	39	7	0
319		295	104	17	23	126	0	24	1	273	10	0	12	26	56	21	2	0	0
772		712	494	62	26	103	1	25	1	649	35	1	25	30	102	29	2	0	0
16,222		15,205	5,874	881	661	6,393	32	1,284	80	13,973	509	160	492	1,713	2,257	1,834	283	13	1
1,866		1,607	1,294	83	28	172	2	23	5	1,526	39	2	30	68	170	45	8	0	0
9,790		3,700	1,512	114	203	1,610	5	241	15	3,275	308	9	91	462	665	167	17	2	0
3,612		3,151	2,113	521	85	376	0	48	8	2,878	174	3	82	349	572	124	28	4	0
99		58	18	2	2	34	0	1	1	48	6	0	3	31	23	7	5	2	0
920		628	185	27	27	346	0	37	6	445	131	1	43	198	125	61	39	16	3
26		22	14	0	1	5	0	2	0	18	1	0	2	2	3	2	0	0	0
1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age					Reason				Treated in		Outcome					
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Antispasmodics	2,641	1,078	283	85	126	529	0	51	4	783	224	4	59	434	241	174	105	4	0
	603	250	28	5	4	193	0	20	0	236	3	0	9	27	35	7	4	0	0
	Antispasmodics: anticholinergic containing																		
Miscellaneous gastrointestinal preparations	15,458	13,241	8,039	784	520	3,227	26	593	52	12,039	504	41	621	1,345	1,680	1,215	162	3	1
	10,450	8,422	6,119	404	129	1,444	12	286	28	7,853	220	10	310	574	1,293	392	66	4	0
	Other types of gastrointestinal preparations																		
Serotonin 5-HT3 receptor antagonists	17	6	5	0	0	1	0	0	0	6	0	0	0	0	2	0	0	0	0
	Unknown types of gastrointestinal preparation																		
	Serotonin 5-HT3 receptor antagonists																		
Hormones and hormone antagonists	4,277	2,609	1,511	241	165	609	4	74	5	2,264	255	0	77	667	719	185	59	11	0
	15	3	1	0	0	2	0	0	0	1	0	0	2	0	1	0	0	0	0
	Serotonin 5-HT3 receptor antagonists other or unknown																		
Hypoglycemic, combination	49,775	34,776	21,123	2,266	1,290	8,548	49	1,376	124	31,373	1,865	70	1,329	4,157	5,529	2,379	493	46	4
	Category total:																		
	Hormones and hormone antagonists																		
Hypoglycemic, combination	381	176	24	2	6	135	0	8	1	156	13	0	7	28	38	7	7	2	1
	Hypoglycemic: biguanide combinations (excluding sulfonylurea)																		
	Hypoglycemic: other or unknown oral hypoglycemic combination																		
Hypoglycemic, single agent	54	27	3	2	1	21	0	0	0	17	4	0	6	21	7	1	5	4	0
	Hypoglycemic: sulfonylurea combinations																		
	Hypoglycemic, single agent																		
Hypoglycemic, single agent	8,054	7,341	114	59	157	6,259	3	699	50	6,105	448	5	748	2,303	623	1,697	567	19	2
	Hypoglycemic: glucagon-like peptide-1 (glp-1) receptor agonists																		
	Hypoglycemic: other or unknown																		
Hypoglycemic, combination	449	406	14	4	5	331	0	52	0	340	20	6	40	72	34	67	31	1	0
	6,752	5,553	129	90	192	4,814	0	304	24	4,538	891	13	74	2,779	1,923	450	1,024	108	2
	32	20	8	0	1	11	0	0	0	18	2	0	0	5	8	1	0	0	0
Hypoglycemic, combination	9,792	3,936	607	105	331	2,660	4	217	12	3,048	702	4	150	1,217	661	420	276	83	30
	716	214	55	5	1	135	0	18	0	201	6	1	5	48	55	10	2	0	0
	Oral hypoglycemics: biguanides																		
Hypoglycemic, combination	82	37	14	1	0	20	0	2	0	34	2	0	1	22	9	5	2	0	0
	3,428	1,374	230	24	20	999	0	96	5	1,275	59	0	35	214	314	50	28	1	0
	Oral hypoglycemics: sodium glucose co-transporter 2 inhibitor (sglt2) inhibitors																		
Hypoglycemic, combination	2,707	985	373	32	22	532	1	24	1	778	104	2	75	760	308	78	271	55	1
	364	90	32	3	3	45	0	6	1	85	4	0	1	20	27	6	2	0	0
	Oral hypoglycemics: sulfonylureas																		
Miscellaneous hormones and hormone antagonists	481	370	58	6	29	235	1	38	3	298	28	2	40	73	67	58	17	2	0
	12,204	9,981	3,410	827	414	4,515	11	741	63	9,279	183	11	480	662	1,239	407	53	2	0
	1,294	902	258	20	33	507	1	76	7	823	39	8	27	88	133	56	7	0	0
Miscellaneous hormones and hormone antagonists	2,022	1,454	900	49	181	271	6	43	4	1,253	163	3	21	170	222	82	7	0	0
	845	629	129	28	21	401	0	46	4	596	18	0	13	76	114	19	8	1	0
	949	705	144	91	58	361	2	46	3	638	34	2	31	173	144	66	23	1	1
Miscellaneous hormones and hormone antagonists	1,555	1,124	519	54	53	423	2	64	9	1,012	41	0	64	121	162	68	10	1	0
	217	126	23	8	6	82	0	6	1	119	3	0	4	19	22	7	1	0	0
	Selective estrogen receptor modulators																		

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason			Treated in			Outcome						
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Thyroid preparations (including synthetics and extracts)	14,711	10,137	3,285	331	264	5,574	10	628	45	9,760	272	8	59	1,081	1,528	189	70	3	1
Unknown hormones or hormone antagonists	9	8	0	0	0	7	0	1	0	3	2	1	2	5	0	2	0	0	0
Category total:	67,117	45,604	10,331	1,741	1,798	28,344	41	3,116	233	40,384	3,038	66	1,884	9,957	7,639	3,746	2,411	283	38
Miscellaneous drugs																			
Alzheimer drugs																			
Miscellaneous alzheimer drugs	3,077	879	80	10	23	730	0	34	2	785	47	0	31	340	133	167	105	9	0
Antidotes																			
Miscellaneous antidote drugs	523	293	53	14	21	187	1	13	4	227	21	3	40	92	54	37	15	0	0
Other miscellaneous drugs																			
Allopurinol	1,072	273	105	9	8	139	0	10	2	251	15	1	6	37	49	9	4	2	0
Bisphosphonates (including combinations)	739	666	4	0	2	612	1	45	2	639	2	0	24	161	118	61	16	0	0
Disulfiram	160	42	5	3	0	31	0	3	0	17	13	0	8	18	6	5	9	0	0
Ergot alkaloids	32	16	6	0	1	8	0	1	0	12	2	0	2	8	6	2	0	0	0
Hematopoietics	48	38	7	2	2	24	0	2	1	34	1	0	3	11	6	1	0	0	0
Monoclonal antibodies (including fragments)	480	444	34	17	10	330	0	50	3	376	5	0	60	73	42	43	17	1	1
Neuromuscular blocking agents (succinylcholine, curare, etc)	31	16	1	0	0	14	1	0	0	12	0	0	4	12	2	2	2	3	0
Nicotine pharmaceuticals	1,138	1,071	559	99	42	316	0	52	3	910	96	4	58	195	205	152	22	2	0
Other types of miscellaneous prescription or over the counter drug	14,644	9,729	2,709	656	754	4,764	12	772	62	8,553	622	22	465	1,950	1,741	945	333	40	3
Parkinson drugs																			
Decarboxylase inhibitor, alone	67	30	7	0	1	19	0	3	0	26	2	0	1	6	6	1	3	0	0
Levodopa (alone or with decarboxylase inhibitor)	1,686	955	106	10	3	774	0	56	6	852	63	1	33	213	178	124	50	2	0
Levodopa and carbidopa with other drugs	59	43	5	0	0	36	0	2	0	41	1	0	1	10	9	5	2	0	0
Other parkinson drugs (including combinations)	1,469	548	151	18	9	342	0	27	1	439	74	0	31	204	88	129	63	5	1
Category total:	25,225	15,043	3,832	838	876	8,326	15	1,070	86	13,174	964	31	767	3,330	2,643	1,683	641	64	5
Muscle relaxants																			
Miscellaneous muscle relaxants																			
Baclofen	4,195	1,760	188	56	92	1,388	1	29	6	675	856	29	101	1,362	217	287	505	328	3
Carisoprodol (formulated alone)	553	235	10	2	9	206	0	5	3	51	164	2	6	203	21	56	88	17	1
Cyclobenzaprine	7,623	3,062	576	151	220	1,996	0	106	13	1,608	1,356	3	60	1,774	573	664	538	120	1
Metaxalone	185	87	10	3	4	63	1	6	0	49	32	0	6	38	9	19	10	3	0
Methocarbamol	3,190	1,170	155	19	118	830	0	44	4	564	562	0	40	657	256	269	160	28	0
Other types of muscle relaxant	289	90	9	3	9	63	0	5	1	49	36	0	4	55	17	25	19	1	0
Tizanidine	4,892	1,977	206	49	84	1,564	0	66	8	916	922	7	94	1,397	222	431	695	93	2
Unknown types of muscle relaxant	119	22	0	0	2	17	0	3	0	5	16	0	1	20	2	5	8	0	0
Category total:	21,046	8,403	1,154	283	538	6,127	2	264	35	3,917	3,944	41	312	5,506	1,317	1,756	2,023	590	7
Opioid antagonists																			
Miscellaneous opioid antagonists																			
Miscellaneous opioid antagonist	1,822	706	65	18	51	500	1	62	9	402	140	26	126	243	77	116	77	25	0
Category total:	1,822	706	65	18	51	500	1	62	9	402	140	26	126	243	77	116	77	25	0
Radiopharmaceuticals																			
Miscellaneous radiopharmaceutical																			
Specific pharmaceutical radionuclides	46	32	5	2	0	20	1	3	1	18	1	1	10	7	4	2	0	1	0
Category total:	46	32	5	2	0	20	1	3	1	18	1	1	10	7	4	2	0	1	0
(Continued)																			

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in			Outcome							
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death		
Sedative/hypnotics/antipsychotics	Barbiturates	1,085	620	101	23	27	437	0	30	2	507	69	3	29	173	118	53	54	15	0	
		123	62	6	1	3	46	0	6	0	48	11	1	0	24	16	14	3	2	0	
		20	5	1	0	0	4	0	0	0	1	1	0	0	5	0	0	1	1	0	
	Miscellaneous sedative/hypnotics/antipsychotics	44,029	16,450	1,536	848	2,745	10,837	7	391	86	5,735	9,734	46	692	12,228	2,826	4,300	4,061	665	9	
		40,288	14,363	1,473	380	1,330	10,508	12	527	133	4,233	9,302	267	299	10,620	2,573	4,152	2,866	535	12	
		8,963	2,490	401	106	394	1,490	3	87	9	1,155	1,235	1	77	1,401	615	617	260	16	0	
		10	6	0	0	1	4	0	1	0	1	3	0	2	4	0	1	3	0	0	
		5	5	2	1	0	1	0	1	0	5	0	0	0	1	0	1	1	0	0	
		2	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	
		3	3	1	0	0	2	0	0	0	2	1	0	0	2	0	0	0	1	0	
Miscellaneous sedative/hypnotics	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	7,351	2,947	179	72	139	2,436	2	108	11	1,082	1,730	12	49	2,026	376	840	640	112	2		
	2,320	819	39	23	48	663	0	36	10	283	340	27	149	593	90	139	262	37	1		
	Serums, toxoids, vaccines	Phenothiazines	1,523	618	56	20	71	439	2	28	2	319	227	3	62	371	109	122	127	14	2
			2,016	1,470	859	12	119	450	0	25	5	973	474	3	11	664	396	175	211	26	1
134			45	1	0	3	33	0	7	1	2	34	5	3	41	5	16	12	1	0	
Category total:		107,873	39,904	4,655	1,487	4,880	27,350	26	1,247	259	14,347	23,161	368	1,373	28,153	7,124	10,450	8,501	1,425	27	
		Miscellaneous serums, toxoids, vaccines	1,588	1,335	198	65	94	758	2	199	19	1,136	8	4	182	426	191	264	57	3	1
Category total:			1,588	1,335	198	65	94	758	2	199	19	1,136	8	4	182	426	191	264	57	3	1
			Stimulants and street drugs	Cannabinoids and analogs	2,095	1,707	835	200	138	469	6	52	7	1,213	290	8	173	800	298	361	254
8		4			1	0	0	3	0	0	0	6	2	0	0	6	0	1	6	0	0
1,092		842			367	57	324	76	6	11	1	421	368	5	38	519	79	312	183	18	0
Ecigarettes: marijuana device with added flavors		65		55	33	4	15	3	0	0	0	35	20	0	0	26	7	21	10	0	0
	44	40		26	1	8	3	0	1	1	28	11	0	1	18	2	11	6	1	0	
Ecigarettes: marijuana liquid	71	53		32	4	13	4	0	0	0	35	16	0	1	34	3	17	13	4	0	
	5	5		4	0	1	0	0	0	0	4	1	0	0	2	0	3	0	0	0	
Ecigarettes: marijuana liquid with added flavors	7	7		4	0	2	0	0	1	0	4	3	0	0	5	0	1	0	0	0	
	1,015	679		251	40	144	196	7	36	5	373	251	6	39	434	60	192	169	18	0	
Marijuana: other or unknown preparation	6,550	3,282		1,273	335	586	885	8	146	49	1,839	1,088	63	205	2,196	226	872	900	142	0	
	13,044	11,450	4,736	1,884	1,957	2,591	20	214	48	7,197	3,421	92	562	8,356	719	3,802	3,264	275	1		
	117	99	30	14	12	36	1	6	0	52	37	0	10	64	6	30	24	0	0		
	2,238	840	437	100	112	153	1	33	4	538	186	15	35	677	33	227	336	52	1		

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in		Outcome							
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death	
Marijuana: pharmaceutical preparation	40	19	5	0	2	12	0	0	0	8	4	1	6	13	0	7	6	0	0	
	38	35	13	2	1	16	0	3	0	27	3	0	5	5	6	6	2	0	0	
	33	18	8	0	3	4	0	2	1	10	6	1	1	11	4	4	1	0	0	
	3,972	3,459	1,212	452	607	1,084	4	80	20	1,831	1,346	21	219	2,583	190	984	1,141	98	0	
	684	427	39	14	62	296	1	12	3	87	315	7	7	360	16	119	168	47	1	
Diet aids																				
Diet aids: phenylpropanolamine and caffeine combinations																				
Diet aids: phenylpropanolamine only																				
Other types of diet aid, over the counter only																				
Other types of diet aid, prescription only																				
Unknown types of diet aid																				
Miscellaneous stimulants and street drugs																				
Amphetamines and related compounds																				
Amyl or butyl nitrites (street drugs)																				
Caffeine																				
Cocaine																				
Ephedrine																				
Gamma-hydroxybutyric acid including analogs or precursors																				
Hallucinogenic amphetamines																				
Kratom																				
Lysergic acid diethylamide (lsd)																				
Mescaline/Peyote																				
Methamphetamines																				
Methylphenidate																				
Other hallucinogens																				
Other stimulants (excluding amphetamines)																				
Other street drugs																				
Other synthetic street drugs																				
Phencyclidine (pcp)																				
Synthetic cathinones, analogs and precursors																				
Synthetic phenethylamines, analogs, and precursors																				
Synthetic tryptamines, analogs and precursors																				
Unknown hallucinogens																				
Unknown stimulants or street drugs																				
Mushrooms, processed preparations																				
Mushrooms, processed preparations: amanita containing																				
Mushrooms, processed preparations: psilocybin containing																				
Category total:																				

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in			Outcome						
			≤ 5	6-12	13-19	≥ 20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death	
Topical preparations Miscellaneous topical preparations	1,960	1,827	723	142	287	556	5	105	9	1,696	54	5	56	169	229	203	32	0	0	
	7,087	7,042	272	18	782	5,036	4	851	79	6,949	31	7	51	499	1,042	331	25	0	0	
	1,112	1,082	616	56	17	357	0	34	2	1,056	10	1	10	114	110	133	7	0	0	
	8,815	8,634	6,223	295	219	1,593	19	261	24	8,338	146	14	102	1,222	2,031	1,036	78	13	0	
	901	886	503	46	15	257	2	56	7	836	22	1	25	112	168	125	5	0	0	
	16,110	15,782	14,743	218	97	585	40	91	8	15,701	29	3	18	408	1,944	625	27	0	0	
	23	21	9	1	1	9	0	1	0	19	0	0	2	2	2	3	0	0	0	
	6,560	6,107	1,543	319	378	3,323	11	509	24	5,782	208	38	53	661	558	1,143	80	1	0	
	1,367	1,290	192	85	97	809	1	99	7	1,121	91	1	67	223	214	155	32	2	0	
	33	29	16	2	2	7	0	2	0	26	1	0	2	4	10	3	0	0	0	
	2,168	2,125	1,208	85	57	613	4	146	12	2,027	32	4	60	226	367	256	8	1	0	
	301	285	85	7	11	285	0	153	1	248	12	2	21	74	50	39	15	3	0	
	2,701	2,640	1,818	83	44	569	8	114	4	2,480	29	0	122	147	340	313	12	0	0	
	1,932	1,866	552	77	77	992	4	152	12	1,725	64	4	49	284	255	270	23	1	1	
	53	53	22	5	3	19	0	4	0	42	3	1	7	13	11	6	2	0	0	
	102	82	23	4	17	29	0	8	1	67	4	1	9	18	14	17	2	1	0	
	6,962	6,766	3,558	453	160	2,119	14	431	31	6,613	41	5	86	140	837	242	13	0	0	
	1,315	1,275	578	87	36	470	3	92	9	1,222	7	0	36	43	136	141	6	0	0	
	950	933	437	99	39	289	2	58	9	875	13	3	39	139	149	128	26	0	0	
	60,452	58,725	33,121	2,082	2,339	17,785	117	3,042	239	56,823	797	90	815	4,498	8,467	5,169	393	22	1	
	Unknown drug																			
	Miscellaneous unknown drug																			
	25,889	16,617	3,020	782	2,469	9,039	49	882	376	5,325	6,830	1,063	557	12,799	2,448	2,468	3,961	2,074	129	
	25,889	16,617	3,020	782	2,469	9,039	49	882	376	5,325	6,830	1,063	557	12,799	2,448	2,468	3,961	2,074	129	
	Veterinary drugs																			
	Miscellaneous veterinary drugs																			
	9,016	7,981	1,502	117	157	5,295	7	860	43	7,746	133	26	48	661	1,243	627	108	24	1	
9,016	7,981	1,502	117	157	5,295	7	860	43	7,746	133	26	48	661	1,243	627	108	24	1		
Vitamins																				
Miscellaneous vitamins																				
655	434	292	38	13	69	3	17	2	401	16	0	17	41	51	14	4	0	0		
618	402	258	60	16	46	3	13	6	347	30	2	19	53	44	20	7	0	0		
Multiple vitamin liquids: adult formulations																				
10	7	6	1	0	0	0	0	0	7	0	0	0	0	2	1	0	0	0		
Multiple vitamin liquids: adult formulations with iron (no iron)																				
260	216	120	13	6	66	0	11	0	180	9	6	19	32	34	15	0	0	0		
Multiple vitamin liquids: adult formulations with iron (no fluoride)																				

(Continued)

Table 22B. Continued.

No. of case mentions	No. of single exposures	Age					Reason			Treated in health care facility		Outcome					
		≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	None	Minor	Moderate	Major	Death
Multiple vitamin liquids: adult formulations with iron and fluoride	10	9	6	1	0	0	0	0	9	0	0	0	1	1	0	0	0
Multiple vitamin liquids: adult formulations without iron or fluoride	699	577	361	68	106	0	17	0	515	22	4	32	87	52	1	0	0
Multiple vitamin liquids: pediatric formulations																	
Multiple vitamin liquids: pediatric formulations with fluoride (no iron)	89	83	78	4	1	0	0	0	80	1	0	2	11	2	1	0	0
Multiple vitamin liquids: pediatric formulations with iron (no fluoride)	584	539	511	16	9	0	1	0	529	2	0	8	92	28	1	0	0
Multiple vitamin liquids: pediatric formulations with iron and fluoride	15	13	0	0	0	0	0	0	13	0	0	0	0	0	0	0	0
Multiple vitamin liquids: pediatric formulations without iron or fluoride	653	590	506	67	8	0	1	0	575	7	0	5	79	15	2	0	0
Multiple vitamin tablets: adult formulations																	
Multiple vitamin tablets: adult formulations with fluoride (no iron)	67	60	50	5	1	2	1	0	60	0	0	0	10	2	0	0	0
Multiple vitamin tablets: adult formulations with iron (no fluoride)	4,366	3,354	2,465	101	88	599	90	6	3,167	106	3	70	598	133	18	0	0
Multiple vitamin tablets: adult formulations with iron and fluoride	31	19	14	0	1	1	1	2	18	1	0	0	2	1	0	0	0
Multiple vitamin tablets: adult formulations with iron carbonyl (no fluoride)	117	101	71	8	5	12	5	0	95	1	0	5	17	2	1	0	0
Multiple vitamin tablets: adult formulations without iron or fluoride	11,120	8,901	6,535	1,068	237	903	133	17	8,398	322	3	148	1,431	270	26	0	0
Multiple vitamin tablets: pediatric formulations																	
Multiple vitamin tablets: pediatric formulations with fluoride (no iron)	104	102	95	4	2	1	0	0	101	1	0	0	4	1	0	0	0
Multiple vitamin tablets: pediatric formulations with iron (no fluoride)	3,625	3,376	2,949	323	36	52	11	1	3,293	62	1	17	608	230	18	0	0
Multiple vitamin tablets: pediatric formulations with iron and fluoride	13	10	7	2	1	0	0	0	10	0	0	0	0	0	0	0	0
Multiple vitamin tablets: pediatric formulations with iron carbonyl (no fluoride)	32	27	25	2	0	0	0	0	26	0	0	1	3	3	0	0	0
Multiple vitamin tablets: pediatric formulations without iron or fluoride	15,973	15,028	12,598	2,104	194	89	16	2	14,589	390	0	21	2,002	372	7	1	0
Multiple vitamins, unspecified adult formulations																	
Multiple vitamins, unspecified adult formulations with fluoride (no iron)	6	5	5	0	0	0	0	0	5	0	0	0	3	0	0	0	0

(Continued)

Table 22B. Continued.

	No. of case mentions	No. of single exposures	Age				Reason				Treated in				Outcome				
			≤5	6-12	13-19	≥20	Unknown child	Unknown adult	Unknown age	Unint	Int	Other	Adv Rxn	Health care facility	None	Minor	Moderate	Major	Death
Multiple vitamins, unspecified adult formulations with iron (no fluoride)	719	533	363	25	16	105	1	21	2	506	18	0	7	57	91	41	4	0	0
Multiple vitamins, unspecified adult formulations with iron and fluoride	8	3	1	1	1	0	0	0	0	2	1	0	0	1	0	1	0	0	0
Multiple vitamins, unspecified adult formulations without iron or fluoride	467	388	281	45	26	31	0	4	1	350	32	0	6	35	45	13	4	0	0
Multiple vitamins, unspecified pediatric formulations																			
Multiple vitamins, unspecified pediatric formulations with fluoride (no iron)	16	16	14	1	0	0	0	1	0	15	0	1	0	1	1	2	0	0	0
Multiple vitamins, unspecified pediatric formulations with iron (no fluoride)	64	59	50	7	2	0	0	0	0	52	3	0	2	6	7	2	0	1	0
Multiple vitamins, unspecified pediatric formulations with iron and fluoride	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
Multiple vitamins, unspecified pediatric formulations without iron or fluoride	665	622	533	73	8	5	3	0	0	607	12	0	2	30	83	22	1	0	0
Other vitamins																			
Other B complex vitamins	4,928	2,800	2,001	183	80	454	5	73	4	2,566	113	4	102	275	419	74	12	0	0
Vitamin A	446	357	166	24	10	126	1	30	0	320	10	1	21	50	58	23	4	0	0
Vitamin B3 (niacin)	497	349	86	9	14	220	0	17	3	184	20	1	140	78	28	101	18	0	0
Vitamin B6 (pyridoxine)	439	262	211	12	3	33	0	3	0	250	4	0	7	19	36	6	3	0	0
Vitamin C	2,053	1,247	920	134	40	130	2	18	3	1,164	50	2	24	75	173	52	5	0	0
Vitamin D	10,744	7,292	4,415	424	307	1,935	9	187	15	6,949	208	3	104	927	1,178	308	42	4	0
Vitamin E	557	347	222	16	10	77	1	18	3	325	12	0	9	25	54	15	3	0	0
Category total:	60,651	48,129	36,229	4,839	1,153	5,081	71	688	68	45,709	1,453	31	788	3,703	7,259	1,822	182	6	0
Pharmaceuticals total:	1,513,927	931,135	372,017	67,002	95,941	357,416	1,074	33,511	4,174	689,853	196,687	4,454	29,456	307,943	166,968	115,439	76,618	15,576	968
Grand total (nonpharmaceuticals + pharmaceuticals):	2,531,739	1,831,206	790,612	138,359	139,309	651,090	4,581	95,123	12,132	1,514,592	237,219	17,826	44,061	474,558	295,853	260,757	108,130	18,594	1,274

Appendix C

Narratives of selected fatality cases

Selection of fatality narratives for publication

The narratives included in Appendix C were selected for publication using a 3-stage process consisting of qualifying, ranking, and reading. Changes in place since the 2014 report for the selection of the top 200 cases: include all pregnant subjects with RCF 1–3, all pediatric cases aged 0–2 years old, double the weighting for all cases with autopsy reports, add a weighting for age of subject (1/age in years), and a weighting for infrequency of substance category (Generic Code).

Qualifying cases were thus: Age 0–2 y/o, Pregnant, or RCF 1–Undoubtedly Responsible, 2–Probably Responsible or 3–Contributory. Fatalities by Indirect report were excluded beginning with the 2008 annual report. The ranking was based on Final Case Weighting (FCW).

FCW = $f [1/(\text{num substances in this case}), \text{WCS}, 1/\text{Age (years)}, 1/(\text{num cases in that generic code this year})]$

Where:

Weighted Case Score (WCS) = Hospital records * 8.8
+ Postmortem * 15.2
+ Blood levels * 6.9
+ Quality/Completeness * 6.4
+ Novelty/Educational value * 13.2

WCS Scores were normalized (z-score) within each America's Poison Centers reviewer before the final weighting: 25% for each (1/NumSubstances, WCS, 1/Age, 1/NumCodes).

The WCS weighting factors were the averages of review team recommendations gathered in 2006.

The top ranked narratives (201) were each read by individual reviewers who volunteered (See Appendix E) and the 2 managers (DAS and JBM). Each reader judged each narrative as "publish" or "omit" and all narratives receiving 11 or more of 15 publish votes were selected, further edited, cross-reviewed by the 2 managers and DEH (See Appendix E), and published in this report.

Fatality narratives

Narratives of the cases were selected (see Selection of Narratives for Publication, above) from the human fatalities judged related to an exposure as reported to US PCs in 2024. A structured format for narratives was required in the PC preparation of the narratives and used in the narratives presented. Abbreviations, units and normal ranges omitted from the narratives are given at the end of this appendix.

Case 107. *Acute Heloderma suspectum bite: undoubtedly responsible*

Scenario/Substances: A 34y/o male received a bite from his pet Gila monster with attachment to his hand x 4min. He had 3 syncopal events and intermittent consciousness x 1.5–2h witnessed by his girlfriend prior to calling EMS. They found him minimally responsive, lying supine in bed in apparent severe distress with pulse and respirations and ability to track paramedics. He was tachycardic with SBP 68–78. IVF, O₂ via NC and IM epinephrine administered en route to hospital without clinical response.

Past Medical History: prescription opioid/fentanyl and alcohol use, remote cocaine use, sleeping disorders, obsessive compulsive disorder, ADHD, depression.

Physical Exam (HCF): altered mental status, soft tissue wound to right hand with associated puncture, abrasion, swelling and bruising,

mottled skin, cool extremities, mydriasis with roving eye movements. BP 117/79, HR 160s, notably tachypneic, afebrile.

Laboratory/Diagnostic Findings: ABG-pH 7.1, Na 148 / K 2.6 / Cr 1.74 / Glu 225, lactate 8.3mmol/L, troponin 117 to 511 to 26,000 to 53,000, WBC 23.2. UDS was negative. CxR: borderline heart size, low lung volumes with bibasilar atelectasis.

Clinical Course: On arrival to the ED, he was in acute respiratory failure, hypotensive and having seizure-like activity. Lorazepam and levetiracetam resolved seizures. He was intubated for airway protection, sedated with propofol, midazolam and fentanyl, and vasopressors started. No mention of laryngeal edema/swelling or other procedural difficulties/complications during intubation. Empiric cefepime and vancomycin were started for T 38.4°C. Inferior lead ST elevations on ECG and persistent troponin elevations led to cardiac catheterization with normal results. DIC (INR 1.6, fibrinogen 52, D-dimer >30) ensued and cardiac ECHO showed severely reduced LVEF at 25–30%. Rotational thromboelastometry showed reduced amplitude, prolonged clot formation time and decreased max clot firmness. CT head was consistent with global ischemia, and MRI brain with multiple thrombotic events, massive middle cerebral artery stroke and herniation. Based on the prognosis, the family opted for institution of comfort measures, and he died on Day 4.

Autopsy Findings: Diagnoses: status post Gila monster bite of right hand. Status post hospitalization due to mixed anaphylactic/distributive shock, microscopic findings of the larynx and epiglottis including submucosal edema with associated scattered/increased mast cells via tryptase stain, moderately dilated cardiomyopathy, microscopic findings of severe hepatic steatosis. Cause of death: complications of Gila monster envenomation. Dilated cardiomyopathy and hepatic steatosis are contributing factors. Manner of death: accident.

Case 111. *Acute Hymenoptera envenomation: undoubtedly responsible*

Scenario/Substances: An 87y/o male was found unresponsive after sustaining >500 bee stings. EMS was called, began CPR, and transported him to the ED.

Physical Exam (HCF): intubated, ventilated and being treated for anaphylaxis, anuria. BP 40/0.

Laboratory/Diagnostic Findings: BUN 17 / Cr 1.21, AST 3602, ALT 1515, bili (indirect) 0.4, bili (direct) <0.01, CK 55,593, CK-MB 343, troponin-HS 13,000, Hgb 8.1 / Hct (50% reduced from initial). CxR: bilateral interstitial opacities.

Clinical Course: The patient was ventilated and on epinephrine, norepinephrine, vasopressin, phenylephrine, and sodium bicarbonate. One unit PRBC was given for suspected hemolytic anemia. Asystole developed and 2–3 rounds CPR were provided without ROSC and he died on Day 1.

Autopsy Findings: not available.

Case 113. *Acute calcium hydroxide inhalation: probably responsible*

Scenario/Substances: A 31y/o male was pouring concrete at home without respiratory protection and presented to the ED with acute hypoxic respiratory failure.

Past Medical History: HTN. Home medications: none.

Physical Exam (HCF): pink frothy respiratory secretions. BP 138/78, HR 65, RR 20, T 36.6°C, O₂ sat 96% on 45% FiO₂.

Laboratory/Diagnostic Findings: ABG-pH 6.8 / pCO₂ 90 / HCO₃ 15.6, Na 139 / K 3 / Cl 107 / CO₂ 20 / BUN 14 / Cr 1.25, AST 40, ALT 39, lactate 8mmol/L. CxR: bilateral airway opacities.

Clinical Course: The patient was intubated upon arrival and sedated. Broad spectrum antibiotics and corticosteroids were started, and he was placed on norepinephrine. Over the next 24 h norepinephrine was titrated off and he was successfully extubated on Day 2. He developed worsening hypoxia and was re-intubated on Day 3. CT head was reported as normal. He was placed on a heparin infusion and sedated with ketamine, midazolam, propofol, and fentanyl. Norepinephrine was restarted for hypotension. CxR on Day 4 was described as “complete white out.” Epinephrine infusion was added, and a sodium bicarbonate infusion was started. The patient was placed on ECMO and transferred to a tertiary care center. CT chest showed extensive ground glass opacities. Over the next 5 days he remained on ECMO receiving milrinone and clevidipine. On Day 8, he was extubated again but remained disoriented and still required ECMO. On Day 12, he developed recrudescence of his hypoxia and received a tracheostomy. CRRT was started and his electrolytes were aggressively repleted. On Day 20, he developed escalating ventilatory requirements with 100% FiO₂. Over the next 2 months he remained on ECMO awaiting a lung transplant. On Day 70, he died from refractory hypoxic respiratory failure.

Autopsy Findings: not performed.

Case 115. Acute hydrofluoric acid, phosphoric acid ingestion: probably responsible

Scenario/Substances: A 21m/o male ingested wheel cleaner while briefly unattended in the garage with his father. He was brought to the ED by private vehicle 22min after the ingestion. Parents confirmed the product contained hydrofluoric acid 3–7% and phosphoric acid 10–30%.

Past Medical History: unremarkable.

Physical Exam (HCF): vomiting with red streaks in vomitus, pharyngeal swelling, excessive oral secretions, stridor. BP 80/53, HR 128, RR 32, O₂ sat 99% RA, GCS 14 (E4/V4/M6).

Laboratory/Diagnostic Findings: Attempts at IV access failed, so no labs provided.

Clinical Course: In the ED, IO access was placed after failed IV attempts. The patient was intubated, mechanically ventilated, sedated and chemically paralyzed. He was transferred to a tertiary care center via helicopter, arriving at Hour 3. His left leg was cool and pulseless; IO infiltration was suspected. The patient went into VF but after CPR, ROSC was achieved. IO access was re-established with calcium chloride administered and calcium gluconate infusion initiated. The patient developed VF again, but despite PALS measures he died at Hour 4.

Autopsy Findings: not available.

Case 118. Acute ethylene glycol ingestion: probably responsible

Scenario/Substances: A 17y/o male was last seen well the previous night at 10pm. In the morning, he was sleeping. He was found at 5pm, obtunded and covered in vomit. No empty medication bottles were found.

Past Medical History: hyperlipidemia, treated with a statin, depression, prior suicide attempt.

Physical Exam (HCF): intubated, pupils miotic (received fentanyl en route), intermittently bucking ventilator. BP 145/96.

Laboratory/Diagnostic Findings: ABG-pH 6.9 / PCO₂ 18 / HCO₃ <10, Na 136 / K 6.6 / Cl 105 / CO₂ <10 / BUN 11 / Cr 1.81 / Glu 334 (after dextrose) / AG >21, lactate >16mmol/L, Serum osm 336, OG 41, AST and ALT WNL, bilirubin 2.1, lipase 114, WBC 39. Serum APAP, ethanol and salicylate not detected. UDS negative. Urine: ketones negative, tea colored/blood tinged, positive for calcium oxalate

crystals. CT head and c-spine WNL, CT chest/abd/pelvis WNL, lumbar puncture WNL.

Clinical Course: He was given calcium gluconate, sodium bicarbonate, intubated and put on propofol. Patient's father later found an empty bottle of antifreeze containing ethylene glycol in patient's room. CT prior to HD had borderline cerebral edema for which he was given hypertonic saline. 2h into HD, he became hypertensive (SBP 250s) and bradycardic, had VT arrest lasting 10min. After ROSC, there was concern for herniation with pupils fixed and dilated, cough and gag reflexes now absent. Repeat CT again showed cerebral edema but no definite herniation. Planned to start CRRT and continue fomepizole. The patient did not have full brain herniation based on imaging but seemed to have had cardiac arrest around the time cerebral pressures were increasing. He did not regain brainstem reflexes with bilateral blown pupils. Continued fomepizole but only received 1 session of HD. Based on the prognosis, his family decided on organ donation. Both lungs, both kidneys and the liver were harvested for transplant on Day 3.

Autopsy Findings: ethylene glycol 56.1 mg/dL.

Case 135. Acute hydrochloric acid ingestion: undoubtedly responsible

Scenario/Substances: A 26y/o male drank 2 sips of muriatic acid in a self-harm attempt after getting in an argument with his partner. The concentration and brand of product was unknown.

Physical Exam (HCF): drooling, no stridor, abdominal tenderness. BP 120/58, HR 68, O₂ sat 100% on 4L NC.

Laboratory/Diagnostic Findings: BUN 7.8 / Cr 1.2, AST 26, ALT 13, lactate 3.0mmol/L. Serum APAP, salicylate and ethanol not detected. CxR: elevation of left hemidiaphragm. CT chest/abdomen/pelvis: perforation of esophageal-gastric junction, pneumoperitoneum.

Clinical Course: In the ED, he complained of abdominal pain and was vomiting for which he was given morphine and metoclopramide. He was intubated due to worsening pain and concern for airway control and sedated with propofol and fentanyl. General surgery recommended transferring to a tertiary care center for thoracic surgery. The patient became hypotensive and was initially too unstable to transfer. His Cr rose to 3, BUN 34.9, AST 920, ALT 676, WBC 16 / Plt 78 / Hgb 9.3, INR 3.5. His pH dropped to 7.1 and he was started on norepinephrine and vasopressin for hypotension and tachycardia. He was also given insulin and dextrose for a potassium of 7.3, IVF for hypotension, blood products, FFP, and sodium bicarbonate. He was transferred the next day for surgery but died shortly after arrival on Day 3.

Autopsy Findings: not performed.

Case 137. Acute sodium azide ingestion: undoubtedly responsible

Scenario/Substances: A 28y/o male ingested an unknown amount of a sodium azide solution in a self-harm attempt. EMS was called and transported him to the ED.

Physical Exam (HCF): BP 140/71, HR 109, O₂ sat 98% on RA.

Laboratory/Diagnostic Findings: VBG-pH 7.4 / pCO₂ 18.6 / pO₂ 161 / HCO₃ 11.3 / AG 27, Na 143 / K 4.8 / Cl 107 / CO₂ 9 / BUN 15 / Cr 1.8, Ca 7.4, AST 19, ALT 12, lactate 17mmol/L, MetHgb 1.9%. Serum APAP, salicylate and ethanol not detected. ECG: QRS 90, QTc 523.

Clinical Course: In the ED he was intubated for airway protection and gastric lavage was performed. Hydroxocobalamin was administered. He was admitted to the ICU, sedated with propofol, fentanyl and dexmedetomidine, and a central line was placed for HD. Later that evening he required vasopressors to maintain his blood pressure. He developed bradycardia then cardiac arrest. ECMO was initiated early on Day 2, and he had 2 further cardiac arrests, 1

with bradycardia and the second with hypotension. His blood clotted rapidly precluding further lab work. A dose of methylene blue was administered. Based on the prognosis, the family opted for institution of comfort measures, and he died on Day 2.

Autopsy Findings: not available.

Case 143. Acute cyanide ingestion: undoubtedly responsible

Scenario/Substances: A 44y/o male chemist ingested 10g sodium cyanide in a suicide attempt. He regretted the action and called EMS who found him alert and anxious.

Physical Exam (HCF): BP 132/79, HR 160, RR 31, O₂ sat 85% on 15L via NRB.

Laboratory/Diagnostic Findings: pH 7.32 / HCO₃ 10.5 / pCO₂ 21.3, lactate 16 mmol/L, MetHgb 2.1%.

Clinical Course: In the ED he became progressively more sedated, was intubated, and a nasogastric tube returned 800mL of stomach contents. Activated charcoal was given. The patient received 4 doses of hydroxocobalamin 5g IV. He also received sodium thiosulfate 12.5g IV. The treating team also elected to administer fomepizole. He became progressively more hypotensive with SBP in the 60s despite epinephrine, norepinephrine, vasopressin, calcium chloride, and sodium bicarbonate. Insulin infusion was given for hyperglycemia. ~3h after exposure, pH 6.96, lactate 19 mmol/L. Lactate peaked at 26 mmol/L. CRRT was initiated for acidosis. About 9h after the ingestion he developed refractory shock, acidosis and died.

Autopsy Findings: There was pinkish discoloration of the skin and pulmonary edema. Hour 1 cyanide (hospital blood) 0.5 mcg/mL. Cause of death: cyanide toxicity. Manner of death: suicide.

Case 148. Acute potassium thiocyanate ingestion: undoubtedly responsible

Scenario/Substances: A 54y/o male was in court and remanded directly into custody. He arrived at the correctional facility 2h later stating he wanted to die. He had projectile vomiting, 1 seizure, and was incontinent of urine. EMS was called, administered lorazepam and transported him to the ED.

Past Medical History: lower back pain, depression. Home medications: finasteride, methocarbamol, cyclobenzaprine, pantoprazole.

Physical Exam (HCF): aphasic, not responding to commands, eyes open with roving eye movements, localizing to pain. He was flailing at times and moaning incoherently. He was diaphoretic with mild abdominal tenderness. BP 184/104, HR 86, RR 28, O₂ sat 97% on RA, T 37.5°C.

Laboratory/Diagnostic Findings: ABG-pH 7.46 / pCO₂ 48 / pO₂ 52 / HCO₃ 27 / Na 141 / K 5.2 / Cl >200 / CO₂ 25 / BUN 14 / Cr 1.0 / Glu 123, lactate 3.1 mmol/L, AST 26, ALT 24, bilirubin 0.6. INR 1.1, ammonia 31 mcmol/L, eGFR WNL, WBC 14.3 / Hgb 15.6 / Plt 265. Serum APAP, ethanol, salicylate and lithium not detected. UDS was positive for THC. CxR: WNL. CT head: WNL. CT abd/pelvis: fatty liver, inguinal hernia, diverticular disease. ECG: HR 86, QRS 90, QTc 402.

Clinical Course: In the ED he had a brief seizure, was intubated and placed on a ventilator. The serum chloride was re-drawn in different test tubes and re-measured; all measurements were >200. He received fluids and labetalol for blood pressure management. He was admitted to the ICU for toxic encephalopathy. In the ICU, he had waxing and waning encephalopathy and seizures. Blood pressure remained elevated. Nephrology recommended HD for unknown toxins, but family refused. Day 2 EEG: severe encephalopathy with burst suppression. Day 2 EGD showed gastritis. Day 3 MRI: acute right thalamic infarction. His blood pressure was elevated throughout his hospital stay with the lowest recorded blood pressure 126/61. Daily serum chloride over time: 199, 196, 164, 150, 142, 128. Other halides were measured. Serum iodide 104 mcg/mL, serum bromine: 1.4 mg/L

(normal 0.9–7.3 mg/L). On Day 6 he was extubated. He was following commands but had poor cognition. On Day 7 he developed new right-sided weakness and right-sided gaze; EEG showed non-convulsive status epilepticus. He was re-intubated. Family again refused HD. Day 8, he developed fevers (T 39.2°C). Comprehensive infectious disease work up was negative. Based on the prognosis, the family opted for comfort measures, and he died on Day 8.

Autopsy Findings: Investigators found an internet purchase for laboratory grade potassium thiocyanate 4 days before ingestion on the victim's personal computer. Autopsy found hypoxic-ischemic changes in the brain. Antemortem blood (95h post-ingestion) thiocyanate 260g/mL and cyanide 0.21g/mL. Toxicology on postmortem specimens: thiocyanate 78g/mL cardiac blood, 75g/mL peripheral blood, 16g/mL brain, 49g/mL kidney, 37g/mL liver and 21g/mL skeletal muscle.

Assessment/Discussion: The auto-analyzer used to measure chloride did not list thiocyanate as a possible interfering substance. However, methods that use ferric thiocyanate as reactant, such as ion selective analyzers, can falsely report an elevated chloride when thiocyanate is present. Other substances known to interfere with chloride measurement are halides such as bromide, iodide, and sodium nitroprusside. Bromine and iodide were measured and were within the limits of normal. The patient did not receive sodium nitroprusside during his hospital stay. Thiocyanate toxicity has historically been associated with hypocalcemia, but this patient's calcium remained normal. It has also been associated with hypotension, but this patient remained hypertensive throughout his hospital stay. The PC spoke with the manufacturer to correct their list of substances that can interfere with chloride measurement.

Case 168. Acute phosphoric acid cleaner and paint additive ingestion: undoubtedly responsible

Scenario/Substances: A 66y/o male ingested 16oz of phosphoric acid-based cleaner and an unknown amount of a latex paint additive 10min prior to calling EMS who found him asymptomatic.

Past Medical History: suicide attempts, Type 2DM, HTN, hyperlipidemia, cirrhosis.

Physical Exam (HCF): awake, alert and oriented, no respiratory distress. BP 133/89, HR 100, O₂ sat 95% RA.

Laboratory/Diagnostic Findings: Na 136 / K 4.3 / Cl 97 / CO₂ 17 / BUN 21 / Cr 1.27 / Glu 211, AST 27, ALT 26, ALP 99, WBC 12.3 / Hgb 13.0 / Plt 248. Serum APAP, ethanol and salicylate not detected. EKG: Multifocal atrial tachycardia, QTc 462, QRS 98.

Clinical Course: The patient arrived in the ED 25min after ingestion. Soon after he began vomiting with complaints of burning sensation in his throat and upper abdomen. He was started on a proton pump inhibitor. Gastroenterology was consulted and EGD showed Grade 2A esophageal injury with diffuse gastritis and possible deeper mucosal injury at the gastroesophageal junction. Over the next 24h, he developed worsening metabolic acidosis (CO₂ 9–11, AG 26–29), AKI (Cr 1.4 → 1.94, BUN 32 → 42), hyperphosphatemia (Phos 15.6), hyperglycemia (Glu 469) and hypocalcemia (Ca 6.0–7.0), despite treatment with IV calcium gluconate and sodium bicarbonate. ABG-pH 7.16 / pO₂ 68 / pCO₂ 33 / HCO₃ 11 / BE -17. He developed hypotension, was started on vasopressors, intubated and started on HD. He developed progressive shock while on HD and died on Day 2.

Autopsy Findings: not performed.

Case 179. Acute hydrochloric acid/cationic detergent cleaner ingestion: undoubtedly responsible

Scenario/Substances: A 94y/o female ingested a food service sanitizer containing a mixture of hydrochloric acid and concentrated cationic detergents while riding in the back seat of her daughter's

vehicle. She had a history of drinking liquids from any available containers. The daughter tried to get the patient to drink water, but the patient started vomiting up blood and then defecated. She was first driven home to get cleaned up before being driven to the ED ~90 min post ingestion.

Past Medical History: dementia, coronary artery disease, hypertension.

Physical Exam (HCF): She was in acute distress and ill-appearing. Bloody sputum in pharynx with no lacerations noted. Unclear origin of bleeding. No scleral icterus, no discharge, EOMI. BP 171/68, HR 63, RR 29, O₂ sat 94% on RA, T 35.6°C.

Laboratory/Diagnostic Findings: VB_G-pH 7.26 / pCO₂ 43 / pO₂ <49 / HCO₃ 18.9 / BE -8.2, Na 135 / K 3.3 / Cl 101 / CO₂ 14 / BUN 31 / Cr 1.32 / Glu 189 / AG 20, lactate 6.5 mmol/L, AST 41, ALT 6, bilirubin 0.8, INR 1.37, WBC 15.4 / Hgb 13.2 / Hct 42 / Plt 286. Serum APAP, ethanol and salicylate not detected. CxR: Minimal bibasilar infiltrate.

Clinical Course: In the ED, she had blood in her mouth and demonstrated dysphagia, tachypnea, and was more altered than baseline. Verbal response was limited due to blood pooling in mouth, but she was resistant to suctioning. She was agitated, pulling at lines, attempting to get out of bed, not following commands. At least 200 mL of blood was produced since arrival and more in sheets and on floor, but she was not in respiratory distress. She received antiemetics, IV fluids and antipsychotics. In the morning she was admitted to the medical ICU and was sedated, intubated and mechanically ventilated after aspiration of bloody secretions. Bronchoscopy revealed no active bleeding, or significant airway edema. EGD showed severe necrosis in oropharynx and hypopharynx. CT chest with contrast showed esophageal dilation and bilateral airspace opacities. Based on the prognosis, the family opted for institution of comfort measures, and she died on Day 3.

Autopsy Findings: not performed.

Case 181. Acute heptamethylnonane cleaner ingestion and aspiration: undoubtedly responsible

Scenario/Substances: A 14 m/o female ingested an estimated 40 mL from a 150 mL bottle of makeup brush cleaner resulting in coughing followed by drowsiness and crying, but no vomiting. The parents initially went to an urgent care and were instructed to take child to the ED. The product's labeled bottle was brought to the ED.

Past Medical History: healthy, normally developed, immunizations up-to-date, no daily medications.

Physical Exam (HCF): awake, active, crying but consolable, intermittent cough, normal breath sounds, negative wheeze, stridor, nasal flaring, or retractions. Skin pink, warm, dry, no cyanosis. BP 94/63, HR 79, RR 28, O₂ sat 96% on RA, T 36.2°C (rectal).

Laboratory/Diagnostic Findings: ABG-pH 6.82 / CO₂ 76 / PO₂ 27 / HCO₃ 12.6, Na 142 / K 3.7 / Cl 108 / CO₂ 20 / BUN 15 / Cr 0.23 / Glu 169 / AG 14, Ca 10.1 AST 72, ALT 19, WBC 4.4 / Hgb 11.6 / Hct 35.2. CxR: Lungs clear, no effusion.

Clinical Course: She was initially well appearing upon arrival to the ED 1 h post exposure. ~1.5 h post exposure, she was minimally responsive with snoring respirations. IV access was obtained and O₂ administered. Her O₂ sat continued to decrease into the 80s, and she was emergently intubated. She was difficult to oxygenate and had subcutaneous emphysema in her neck. Serial CxRs showed worsening upper lobe opacities, extensive pneumomediastinum, a newly rounded lucency in lateral left lung base, hazy and worsening interstitial opacities, but no pneumothorax. Epinephrine and norepinephrine were given via right tibia IO. A flight team and pediatric critical care provider arrived in the ED for emergent transfer to a tertiary care center. Multiple needle decompressions without rush of air or clinical improvement were attempted. Bilateral chest tubes were inserted with rush of air from left side, not right. CxR showed

diffuse opacification, ETT in right mainstem bronchus, bilateral chest tubes, enteral tube in stomach, extensive pneumomediastinum, small left pneumothorax, and no pneumothorax on right. ~4.5 h post exposure, the child had a cardiac arrest. CPR was started and multiple rounds epinephrine and sodium bicarbonate were given. ROSC was not obtained, and she died in the ED 5 h post exposure.

Autopsy Findings: Severe pulmonary congestion with hyperemia but no gross edema or sloughing of epiglottis or trachea. Active ingredient: 2,2,4,4,6,8,8-heptamethylnonane. Cause of death: toxicity of 2,2,4,4,6,8,8-heptamethylnonane. Manner of death: accident.

Case 183. Acute hydrogen peroxide and ethanol ingestion: undoubtedly responsible

Scenario/Substances: A 47 y/o female accidentally drank an estimated 30 mL of 35% food grade hydrogen peroxide to swallow her medications and then called her boyfriend. When he arrived home, she was on the floor agitated and confused. EMS was called and found her obtunded.

Past Medical History: ADHD, alcohol use. Home medications: lisdexamfetamine, clonazepam.

Physical Exam (HCF): obtunded with right gaze deviation, intermittently agitated/combatative. BP 132/79, HR 108, RR 30's, O₂ sat 96% on RA.

Laboratory/Diagnostic Findings: ABG-pH 7.26, lactate 6.9 mmol/L, WBC 31 / Hgb 15.3 / Hct 48 / Plt 265. Ethanol 39 mg/dL. UDS positive for amphetamines, benzodiazepines, THC. ECG: QRS 102, QTc 542. CT chest/abdomen/pelvis: extensive consolidation in upper lobes and superior segments of lower lobes concerning for pneumonitis; reduced portal venous gas in liver and portal vein; moderate thickening of gastric wall and mild gastric wall pneumatosis.

Clinical Course: On Days 1 and 2, the patient had several episodes of vomiting, became rapidly more obtunded and was intubated. She received naloxone, lorazepam, dexmedetomidine, propofol, norepinephrine, and pantoprazole. She was transported to a tertiary care center for HBO by air ambulance. During transport, she had frothy sputum which then turned to blood. ~400 mL of fluid was suctioned from her lungs. She had coffee ground emesis, her stomach became more distended, and she was over breathing the ventilator. On arrival at the tertiary care facility, she was on epinephrine and phenylephrine. She was deemed too unstable for HBO and admitted to the cardiac ICU. ABG-pH 7.01 / pCO₂ 69 / pO₂ 57 / HCO₃ 17.5 / BE -14, Na 142 / K 3.2 / Cl 120 / CO₂ 12 / BUN 8 / Cr 0.79 / Glu 271 / AG 10, AST 50, ALT 20, bilirubin 0.8, INR 1.4, lactate: 7.9 mmol/L, troponin 4.8, D-Dimer >40 mcg/mL. CxR showed increased bilateral lung opacities. Endoscopy showed a normal esophagus with patchy mild erythematous mucosa without bleeding in the gastric fundus and gastric antrum, duodenum was normal. She was switched to norepinephrine and propofol only, and antibiotics were added. She was cannulated for VV ECMO. Treatment included vasopressin, dobutamine, FFP, and sodium bicarbonate. BP 115/85, HR 139, RR 47, O₂ sat 95% on ventilator. Hgb/Hct decreased to 11.3/32. On Day 3, she demonstrated cardiomyopathy with a 25% LVEF. CT head showed progression of cerebral ischemia from anoxia, loss of gray-white matter differentiation and evidence of downward tentorial herniation. Hypertonic saline and mannitol were started. She had some shoulder shaking but an EEG showed no seizure activity. AST 518, ALT 123. Sedated with midazolam and hydromorphone. On Day 4, she was weaned from ECMO and decannulated. On Day 5, increased posturing, agonal respirations, and fighting the ventilator were noted. Repeat CT head showed evolving multifocal embolic strokes. On Day 6, based on the prognosis, comfort measures were instituted, and she died on Day 9.

Autopsy Findings: not available.

Case 206. Acute hydrogen sulfide inhalation/dermal exposure: undoubtedly responsible

Scenario/Substances: A 21 y/o female paramedic was found unresponsive for an unknown time in asystole in her home bathroom. The bathtub was “full of chemicals” with a “strong smell of sulfur”; and Fire Department responders noted a cyanide measurement of 9ppm. CPR and intubation were performed on scene, and epinephrine, sodium bicarbonate, and norepinephrine administered en route to the ED.

Past Medical History: Wilson’s disease, PTSD, methamphetamine use, previous self-injurious behaviors and suicidal ideation.

Physical Exam (HCF): intubated, unresponsive (without sedation). BP 135/97, HR 126, RR 34, O₂ sat 94% on ventilator (unknown FiO₂).

Laboratory/Diagnostic Findings: (post-intubation) ABG-pH 6.6 / pCO₂ 100 / pO₂ 62 / HCO₃ 11.7 / BE -27, Na 142 / K 4.2 / Glu 386, WBC 7500 / Hct 48. UDS was negative. ECG: QRS 96, QTc 513.

Clinical Course: Imaging revealed anoxic brain injury. Based on the prognosis, the family opted for institution of comfort measures, and she died 4 h after arrival.

Autopsy Findings: Postmortem investigative report stated containers of soil sulfur, sodium hydroxide, and muriatic acid were found. Lungs: markedly edematous lungs with approximately 100mL of serosanguinous fluid within the right and left pleural cavities. Cut surfaces exude a copious amount of blood and frothy fluid. CNS: severe cerebral edema with slight blurring of the gray-white junction. Marked pinkish-red discoloration of the skin, viscera, and urine (likely from hydroxocobalamin). Toxicology: antemortem blood thiocyanate 0.48mcg/mL, thiosulfate 11 mcg/mL. Cause of death: acute hydrogen sulfide intoxication. The amount of thiocyanate is negligible and in combination with a negative blood cyanide level excludes acute cyanide intoxication. Manner of death: suicide.

Case 213. Acute hydrogen sulfide inhalation/dermal exposure: undoubtedly responsible

Scenario/Substances: A 27 y/o male was performing routine maintenance in a sugar processing factory utility building with coworkers. He fell into an underground wastewater pit or entered it to rescue someone else. When he became less responsive and combative, the Fire Department was called to assist with the rescue. High concentrations of hydrogen sulfide gas were detected. The patient was in asystolic cardiac arrest and received ACLS on scene and en route to hospital.

Physical Exam (HCF): pulseless, thermal burns.

Clinical Course: In the ED, ACLS was continued, he was intubated and received IVF and methylene blue, and numerous rounds of CPR without achieving ROSC. The patient died shortly after arrival.

Autopsy Findings: Evidence of fall: abrasions and scratches on shoulder, upper arm, hand, forehead. Small contusion in the temporalis muscle on left side of head. 7 cm sub scalp/galeal hematoma extending from the vertex of the head. Respiratory system: upper airway with some granular material consistent with aspiration of gastric contents. Pulmonary parenchyma exuding moderate amounts of blood and frothy fluid with a pattern of blood aspiration. Microscopic exam not significant for any abnormalities. Toxicology: postmortem blood thiosulfate 4.2mcg/mL. Cause of death: hydrogen sulfide gas poisoning. Manner of death: accident.

Case 227. Acute acetylene dermal and inhalation: contributory

Scenario/Substances: A barrel containing acetylene gas exploded in the face of a 47 y/o male in an occupational incident. He was found unconscious and intubated on scene by EMS. Rhythm was PEA, ROSC after 1 round of epinephrine. He also underwent needle

decompression in the field. The acetylene barrel was noted to be non-standard for the material.

Past Medical History: recent appendectomy (1 week prior). Home medications: none.

Physical Exam (HCF): burns involving hands, forearms, back, groin, and abdomen (15% TBSA), pupils nonreactive initially, occasional weak cough reflex noted later. BP 115/77, HR 77, RR 16, O₂ sat 94% intubated on 100% FiO₂, T 34.7°C.

Laboratory/Diagnostic Findings: ABG-pH 7.27 / pO₂ 177 / HCO₃ 19.0, Na 140 / K 4.0 / Cl 139 / CO₂ 18 / Cr 1.17 / Glu 281 / AG 15, lactate 4.2mmol/L, CK 590. CT head/neck: Subdural and subarachnoid hemorrhages, multifocal hemorrhagic contusions, and complex facial fractures.

Clinical Course: In the ED he required norepinephrine and vasopressin for hypotension. Hypothermia was treated with a warming blanket. On Day 2, his neurologic status remained poor, with absent gag and cough reflexes. Vasopressor doses were reduced. Lactate 5.2mmol/L, no significant improvement in hemodynamics or neurological function. On Day 3, neurological decline continued with minimal reactivity, COHB 0.9%, pupils became sluggishly reactive. On Day 5, an MRI confirmed multifocal hemorrhagic contusions and bilateral occipital infarcts. Based on the prognosis, the family opted for institution of comfort measures and on Day 7 the family decided to proceed with organ donation after circulatory death. The patient remained intubated and hemodynamically stable without vasopressors and underwent the donation process on Days 9 and 10.

Autopsy Findings: not available.

Assessment/Discussion: Traumatic injury from the explosion, and not the acetylene itself was responsible for the death, though initial hypoxia from byproducts of combustion may have contributed to hypoxia.

Case 254. Acute nitrogen inhalation: probably responsible

Scenario/Substances: An 82 y/o male was found unresponsive at home after sending a suicide note via email to family. EMS found him unresponsive, O₂ sat 78%. An empty bottle of compressed nitrogen was found beside his CPAP and was believed to have been fed into the CPAP air intake.

Past Medical History: obstructive sleep apnea on CPAP.

Physical Exam (HCF): unresponsive. SBP 125 palp, HR 110, RR 27, O₂ 95% intubated, T 38.8°C.

Laboratory/Diagnostic Findings: VBG-pH 7.18 / pCO₂ 44 / pO₂ 60, AST 103, ALT 85, lactate 4.5mmol/L, troponin-HS 2430, BNPT 804, MethHgb <0.7. Serum APAP and salicylate not detected.

Clinical Course: The patient was transferred to the ED, intubated, and placed on vasopressors for hypotension. CT head revealed a prior infarction. The patient remained unresponsive. Based on the prognosis, the family opted for institution of comfort measures, and he died on Day 4.

Autopsy Findings: not performed.

Case 278. Acute thallium ingestion: undoubtedly responsible

Scenario/Substances: A 39 y/o male chemistry professor ingested 2–3 g of thallium powder and later induced vomiting. He presented to the ED an hour later.

Laboratory/Diagnostic Findings: Na 131 / K 4.0 / CO₂ 19, lactate 4.8mmol/L. Serum APAP, ethanol and salicylate not detected. Abd X-ray: Radiopaque material lining stomach.

Clinical Course: In the ED, nausea was managed with ondansetron and multiple dose activated charcoal was started. On day 2 WBI was

started however was not well tolerated due to vomiting and the patient developing restlessness and confusion. His serum Cr began to climb, he became acidotic and developed swelling of his lips and tongue. He was transferred to the ICU where he was intubated and sedated. Multiple dose activated charcoal/WBI were reinitiated for a short period of time before being discontinued due to abdominal distention. Lactate climbed to 4.8 mmol/L, and he became tachycardic and hypotensive requiring vasopressor support with norepinephrine. Prussian blue was obtained, and the first dose was administered about 22 h after ingestion. A spot urine thallium was 68,000 mcg/L. HD was started ~24 h post exposure. On Day 3 he was anuric and hypotension worsened requiring 5 vasopressors, so he was cannulated for VA ECMO and converted to CRRT. Serum thallium returned at >5,000 ng/mL and a CT scan revealed more radiopacities in the stomach and small intestines. Recurrent ventricular dysrhythmias developed, and vasoplegic shock worsened. Despite ECMO and multiple vasopressors, he died on Day 3.

Autopsy Findings: Cause of death: thallium toxicity. Manner of death: suicide.

Case 279. Acute carbon tetrachloride, dolomite and beta-cyclodextrin ingestion: probably responsible

Scenario/Substances: A 12y/o male drank an unknown liquid from a bourbon bottle offered to him by an adult at a skate park. Both drank from the bottle and within 2 min, both became unresponsive, had seizures, and this patient, foaming at the mouth. EMS transported him to the ED.

Past Medical History: drug and ethanol use, including accepting substances from strangers.

Physical Exam (HCF): BP 76/41, HR 50, RR 14, O₂ sat 100% on ventilator.

Laboratory/Diagnostic Findings: VBG-pH 6.99 / pCO₂ 22.2 / pO₂ 152 / HCO₃ 5.4, Na 136 / K 3.8 / Cl 93 / CO₂ 10 / BUN 11 / Cr 0.73 / Glu 306 / AG 33, AST 20.0, ALT <5.0, Serum osm 319, calculated osm 293, OG 26, CK 89, lipase 13, troponin 6.0, lactate 20.3 mmol/L. Serum APAP, methanol, ethylene glycol and salicylate not detected. Ethanol <14 mg/dL. UDS positive for THC. ECG: HR 106, QRS 85, QTc 430, sinus tachycardia.

Clinical Course: He had no cough or gag reflex, was given ketamine and intubated, but no continuous sedation. He was hypotensive and bradycardic, and started on norepinephrine and epinephrine. He was flown to a pediatric tertiary care center, vasopressin was added, and he was admitted to PICU. BP 125/74, HR 113, RR 18, O₂ sat 99% on ventilator, T 33.0°C. His pupils were fixed and dilated. Fomepizole, hydroxocobalamin and CRRT were started. CT head revealed cerebral edema with effacement and cerebellar herniation. On Day 2 he remained intubated, unresponsive, on CRRT and vasopressin. BP 154/71, HR 116, RR 18, O₂ sat 96% on FiO₂ 0.3. He developed polyuria and started on levocarnitine. Additional history revealed that they had found the open whiskey bottle with a white label that was half full of a very dark brown liquid. The liquid was more viscous than alcohol. The other victim drank "a very large gulp, at least 2 shots worth." The other victim suspected it contained something else because "it was very strong" and made him feel "very weird" and unlike any other intoxication he had experienced (known use of methamphetamine, MDMA, ethanol, and synthetic cannabinoids). On Day 3, he remained intubated, unresponsive, on epinephrine and vasopressin, but off CRRT. Flumazenil and naloxone were given with no response. Mass spectrometry on the unknown liquid revealed it was a mixture of carbon tetrachloride 38% v/v, dolomite 34% w/v, and beta-cyclodextrin 26% w/v. On Day 4 a brain death exam was consistent with death by neurological criteria. On Day 5 a second brain death exam was consistent with death, and he died on Day 5.

Autopsy Findings: not available.

Case 298. Acute cyclopeptide mushroom ingestion: undoubtedly responsible

Scenario/Substances: A 69y/o female ingested mushrooms foraged from her yard 5 days prior. Symptoms began ~10h post-ingestion with nausea, vomiting, diarrhea, and abdominal pain.

Physical Exam (HCF): alert. BP 87/60, HR 113, RR 20, O₂ Sat 98% on RA, T 36.1°C.

Laboratory/Diagnostic Findings: VBG-pH 7.25 / pCO₂ 30 / pO₂ 68, Na 129 / K 4.4 / Cl 93 / CO₂ 13 / BUN 60 / Cr 1.2 / Glu 181 / AG 23, AST 3,785, ALT 4,379, ALP 166, bilirubin 5.8, INR 9.4, lactate 9.6 mmol/L. Serum APAP and ethanol not detected. CT abd/pelvis: severe diffuse hepatic steatosis without hepatomegaly, normal gallbladder.

Clinical Course: On Day 1, NAC infusion was started. On Day 2, she was transferred to a tertiary care center. Continued on NAC, received phytonadione, and penicillin was started. The patient was not interested in liver transplantation due to cultural and religious beliefs. She developed lethargy and underwent intubation for airway protection. A stroke code was activated due to neurologic decline. CT head negative for acute stroke. CRRT was started for declining renal function. On Day 3, she remained intubated and requiring norepinephrine, epinephrine, vasopressin, CRRT, and silibinin. Laboratory showed worsening liver function (AST 5,619, ALT 4,989, bilirubin 3.4, INR 6.4). On Day 4, neurologic deterioration with pupillary changes was noted. AST 5,798, ALT 4,706, bilirubin 4.1, ammonia 571 mc-mol/L, and lactate 14 mmol/L. Based on the prognosis, comfort measures were instituted, and she died on Day 4.

Autopsy Findings: not available.

Case 300. Acute diquat parenteral, ingestion: undoubtedly responsible

Scenario/Substances: A 26y/o male intentionally injected himself with 7.5 mL of 37% diquat dibromide aquatic herbicide.

Past Medical History: born female, transitioning to male. Depression, previous suicide attempts. Home medications: lamotrigine, pantoprazole, bupropion.

Physical Exam (HCF): In the ED he stated he felt hot, sweaty, and nauseated. He was agitated and vomiting. BP 137/89, HR 107, RR 24, O₂ sat 99% on RA, T 37°C.

Laboratory/Diagnostic Findings: ABG-pH 7.38 / pCO₂ 21 / pO₂ 99 / HCO₃ 12.4, Na 139 / K 3.6 / Cl 110 / CO₂ 15 / BUN 6 / Cr 0.2 / Glu 89 / AG 14, lactate 7.8 mmol/L, AST 20, ALT 13, ALP 88, bilirubin 0.4, Hgb 17.1 / Hct 50.6 / Plt 348. Serum APAP and salicylate not detected, ethanol 11 mg/dL. UDS was negative, fentanyl was negative. ECG: QRS 92, QTc 450. CxR: no pleural effusion, no pneumothorax.

Clinical Course: He was admitted to the telemetry unit, became more agitated and restless, not following commands, tachypneic, tachycardic, and facial flushing. He was placed in restraints and transferred to the ICU where he was given benzodiazepines, intubated and ventilated. He was then transferred to a tertiary care center. He was started on NAC as LFTs were increasing along with lactate and creatinine. Medications included propofol, fentanyl, norepinephrine, deferoxamine, magnesium and phenobarbital. The patient developed encephalopathy, AKI, anion gap metabolic acidosis and seizures. 200 mL was missing from the container raising concern for ingestion as well as injection. BP 179/85, HR 140, RR 32, O₂ sat 95% intubated, T 38°C. Na 147 / K 4.3 / Cl 110 / CO₂ 13 / BUN 14 / Cr 1.98 / Glu 138 / AG 24, AST 526, ALT 418, PT 13.4, CK 3571, lactate 9.4 mmol/L. Day 2 he remained intubated in multisystem organ failure. EEG showed a suppressed non-reactive background suggestive of severe encephalopathy of non-specific etiology. He completed NAC and received vitamin C, aspirin, calcium gluconate, diazepam, magnesium sulfate, methylene blue, phenobarbital, piperacillin/

tazobactam, vancomycin and was on deferoxamine, norepinephrine, epinephrine, vasopressin, fentanyl, midazolam, propofol, and sodium bicarbonate infusions. He was too unstable to start HD, so CRRT was initiated. ABG-pH 6.98 / pCO₂ 61 / pO₂ 66 / HCO₃ 14, Na 144 / K 5.3 / Cl 104 / CO₂ 12 / BUN 19 / Cr 2.5 / Glu 131 / AG 27, AST 1620, ALT 1140. BP 98/67, HR 122, RR 30, O₂ sat 97% intubated. Serum diquat 23,000 ng/mL. His condition continued to decline, and he died on Day 2.

Autopsy Findings: not available.

Case 301. Acute-on-chronic bromethalin ingestion: probably responsible

Scenario/Substances: A 30y/o female ingested bromethalin-based rodenticide over 2 weeks in a self-harm attempt.

Past Medical History: born male, transitioning to female.

Physical Exam (HCF): normocephalic, atraumatic, PERRL, EOMI, conjunctiva clear, moist mucous membranes, neck supple and nontender, lungs clear to auscultation, no heart murmurs, bowel sounds present, abd soft and nontender, skin warm and dry, no neuro deficits. BP 117/87, HR 89, RR 20, O₂ sat 98%, T 36.6°C.

Laboratory/Diagnostic Findings: Na 139 / K 4.5 / Cl 108 / CO₂ 25 / BUN 8 / Cr 0.7 / Glu 102 / AG 6, AST 23, ALT 15, WBC 6.3 / Hgb 14.3 / Plt 280. Serum APAP, ethanol, and salicylate not detected. Head CT: Cerebral edema. ECG: HR 69, QRS 103, QTc 437.

Clinical Course: In the ED she complained of a severe headache and vomiting, then had a seizure treated with levetiracetam, antiemetics, acetaminophen, and diphenhydramine. Seizure activity did not recur. She remained alert with persistent headache and was admitted to the ICU. Repeat imaging showed persistent cerebral edema. She died on Day 4.

Autopsy Findings: Cause of death: cerebral edema consistent with bromethalin poisoning.

Case 306. Acute-on-chronic dinitrophenol ingestion: undoubtedly responsible

Scenario/Substances: A 43y/o male ingested 1000 mg of dinitrophenol for weight loss and presented to the ED.

Physical Exam (HCF): BP 127/67, P 180, RR 50, O₂ sat 97%, T 40.3°C, GCS 15.

Laboratory/Diagnostic Findings: ABG-pH 7.38 / pCO₂ 28 / HCO₃ 16, Na 133 / K 5.3 / Cl 103 / CO₂ 13 / BUN 12 / Cr 1.59 / Glucose 146, AST 6,817, ALT 2,412, ALP 32, Ca 5.8, bilirubin 1.2, Plt 74, INR 2.8, PTT 40.3, fibrinogen 92, D-Dimer 35.93, lactate 10.6 mmol/L, CK 343,319, troponin 2.11.

Clinical Course: In the ED, he was started on IVF and placed on targeted-temperature management protocol. He was intubated and transferred to a tertiary care center. He was not an ECMO candidate but did receive 12 h of SLED, IV NAC, sodium bicarbonate, norepinephrine, phenylephrine, vasopressin, calcium chloride and sodium zirconium cyclosilicate. On Day 1, he lost arterial flow of bilateral lower extremities and developed compartment syndrome that abated with bilateral fasciotomy. The patient's condition continued to deteriorate, and he died on Day 2.

Autopsy Findings: Cause of death: acute dinitrophenol poisoning.

Case 307. Acute paraquat ingestion: undoubtedly responsible

Scenario/Substances: A 59y/o male drank several ounces of 46% paraquat mixed with chocolate milk in a suicidal attempt and presented to the ED.

Past Medical History: asthma, chronic lower back pain, and suicidal ideation. Not a smoker. Heavy drinker but abstained from drinking 3 weeks prior to this exposure. He denied recreational drug use. Home medications: acetaminophen/hydrocodone.

Physical Exam (HCF): unremarkable. BP 132/91, HR 78, RR 18, O₂ sat 98% on RA.

Laboratory/Diagnostic Findings: Na 141 / K 3.5 / Cl 102 / CO₂ 20 / BUN 20 / Cr 1.3 / Glu 217, AST 27.0, ALT 20, ALP 129, bilirubin 0.40, WBC 7.87 / Hgb 14.6 / Hct 43.3 / Plt 169. Serum APAP and salicylate not detected. UDS negative. ECG: HR 91, QRS 96, QTc 452, normal sinus rhythm.

Clinical Course: In the ED, he was depressed and suicidal. He vomited after ingestion. There were no burns in his mouth. He also had episodes of diarrhea and dry heaving. He was given activated charcoal and started on IV NS. He was transferred to a tertiary care center where he was alert and complained of cramps in legs and feet and continued to have diarrhea. He was hemodynamically stable, BP 119/66, HR 71, sat 99% on RA, T 37°C. ABG-pH 7.33 / pCO₂ 31 / PO₂ 96 / HCO₃ 16.3 / BE -8.4, Na 136 / K 4.0 / Cl 100 / CO₂ 18.0 / BUN 23 to 61.5 / Cr 1.7 to 8.21 / Glu 210 / AG 22, Ca 11.0 to 7.7, bilirubin 0.43 to 6.89, lactate 6.2 mmol/L, Alb 5.46, Tprot 9.20, AST 20, ALT 20, ALP 112, WBC 18.3 / Hgb 17.9 / Hct 55.1 / Plt 288. Serum APAP, ethanol and salicylate not detected. The peripheral infiltrates were concerning for coexisting pneumonia. No definite effusions. ECG: HR 79, QRS 53, QTc 451, sinus rhythm, right atrial enlargement, left anterior fascicular block, abnormal R wave progression, late transition. CxRs on Days 2–4 showed no acute cardiopulmonary process. Lungs clear of infiltrates. Day 7 showed cardiomegaly with interval development of moderate congestive failure. His lactate had decreased upon repeat. On Day 2, Cr 4.13, BUN 39.4 rising to 47.6. He had an episode of nausea and complained of sore throat, dysphagia, and stomach pain treated with ondansetron. Labs revealed significantly decreased Plt, elevated WBC, LFTs, and AKI. He was started on NAC and given morphine. His dysphagia worsened and tolerated only cold soft food and drinks. He continued to decline and developed respiratory distress. His prognosis was discussed with patient and his family. He was placed on a chemical code and transitioned to hospice. On Day 7, he declined rapidly, was on 10L O₂, RR in the 40s, O₂ Sat in the 70s, heart rate 150s with a normal BP, and died.

Autopsy Findings: not performed.

Case 309. Acute phosphine ingestion: probably responsible

Scenario/Substances: A 68y/o male intentionally ingested 6 oz of mole and gopher poison peanuts containing zinc phosphide. EMS initially found RR in the 50s, O₂ sat in the 80s, with improvement to 95% on 15L O₂. Activated charcoal was given en route to the ED.

Past Medical History: chronic back pain, glaucoma, hyperlipidemia, HTN, sleep apnea.

Physical Exam (HCF): Awake but in acute distress with diaphoresis. Able to answer questions and oriented to person. Tachypnea using accessory muscles with decreased breath sounds bilaterally. Cyanosis of the hands and feet. BP 126/53, HR 153, RR 38, O₂ sat 100% on 15L, T 36.2°C.

Laboratory/Diagnostic Findings: ABG-pH 7.33 / pCO₂ 34 / pO₂ 314 / HCO₃ 17, Na 136 / K 4.2 / Cl 100 / CO₂ 23 / BUN 6 / Cr 1.27 / AG 13, AST 17, ALT 13, bilirubin 0.7, INR 1.0, troponin 37, lactate 8.1 mmol/L, WBC 18 / Hgb 14 / Plt 383. Serum APAP, ethanol and salicylate not detected. UDS positive for fentanyl (given to patient in the ED). CxR: Cardiomegaly and mild pulmonary vascular congestion. ECG: QRS 86, QTc 411, supraventricular tachycardia versus sinus tachycardia.

Clinical Course: Upon arrival in the ED, he appeared pale, diaphoretic and complained of dizziness. He was intubated on arrival after

which he developed hypotension and was started on norepinephrine and IVF boluses. He developed worsening lactic acidosis and hypotension refractory to increasing doses of norepinephrine. Based on the prognosis, comfort measures were instituted, and he died on Day 1.

Autopsy Findings: not available.

Case 311. Acute malathion ingestion: probably responsible

Scenario/Substances: An 82y/o male was witnessed by store employees drinking insecticide spray he had just purchased containing 50% malathion and 42.5% xylene. EMS found him initially speaking, then complaining of difficulty swallowing before becoming unresponsive. They transported him to the ED.

Past Medical History: pacemaker.

Physical Exam (HCF): pupils dilated 5–6 mm and unreactive, managing his own secretions, tongue fasciculations, not withdrawing from pain or responding to verbal stimuli, not diaphoretic, occasional arm movements. BP 156/88, HR 65 RR 16, O₂ sat 96% RA.

Laboratory/Diagnostic Findings: ABG-pH 7.32 / pCO₂ 27 / pO₂ 83 / HCO₃ 14, Cl 108 / BUN 26 / Cr 0.7 / AG 19, AST 47, ALT 54, lactate 3 mmol/L. Serum APAP and salicylate not detected.

Clinical Course: In the ED, he was emergently intubated and placed on propofol infusion and decontaminated. Atropine 2 mg was given for increased secretions, and he was transferred to a tertiary care center ICU where pralidoxime infusion was initiated. EEG showed no seizure activity. He became more arousable as sedation was weaned, but fever and increased secretions became evident. On Day 2, as secretions resolved, atropine and pralidoxime were discontinued. On Day 3, without any sedation, he was neither awake nor alert and could not be extubated. On Day 4, he remained unresponsive with non-reactive pinpoint pupils, and SBP >200 for which nicardipine was started. CT angiography showed signs of stroke, but his pacemaker prevented performing an MRI. He remained unresponsive to commands and pain, had absent gag reflex, and only partial intermittent eye opening. On Day 5, he became hypotensive, requiring 2 vasopressors and increased ventilatory support for ARDS. Reflexes were absent and pupils fixed. Based on the prognosis, the family opted for institution of comfort measures, and he died on Day 5.

Autopsy Findings: not performed.

Case 313. Acute Taxus canadensis ingestion: contributory

Scenario/Substances: An 8y/o male was seen by his mother eating mouthfuls of yew plant needles 2 h prior. He vomited 45 min post ingestion. The mother stated that she dispenses his medications to him and that no medications were missing. The child was taken to the ED.

Past Medical History: autism. Home medications: escitalopram, clonidine, tamsulosin.

Physical Exam (HCF): somnolent, not following commands. BP 134/66, HR 40, manual ventilation in progress, O₂ sat 100%.

Laboratory/Diagnostic Findings: Glucose “normal”.

Clinical Course: The child vomited again in the ED. Pacer pads were placed and manual ventilation was continued. He was given atropine with no improvement after which epinephrine, calcium gluconate, atropine, and several amps of sodium bicarbonate were administered. HR remained in the 40s. Hour 3 the patient had a cardiac arrest and was intubated. He remained pulseless and ILE was administered. Transfer for ECMO was not possible and ROSC was not obtained. He died at Hour 4.

Autopsy Findings: not available.

Case 314. Acute ibogaine ingestion: probably responsible

Scenario/Substances: A 49y/o man purchased ibogaine online with the intent of using it for opioid cessation and was found in cardiac arrest at home. CPR was initiated in the field and continued en route to the ED.

Past Medical History: opioid use disorder. Home medications: ibogaine.

Physical Exam (HCF): lethargic, unresponsive to noxious stimuli, pupils 6 mm bilaterally, non-reactive to light, intubated, tachycardic, obtunded, positive cough reflex, no corneal reflex, no clonus. BP 134/68, HR 107, RR 16, O₂ sat 96% on 100% FiO₂, T 35.5°C.

Laboratory/Diagnostic Findings: ABG-pH 7.30 / pCO₂ 36 / pO₂ 273 / HCO₃ 17.7, Na 136 / K 4.0 / Cl 102 / CO₂ 25 / BUN 32 / Cr 1.9 / Glu 150, Ca 8.5, Mg 2.3, AST 105, ALT 60, ALP 70, CK 1050, troponin-HS 26 (repeat 61), osm 301, lactate 9.4 mmol/L. Serum APAP, ethanol and salicylate not detected. Ibogaine 5,330 ng/mL. Benzoylcegonine 2,340 ng/mL.

Clinical Course: The patient was treated with therapeutic hypothermia for 24 h after initial arrest and started on propofol, followed by midazolam for possible neurostorming. He was treated with amiodarone and lidocaine infusions for frequent ectopy, with repeated episodes of VT. On Day 1, he had 3 episodes of sustained VT, was defibrillated, and returned to a sinus rhythm each time. After these cardiac arrests, the patient was unresponsive and had evidence of anoxic brain injury on head CT. Based on the prognosis, comfort measures were instituted, and he died on Day 2.

Autopsy Findings: not performed.

Case 315. Acute Cerbera odollam ingestion: probably responsible

Scenario/Substances: A 62y/o female ingested pong pong seed (*Cerbera odollam*) as a suicide attempt. EMS was called, she became pulseless with dysrhythmias en route, so EMS began CPR, gave magnesium and sodium bicarbonate.

Past Medical History: suicide attempts.

Physical Exam (HCF): BP 80/57, HR 43–59.

Laboratory/Diagnostic Findings: K 7.3. ECG: QRS 102, QTc 305, junctional rhythm, depressed ST segment, non-distinct T waves, and P waves.

Clinical Course: Upon presentation to the ED, ROSC was achieved, and she was intubated. Norepinephrine infusion was initiated, and amiodarone, calcium and 1 vial of digoxin Fab fragments were given. The patient died from recurrent cardiac arrest 2 h later.

Autopsy Findings: not available.

Case 316. Acute nicotine liquid ingestion: undoubtedly responsible

Scenario/Substances: A 31 y/o female drank 2 mouthfuls of nicotine 50 mg/mL vape liquid after a family fight. She was having shortness of breath and nausea. EMS was called, found her pulseless, began CPR and epinephrine, and ROSC was achieved after 25 min. The patient was transferred to a tertiary care center.

Physical Exam (HCF): obtunded. BP 114/65, HR 105, RR 17, O₂ sat 93% on ventilator, T 38.4°C.

Laboratory/Diagnostic Findings: Na 136 / K 3.8 / Cl 105 / CO₂ 13 / BUN 11 / Cr 1.04 / Glu 303 / AG 18, lactate 8.1 mmol/L, AST 129, ALT 81, bilirubin 0.2, troponin 374. Serum APAP, ethanol and salicylate not detected. UDS was negative. ECG: HR 72, QTc 586, QRS 84.

Clinical Course: She was intubated and sedated with propofol. The patient's QTc remained prolonged, and she developed tremors and status myoclonus of bilateral lower extremities requiring increased

sedation with midazolam. Norepinephrine was given for BP support. CT head revealed cerebral herniation and diffuse cerebral edema secondary to anoxic brain injury. Based on the prognosis, the family opted for institution of comfort measures, and she died on Day 3.

Autopsy Findings: peripheral blood nicotine 700 ng/mL and cotinine 9,700 ng/mL.

Case 319. Acute fentanyl (non-prescription) exposure: undoubtedly responsible

Scenario/Substances: A 3y/o male was found apneic by his mother ~20–40 min after she last saw him well. She gave rescue breaths until EMS arrived 15 min later. They found the child pulseless, began CPR, gave epinephrine boluses x9, and transported him to the ED.

Past Medical History: Hospitalized for fentanyl toxicity at 8m/o then placed in grandparents' custody. The mother had resumed physical custody 2 days prior, although the child was still under legal custody of the state. The child was assumed to have a possible toxic exposure; the child's mother denied current substance use. Police investigating the scene reported finding fentanyl pills/powder at the residence.

Physical Exam (HCF): Post ROSC: fixed, dilated pupils, no spontaneous movements. HR 130, on ventilator.

Laboratory/Diagnostic Findings: Post-ROSC: Capillary blood gas-pH <6.8 / pCO₂ 68 / pO₂ 126 / HCO₃ 10 / BE -27.2, Na 147 to 166 / K 4.7 to 2.2 / Cl 114 to 129 / CO₂ 10 to 31 / AG 23 to 11, Glu 341 to 533, BUN 9 to 59 / Cr 0.58 to 3.57, AST 378 to 709, ALT 305 to 407, albumin 2.3, Ca 10.1 to 8.1, lactate 14 to 1.5 mmol/L, INR 1.14, WBC 7.4 to 26.6 / Hgb 10.7 to 6.9 / Plt 146 to 103. UDS positive for fentanyl but below the official reporting cutoff. CxR: Bilateral hazy air space opacities, likely representing pulmonary edema and possibly aspiration. CT head (6h post-ROSC): Diffuse cerebral edema and effacement of the basal cisterns and ventricles, loss of grey-white differentiation. CT chest/abd/pelvis: Ground glass opacities and consolidation bilaterally in dependent lung possibly with superimposed alveolar hemorrhage, hypoperfusion complex, small ascites, periportal and GB edema, no bony trauma.

Clinical Course: On arrival in the ED, he was in PEA or VF, so CPR was continued for 90 min with naloxone, multiple epinephrine, defibrillation, sodium bicarbonate, calcium, and IV fluid bolus given before ROSC. After ROSC he received multiple vasopressors and albumin infusions for several days for hypotension, was comatose with global hypoxic brain injury and without brainstem reflexes or evidence of cortical activity of continuous EEG, and was mechanically ventilated with hypoxemic, hypercarbic respiratory failure. Stress dose steroids and antibiotics were started. Oliguria and AKI, electrolyte abnormalities, hyperglycemia, and severe metabolic acidosis developed requiring potassium, sodium bicarbonate and sodium acetate supplementation, insulin infusion, IVF, furosemide, and PRBC transfusion. After confirmation that his grandmother had medical decision making responsibility, brain death criteria were met, and he was apneic off the ventilator; he died on Day 8.

Autopsy Findings: not available.

Assessment/Discussion: Initial routine UDS and comprehensive UDS performed by a forensic laboratory were negative for all drugs, including fentanyl. The forensics laboratory director reviewed the analysis and determined that fentanyl was present but at a level below the cutoff for reporting as positive. It was thought that the arrest occurred so soon after exposure that there was little time for delivery into the urine. Serum was sent for analysis to a reference laboratory and returned positive for both fentanyl (0.15 ng/mL) and norfentanyl (1.33 ng/mL). It was confirmed that no fentanyl was administered to the patient by healthcare providers.

Case 325. Acute non-prescription fentanyl and methamphetamine: undoubtedly responsible

Scenario/Substances: A 15y/o female went into her bedroom after an argument with her mother. ~15 min later, she was responsive and

conversant. After another 15 min the mother found her unconscious and unresponsive, called EMS who found her apneic and asystolic. Pupils were 7 mm and non-reactive. She was placed on bag-valve-mask ventilation and given multiple rounds of CPR and epinephrine. She received naloxone, IVF, and was defibrillated with ROSC. She was hypotensive and a pressure bag was placed on her IVF. She remained hypotensive during transport to the ED, despite receiving bolus doses of epinephrine and IVF. An unknown white powder was found in the patient's bedroom. Law enforcement also reported evidence of hematemesis and drug paraphernalia in her room.

Physical Exam (HCF): unresponsive, pupils 7 mm and nonreactive. Supraglottic airway in place, coarse lung sounds bilaterally, abdomen nondistended, slight hyperreflexia in lower extremities. BP 83/51, HR 116, RR 26, O₂ sat 42%, T 35.1 °C.

Laboratory/Diagnostic Findings: VBG-pH 6.87 / pCO₂ 86 / pO₂ 105 / HCO₃ 16 / BE -17.0. Na 145 / K 3.7 / Cl 103 / CO₂ 20 / BUN 8 / Cr 1.02 / Glu 424 / AG 22, lactate 12.6 mmol/L, AST 100, ALT 52, bilirubin 0.3, INR 1.3, troponin-HS 47, WBC 16.4 / Hgb 9.6 / Hct 30.5 / Plt 216. Serum APAP, salicylate, and ethanol not detected. UDS was positive for benzodiazepines, THC, and fentanyl. CxR: Central perihilar airspace opacities, small airways disease versus pulmonary vascular congestion. ECG: HR 103, QRS 101, QTc 547, diffuse ST segment depression with elevation in aVR.

Clinical Course: In the ED, she was intubated with ketamine, and given epinephrine, midazolam, and fentanyl. Bloody frothy secretions were noted. A bedside ultrasound showed poor LV function, and her ECHO confirmed a LVEF of 6%. She was given IV sodium bicarbonate, pantoprazole, and furosemide. CT chest showed diffuse pulmonary consolidations bilaterally. CT abdomen showed periportal edema and bowel wall edema. CT head showed diffuse hypoattenuation and sulcal effacement. Despite maximal ventilator support, she developed worsening hypoxia that required manual ventilation to keep her O₂ saturation in the upper 70s to 80s. In the PICU, she started initiating large breaths and became more hypotensive with each inspiration. Cisatracurium was administered. She became more hypotensive and developed PEA. CPR and epinephrine were given, and ROSC was obtained. She was placed on epinephrine and norepinephrine infusions, stress doses of steroids, and antibiotics. Episodes of VT and frequent PVCs were noted. Repeat laboratory values showed a K of 2.8 and an Ca (ionized) of 3.8, and she was given electrolyte replacement. Her troponin-HS increased to 1,296 and lactate to 16.4 mmol/L. On Day 2, repeat head CT showed worsening global cerebral edema with uncal herniation and inferior cerebellar tonsillar herniation. Her pupils were fixed and dilated bilaterally, and she was unresponsive to pain. Sedation and paralytics were discontinued to assess her neurological condition. A comprehensive drug screen by mass spectrometry was positive for APAP, naloxone, methamphetamine, ketamine, norketamine, fentanyl, norfentanyl, midazolam, nicotine, cotinine, caffeine, and theobromine. On Day 3, an initial formal brain death testing was consistent with brain death. On Day 6, a second formal brain death exam was completed. The test confirmed brain death, and she was pronounced dead.

Autopsy Findings: After her death, the quantitative urine results were fentanyl 209 ng/mL and norfentanyl 316 ng/mL (normal reference <0.5 ng/mL for each). Cause of death: anoxic encephalopathy due to presumed mixed drug intoxication of unknown substances. Manner of death: undetermined.

Case 353. Acute colchicine, prednisone, and ethinyl estradiol/norgestimate ingestion: undoubtedly responsible

Scenario/Substances: A 20y/o female took 161 mg of colchicine and 300 mg of prednisone that were prescribed to a family member and 21 tablets of her own oral contraceptive in a suicide attempt.

Physical Exam (HCF): diaphoresis, abdominal pain. BP 148/111, HR 58, RR 16, O₂ sat 97% on RA, T 37.7°C.

Laboratory/Diagnostic Findings: Na 138 / K 3.9 / Cl 105 / CO₂ 24 / BUN 5 / Cr 0.8 / Glu 147, AST 15, ALT 12, total bilirubin 0.4. Serum APAP, ethanol and salicylate not detected. ECG: HR 58, QRS 74, QTc 373, sinus bradycardia.

Clinical Course: She presented to the ED ~3h after the ingestion. The dose of colchicine was estimated to be ~2.56mg/kg. Gastric lavage was tried without any pills visualized, activated charcoal was given, and she was transferred to a tertiary care center for ICU admission. Two hours later a repeat ECG showed sinus bradycardia with peaked T waves. BP 151/106, HR 44, O₂ sat 95% on RA. An NG tube was placed but persistent vomiting refractory to multiple antiemetics prevented administration of additional activated charcoal. ~3h later, she was intubated and sedated with propofol and fentanyl. BP 170/105, HR 50, RR 20, O₂ sat 100% on FiO₂ 0.5. Day 2: Other therapies included IVF and cisatracurium. Labs were unremarkable except for lactate of 2.9mmol/L and a troponin-HS 67. BP 132/82, HR 81, RR 16, O₂ sat 98% on FiO₂ 0.3, T 37.2°C. She was noted to have blood in her stool. ~6h later, she remained intubated and sedated with propofol and fentanyl. ~4.5h later, repeat labs were notable for CO₂ 16, BUN 14, Cr 1.3, AG 12, AST 157, ALT 41. An ECHO was performed that showed 35-40% LVEF. NAC was started. One hour later, the patient became anuric and was started on norepinephrine. Cannulation for ECMO was attempted but unsuccessful. ~7h later, her BP abruptly dropped to 50/30, resuscitation was attempted, but hypotension persisted despite norepinephrine, epinephrine, and vasopressin. On Day 3, ~5.5h later, VBG- pH 7.28 / pCO₂ 41 / pO₂ 65 / HCO₃ 19. Cannulation for ECMO was achieved after >4h in the OR and she was started on ECMO and dialysis. Na 159 / K 3.2 / Cl 110 / CO₂ 22 / BUN 17 / Cr 1.7 / Glu 131 / AG 27, AST 75, ALT 13, PT 30.3, INR 2.76, PTT 158, lactate 11.9mmol/L. Repeat INR about 2h later was 3.9 and troponin-HS was 5,732. Three hours after that, she remained intubated and sedated with midazolam, fentanyl, and ketamine. Other therapies included vasopressors, potassium, IVF, ECMO, albumin, HD, PRBCs, cryoprecipitate, and fresh frozen plasma. Based on the prognosis, the family opted for institution of comfort measures, and she died on Day 4.

Autopsy Findings: not available.

Case 392. Acute salicylate ingestion: undoubtedly responsible

Scenario/Substances: A 26y/o male started taking 300 aspirin 325mg in the midafternoon prior to arrival at the ED. He also stabbed himself multiple times, including the neck possibly involving the trachea.

Physical Exam (HCF): BP WNL, HR tachycardic, RR tachypneic, T 37.94°C.

Laboratory/Diagnostic Findings: ABG-pH 7.41 / pCO₂ 38 / HCO₃ 24, K hemolyzed / Cr 1.4 / Glu 175. Salicylate 23mg/dL. Serum APAP and ethanol not detected. UDS was negative. ECG: sinus tachycardia, QRS 78, QTc 404.

Clinical Course: The patient arrived at the ED and had emergency surgery on his neck wound. Tinnitus and hyperglycemia were reported. BP remained normal, HR 91. The patient was intubated. The patient's salicylate increased from 54 to 63 to 79mg/dL. His pH went from 7.34 to 7.48, pCO₂ from 56 to 37, pO₂ 326, HCO₃ from 30 to 28, BE 2.7 to 4, K 5.7. The patient was on a sodium bicarbonate infusion after receiving 2 amps of sodium bicarbonate and IVFs. He remained intubated and sedated. HR 140-150s, T 37.9-38.1°C. On Day 2, HR 170s, T 39.9°C. He was diaphoretic. Cooling measures were performed. Salicylate 98mg/dL, K 5.4, and Cr. 1.7. The patient had received a dose of activated charcoal and remained on a sodium bicarbonate infusion. CRRT was started, and the patient became hypotensive with MAP 40s, HR 150s. The patient was started on vasopressors and fluids. Salicylate 91mg/dL. He died ~30min later.

Autopsy Findings: Cause of death: acute salicylate toxicity with complications. Other significant conditions contributing to death but not related to the terminal disease condition were sharp force injuries of neck, torso, and extremities.

Case 1266. Acute lidocaine ingestion: undoubtedly responsible

Scenario/Substances: A 19y/o male obtained several lidocaine patches, extracted the liquid from them, and ingested it. He was found down, EMS was called and found him pulseless, initiated CPR, achieved ROSC, and transported him to the ED. His cell phone search history included lethal dose of lidocaine.

Physical Exam (HCF): BP 140-160/70-80, HR 60, O₂ sat 99% on ventilator, T 33°C.

Laboratory/Diagnostic Findings: Serum APAP, salicylate and ethanol not detected. UDS positive for cannabis.

Clinical Course: Upon arrival to the ED, the patient was intubated and sedated with midazolam, then given a dose of ILE. Serum lidocaine (post ILE) was 3.7mg/L (therapeutic range 1.5-5mg/L). He was admitted to the ICU and cooled per the post-arrest hypothermia protocol. His neuro examination worsened, and a head CT indicated cerebral edema and herniation. On Day 2, sedation was stopped, and he was unresponsive to stimulus. On Day 3, he was given sodium polystyrene sulfonate for elevated potassium and mannitol for increased intracranial pressure. He remained hypothermic per protocol. On Day 4, rewarming was initiated, and he received hypertonic saline. BP 158/66, HR 103, RR 26, T 34.5°C. Brain death testing was completed over the next 2 days. Based on the prognosis, comfort measures were instituted, and he died on Day 7.

Autopsy Findings: not available.

Case 1268. Acute-on-chronic nitrous oxide inhalation: contributory

Scenario/Substances: A 33y/o female presented to the hospital with 1 week of ongoing mental status changes and lower extremity weakness after inhaling 4-6 cans of nitrous oxide per day for the previous 2 weeks along with intermittent cocaine and cannabis use. She had been largely bed-bound during this time.

Past Medical History: major depressive disorder, anxiety, GERD, constipation, insomnia, substance use disorder, bilateral PE (2023). Home medications: rivaroxaban.

Physical Exam (HCF): multiple superficial partial thickness, old appearing, burns on the lower abdomen and upper thighs, drowsy but awake, pupils 2mm ERRL, no nystagmus, lagging but intact EOMI, no facial droop or dysarthria, oriented to self/location/year/situation, mildly impaired attention, mild bilateral pronator drift, 4+/5 strength throughout bilateral upper extremities, minimally increased tone in upper extremities, hyporeflexia, no inducible lower extremity clonus, grossly intact proprioception in feet, sensation intact to light touch grossly and symmetric in lower extremities. Acute, occlusive DVT in the right lower extremity. BP 110/76, HR 101, RR 16, O₂ sat 100%, T 36.7°C.

Laboratory/Diagnostic Findings: AST 13, ALT 21, Hgb 10.2, MCV 80.8, MethHgb 0.6%, serum B₁₂ 137pg/mL, methylmalonic acid 4.38mmol/L, troponin negative. UDS positive for THC. CT head unremarkable. CxR: WNL. ECG: HR 90, QRS 84, QTc 475, no evidence of right heart strain.

Clinical Course: The patient experienced cardiac arrest on Day 1 (conflicting reports of PEA versus asystole) with ROSC achieved after 23min of ACLS. She was emergently intubated and transferred to the ICU. She had a second cardiac arrest in the ICU and was administered thrombolytic therapy for presumed PE/obstructive shock in the setting of known DVT. She developed worsening hypotension,

lactic acidosis, and generalized seizures leading to multisystem organ failure. Based on the prognosis, comfort measures were instituted, and she died on Day 2.

Autopsy Findings: not available.

Assessment/Discussion: It was suspected that the patient's cardiac arrest involved massive PE and obstructive shock associated with a history of nitrous oxide exposure and muscle weakness with reduced mobility in the setting of prior thromboembolic events.

Case 1273. Acute mepivacaine parenteral: probably responsible

Scenario/Substances: A 73y/o male underwent spinal anesthesia with mepivacaine at L3/4 for elective bilateral knee replacement. A few minutes later he developed itching and pain in the buttocks for which he received diphenhydramine and midazolam. Shortly after, he developed fasciculations and tonic clonic jerks in his legs. He was intubated and placed under general anesthesia with a propofol infusion and sevoflurane. When sevoflurane was weaned, he had additional tonic clonic jerking. The surgery was aborted, and he was admitted to the ICU.

Past Medical History: psoriatic arthritis, HTN, CAD.

Physical Exam (HCF): Prior to surgery: afebrile, exam unremarkable, BP 145/72, HR 75, RR 16, O₂ sat 100%. In the ICU: Generalized seizures, muscle fasciculations, leg jerking, posturing of arms and tensing of muscles at mouth and jaw. SBP 150–170s, HR 120s, ventilated.

Laboratory/Diagnostic Findings: ABG-pH 6.88 / pCO₂ 46 / pO₂ 406 / HCO₃ 8 / BE -25, Na 125 / K 4.1 / Cl 90 / CO₂ 10 / BUN 11 / Cr 1.57 / Glu 167 / AG 25, AST, ALT, bilirubin: lipemic interference, CK 306, lactate 20.7 mmol/L, WBC 18.5 / Hct 42 / Plt 32. CxR: unremarkable. ECG: QRS 98, QTc 500, left axis deviation. EEG: burst suppression, periods of electrical activity to minimal brain activity; jerking and spikes.

Clinical Course: In the ICU, he had continued seizures and jerking movements. He received ILE, benzodiazepines, ketamine, levetiracetam, lacosamide, fosphenytoin, valproic acid and phenobarbital. MRI did not show any acute findings. He developed hypotension requiring vasopressors, AKI, rhabdomyolysis (CK >42,000), pulmonary edema and possible infection with sputum cultures growing *Streptococcus pneumoniae*. Later he became more hypoxic, acidotic and developed tachyarrhythmias and PEA arrest. He received sodium bicarbonate, calcium, and insulin for hyperkalemia but died on Day 3.

Autopsy Findings: Postmortem toxicology not available. Cause of death: local anesthetic systemic toxicity.

Case 1280. Acute lamotrigine ingestion: undoubtedly responsible

Scenario/Substances: A 16y/o female had complaints of nausea, vomiting and upset stomach, and unusually affectionate behavior on the day prior to being found by her mother unconscious with possible head injury. EMS found she had agonal respirations and was pulseless with VT/VF, CPR was initiated. ROSC was achieved after 3 rounds of epinephrine and naloxone, although she had cardiac arrest on arrival to the hospital. Mother reported 3 empty bottles of lamotrigine 25mg (unknown quantity) and was not able to locate patient's hydroxyzine. Goodbye notes, dated and timed from the day, were found.

Past Medical History: bipolar disorder, schizoaffective disorder, epilepsy, previous suicide attempt. Home medications: lamotrigine, hydroxyzine.

Physical Exam (HCF): obese, ill-appearing, pale and noncommunicative, pupils fixed and dilated (9mm), tongue with bite marks, OG tube with dark reddish brown gastric contents. BP 89/77, HR 103, RR 10, O₂ sat 93%, intubated, T 37.7°C, GCS 3T.

Laboratory/Diagnostic Findings: ABG-pH 6.68 / pCO₂ >110 / pO₂ 48, Na 148 / K 3.6 / Cl 99 / CO₂ 9 / BUN 10 / Cr 1.33 / Glu 352 / AG

40, AST 215, ALT 125, ALP 113, bilirubin 0.4, Ca 9.9, Tprot 6.8, albumin 4.2, globulin 2.6. Serum APAP and ethanol not detected. ECG: HR 85, QRS 266, irregular, wide complex rhythm.

Clinical Course: The patient received 7 boluses and an infusion of sodium bicarbonate, and ILE bolus and infusion. She was transferred after stabilization to a local PICU where she was unresponsive, not on sedation. Aggressive electrolyte repletion and dual vasopressor support were required. On Day 2: pH 7.18 / pCO₂ 46 / pO₂ 37 / HCO₃ 17 / BE -11.6, Na 142 / K 6.9 / Cl 100 / CO₂ 18 / Cr 3.75 / Glu 173 / AG 24, lactate 17 mmol/L, AST 1838, ALT 584, ALP 166, albumin 2.8, procalcitonin 0.16, CRP 0.2, Ca (ionized) 0.93, Mg 3.1, PTT 79.7, PT 66.5, INR 7.7, fibrinogen <60. APAP 8.2. ethanol 23. UDS positive for fentanyl. CT head showed multifocal cortical and subcortical infarctions in both cerebral hemispheres, moderate diffuse cerebral and cerebellar edema, and tonsillar herniation. The patient developed DIC requiring blood products and ARF with anuria. MARS and CVVH were started with limited fluid removal due to hypotension. On Day 3, pH 7.49 / pCO₂ 37 / pO₂ 96 / HCO₃ 28 / BE 4.7, Na 151 / K 3.2 / Cl 103 / CO₂ 29 / AG 19 / Glu 131 / BUN 25 / Cr 3.73, lactate 5.6 mmol/L, Ca 10.6, Mg 1.9, Phos 8.4, ALT 840, lipase 87, amylase 167. On Day 4, lamotrigine 3.5 mcg/mL. Based on the prognosis, the family opted for institution of comfort measures, she was extubated and died on Day 4.

Autopsy Findings: lamotrigine detected. Cause of death: toxic effects of lamotrigine. Manner of death: suicide.

Case 1330. Acute bupropion ingestion: undoubtedly responsible

Scenario/Substances: A 30y/o female ingested ~15 tablets of 300mg bupropion with alcohol. She was found down at home, unresponsive, next to empty bottles of bupropion and alcohol. EMS was called and transported her to the ED.

Past Medical History: anxiety, depression, alcohol use disorder, seizures.

Laboratory/Diagnostic Findings: ABG-pH 7.29 / pCO₂ 43 / pO₂ 67.8 / HCO₃ 20.4 / BE -5.9, lactate 14.3 mmol/L, Na 154 / K 4.4 / Cl 121 / CO₂ 15 / BUN 18 / Cr 2.59 / Glu 106 / AG 18, AST 4729, ALT 2880, bilirubin 4.6, INR 2.6, WBC 13.4 / Hgb 12.3 / Hct 36.3 / Plt 277. CxR: nearly complete opacification of the right lung.

Clinical Course: Upon ED arrival she decompensated and had a seizure treated with levetiracetam and lorazepam. She was intubated for airway protection and sedated with propofol, intubated and sedated. She received ILE. While awaiting ECMO she went into cardiac arrest and received 3 rounds of CPR and cardioversion with ROSC. She was started on vasopressors and transferred to the ICU where she was started on VVA-ECMO and continued on vasopressors. Mechanical breath sounds, abd soft, nontender with mild distension, legs cool to touch with mild mottling. HR 56, RR 14, O₂ Sat 72% on vent, T 36.8°C. She developed worsening cardiogenic shock, metabolic acidosis, worsening lower extremity ischemia, hypoxia with CxR showing pulmonary edema and opacification of the right lung, and had no neurologic activity when sedation was withheld. Based on the prognosis, the family opted for institution of comfort measures, and she died on Day 3.

Autopsy Findings: Internal examination showed pills in the stomach and intestine. Based on the antemortem laboratory findings, a postmortem examination was deferred. Postmortem blood bupropion >2,500 ng/mL. Cause of death: bupropion toxicity.

Case 1336. Acute-on-chronic amitriptyline ingestion: undoubtedly responsible

Scenario/Substances: A 32y/o female ingested 4.2g amitriptyline and cut her wrist 45 min prior to ED presentation.

Past Medical History: substance abuse, bipolar disorder, asthma, non-compliance with psychiatric medications. Home medications: amitriptyline, doxylamine, topiramate, spironolactone, sumatriptan, valacyclovir.

Physical Exam (HCF): somnolent but arousable, soft abdomen, PERRL. BP 129/73, HR 89, RR 16, O₂ sat 100 % RA, T 36.7°C.

Laboratory/Diagnostic Findings: ABG (post-intubation)-pH 7.21 / pCO₂ 59.7 / pO₂ 461 / HCO₃ 24 / BE -4, Na 142 / K 4.0 / Cl 110 / CO₂ 24 / BUN 14 / Cr 0.73 / Glu 99, AST 16, ALT 20, bilirubin <0.2, WBC 12.2 / Hgb 12.5 / Hct 38.2 / Plt 310. Serum APAP, ethanol and salicylate not detected. CxR: unremarkable post intubation. CT head: unremarkable. ECG: (1 h post ingestion) HR 117, QRS 136, QTc 630.

Clinical Course: QRS narrowed to 124 following IV sodium bicarbonate. PO potassium replacement was given. At Hour 4 she developed seizures treated with lorazepam and levetiracetam. Lactate 17.2mmol/L after the seizures. QRS widened, so sodium bicarbonate bolus was given, and an infusion was started; 12 doses IV sodium bicarbonate were given during her stay. She developed hypotension requiring IVF boluses and infusions of epinephrine and norepinephrine. She was intubated for airway protection. Several hours post ingestion she became asystolic. Resuscitation was attempted with epinephrine, sodium bicarbonate, calcium and amiodarone, but she died shortly thereafter on Day 1.

Autopsy Findings: Autopsy demonstrated moderate pulmonary vascular congestion and edema without trauma. Postmortem peripheral blood amitriptyline 0.17mg/L, nortriptyline 0.20mg/L. Cause of death: amitriptyline toxicity. Manner of death: suicide.

Case 1353. Tianeptine, diphenhydramine and caffeine ingestion: undoubtedly responsible

Scenario/Substances: A 40y/o male was found unresponsive by family. EMS was called and found him pulseless and apneic.

Past Medical History: hypertension, diabetes, morbid obesity. Family members said that he had stopped taking his medications 3 months earlier. He had reportedly been misusing tianeptine and caffeine pills.

Physical Exam (HCF): tachycardic, pupils fixed and dilated, skin diaphoretic. Spontaneous but non-purposeful movement of all 4 extremities, not following commands. SBP 140.

Laboratory/Diagnostic Findings: Na 135 / K 3.5 / Cl 103 / CO₂ 21 / BUN 17 / Cr 1.07 / Glu 93 / AG 11, AST 89, ALT 53, bilirubin 0.7, INR 1.0, CK 2213, WBC 16.7 / Hgb 13.6 / Hct 40.5 / Plt 278. Serum APAP and salicylate not detected. ECG: QRS 98, QTc 504.

Clinical Course: In the ED he was intubated and sedated with propofol and fentanyl. He received ceftriaxone for a possible urinary tract infection and was transferred to a tertiary care center. He was initially hypertensive post-resuscitation and was treated with antihypertensives in the ICU. Initial head CT was WNL but an MRI on day 3 showed changes consistent with diffuse hypoxic brain injury. Based on the prognosis, the family opted for institution of comfort measures, and he died on Day 9.

Autopsy Findings: hospital blood tianeptine 600ng/mL, diphenhydramine 170ng/mL. Caffeine was also detected but not quantified. Cause of death: toxic effects of tianeptine and diphenhydramine. Manner of death: accident.

Case 1414. Acute diphenhydramine: undoubtedly responsible

Scenario/Substances: A 3y/o male was reported to be playing with his brother when he had a seizure and collapsed. Bystander CPR was provided with guidance from 911. EMS arrived and found him in full arrest and gave naloxone without response. EMS placed a

supraglottic airway, started an IO line, performed CPR and transported him to the ED.

Past Medical History: asthma, possible autism, and 3 recent serious abscesses.

Physical Exam (HCF): The child arrived with CPR in progress, pupils fixed and dilated, no findings suggestive of injury. T 34°C.

Laboratory/Diagnostic Findings: ABG-pH <6.8 / pCO₂>98 / pO₂<15 / HCO₃<5, Na 144 / K>8 / Cl 107 / CO₂<5 / BUN 15 / Cr 1.22 / Glu 588 / AG >32, lactate >20mmol/L, AST 1208, ALT 422, bilirubin 0.2, INR 2.6, WBC 46 / Hgb 8.2 / Hct 278.8 / Plt 559. UDS negative on 2 different specimens. Serum drug screen was sent on admission, and later reported: naloxone present, diphenhydramine present, diphenhydramine serum concentration >5,000ng/mL. CxR: endotracheal tube in position; multiple hazy airspace opacities consistent with atelectasis, pneumonia, edema, or hemorrhage. ECG 1h after event, CPR pause: PEA; A-flutter variable ventricular response, ventricular rate 121, QRS 74, QTc 474; ECG 4.5h after event: HR 183 QRS 68 QTc 547, atrial flutter with RVR.

Clinical Course: After 20–30min pre-hospital downtime, the child had CPR continued in the ED where an endotracheal tube was placed, multiple rounds of epinephrine were given and an epinephrine infusion started. Sodium bicarbonate bolus, dextrose, and insulin bolus and infusion were given. ROSC was achieved after 28min in-hospital CPR, and the child was transported for head CT and then to the PICU; en route to PICU pulses were lost and CPR was resumed. ROSC again achieved in the PICU. Stress-dose steroids were given and EEG monitoring was started. Head CT showed central herniation, severe hypoxic-ischemic insult, and dural sinus thrombosis. Based on the prognosis, the family opted for institution of comfort measures, and he died on Day 1.

Autopsy Findings: postmortem diphenhydramine 5,707ng/mL. Cause of death: complications of acute diphenhydramine intoxication. Manner of Death: homicide.

Assessment/Discussion: Because of the unexplained serious illness and fatality (diphenhydramine detection from ante-mortem blood was not reported to clinicians until after death), child protective services (CPS) and law enforcement were notified. The mother reported the child had been living with a male family friend for the past month. News media reported he had prior felony convictions for child abuse and had served an 8-year prison term. The man has been arrested.

Case 1453. Acute acyclovir parenteral: probably responsible

Scenario/Substances: An 88y/o female admitted to the hospital was receiving acyclovir intravenous therapy for viral encephalitis. She had been improving cognitively and was conversant at the time of the exposure. She was given 5,000mg of acyclovir instead of the intended dose of 500mg.

Past Medical History: viral encephalitis.

Physical Exam (HCF): BP 162/140, HR 63, RR 17, O₂ sat 96% on 2L O₂, T 37.9°C.

Clinical Course: She developed confusion, became disoriented and somnolent, and her kidney function decreased with her serum creatinine doubling to 1.8 from 0.9. The patient's family refused further invasive procedures, and she was placed on comfort measures, receiving hydromorphone and midazolam IV infusions, and died on Day 3.

Autopsy Findings: not performed.

Case 1455. Acute methotrexate parenteral: undoubtedly responsible

Scenario/Substances: A 66y/o female was admitted for initiation of high-dose methotrexate. She was given her first and only methotrexate/cytarabine/rituximab infusion on Day 2. She developed AKI (Cr 2.6

from 0.59 and oliguric despite IVF and furosemide). She reported dyspnea with O₂ sats that dropped to 80%, requiring additional O₂.

Past Medical History: CNS lymphoma, lung cancer, HTN, syphilitic ophthalmopathy, asthma, GERD.

Physical Exam (HCF): BP 162/82, HR 93, RR 18, O₂ sat 91% on 2L NC, T 36.6°C.

Laboratory/Diagnostic Findings: Na 137 / K 3.7 / Cl 95 / CO₂ 37 / BUN 55 / Cr 3.45 / Glu 131 / AG 5.0, AST 20, ALT 18, total bilirubin 0.5.

Clinical Course: On Day 3 she was admitted to a tertiary care center ICU on IV leucovorin and in stable condition. Creatinine increased to 3.45. Hour 24 methotrexate 34 mcmol/L. Hour 48 methotrexate 16 mcmol/L for which she received glucarpidase on Day 4 while continuing on leucovorin and sodium bicarbonate infusions. Her dyspnea was noted to be improving and her O₂ requirements decreased to 1.5L/min NC. Urine output was improved. She was transferred to an oncology floor at a different tertiary care center. Hour 64 methotrexate 8.2 mcmol/L. BP 166/93, HR 89, RR 23, O₂ sat 93% on 2L NC. On Day 5 mucositis was noted. On Day 6 sodium bicarbonate was held due to dyspnea, volume overload, and alkalosis. Hour 108 methotrexate 5.8 mcmol/L, Cr 3.78. BP 170/86, HR 87, RR 16, O₂ sat 96% on 2L NC. On Day 7 she had increased SOB and hypoxia, Cr 4.08. Sodium bicarbonate stopped but continued on leucovorin and furosemide. She developed a rash with blisters on upper and lower extremities. Lesions were itchy and red-purple on legs, necrotic papules on arms, and bullae and epidermal necrosis were noted on left foot consistent with methotrexate toxicity. Hour 141 methotrexate 2.55 mcmol/L. In the evening, she became more hypoxic (O₂ sat 77% on 40L/min 100% O₂), confused, and tachypneic (RR 40), and was transferred back to ICU where she was intubated for pulmonary edema. On Day 8 she remained intubated with no significant change and antibiotics were started for possible pneumonia. Hour 156 methotrexate 2.88 mcmol/L, Cr 4.25, WBC 1.8, Hgb 9.8, Plt 77. On Day 9 she briefly required norepinephrine for hypotension. Urine output was down from 3.3L the previous day to 2.4L. She was started on filgrastim for bone marrow suppression and fentanyl and propofol for sedation. Hour 181 methotrexate 2.16 mcmol/L, Cr 4.76, WBC 0.7, Hgb 7.6, Plt 56. On Day 10 her urine output continued to decrease to 750mL. She remained on fentanyl and propofol, received 1 unit PRBCs for Hgb 6.6. Her leucovorin dose was increased. Hour 206 methotrexate 1.82 mcmol/L, Cr 5.08, WBC 0.4, Hgb 7.7, Plt 31. By Day 11 she failed extubation due to excess secretions. Urine output down to 665mL. Hour 228 methotrexate 1.53 mcmol/L, BP 153/76, HR 111, RR 31, O₂ sat 95% on FiO₂ 0.45. On Day 12 her urine output was down to 490mL, and she received her first session of HD for volume overload. Hour 252 methotrexate 1.54 mcmol/L. BP 124/82, HR 116, RR 24, O₂ sat 93% on FiO₂ 0.55. On Day 13 she developed a fever and tachypnea, was no longer responsive to commands, developed abdominal distension, and became oliguric. She was started on CRRT. Hour 276 methotrexate 1.12 mcmol/L, BP 121/40, HR 128, RR 36, O₂ sat 93% on FiO₂ 0.4. On Day 14 she became hypotensive and anuric and was started on norepinephrine and vasopressin. Sedation changed to dexmedetomidine and fentanyl. Filgrastim was stopped. Hour 306 methotrexate 1.32 mcmol/L, WBC 17.4, Hgb 8.1, Plt 119. BP 99/49, HR 113, RR 19, O₂ sat 92% on FiO₂ 0.5. On Day 15 she remained intubated on fentanyl, dexmedetomidine, norepinephrine, and vasopressin. She developed worsening cerebral edema and acidosis overnight and remained anuric. Hour 330 methotrexate 0.91 mcmol/L, BP 120/75, HR 76, RR 27, O₂ sat 74% on FiO₂ 0.75. She became hypothermic to 34.7°C and antifungals were added. Based on the prognosis, the family opted for institution of comfort measures, and she died on Day 15.

Autopsy Findings: not available.

Case 1461. Chronic theophylline ingestion: undoubtedly responsible

Scenario/Substances: A 61 y/o male presented to an ED with complaints of nausea, vomiting, and generally feeling unwell. He was non-compliant on theophylline and had only taken a half tablet on the day of presentation.

Physical Exam (HCF): BP 97/55, HR 121.

Laboratory/Diagnostic Findings: K 2, Mg 1.4, troponin-I 0.07. Ethanol not detected. Theophylline 150mcg/mL (therapeutic 10–20mcg/mL), ECG: AF, ventricular rate 185. Repeat ECG: HR 133, QRS 92, QTC 577, sinus tachycardia.

Clinical Course: The patient was treated with amiodarone, metoprolol and subsequently esmolol. A few hours into his course he was transported to a tertiary care center for HD. Just prior to arrival at the tertiary care center, he had a seizure, developed respiratory distress, and cardiac arrest with bradycardia, PEA, and asystole. He was intubated and resuscitated. He then required the use of 4 vasopressors. CRRT was initiated as his blood pressure was too low to perform HD at any point during his hospital course. Additional medications at this point included sodium bicarbonate, midazolam, and propofol. Continuous EEG monitoring showed no seizure activity. Two days after presentation, he was noted to have rhabdomyolysis with CPK >200,000. BUN and creatinine remained unremarkable. He was also treated with NAC (AST 6,300, ALT 2,316) and phytonadione (INR 4.05). Theophylline 3.4mcg/mL on Day 5. His neurologic response remained poor and based on the prognosis, comfort measures were instituted, and he died on Day 7.

Autopsy Findings: not performed.

Case 1463. Acute flecainide and atenolol ingestion: undoubtedly responsible

Scenario/Substances: A 2y/o female got into sibling's liquid flecainide and atenolol. When EMS arrived, she was unresponsive and bradycardic (<60) with a wide complex rhythm. She then went into cardiac arrest and was intubated. She received multiple rounds of epinephrine, sodium bicarbonate, and dextrose.

Physical Exam (HCF): arrived in active CPR, unresponsive, fixed and dilated pupils.

Laboratory/Diagnostic Findings: ABG-pH 6.95 / pCO₂ 84 / pO₂ 18 / HCO₃ 18.7 / BE -13.7, Na 143 / K 3.9 / Glu >700, Hgb 10.4 / Hct 31, lactate 14.5mmol/L, Ca (ionized) 1.77.

Clinical Course: She was noted to be in asystole for nearly the entirety of resuscitation, with PEA noted twice on pulse checks that then returned to asystole. She was given dextrose, insulin, IVF, sodium bicarbonate, calcium chloride, hypertonic saline, and ILE. She received a total of 45min of CPR with epinephrine every 3min After a total of 75+ minutes of CPR and resuscitative measures, death was declared.

Autopsy Findings: not available.

Case 1476. Acute flecainide ingestion: undoubtedly responsible

Scenario/Substances: A 26y/o female ingested 53 tablets of her mother's flecainide. EMS was called and transported her to the ED within 15min after the ingestion.

Past Medical History: autism, epilepsy, pica.

Physical Exam (HCF): awake, alert and ambulating initially, then somnolent but arousable, ill-appearing, coarse breath sounds bilaterally, soft nondistended abdomen and no active vomiting. BP 122/97, HR 48, RR 27, O₂ sat 98% RA, afebrile.

Laboratory/Diagnostic Findings: Na 137 / K 3.3 / Cl 112 / CO₂ 21 / BUN 14 / Cr 1.06 / Glu 110 / AG 4, AST 27, ALT 28, bilirubin

0.4, INR <0.8, WBC 13.5 / Hgb 12.4 / Hct 36.4 / Plt 384. Serum APAP, ethanol and salicylate not detected. ECG: HR 50, QRS 162, QTc 532.

Clinical Course: Within 2–3 min of ED arrival, she developed wide complex bradycardia and generalized tonic-clonic seizures. She was intubated and received sodium bicarbonate and lorazepam. Bradycardia worsened and she became pulseless. She received ACLS and had ROSC briefly. Transcutaneous overdrive pacing was attempted with intermittent capture. She subsequently became pulseless and could not be resuscitated despite multiple doses of sodium bicarbonate, calcium, epinephrine, and ILE.

Autopsy Findings: Tracheal bronchial tree filled with brown, thick fluid containing pieces of chicken and apple. Small and large intestine filled with multiple foreign bodies including oxidized coins (1 quarter, 3 dimes, 2 pennies), a yellow metal zipper pull, a yellow metal flat drill bit, 2 white metal screws, a copper metal cylinder, a green 2-bead charm, a black irregularly shaped rock, a metal screw embedded in a spring, a black silicone or rubber O-ring. Postmortem blood analysis: flecainide >5.0 mg/L (therapeutic range 0.5–1 mg/L). Cause of death: aspiration of gastric contents due to acute mixed drug intoxication.

Case 1583. Acute propranolol ingestion: undoubtedly responsible

Scenario/Substances: A 63y/o female was found unresponsive in bed by her husband who called EMS who found her unresponsive with respiratory depression, and pinpoint pupils. EMS performed CPR, ventilated her with bag-valve-mask and administered 4 mg naloxone intranasally x2 with ROSC. She admitted to ingesting pills in suicide attempt and was transported to the ED.

Past Medical History: DM, hyperlipidemia. Home medications: propranolol, clonazepam, gabapentin, empagliflozin, metformin, mirazapine, quetiapine, venlafaxine, tramadol, atorvastatin, omeprazole, sumatriptan.

Physical Exam (HCF): awake and talking, BP 80/54, HR 85, RR 21, O₂ sat 96% on RA, T 36°C, Glu (fingerstick) 130s.

Laboratory/Diagnostic Findings: ABG-pH 7.2 / pCO₂ 57 / pO₂ 50 / HCO₃ 22.3 Na 139 / K 4.6 / Cl 102 / CO₂ 24 / BUN 18 / Cr 1.1 / Glu 116 / AG 13, lactate 1.9 mmol/L, AST 85, ALT 65, WBC 11 / Hgb 15.7 / Plt 450. Serum APAP, ethanol and salicylate not detected. UDS positive for benzodiazepines, PCP. CxR: unremarkable, ECG: QRS 90, QTc 491, sinus rhythm.

Clinical Course: In the ED she was awake and hypotensive with normal heart rate. She became more sedated but did not respond to naloxone 2 mg IV. She received vasopressors and fluids. In the ICU, BP 114/61, HR 89/min, afebrile. She was lethargic. Labs remained unremarkable except for lactate 3.1 mmol/L. She became progressively more hypotensive. She did not respond to glucagon 1 mg IM. She became pulseless, was intubated and CPR was performed. Multiple doses of epinephrine were administered. She died 24 h after initial presentation.

Autopsy Findings: Pathology was unremarkable except for coronary artery atherosclerosis. Toxicology on premortem blood: propranolol 1,800 ng/mL (severe toxicity >2000 ng/mL). Cause of death: acute intoxication of propranolol. Contributory: coronary artery atherosclerosis, hypertension, aortic atherosclerosis. Manner of death: suicide.

Case 1595. Acute-on-chronic diltiazem ingestion: undoubtedly responsible

Scenario/Substances: A 64y/o male ingested 90×120 mg diltiazem XR in a self-harm attempt. He was being discharged from the ED after a 9-hour stay when his BP dropped to 80s/60s. IVF were started with minimal BP response.

Past Medical History: type 2 DM, HTN, anxiety. Home medications: diltiazem, gabapentin, rosuvastatin.

Physical Exam (HCF): BP 161/110, HR 75, RR 23, O₂ sat 94% RA, afebrile.

Laboratory/Diagnostic Findings: Complete metabolic panel WNL, Ca (ionized) 1.49. Serum APAP, ethanol and salicylate not detected. ECG: QRS 90, QTc 406.

Clinical Course: On Day 2, he became lethargic and required supplemental O₂ to maintain saturations. Hypotension required norepinephrine, epinephrine, vasopressin, insulin and calcium gluconate. Lactate 10.8 mmol/L, Glu 1043 despite HIE. Na 125, Cr 2.6 with no reported urine output. He was intubated, and transcutaneous pacing was initiated. Bedside echo showed an LVEF of 65%. Methylene blue 145 mg and ILE 75 g and 25 g were given. Sodium bicarbonate infusion was started for persistent acidosis, and warming measures were provided for hypothermia. Pupils were dilated and non-reactive. He was not a candidate for ECMO due to poor SVR. CPR was attempted without response, and he died on Day 2.

Autopsy Findings: postmortem femoral blood cocaine 0.010 mg/L, benzoylecgonine 0.26 mg/L and diltiazem 8.7 mg/L. Cause of death: complications of diltiazem toxicity. Manner of death: suicide.

Assessment/Discussion: Cocaine was not considered an involved substance because the clinical course did not suggest cocaine toxicity, the history did not suggest recent use as he had been in the ED for 9 h without evident use, the ME did not include it as a substance implicated in death, and the levels are consistent with distant use (i.e., very low parent compound) and not acute toxicity.

Case 1645. Acute amlodipine ingestion: undoubtedly responsible

Scenario/Substances: A 71y/o male was putting his weekly meds together and instead of ingesting his meds for the night, he accidentally picked up his 5 mg amlodipine bottle and ingested all 80 tabs. He was brought to the ED by air ambulance.

Past Medical History: cardiomyopathy, HTN, hyperlipidemia, diabetes, arrhythmia, pacemaker, prostate cancer. Home medications: amlodipine, rosuvastatin.

Physical Exam (HCF): alert and oriented. BP 131/95, HR 91, RR 20, O₂ sat 96 on RA, T 37°C.

Laboratory/Diagnostic Findings: ABG-pH 7.4 / pCO₂ 34 / pO₂ 75 / HCO₃ 21.1, Na 142 / K+3.7 / Cl 111 / CO₂ 22 / BUN 14 / Cr 1.0 / Glu 142 / AG 13, lactate 1.1, AST 18 / ALT 13, ALP 80, Hgb 12.3 / Hct 36.5 / Plt 252. Serum APAP, ethanol, and salicylate not detected. UDS negative. ECG: QRS 93, QTc 446. CxR: no pneumothorax, L subclavian pacer.

Clinical Course: In the ED he was given calcium gluconate and IVFs for BP 90/55. He was admitted to ICU and a central line was placed. He developed hypotension and was intubated and ventilated. He received a single dose of activated charcoal via NG tube and started on norepinephrine, vasopressin, epinephrine, phenylephrine, fentanyl, and propofol infusions. Insulin and D5W infusions and ILE were started. Methylene blue was given for refractory vasoplegic shock. He was placed on ECMO. BP 99/54, RR 10, O₂ sat 95% / intubated, T 36.8°C. Day 2 he remained intubated, on ECMO. ABG-pH 7.28 / pCO₂ 38 / pO₂ 61 / HCO₃ 17.9, Na 138 / K 3.6 / Cl 117 / CO₂ 19 / BUN 6 / Cr 0.75 / Glu 282 / AG 6, AST 30, ALT 28, INR 1.4, lactate 1.8. Day 3 he was started on a sodium bicarbonate infusion for acidosis. He remained intubated and on ECMO. BP 109/64, HR 68, RR 20, O₂ sat 94% intubated. Day 4 he was weaned from ECMO and decannulated. Based on the prognosis, the family opted for institution of comfort care measures, and he died on Day 5.

Autopsy Findings: not available.

Case 1733. Benzonatate ingestion: undoubtedly responsible

Scenario/Substances: A 12m/o female suddenly went unresponsive with possible convulsions while sitting with mother and eating a snack. She became cyanotic and CPR was initiated. EMS arrived and continued CPR for PEA while transporting to the ED. Police investigation found several loose benzonatate capsules at the residence.

Past Medical History: Normal childhood development. No reported fever, nausea, vomiting, or fatigue prior to current presentation. A case was open with the Department of Social Services prior to this incident.

Physical Exam (HCF): Pupils fixed 5–6mm bilaterally, abdominal distention, skin cool and mottled, unresponsive. BP 113/80, HR 130, RR 24, O₂ sat 98%, T 31.8°C (rectal), GCS 3.

Laboratory/Diagnostic Findings: ABG-pH <6.77 / pCO₂ 80.8 / pO₂ 102 (during code), Na 145 / K 3.0 / Cl 104 / CO₂ 12 / BUN 14 / Cr 0.52 / Glu 408 / AG 29, AST 122, ALT 57, bilirubin 0.2, WBC 9.8 / Hgb 12.0 / Hct 38.7 / Plt 333. Serum APAP, ethanol and salicylate not detected. UDS was negative. Ultrasound ruled out retroperitoneal bleed during admission. CxR: Increased density bilaterally, greater on right; multifocal pneumonia versus pulmonary edema. ECG: HR 138, QRS 66, QTc 388.

Clinical Course: In the ED, the patient was asystolic, was intubated, and given several doses of epinephrine and sodium bicarbonate, as well as calcium, amiodarone, and naloxone. After ~45min of CPR, the family requested no further resuscitation efforts, and CPR was terminated. After stopping CPR, her cardiac rhythm demonstrated PEA at a rate in the 40s, which increased to a rate in the 120s, and she regained a pulse. Blood glucose increased to the 200s. ECG showed narrow complex tachycardia. She was treated with vasopressors, antibiotics, steroids, midazolam and levetiracetam. On Day 2, the length of downtime, poor exam and imaging prompted an initial brain death examination. Her urine output increased, suggestive of diabetes insipidus. FFP was given for increased PT/PTT. On Day 3, confirmation examination verified brain death, comfort measures were provided, and she died later that day.

Autopsy Findings: anoxic injury, pulmonary edema, gastric ulceration with perforation without hemorrhage or gastric contents in the abdominal cavity and not believed to be a proximate cause of death. Antemortem urine collected ~2h after onset of illness demonstrated benzonatate 7.3mg/L and methamphetamine (slightly above the reference range). Postmortem heart blood demonstrated 4-butylaminobenzoic acid (a major metabolite of benzonatate) and methamphetamine. Cause of death: hypoxic-ischemic encephalopathy due to complication of prolonged cardiopulmonary arrest due to benzonatate toxicity, methamphetamine non-contributory. Manner of death: accident.

Case 1735. Chronic dietary supplement ingestion: probably responsible

Scenario/Substances: A 45y/o female presented to the ED with lower abdominal pain and vomiting after chronic ingestion of mushroom coffee and an unknown Brazilian weight loss drug for the previous 15 days.

Past Medical History: chronic alcohol use.

Physical Exam (HCF): alert, oriented. BP 102/69, HR 110, RR 18, O₂ sat 100% RA, T 37°C.

Laboratory/Diagnostic Findings: ABG-pH 7.02 / pCO₂ 47, Na 136 / K 3.2 / Cl 107 / CO₂ 17 / BUN 40 / Cr 2.87 / Glu 121 / AG 12, AST ≥15,000, ALT ≥7,500, ALP 200, albumin 2.5, bilirubin 5.1, INR 1.8, lactate 3.1mmol/L, ammonia 112mcmmol/L. Serum APAP, ethanol and salicylate not detected.

Clinical Course: The patient was started on IV NAC. On Day 2, she was transferred to a liver transplant center for acute decompensated

liver failure. Abdominal ultrasound showed echogenic liver suggestive of hepatic steatosis. On Day 4, transthoracic ECHO showed increased right-sided atrial arterial pressure with reduced AST, ALT and ammonia concentrations. On Day 5, the patient had worsening mental status and work up concerning for sepsis with anemia, thrombocytopenia, and elevated lactate. Antibiotics were initiated. She was intubated for respiratory and CNS depression, and became hypotensive, developed VF, then became pulseless. ROSC was achieved with initial CPR, but PEA and asystole subsequently developed, and she died on Day 6.

Autopsy Findings: not available.

Assessment/Discussion: Initial ingestion led to liver failure and coagulopathy with resultant multisystem organ failure, later contributing to death.

Case 1740. Acute selenium and ethanol ingestion: undoubtedly responsible

Scenario/Substances: A 29y/o male ingested ~15g of sodium selenite that he had procured at work and suspended in an alcoholic beverage. He self-presented to the ED after reporting a suicide attempt. In his car his family found a bottle of beer and an empty clear plastic bag with white residue. Patient worked at a pharmaceutical/medical supplement compounding facility and obtained the materials from his workplace.

Past Medical History: major depressive disorder.

Physical Exam (HCF): normal neurologic exam and mentation. Vomiting, soft, nondistended abdomen with diffuse tenderness to palpation, testicular pain. BP 109/60, HR 102, RR 20, O₂ sat 100% on RA.

Laboratory/Diagnostic Findings: Na 139 / K 4.0 / Cl 104 / CO₂ 17 / Cr 1.3, lactate 4mmol/L, AST 27, bilirubin 2.0, WBC 14. Serum APAP, ethanol and salicylate not detected. ECG: WNL.

Clinical Course: On Day 1 he developed CK of 800, and was transferred to the ICU, where he showed signs of cardiac injury with troponins in the 4,000s as well as progressive hypoxia and findings of diffuse pulmonary edema on CxR. He was intubated on Day 2. He developed cardiogenic shock and was placed on vasopressor support. His pulmonary injury progressed, leading to hypoxia despite optimized ARDS management, and he was placed on VV ECMO on Day 3. The patient was then transferred to the coronary ICU at a tertiary care center for ongoing management. He developed progressive kidney injury despite aggressive fluid resuscitation and was placed on CRRT on Day 3. He was treated with NAC on Days 3–8 and methionine Days 5–10. His early ICU course was complicated by multiorgan failure, thrombocytopenia treated with intravenous immunoglobulin, cerebral hemorrhages, shock liver, ATN requiring HD, and right internal jugular deep vein thrombosis. His initial selenium concentration obtained on Day 1 was 7,280mcg/L (nl: 23.0–190.0mcg/L), Day 5 195mcg/L and Day 7 203mcg/L. With HD and continued aggressive supportive care, he was eventually weaned from VV ECMO and successfully decannulated. He was extubated but remained significantly encephalopathic, following commands intermittently. He was transferred to the medical ICU following a potential mucous plugging/aspiration event requiring high-flow nasal cannula, and reintubation. He was again extubated and transferred to the medical floor. He exhibited some neurologic improvement and was transiently able to follow simple commands. He continued to have impaired GI motility and had a large volume aspiration event leading to sepsis from aspiration pneumonia on Day 48. He was started on IV antibiotics but later that day was noted to be in PEA arrest. He received a course of CPR with all potential reversible causes addressed. He was deemed not to be a candidate for ECMO and pronounced dead on Day 51.

Autopsy Findings: not available.

Case 1743. Acute magnesium ingestion: probably responsible

Scenario/Substances: A 77y/o female admitted to taking a bottle of a laxative containing magnesium earlier in the day to treat constipation.

Physical Exam (HCF): altered, hypotensive, hypoxic.

Laboratory/Diagnostic Findings: Mg >9.7, lactate >9mmol/L. ECG: HR 89, PR 204, QRS 104, QTc 511.

Clinical Course: She received IVF in the ED and her lactate trended down and MAP improved to 70s. She briefly improved following IVF and furosemide, becoming more awake and requiring less oxygen. However, a short time later, she developed respiratory distress and became unresponsive, requiring intubation in the ED. She was admitted to the ICU without sedation, requiring norepinephrine and vasopressin for BP support. She was making urine. Repeat Mg 8h after initial level remained >9.7. Repeat lactate increased to 8.7mmol/L. Shock worsened, requiring addition of epinephrine for BP support, and a sodium bicarbonate infusion to manage developing acidosis. O₂ sat was around 90% on maximal O₂. The care team discussed HD with family. Based on the prognosis, the family opted for institution of comfort measures, and she died on Day 2.

Autopsy Findings: not performed.

Case 1745. Acute loperamide ingestion: probably responsible

Scenario/Substances: An EMS call was placed by the 9y/o son about his 28y/o father needing help. On arrival to the home, EMS could not enter the locked door, and police had to forcibly gain entry. EMS found the patient unresponsive and pulseless, initiated CPR and ACLS. He was given epinephrine, atropine, sodium bicarbonate and defibrillated once after EKG showed VF. ROSC was achieved ~10min after arrival, and he was intubated in the field. Multiple boxes of loperamide 2mg were found on the bed and 73 tablets were reported missing. He was transported to the ED.

Past Medical History: opioid use disorder history with heroin and fentanyl. Previously sober for 2years.

Physical Exam (HCF): pulseless and apneic, T 35.8°C.

Laboratory/Diagnostic Findings: ABG-pH 7.03 / pCO₂ 76 / pO₂ 128 / HCO₃ 20.1 / BE -11.9, Na 138 / K 3.3 / Cl 104 / CO₂ 15 / BUN 18 / Cr 1.0 / Glu 290 / AG 19, lactate 10.7mmol/L AST 88, ALT 106, bilirubin 0.5, INR 1.1, WBC 10.8 / Hgb 12.2 / Hct 39.9 / Plt 221. Serum APAP, ethanol and salicylate not detected. UDS negative. ECG: HR 115, QRS 126, QTc 405; Repeat ECG in-hospital: QTc 500–600s.

Clinical Course: He was hypotensive with QTc prolongation prompting administration of vasopressors and magnesium and acidotic requiring sodium bicarbonate. ILE was also given. Persistent seizure activity was not controlled on midazolam, fentanyl, propofol, and cisatracurium. Additional antiepileptic therapy with levetiracetam and valproic acid were added. Norepinephrine was administered for hemodynamic support and broad spectrum antibiotics were given. He was transferred to a tertiary care center for continuous EEG monitoring. Neurologic testing showed a probable significant hypoxic injury with low likelihood of meaningful recovery. Based on the prognosis, the family opted for institution of comfort measures and organ donation. He died on Day 7.

Autopsy Findings: not performed.

Case 1771. Acute-on-chronic metformin ingestion: undoubtedly responsible

Scenario/Substances: A 52y/o male ingested 100 metformin 500mg tablets as a suicide attempt. EMS was called and transported him to the ED.

Past Medical History: Home medications: lamotrigine, lithium, mirtazapine; patient denied ingesting these.

Physical Exam (HCF): alert and oriented x3, nausea, vomiting, diarrhea. BP 170/110, HR 115, RR/O₂ sat/T WNL.

Laboratory/Diagnostic Findings: ABG-pH 7.1 / pCO₂ 28, HCO₃ 8.5, Na 138 / K 5 / Cl 102 / CO₂ 9.7 / BUN 14 / Cr 1.49 / Glu 349, AST 19, ALT 35, lactate 13.3mmol/L. Serum APAP and salicylate not detected. UDS was negative. ECG: QRS 94, QTc 436.

Clinical Course: In the ED, sodium bicarbonate infusion was initiated and Nephrology agreed to dialysis. Patient's mental status deteriorated, he became tachypneic with pH 6.9 and lactate 17mmol/L and was intubated. The patient died 10h after presentation.

Autopsy Findings: qualitative peripheral blood: metformin 67mcg/mL, tadalafil 88ng/mL, lorazepam 10ng/mL, lidocaine presumed positive (>0.2mcg/mL).

Case 1817. Acute-on-chronic baclofen, zolpidem and marijuana ingestion: undoubtedly responsible

Scenario/Substances: A 67y/o male ingested an unknown amount of his own baclofen 10mg from 2 prescription bottles each containing 270 tablets, both dispensed more than a year prior. The patient was last seen well the night prior but was found down "completely out of it" on the morning of presentation.

Past Medical History: CVA. Home medications: baclofen.

Physical Exam (HCF): intubated, sedated. BP 103/61, HR 80, RR 16, O₂ sat 100% on ventilator.

Laboratory/Diagnostic Findings: Na 130 / K 3.7 / Cl 95 / CO₂ 23 / BUN 14 / Cr 0.81 / Glu 155 / AG 12, Ca 8.8, AST 69, ALT 28, CK 511. Serum APAP, ethanol and salicylate not detected. ECG: HR 95, PR 170, QRS 102, QT 438, QTc 550.

Clinical Course: In the ED he required endotracheal intubation for airway protection. He did not require sedation for intubation, but post-intubation he was placed on fentanyl and midazolam. Post-intubation BP was "soft", requiring vasopressor support with norepinephrine. On Day 2, he had 2 generalized seizures that resolved with lorazepam. QTc remained prolonged despite optimization of electrolytes. On Day 3, based on the prognosis, the family and legal surrogate opted for institution of comfort measures, and he died on Day 4.

Autopsy Findings: An external examination was performed by the ME. Antemortem blood: baclofen 6.8mcg/mL, zolpidem 71ng/mL, delta-9 carboxy THC 5.8ng/mL, midazolam 17ng/mL, acetone 6.0mg/dL. Cause of death: toxic effects of baclofen and zolpidem. Manner of death: suicide.

Case 1822. Chronic propofol parenteral exposure: contributory

Scenario/Substances: An 11y/o female developed breakthrough seizures secondary to being unable to refill her rufinamide, prompting presentation to the ED. She had no prodrome of illness or missed medication doses other than rufinamide.

Past Medical History: Lennox-Gastaut Syndrome with refractory epilepsy s/p vagal nerve stimulator implant. Baseline 3–4 seizures/day. Home medications: rufinamide, cannabidiol, clobazam.

Physical Exam (HCF): hemodynamically unstable, seizure in progress. HR 150–160s.

Laboratory/Diagnostic Findings: refractory metabolic acidosis, Cr 0.52, AST 154, ALT 32, CK 5,318. ECG: VT.

Clinical Course: In the ED, she was in status epilepticus with acute respiratory failure, was intubated and given levetiracetam and fosphenytoin before transfer to a pediatric care center. Seizures continued, requiring escalation to midazolam, propofol and ketamine. On Day 2, propofol wean led to EEG-confirmed breakthrough seizures, so it was re-instituted. Worsening hypotension (SBP 60–80s) requiring epinephrine and hypercarbia (CO₂ 70s) requiring mechanical ventilation led to transfer to higher level of care pediatric tertiary care center on Day 3. Broad spectrum antibiotics were administered. On arrival to the pediatric tertiary care center, hemodynamic instability (BP 59/40) necessitated IVF and multiple vasopressors, but given concern for propofol-related infusion syndrome

(PRIS), propofol was discontinued and 1 mEq/kg sodium bicarbonate resulted in marked BP and HR improvements. Lidocaine was given for runs of VT. Other complications included metabolic acidosis (ABG 7.23/36/74/16/-13, lactate 7 mmol/L), rhabdomyolysis, hyperthermia, transaminitis (AST 2,453, ALT 503) and mild kidney injury (Cr 1.09). CRRT was initiated for worsening clinical status. On Day 4, she suffered a PEA cardiac arrest and was cannulated for ECMO. Lactate was >16 mmol/L following cannulation. Based on the prognosis, the family opted for institution of comfort measures, and she died on Day 5.

Autopsy Findings: The likely acute cause of demise in this case was rhabdomyolysis in the clinical setting of PRIS with resulting cardiac arrhythmias. The autopsy findings support the clinical impressions of rhabdomyolysis. Other findings include hepatomegaly, which is a recognized finding in PRIS, and RBC casts within renal tubules are compatible with acute kidney injury. However, PRIS is a clinical diagnosis with no specific pathologic correlates and cannot be further suggested on this examination. Additional findings included hypocellular bone marrow in the setting of prior valproic acid use, mild acute bronchopneumonia with chronic bronchiolitis, and no definite structural/developmental brain abnormalities in the setting of Lennox-Gastaut Syndrome.

Case 1884. Acute methamphetamine ingestion: undoubtedly responsible

Scenario/Substances: A 14y/o female had been running around wildly, collapsed and was found supine on the floor with arms flexed and jaw clenched. EMS was called and transported her to the ED.

Past Medical History: custodial aunt was incarcerated; patient had no permanent home.

Physical Exam (HCF): unresponsive with tremors and lip quivering, arms flexed, and jaw clenched. SBP 150s, HR 190–205, T 40°C, GCS 4.

Laboratory/Diagnostic Findings: Cr 1.6, lactate 6.1 mmol/L, WBC 8. UDS positive for amphetamine, methamphetamine, MDMA, and cannabinoids. Head CT: unremarkable.

Clinical Course: The patient was intubated. Pupils were reactive and she was breathing over the ventilator. Midazolam and ice packing were initiated. She was transferred to a tertiary care hospital where T 37.6°C. She was noted to be “oozing”. Norepinephrine was added for pressure support. BP 136/84, HR 112, RR 20. Day 1 fibrinogen 60, INR 4, lactate 6.5, and troponin-HS 1,200. Seizure activity developed. She was given additional benzodiazepines, levetiracetam, pentobarbital, and zonisamide. Repeat troponin-HS 3,600. By midday Day 2, the patient had continued seizure activity and developed heart, renal and liver failure in addition to DIC. CK 157,100, AST 4,197, ALT 1,390, BUN 41, Cr 5.23, PT 39.1, INR 3.8, pH 7.35. She was on epinephrine, norepinephrine and continuous calcium infusions. She developed brain herniation. On Day 4 after a second brain death study, she was pronounced dead.

Autopsy Findings: Autopsy toxicology results positive for THC, methamphetamine 1,300 ng/mL, and amphetamine 150 ng/mL. Cause of death: complications from acute methamphetamine toxicity.

Case 1907. Acute methamphetamine ingestion: undoubtedly responsible

Scenario/Substances: A 28y/o male inmate was found with a white crystalline substance. As a department of corrections (DOC) officer approached the patient, he attempted to ingest “as much of the substance as possible.” DOC identified the substance as methamphetamine. EMS was called and found the patient anxious, diaphoretic, and tachycardic (HR 170s). He was given midazolam and placed on bag-valve-mask ventilation. En route to the ED, he became apneic and was intubated. The DOC estimated he had ingested 45 g of methamphetamine. They recovered 28 g that were dropped when the DOC officer approached him.

Past Medical History: hepatitis C, substance use disorder. Home medications: none.

Physical Exam (HCF): unresponsive, pupils dilated and nonreactive, lungs clear to auscultation, abdomen non-distended, not moving extremities, skin warm, dry, pink; no purposeful movement. BP 129/57, HR 111, RR 2, O₂ sat 97%.

Laboratory/Diagnostic Findings: Initial (post-cardiac arrest): ABG-pH 7.22 / pCO₂ 51 / pO₂ 425 / HCO₃ 21 / BE < 5, Na 148 / K 3.9 / Cl 114 / CO₂ 18 / BUN 18 / Cr 1.0 / Glu 58 / AG 16 / lactate 9.52 mmol/L, AST 16, ALT 13, bilirubin 0.2, CK 137, troponin-I <0.01, WBC 12.6 / Hgb 12.4 / Hct 38.2 / Plt 238. Serum APAP and ethanol not detected. CxR: unremarkable. ECG: HR 112, QRS 108, QTc 496.

Clinical Course: In the ED, he had rapid, “thready, weak pulses” initially. On recheck of his vital signs within 2 min of arrival, he was pulseless. ROSC was achieved after several rounds of CPR and epinephrine. The monitor showed sinus tachycardia, and a bedside ultrasound showed good ventricular motility. His pupils were fixed and dilated. He remained intubated without sedation and was making no purposeful movement. Naloxone was given without response. Head CT was unremarkable. He was given IVF and IV sodium bicarbonate. He became hypotensive and norepinephrine was administered, and ~1 h after arrival he was flown to a tertiary care center. On arrival at the tertiary care center he continued to require vasopressor support (BP 103/55). Laboratory (~4h): ABG-pH 7.30 / pCO₂ 42.2 / pO₂ 121 / HCO₃ 21.0 / BE -5, Na 142 / K 3.3 / Cl 112 / CO₂ 21 / BUN 18 / Cr 1.0 / Glu 115 / AG 9, lactate 1.4 mmol/L, AST 33, ALT 18, bilirubin 0.3, INR 1.1, troponin-HS 297, WBC 20.0 / Hgb 12.2 / Hct 35.8 / Plt 203. Serum APAP and ethanol not detected. UDS was positive for amphetamines, benzodiazepines. CxR: mild atelectasis. On Day 2, his sedation was weaned off, but he remained unresponsive and had no gag or cough reflex. Sharp waves were observed on his EEG, and levetiracetam and midazolam were given. On Day 3, his EEG showed alpha coma, and his MRI revealed diffuse hypoxic ischemic injury. Pupils were 3 mm and sluggish. On Day 4, he was off sedation and areflexic. On Day 5, his condition deteriorated, he became more hypotensive (SBP in the 40s) and tachycardic (HR 200), developed severe electrolyte abnormalities, respiratory acidosis, and central diabetes insipidus. Multiple vasopressors were administered, and his electrolytes were replaced. In the afternoon of Day 5, his EEG showed no electrical activity in the brain. On Day 6, four-vessel cerebral angiogram showed no intracranial blood flow, and he was pronounced dead.

Autopsy Findings: Hospital blood (ED arrival): methamphetamine 27,000 ng/mL (RL 50 ng/mL), amphetamine 180 ng/mL (RL 5 ng/mL), norbuprenorphine 7.4 ng/mL (RL 0.5 ng/mL), midazolam 89 ng/mL (RL 5 ng/mL), and naloxone presumptive positive. Cause of death: methamphetamine toxicity. Manner of death: accident.

Case 1996. Acute-on-chronic methamphetamine: undoubtedly responsible

Scenario/Substances: A 43y/o male was found down outside of a service station, lethargic, and unwilling to answer questions. EMS was called due to suspicion of fentanyl overdose.

Past Medical History: anxiety, schizophrenia, bipolar disorder, poly-substance use disorder. He had been evaluated in the ED x2 previously in the last year for motor vehicle accidents where he was displaying “erratic behavior.” His UDS on these visits were both positive for methamphetamine/amphetamine. Home medications: clonazepam, oxybutynin, quetiapine, solifenacin.

Physical Exam (HCF): lethargic, following commands. Tachypneic. Abdomen soft, diffusely tender to palpation, non-distended. BP 77/48, HR 72, RR 35, O₂ sat 65% on RA, T 35.3°C.

Laboratory/Diagnostic Findings: VBG-pH 7.0 / pCO₂ 24.8 / pO₂ 40.4 / HCO₃ 6.1 / BE -23.8, Na 137 / K 7.4 / Cl 89 / CO₂ 5 / BUN 70 / Cr 9.83 / Glu 21 / AG 43, lactate 12.8 mmol/L, AST 12,550, ALT 5,107, bilirubin 1.3, INR 1.6, ammonia 155 mcmol/L, troponin-HS 843, CK 296,500, WBC 16.5 / Hgb 11.9 / Hct 35.3 / Plt 151. Serum APAP and ethanol not detected. Serum salicylate 2.2 mg/dL. CxR: unremarkable. ECG: HR 84, QRS 142, QTc 607, “nonspecific intraventricular conduction delay”.

Clinical Course: In the ED, he was lethargic with incomprehensible speech. He was placed on 15L O₂ via NRB, given IVF, and several amps of D50W. His mental status improved, and he was able to speak in full sentences. A central line was placed for emergent CRRT, he received NAC, calcium, insulin, D10W, several more amps of D50W, sodium bicarbonate, ondansetron, antibiotics, and was admitted to the ICU where dexmedetomidine, phytonadione, IV hydrocortisone, and lactulose were given. His CT chest/abdomen/pelvis showed bilateral renal cortical ischemia and necrosis, acute splenic infarction and ischemia, liver ischemia and infarction, and findings suggestive of bowel ischemia. He was tolerating CRRT, but during the evening of Day 1, he became acutely agitated and combative, and tried to pull the central line. He was given haloperidol and midazolam, then intubated and sedated with propofol and fentanyl. On Day 2, he remained intubated and sedated and continued to receive CRRT and NAC. He was anuric, became hypotensive, and received norepinephrine. Day 2 his vasopressor requirements increased. An ECHO showed an LVEF of 23%. On Day 3, he was requiring multiple vasopressors for BP support. An ECG showed ST elevation and troponin-HS >10,000. A bedside TTE revealed worsening LVEF. Heparin was given due to concern for possible STEMI. He remained persistently hyperkalemic (6.3) and acidotic (pH 7.20, HCO₃ 7.8, lactate 18 mmol/L, AST 14,200, ALT 7,450, bilirubin 4.5, and Plt 68. His pupils became fixed and dilated, and his body more diffusely rigid. Based on the prognosis, the family opted for institution of comfort measures, and he died on Day 3.

Autopsy Findings: Hospital blood Day 1: methamphetamine 780 ng/mL (reporting limit 5.0 ng/mL), amphetamine 62 ng/mL (reporting limit 5.9 ng/mL). Cause of death: complications of methamphetamine toxicity. Manner of death: accident.

Abbreviations & normal ranges

~:	Approximately
abd:	Abdomen
ABG-pH/pCO ₂ /pO ₂ /HCO ₃ /BE	
ABG:	Arterial blood gases
pH:	Hydrogen ion concentration [7.38–7.42 mmHg]
pCO ₂ :	Partial pressure of carbon dioxide [38–42 mmHg]
pO ₂ :	Partial pressure of oxygen [90–100 mmHg]
HCO ₃ :	Bicarbonate [22–28 mmol/L or mEq/L]
BE:	Base excess [±2 mEq/L or mmol/L]
ACLS:	Advanced cardiac life support, protocol for the provision of cardiac resuscitation
ADHD:	Attention deficit hyperactivity disorder
AF:	Atrial fibrillation
AG:	Anion gap Na – (Cl + HCO ₃) [12 ± 4 mEq/L or mmol/L]
AICD:	Automatic implanted cardioverter-defibrillator
AKI:	Acute kidney injury
ALP:	Alkaline phosphatase [13–100] U/L
ALT:	Alanine aminotransferase [7–41] U/L (SGPT)
AMA:	Against medical advice

Ammonia:	[15–47] mmol/L, [25–80] mcg/dL
amp:	Ampoule
amphetamines (hallucinogenic):	One or more of the products (6-APB, bath salts, plant food, Bliss, Ivory Wave, Purple Wave, Vanilla Sky, et al) or chemicals (3,4-methylenedioxypyrovalerone [MDPV], 6-(2-aminopropyl)benzofuran [6-APB], butylone, desoxypipradrol [2-DPMP], ethylone, flephedrone, naphyrone, mephedrone, methylenedioxypyrovalerone, methylone, methcathinone, et al.)
AMS:	Altered mental status
APAP:	Acetaminophen (acetyl-para-aminophenol), therapeutic [10–20] mcg/mL
APLS:	Advanced pediatric life support, protocol for the provision of cardiac resuscitation
aPTT:	Activated partial thromboplastin time [30–40] sec
ARDS:	Acute respiratory distress syndrome
AST:	Aspartate aminotransferase [12–38] U/L = (SGOT)
AV block:	Atrio-ventricular block
BAL:	British anti-Lewisite
BBB:	Bundle branch block
BE:	Base excess [±2 mEq/L or mmol/L]
Bicarbonate:	22–28 mEq/L or mmol/L
bili (direct):	Direct bilirubin [0.1, 0.4] mg/dL
bili (indirect):	Indirect bilirubin [0.2, 0.9] mg/dL
bilirubin:	Total [0.3–1.3] mg/dL
BiPAP:	Bilevel positive airway pressure, pressure support with 2 levels of continuous positive airway pressure
BLQ:	Below the limit of quantitation
BMI:	Body mass index [18.5–29.4] kg/m ²
BNPT:	Prohormone with a 76 amino acid N-terminal inactive protein that is cleaved from the molecule to release brain natriuretic peptide. CHF is likely if BNPT >125 pg/mL (<75 y/o), >450 pg/mL (>75 y/o)
body packing:	Insertion of drugs into body orifices to evade law enforcement
body stuffing:	The ingestion of drugs to evade law enforcement
BP:	Blood pressure, systolic/diastolic, (Torr)
BPH:	Benign prostatic hypertrophy
BUN:	See urea nitrogen
BVM:	Bag-valve mask
C:	Degrees centigrade
Ca (ionized):	Ionized calcium, [4.5–5.6] mg/dL
Ca:	Calcium [8.7–10.2] mg/dL
CABG:	Coronary artery bypass graft

CAD:	Coronary artery disease	EGD:	Esophagogastroduodenoscopy
CHF:	Congestive heart failure	eGFR:	Estimated glomerular filtration rate
CIWA:	Clinical Institute Withdrawal Assessment for Alcohol	ELISA:	Enzyme-linked immunosorbent assay
CK:	Creatine kinase (CPK), total: [39–238] U/L females, [51–294] U/L males	EMS:	Emergency medical services, paramedics, the first responders
CKMB:	MB fraction of CK [0.0–5.5 mcg/L: 0.0–5.5 ng/mL] Fraction of total CK activity [0–0.04 = 0–4.0%]	EOMI:	Extraocular movements intact.
Cl:	Chloride [102–109] mEq/L	ER:	Extended-release medication
CMV:	Cytomegalovirus	ERCP:	Endoscopic retrograde cholangiopancreatography
CNS:	Central nervous system	ETT:	Endotracheal tube
COHb:	Carboxyhemoglobin (<3%)	Fe:	Serum iron [60–170] mcg/dL
COPD:	Chronic obstructive pulmonary disease	FFP:	Fresh frozen plasma
CPAP:	Continuous positive airway pressure	FIO ₂ :	Fraction of inspired oxygen (%)
CPR:	Cardiopulmonary resuscitation	g:	Grams
Cr:	Creatinine [0.5–0.9] mg/dL females, [0.6–1.2] males	g/dL:	Grams per deciliter
CRRT:	Continuous renal replacement therapy	GCS:	Glasgow Coma Score, ranges from 3 to 15
CSF:	Cerebrospinal fluid	GERD:	Gastroesophageal reflux disease
CT:	Computed tomography (CAT scan)	GI:	Gastrointestinal
CVA:	Cerebrovascular accident	Glu:	Glucose, fasting [75–110] mg/dL
CVP:	Central venous pressure	h:	Hours
CVVH:	Continuous venovenous hemodiafiltration	HBO:	Hyperbaric oxygen treatment/therapy
CxR:	Chest radiograph, chest x-ray	HCF:	Health care facility
c/w:	Consistent with	HCG:	Human chorionic gonadotropin test for pregnancy
D-dimer:	Protein released from blood clot break down, [0–0.50] mg/L	HCO ₃ :	Bicarbonate [22–28 mEq/L or mmol/L]
D10W:	10% Dextrose in water	HCP:	Health care provider
D50W:	50% Dextrose in water	Hct:	Hematocrit [35.4–44.4]% females, [38.8–46.4]% males
D5NS:	5% Dextrose in normal saline	HD:	Hemodialysis
D5W:	5% Dextrose in water	Hgb:	Hemoglobin [12.0–15.8] g/dL females, [13.3–16.2] g/dL males
Day	when capitalized, Day = Hospital day, i.e., days since admission to the initial hospital admission for this exposure, first hospital day = Day 1.	HIE:	Hyperinsulinemia-euglycemia therapy
DIC:	Disseminated intravascular coagulation	HIV:	Human immunodeficiency virus
DM:	Diabetes mellitus	Hour:	When capitalized, Hour = hours since admission or since exposure as specified in the narrative
DNI:	Do not intubate	HR:	HR, beats per min
DNR:	Do not resuscitate	HTN:	Hypertension
Dx:	Diagnosis	IABP:	Intra-aortic balloon pump
ECG:	Electrocardiogram (EKG), leads = I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6	ICP:	Intracranial pressure
ECHO:	Echocardiogram	ICU:	Intensive care unit
ECMO:	Extracorporeal membrane oxygenation.	IDDM:	Insulin dependent diabetes mellitus
VV:	Venous to venous	IgE:	Immunoglobulin E
ED:	Emergency department, in these abstracts refers to the initial health care facility	ILE:	Intravenous lipid emulsion (20%)
EDDP:	Principal methadone metabolite, 2-ethylidene-1,5-dimethyl-3,3-diphenylpyrrolidine	IM:	Intramuscular
EEG:	Electroencephalogram	INR:	International normalized ratio (PT to control) [0.8–1.2]
		IO:	Intraosseous
		IU/L:	International units per Liter
		IV:	Intravenous
		IVF:	Intravenous fluid(s)
		K:	Potassium [3.5–5] mEq/L
		kg:	Kilogram
		L:	Liter

lactate:	Lactic acid [0.5–1.6] mmol/L arterial/[0.5–2.2] mmol/L venous[4.5–14.4] mg/dL arterial/[4.5–19.8] mg/dL venous	nl:	Normal
LBBB:	Left bundle branch block on ECG	NOS:	Not otherwise specified
LFT:	Liver function tests	not detected:	Analyte below the level of quantitation/negative
LV:	Left ventricle	NPO:	Nil per os/nothing by mouth
LVEF:	Left ventricular ejection fraction	NRB:	Non rebreathing mask for O ₂ delivery
m/o:	Months old	NS:	Normal saline
MAP:	Mean arterial pressure	NSR:	Normal sinus rhythm
MARS:	Molecular adsorbent recirculating system	NSTEMI:	Non-ST segment elevation myocardial infarction
mcg/dL:	Micrograms per deciliter	O ₂ sat:	Oxygen percent saturation [94–100]% at sea level
mcg/L:	Micrograms per liter	OG:	Serum osmol gap=measured serum osmolality – calculated serum osmolality [0±10 mOsm/kg]
mcg/min:	Micrograms per minute	OR:	Operating room
mcg/mL:	Micrograms per milliliter	Osm:	Osmole
mcmol/L:	Micromoles per liter	OTC:	Over the counter
MDA:	3,4-Methylenedioxyamphetamine	ODU:	Opioid use disorder
MDMA:	Methylenedioxymethamphetamine (ecstasy/molly)	PALS:	Pediatric advanced life support
ME:	Medical examiner	PC:	Poison Center (or Poison Control Center)
MetHgb:	Methemoglobin (RR <1%)	PCC:	Prothrombin complex concentrate
Mg:	Magnesium [1.5–2.3] mg/dL	PCP:	Primary care provider
mg:	Milligrams	PEA:	Pulseless electrical activity
mg/dL:	Milligrams per deciliter	PEEP:	Positive end expiratory pressure
mg/kg:	Milligrams per kilogram	PERRLA:	Pupils/equal/round and reactive to light and accommodation
mg/L:	Milligrams per liter	Phos:	Phosphate (phosphorous) [2.5–4.5] mg/dL
min:	Minutes	PICU:	Pediatric intensive care unit
mL:	Milliliter	Plt:	Platelet count [150–400] × 10 ⁹ /L
mmol:	Millimoles	PO:	Per os (“by mouth” in Latin)
mEq/L:	Milliequivalents per liter	POC:	Point of care
mmol/L:	Millimoles per liter	PPE:	Personal protective equipment
mosm/kg:	Milliosmoles per kilogram	ppm:	Parts per million
mosm/L:	Milliosmoles per liter	PR:	P-R interval [120–200] msec on the ECG
MRI:	Magnetic resonance imaging	PRN:	as needed
MRSA:	Methicillin-resistant <i>Staphylococcus aureus</i>	PT:	prothrombin time
ms:	Milliseconds	PTA:	Prior to admission
MSDS:	Material safety data sheet	PTSD:	post-traumatic stress disorder
myoglobin:	Measure of muscle injury [5–70] mcg/mL	PTT:	Partial thromboplastin time [26.3–39.4] sec
Na:	Sodium [136–146] mEq/L	PVC:	Premature ventricular contraction
NAC:	N-Acetylcysteine	QRS:	ECG QRS complex duration [60–100] msec
Narrative Headers:		QT:	Q to T interval on the ECG wave-form/ varies with HR
• Scenario/Substances: concise narrative of EMS & pre-HCF events		QTc:	QT interval corrected for HR/usually QTcB=QT/RR ^{1/2} (Bazett correction) 1–15 y/o [<440] msec/adult male [<430] msec/adult female [<450] msec
• Past Medical History: available relevant past medical history		RA:	On room air
• Physical Exam: initial physical exam if available		RBBB:	Right bundle branch block on ECG
• Laboratory/Diagnostic Findings: initial results		RBC:	Red blood cell(s)
• Clinical Course: concise narrative of HCF and beyond with outcomes		RL:	Reporting limit
• Autopsy Findings: medical examiner and/or autopsy results			
NC:	Nasal cannula		
NG:	Nasogastric		
ng/mL:	Nanograms per milliliter		

RLL:	Right lower lobe
ROSC:	Return of spontaneous circulation
RPC:	Regional Poison Center
RR:	Respiratory rate/breaths per minute
s/p:	Status post
salicylate:	Aspirin/acetylsalicylic acid/therapeutic [15–30] mg/dL
SBP:	Systolic blood pressure
sec:	Seconds
SL:	Sublingual
SLED:	Sustained low-efficiency dialysis
STEMI:	ST segment elevation myocardial infarction
SVR:	Systemic vascular resistance [700–1500] dynes-sec/cm ⁵
SVT:	Supraventricular tachycardia
T (oral):	Temperature (oral) [36.4/37.2]°C
T (rectal):	Temperature (rectal) [36.4/37.2]°C
T (tympanic):	Temperature (tympanic) [36.4/37.2]°C
TBSA:	Total body surface area
THC:	Tetrahydrocannabinol
THC Homolog:	One or more of the products (Blaze/Dawn/herbal incense/K2/Red X/spice/et al) or chemicals (cannabicyclohexanol/CP-47/497/JWH-018/JWH-073/JWH-200/et al.)
TPN:	Total parenteral nutrition
Tprot:	Total protein
Troponin-HS:	High-sensitivity cardiac troponin T (hs-cTnT) [<14 females, <22 males] ng/mL
Troponin-I:	Troponin I/normal range [0–0.08] ng/mL/Cut-off for MI >0.04 ng/mL
TTE:	Transthoracic echocardiogram
U:	Units
U/dL:	Units per deciliter
U/L:	Units per liter
U/mL:	Units per milliliter
UA:	Urinalysis

UDS:	Urine drug screen
Urea nitrogen (BUN):	[6–17] mg/dL
VBG:	Venous blood gases
pH:	Hydrogen ion concentration [7.31–7.41 mmHg]
pCO ₂ :	Partial pressure of carbon dioxide [41–51 mmHg]
pO ₂ :	Partial pressure of oxygen [30–40 mmHg]
HCO ₃ ⁻ :	Bicarbonate [22–28 mEq/L or mmol/L]
BE:	Base excess [±2 mEq/L or mmol/L]
VF:	Ventricular fibrillation
VSD:	Ventricular septal defect
VT:	Ventricular tachycardia
WBC:	White blood cell (leukocyte) count [3.54–9.06] 103/mm ³
WBI:	Whole bowel irrigation
WNL:	Within normal limits
y/o:	Year old

Disclaimer – all laboratories are different and provide their own normal ranges. Units and normal ranges are provided here for general guidance only. These values were taken from Harrison's^a, Goldfrank's^b or Dart^c.

^aTable S10-1 Selected Examples of Laboratory Critical Values, *Harrison's Principles of Internal Medicine* 21e. McGraw-Hill Professional, 2022. Available from: <https://accesspharmacy.mhmedical.com/> Sep 7, 2025.

^b*Goldfrank's Toxicologic Emergencies*, 11e, McGraw-Hill Education, 2019. Available from <https://accesspharmacy.mhmedical.com/> Sep 7, 2025.

^c*Dart RC, ed. Medical Toxicology, 3rd Edition*, Lippincott, Williams & Wilkins, 2004.

Typical laboratory panels

ABG-pH/pCO₂/pO₂/HCO₃⁻/BE

Basic metabolic panel: Na/K/Cl/CO₂/BUN/Cr/Glu/AG

Complete blood count: WBC/Hgb/Hct/platelets

Appendix D

PoisonHelp.org

Methods: Raw data from PoisonHelp.org was extracted into Microsoft Excel 365 in sequential contact format. Data regarding annual contacts, age distribution of person involved in online contact, and substances involved in contact, were displayed by year. On review of each year, it was evident that some sequential contacts appeared to be repeat entries due to similarities of nearly every data field over short time periods (i.e., minutes) between contacts. The reason for these repeats is unknown but could be due to the pressing of the submission button multiple times. To account for these apparent repeat values, criteria were put in place to identify and exclude this data from final analysis. The criteria included adjacent sequential contacts or those separated by only a single contact, line items with identical age category, state, the first three characters of the verbatim substance input, browser version and originating operating system.

Results: Table D-1 shows ages of contacts reported to PoisonHelp.org Virtual Poison Center during the past 3 years of operation.

Table D-1. Age distribution of PoisonHelp.org contacts.

Year	2022	2023	2024
Age ≤5	2,614	3,340	3,966
6–12	884	977	1,158
Teen 13–19	2,058	2,134	2,253
Adult 20–69	8,687	10,263	10,761
Adult 70+	1,093	1,214	1,315
Total counts	15,336	17,928	19,453

Table D-2. Substance frequency of the 15 most common substances (when known).

Substance generic code	Contact counts	Percentage of contacts (%)
Systemic antibiotic preparations (oral, intravenous, intramuscular)	718	3.84
Benzodiazepines	624	3.34
Sertraline	551	2.95
Other types of miscellaneous prescription or over the counter drug	518	2.77
Amphetamines and related compounds	420	2.25
Ibuprofen	406	2.17
Hypoglycemics: glucagon-like peptide-1 (GLP-1) Receptor Agonists	346	1.85
Cetirizine	321	1.72
Melatonin alone	320	1.71
Acetaminophen alone, adult	270	1.44
Hand sanitizers: ethanol based	236	1.26
Antihyperlipidemic alone	232	1.24
Bupropion	223	1.19
Miscellaneous unknown drugs	216	1.16
Antacids: proton pump inhibitors	214	1.15

Table D-2 lists the 15 most frequent substances involved in contacts when the substance was known ($n=18,687$) reported to PoisonHelp.org in 2024. The category “Exact Substance Match not Found” was the most frequent substance ($n=766$).

Total number of entries and excluded entries for the past three years of contacts are expressed in Table D-3.

An analysis was performed to determine the percentage of cases where the initial guidance was sufficient for managing the case without recommending immediately calling a PC using the 1-800-222-1222 number. There were 129 unique recommendations in the entire 2024 data set, and each could be categorized into either requesting that they immediately call, or instructions when to call a PC based on the exposure and symptoms (categorization performed by authors MCB, RF, JBM). For all cases included ($n=19,453$), 89.6% of cases were provided recommendations that instructed them to call a PC. Analysis without the information contacts ($n=7,969$) demonstrated a similar percentage of 89.3% being instructed to call a PC.

The total number of contacts for the initial eight years of PoisonHelp.org are presented in Figure D-1.

Table D-3. PoisonHelp.org contacts per year.

Year	2022	2023	2024	3 Year totals
Original counts	18,649	21,675	23,788	64,112
Excluded repeat entries	3,313	3,747	4,335	11,395
Final counts	15,336	17,928	19,453	52,717
Information contacts*	6,283	8,646	11,484	26,413
Final count minus information counts*	9,053	9,282	7,969	26,304

*Information contacts are those in which the user indicated there was no exposure to the queried substance.

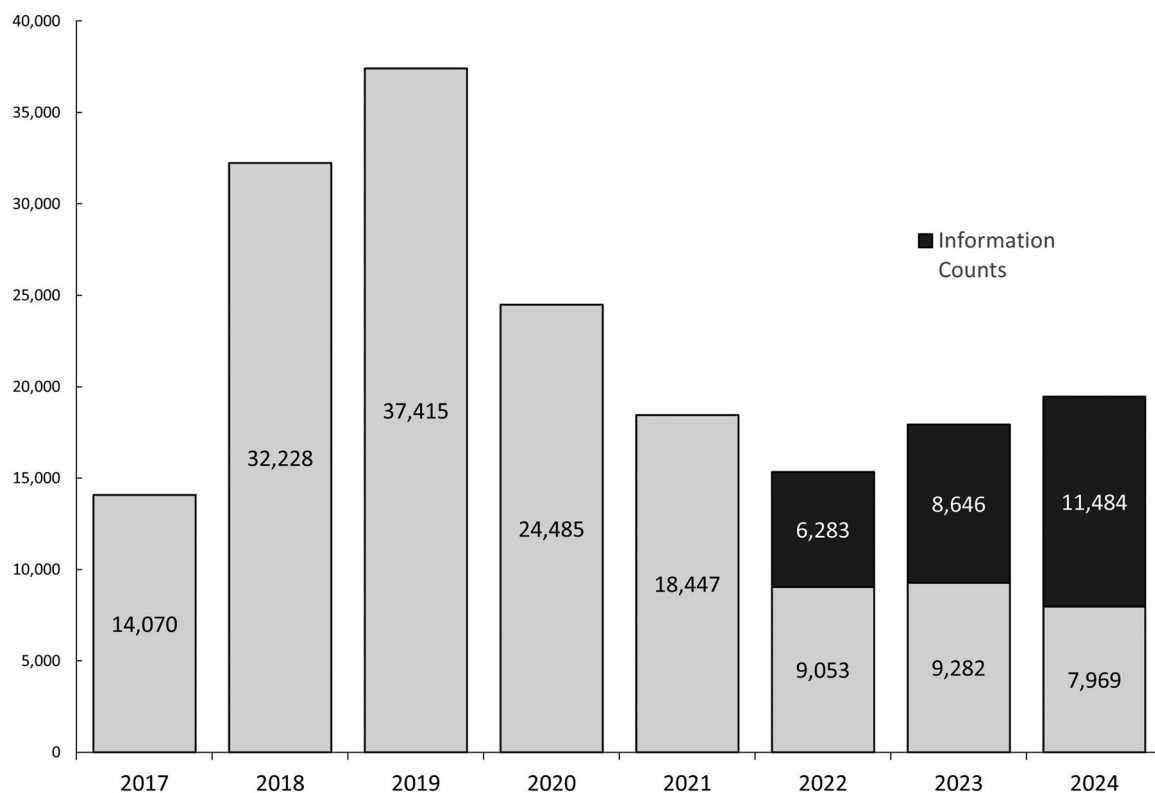


Figure D-1. Contacts per year for PoisonHelp.Org.

Appendix E

Acknowledgments

The compilation of the data presented in this report was supported in part through the US Centers for Disease Control and Prevention America's Poison Centers Cooperative Agreement NUE1EH001395. The authors express their sincere gratitude to the staff at the America's Poison Centers Central Office for their support during the preparation of the manuscript: Richard Fogelson, JD, CEO, and the entire staff.

Poison Centers (PCs)

We gratefully acknowledge the extensive contributions of each participating PC and the assistance of the many health care providers who report comprehensive data to the PCs for inclusion in this database. We especially acknowledge the dedicated efforts of the Specialists in Poison Information (SPIs) who meticulously managed and coded 2,418,426 encounters at US PCs in 2024.

As in previous years, the initial review of reported fatalities and development of the abstracts and case data for NPDS was the responsibility of the staff at the 55 participating PCs. Many individuals at each center participated in the fatality case preparation. These toxicology professionals and their centers are:

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The Lead and Peer review of the 2024 fatalities was carried out by the 53 individuals listed here including 4 who reviewed the pediatric cases [Peds]. The authors and the America's Poison Centers wish to express our appreciation for their volunteerism, dedication, hard work and good will in completing this task in a limited time frame.

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NPDS surveillance anomalies are analyzed daily by a team of 9 medical and clinical toxicologists working across the country in a distributed system. These dedicated professionals interface with the Emerging Environmental Hazards and Health Effects Branch, Division of Environmental Health Science and Practice, Centers for Disease Control and Prevention (EEHHEB/DEHSP/CDC) and the PCs on a regular basis to identify anomalies of public health significance and improve NPDS surveillance systems:

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Regional Poison Center fatality awards

Each year, America's Poison Centers and the Fatality Review team recognize several regional PCs for their extra effort in their preparation of fatality reports and prompt responses to reviewer queries. The awards are presented each year at the North American Congress of Clinical Toxicology Annual meeting.

First Center to Complete all Cases (11/28/2024)

Blue Ridge Poison Center (Charlottesville)

Largest Number with Autopsy Reports (27 of 52 cases; 52%)

Illinois Poison Center (Chicago)

Highest Percentage with Autopsy Reports (81% of 26 cases)

Connecticut Poison Control Center (Farmington)

Largest Number of Indirect cases (n= 831; 82% of all Indirect cases)

Banner Poison & Drug Information Center (Phoenix)

Highest Overall Quality of Reports (6.09 out of possible 12 for 23 cases)

Arizona Poison & Drug Information Center (Tucson)

Greatest improvement in Overall Quality of Reports (1.19 increase from last year)

Connecticut Poison Control Center (Farmington)

Most Narratives Published in the 2024 Annual report (4 of the 61 published narratives)

Rocky Mountain Poison & Drug Safety (Denver)

Most Responsive Regional Poison Center (48 hours 90% responses to 10 initial queries)

Connecticut Poison Control Center (Farmington)

Honorable Mention (60 hours 91% responses to 23 initial queries)

Upstate New York Poison Center (Syracuse)

Teaching Award for outstanding contributions to case quality through Reviewer teaching

Curtis Elko, PharmD, CSPI, Washington Poison Center, Seattle, WA

Honorable Mention (with 96% response rate)

Steven Marcus, MD, Professor Emeritus, NJ Medical School, Rutgers University, Newark, NJ

Christine Murphy, MD, Atrium Health's Carolinas Medical Center, Charlotte, NC

Special Award for consistently outstanding Narrative Quality

Missouri Poison Center (St. Louis) and Rebecca Tominack, MD

Appendix F

Data definitions

Reason for exposure

NPDS classifies all encounters as either an Exposure case (concern about an exposure to a substance) or an Information request (no exposed human or animal). A contact may provide information about one or more exposed person or animal, however each exposed person should have a separate case entry.

SPIs coded the reasons for exposure reported by callers to PCs according to the following definitions:

Unintentional general: All unintentional exposures not otherwise defined below. This includes most unintentional exposures in children ≤ 5 years of age.

Environmental: Any passive, non-occupational exposure that results from contamination of air, water, or soil. Environmental exposures are usually caused by manmade contaminants.

Occupational: An exposure that occurs as a direct result of the person being on the job or in the workplace.

Therapeutic error: An unintentional deviation from a proper therapeutic regimen that results in the wrong dose, incorrect route of administration, administration to the wrong person, or administration of the wrong substance. Only exposures to medications or products used as medications are included. Drug interactions

resulting from unintentional administration of drugs or foods which are known to interact are also included.

Unintentional misuse: Unintentional, improper or incorrect use of a nonpharmaceutical substance. Unintentional misuse differs from intentional misuse in that the exposure was unplanned or not foreseen by the patient.

Bite/sting: All animal bites and stings, with or without envenomation, are included.

Food poisoning: Suspected or confirmed food poisoning; ingestion of food contaminated with microorganisms is included.

Unintentional unknown: An exposure determined to be unintentional, but the exact reason is unknown.

Suspected suicidal: An exposure resulting from the inappropriate use of a substance for reasons that are suspected to be self-destructive or manipulative.

Intentional misuse: An exposure resulting from the intentional improper or incorrect use of a substance for reasons other than the pursuit of a psychotropic effect.

Intentional abuse: An exposure resulting from the intentional improper or incorrect use where the patient was likely attempting to gain a high, euphoric effect or some other psychotropic effect, including recreational use of a substance for any effect.

Intentional unknown: An exposure that is determined to be intentional, but the specific motive is unknown.

Contaminant/tampering: The patient is an unintentional victim of a substance that has been adulterated (either maliciously or unintentionally) by the introduction of an undesirable substance.

Malicious: Patients who are victims of another person's intent to harm them.

Withdrawal: Inquiry about or experiencing of symptoms from a decline in blood concentration of a pharmaceutical or other substance after discontinuing therapeutic use or abuse of that substance, or withdrawal triggered by administration of another drug.

Adverse Reaction Drug: Unwanted effects due to an allergic, hypersensitivity, or idiosyncratic response to the active ingredient(s), inactive ingredient(s) or excipient of a drug, chemical, or other drug substance when the exposure involves the normal, prescribed, labeled or recommended use of the substance.

Adverse Reaction Food: Unwanted effects due to an allergic, hypersensitivity, or idiosyncratic response to a food substance.

Adverse Reaction Other: Unwanted effects due to an allergic, hypersensitivity, or idiosyncratic response to a substance other than drug or food.

Unknown Reason: Reason for the exposure cannot be determined or no other category is appropriate.

Medical outcome

No effect: The patient did not develop any signs or symptoms as a result of the exposure.

Minor effect: The patient developed some signs or symptoms as a result of the exposure, but they were minimally bothersome and generally resolved rapidly with no residual disability or disfigurement. A minor effect is often limited to the skin or mucus membranes (e.g., self-limited gastrointestinal symptoms, drowsiness, skin irritation, first degree dermal burn, sinus tachycardia without hypotension, and transient cough).

Moderate effect: The patient exhibited signs or symptoms as a result of the exposure (including complications of the exposure) that were more pronounced, more prolonged, or more systemic in nature than minor symptoms. Usually, some form of treatment is indicated. Symptoms were not life-threatening, and the patient had no residual disability or disfigurement (e.g., corneal abrasion, acid-base disturbance, high fever, disorientation, hypotension that is rapidly responsive to treatment, and isolated brief seizures that respond readily to treatment).

Major effect: The patient exhibited signs or symptoms as a result of the exposure (including complications of the exposure) that were life-threatening or resulted in significant residual disability or disfigurement (e.g., repeated seizures or status epilepticus, respiratory compromise requiring intubation, ventricular tachycardia with hypotension, cardiac or respiratory arrest, esophageal stricture, and disseminated intravascular coagulation).

Death: The patient died as a result of the exposure or as a direct complication of the exposure.

Not followed, judged as nontoxic exposure: No follow-up calls were made to determine the outcome of the exposure because the substance implicated was nontoxic, the amount implicated was insignificant, or the route of exposure was unlikely to result in a clinical effect.

Not followed, minimal clinical effects possible: No follow-up calls were made to determine the patient's outcome because the exposure was likely to result in only minimal toxicity of a trivial nature. (The patient was expected to experience no more than a minor effect.).

Unable to follow, judged as a potentially toxic exposure: The patient was lost to follow-up, refused follow-up, or was not followed, but the exposure was significant and may have resulted in a moderate, major, or fatal outcome.

Unrelated effect: The exposure was probably not responsible for the effect.

Confirmed nonexposure: This outcome option was coded to designate cases where there was reliable and objective evidence that an exposure initially believed to have occurred actually never occurred (e.g., all missing pills are later located). All cases coded as confirmed nonexposure are excluded from this report.

Death, indirect report: Death, indirect report are deaths that the Poison Center acquired from medical examiner or media, but did not manage nor answer any questions about the death.

Relative contribution to fatality (RCF)

The Case Review Team (CRT) includes the Author and Reviewer from the RPC, the America's Poison Centers Lead Reviewer, Peer Reviewer and Manager.

The definitions used for the Relative Contribution to Fatality (RCF) classification were as follows:

1. **Undoubtedly responsible** - In the opinion of the CRT, the Clinical Case Evidence establishes beyond a reasonable doubt that the substances actually caused the death.
2. **Probably responsible** - In the opinion of the CRT, the Clinical Case Evidence suggests that the substances caused the death, but some reasonable doubt remained.
3. **Contributory** - In the opinion of the CRT, the Clinical Case Evidence establishes that the substances contributed to the death, but did not solely cause the death. That is, the substances alone would not have caused the death, but combined with other factors, were partially responsible for the death.
4. **Probably not responsible** - In the opinion of the CRT, the Clinical Case Evidence establishes to a reasonable probability, but not conclusively, that the substances associated with the death did not cause the death.
5. **Clearly not responsible** - In the opinion of the CRT, the Clinical Case Evidence establishes beyond a reasonable doubt that the substances did not cause this death.
6. **Unknown** - In the opinion of the CRT, the Clinical Case Evidence is insufficient to impute or refute a causative relationship for the substances in this death.